

Actividad - Proyecto práctico

Actividad en grupos de 4 alumnos. Nombres en orden alfabético (de apellidos):

- **Alumno 1:** Vara Rodríguez, Jorge
- **Alumno 2:** Larios Pérez, José María
- **Alumno 3:** Fernández Gordo, Javier
- **Alumno 4:** Carvajal Benítez, Pablo

Entorno: SpaceInvaders-v0 · **Algoritmo:** DQN (Stable-Baselines3) · **Requisito mínimo:** media de recompensa en modo test por encima del umbral indicado en el listado del grupo.

Proyecto 10 - DQN con Stable-Baselines3

Entrenamiento de un agente DQN para SpaceInvaders-v0 usando Stable-Baselines3.

El notebook mantiene la idea de la práctica anterior:

- imágenes de entrada de 84x84,
- pila de 4 frames,
- red convolucional tipo DQN Atari,
- recompensa clipped,
- conteo de episodios consecutivos con recompensa clipped superior a 20,
- guardado de métricas completas en CSV/JSON, TensorBoard y guardado manual del modelo.

```
In [ ]: # =====  
# 1. Instalación de Librerías  
# =====  
  
# Instalación completa para usar SpaceInvaders-v0 con Gym clásico  
  
%pip install gym==0.17.3  
%pip install git+https://github.com/Kojoley/atari-py.git  
%pip install stable-baselines3==0.10.0
```

Collecting gym==0.17.3

Downloading gym-0.17.3.tar.gz (1.6 MB)

```
0.0/1.6 MB ? eta -:--:--
1.6/1.6 MB 111.2 MB/s eta 0:00:01
1.6/1.6 MB 39.0 MB/s eta 0:00:00
```

Preparing metadata (setup.py) ... done

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from gym==0.17.3) (1.16.3)

Requirement already satisfied: numpy>=1.10.4 in /usr/local/lib/python3.12/dist-packages (from gym==0.17.3) (2.0.2)

Collecting pygamelet<=1.5.0,>=1.4.0 (from gym==0.17.3)

Downloading pygamelet-1.5.0-py2.py3-none-any.whl.metadata (7.6 kB)

Collecting cloudpickle<1.7.0,>=1.2.0 (from gym==0.17.3)

Downloading cloudpickle-1.6.0-py3-none-any.whl.metadata (4.3 kB)

Requirement already satisfied: future in /usr/local/lib/python3.12/dist-packages (from pygamelet<=1.5.0,>=1.4.0->gym==0.17.3) (1.0.0)

Downloading cloudpickle-1.6.0-py3-none-any.whl (23 kB)

Downloading pygamelet-1.5.0-py2.py3-none-any.whl (1.0 MB)

```
1.0/1.0 MB 78.0 MB/s eta 0:00:00
```

Building wheels for collected packages: gym

Building wheel for gym (setup.py) ... done

Created wheel for gym: filename=gym-0.17.3-py3-none-any.whl size=1654617 sha256=0b5e6520baaf7a136e63263f1b39b2976e202d04f4f03d4a54b66d3a3d7ecc48

Stored in directory: /root/.cache/pip/wheels/99/7e/03/69cdeb432380e05bc9a56ff68451d8063e41aa06db508f4be5

Successfully built gym

Installing collected packages: pygamelet, cloudpickle, gym

Attempting uninstall: cloudpickle

Found existing installation: cloudpickle 3.1.2

Uninstalling cloudpickle-3.1.2:

Successfully uninstalled cloudpickle-3.1.2

Attempting uninstall: gym

Found existing installation: gym 0.25.2

Uninstalling gym-0.25.2:

Successfully uninstalled gym-0.25.2

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

bigframes 2.42.0 requires cloudpickle>=2.0.0, but you have cloudpickle 1.6.0 which is incompatible.

distributed 2026.1.1 requires cloudpickle>=3.0.0, but you have cloudpickle 1.6.0 which is incompatible.

dask 2026.1.1 requires cloudpickle>=3.0.0, but you have cloudpickle 1.6.0 which is incompatible.

Successfully installed cloudpickle-1.6.0 gym-0.17.3 pygamelet-1.5.0

Collecting git+https://github.com/Kojoley/atari-py.git

Cloning https://github.com/Kojoley/atari-py.git to /tmp/pip-req-build-imub_2zg

Running command git clone --filter=blob:none --quiet https://github.com/Kojoley/atari-py.git /tmp/pip-req-build-imub_2zg

Resolved https://github.com/Kojoley/atari-py.git to commit 86a1e05c0a95e9e6233c3a413521fdb34ca8a089

Preparing metadata (setup.py) ... done

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from atari-py==1.2.2) (2.0.2)

Building wheels for collected packages: atari-py

Building wheel for atari-py (setup.py) ... done

Created wheel for atari-py: filename=atari_py-1.2.2-cp312-cp312-linux_x86_64.whl size=4739135 sha256=48f93941e5a0515e594f3c782b419f2cb4ff36dda52493963af0baed20d5dc61

Stored in directory: /tmp/pip-ephem-wheel-cache-4sqy5dcp/wheels/cb/1f/56/35b393

898f74ae3d978063738e8eeb4b3d7357d0d7eeec6b9f

Successfully built atari-py

Installing collected packages: atari-py

Successfully installed atari-py-1.2.2

Collecting stable-baselines3==0.10.0

Downloading stable_baselines3-0.10.0-py3-none-any.whl.metadata (3.5 kB)

Requirement already satisfied: gym>=0.17 in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (0.17.3)

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (2.0.2)

Requirement already satisfied: torch>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (2.11.0+cu128)

Requirement already satisfied: cloudpickle in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (1.6.0)

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (2.2.2)

Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (3.10.0)

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from gym>=0.17->stable-baselines3==0.10.0) (1.16.3)

Requirement already satisfied: pygame<=1.5.0,>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from gym>=0.17->stable-baselines3==0.10.0) (1.5.0)

Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (3.29.4)

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (4.15.0)

Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (75.2.0)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (3.6.1)

Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (3.1.6)

Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (2025.3.0)

Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.1)

Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (12.9.7)

Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (9.19.0.56)

Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (0.7.1)

Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (2.28.9)

Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (3.4.5)

Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (3.6.0)

Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.4.1)

Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.90)

Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (11.3.3.83)

Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (1.13.1.3)

Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.90)

Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (10.3.9.90)

Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (11.7.3.90)

Requirement already satisfied: nvidia-cusparsenvjitlink-cu12==12.5.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.5.8.93)

Requirement already satisfied: nvidia-cusolver-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.93)

Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.93)

Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.90)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (4.63.0)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (1.5.0)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (26.2)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (3.3.2)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->stable-baselines3==0.10.0) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->stable-baselines3==0.10.0) (2026.2)

Requirement already satisfied: cuda-pathfinder~1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings<13,>=12.9.4->torch>=1.4.0->stable-baselines3==0.10.0) (1.5.5)

Requirement already satisfied: future in /usr/local/lib/python3.12/dist-packages

(from pygame<=1.5.0,>=1.4.0->gym>=0.17->stable-baselines3==0.10.0) (1.0.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages
(from python-dateutil>=2.7->matplotlib->stable-baselines3==0.10.0) (1.17.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages
(from sympy>=1.13.3->torch>=1.4.0->stable-baselines3==0.10.0) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages
(from jinja2->torch>=1.4.0->stable-baselines3==0.10.0) (3.0.3)
Downloading stable_baselines3-0.10.0-py3-none-any.whl (145 kB)

146.0/146.0 kB 14.5 MB/s eta 0:00:00

Installing collected packages: stable-baselines3

Successfully installed stable-baselines3-0.10.0

```
In [ ]: import gym
        from stable_baselines3 import DQN

ENV_ID = "SpaceInvaders-v0"
env = gym.make(ENV_ID)

print("Gym OK")
print("SB3 DQN OK")
```

Gym OK

SB3 DQN OK

```
In [ ]: # =====
        # 2. Imports
        # =====

        from pathlib import Path
        import json
        import time
        from collections import deque

        import numpy as np
        import pandas as pd
        import matplotlib.pyplot as plt

        import gym
        import torch as th
        import torch.nn as nn

        from google.colab import drive

        from stable_baselines3 import DQN
        from stable_baselines3.common.env_util import make_atari_env
        from stable_baselines3.common.vec_env import VecFrameStack
        from stable_baselines3.common.callbacks import BaseCallback
        from stable_baselines3.common.torch_layers import BaseFeaturesExtractor
```

```
In [ ]: # =====
        # 3. Google Drive y rutas
        # =====

        drive.mount('/content/gdrive')

        drive_root = '/content/gdrive/My Drive/08_MIAR/sb3/conlogs'
        PROJECT_NAME = 'sb3_1_spaceinvaders_v0'

        OUTPUT_DIR = Path(drive_root) / 'datos' / PROJECT_NAME
        OUTPUT_DIR.mkdir(parents=True, exist_ok=True)
```

```

# Modelos
MODEL_LATEST = OUTPUT_DIR / 'dqn_spaceinvaders_v0_sb3_latest.zip'
POLICY_LATEST = OUTPUT_DIR / 'dqn_spaceinvaders_v0_policy_latest.pt'

# Logs completos propios
LOG_FILENAME = OUTPUT_DIR / 'dqn_SpaceInvaders-v0_sb3_log.json'
CSV_LOG_FILENAME = OUTPUT_DIR / 'dqn_SpaceInvaders-v0_sb3_log.csv'

# Logs oficiales de SB3
TENSORBOARD_DIR = OUTPUT_DIR / 'tensorboard'
MONITOR_DIR = OUTPUT_DIR / 'monitor'
TENSORBOARD_DIR.mkdir(parents=True, exist_ok=True)
MONITOR_DIR.mkdir(parents=True, exist_ok=True)

print('OUTPUT_DIR:', OUTPUT_DIR)
print('MODEL_LATEST:', MODEL_LATEST)
print('POLICY_LATEST:', POLICY_LATEST)
print('LOG_FILENAME:', LOG_FILENAME)
print('CSV_LOG_FILENAME:', CSV_LOG_FILENAME)
print('TENSORBOARD_DIR:', TENSORBOARD_DIR)
print('MONITOR_DIR:', MONITOR_DIR)

```

Mounted at /content/gdrive

```

OUTPUT_DIR: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0
MODEL_LATEST: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_spaceinvaders_v0_sb3_latest.zip
POLICY_LATEST: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_spaceinvaders_v0_policy_latest.pt
LOG_FILENAME: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.json
CSV_LOG_FILENAME: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.csv
TENSORBOARD_DIR: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/tensorboard
MONITOR_DIR: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/monitor

```

```

In [ ]: # =====
# 4. Parámetros principales
# =====

ENV_ID = 'SpaceInvaders-v0'
SEED = 42

TOTAL_TIMESTEPS = 10_000_000

LEARNING_RATE = 1e-4
BUFFER_SIZE = 100_000
LEARNING_STARTS = 50_000
BATCH_SIZE = 32
GAMMA = 0.99
TRAIN_FREQ = 4
TARGET_UPDATE_INTERVAL = 10_000

EXPLORATION_INITIAL_EPS = 1.0
EXPLORATION_FINAL_EPS = 0.001
EXPLORATION_FRACTION = 0.30

```

```
CONSECUTIVE_THRESHOLD = 20.0
TARGET_CONSECUTIVE_EPISODES = 100

SAVE_FREQ_STEPS = 100_000
PRINT_EVERY_EPISODES = 10

LOAD_MODEL = False
```

```
In [ ]: # =====
# 5. Comprobación del entorno
# =====
# Si SpaceInvaders-v0 no existe en el entorno de Colab, esta celda dará error.

test_env = gym.make(ENV_ID)
print('Entorno creado correctamente:', ENV_ID)
print('Action space:', test_env.action_space)
print('Observation space:', test_env.observation_space)
test_env.close()
```

Entorno creado correctamente: SpaceInvaders-v0
Action space: Discrete(6)
Observation space: Box(0, 255, (210, 160, 3), uint8)

```
In [ ]: # =====
# 6. Mostrar imagen original del entorno
# =====

def reset_env_compat(env):
    result = env.reset()
    if isinstance(result, tuple):
        return result[0]
    return result

raw_env = gym.make(ENV_ID)
obs = reset_env_compat(raw_env)

try:
    frame_rgb = raw_env.render(mode='rgb_array')
except TypeError:
    frame_rgb = raw_env.render()

plt.figure(figsize=(6, 6))
plt.imshow(frame_rgb)
plt.title('Imagen original del entorno')
plt.axis('off')
plt.show()

raw_env.close()
```

Imagen original del entorno



In []:

```
# =====
# 7. Crear entorno vectorizado para SB3
# =====
# make_atari_env aplica los wrappers habituales de Atari.
# clip_reward=True hace que la recompensa sea clipped [-1, 0, 1].
# VecFrameStack apila 4 imágenes consecutivas.

def create_env():
    env = make_atari_env(
        ENV_ID,
        n_envs=1,
        seed=SEED,
        monitor_dir=str(OUTPUT_DIR / 'monitor'),
        wrapper_kwargs=dict(clip_reward=True)
    )
    env = VecFrameStack(env, n_stack=4)
    return env

env = create_env()
print('Observation space SB3:', env.observation_space)
print('Action space SB3:', env.action_space)
```

Observation space SB3: Box(0, 255, (84, 84, 4), uint8)
Action space SB3: Discrete(6)

In []:

```
# =====
# 8. Mostrar imágenes procesadas 84x84 y stack de frames
# =====

obs = env.reset()
```



```

print('Shape observación:', obs.shape)

# Normalmente con CnnPolicy/SB3 queda en formato: (n_envs, channels, height, width)
obs0 = obs[0]

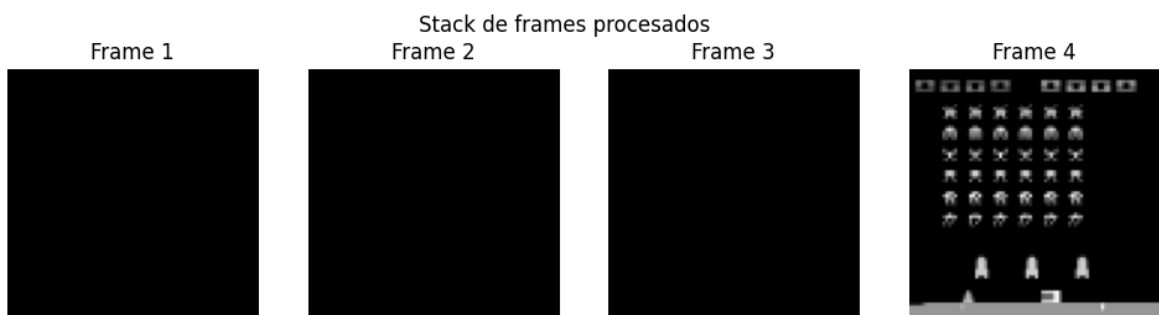
if obs0.shape[0] in [1, 3, 4, 12]:
    # channels first
    frames = obs0
else:
    # channels last
    frames = np.transpose(obs0, (2, 0, 1))

num_frames = min(4, frames.shape[0])

plt.figure(figsize=(12, 3))
for i in range(num_frames):
    plt.subplot(1, num_frames, i + 1)
    plt.imshow(frames[i], cmap='gray')
    plt.title(f'Frame {i + 1}')
    plt.axis('off')
plt.suptitle('Stack de frames procesados')
plt.show()

```

Shape observación: (1, 84, 84, 4)



```

In [ ]: # =====
# 9. Red convolucional personalizada 84x84
# =====
# Arquitectura tipo DQN Atari:
# Entrada: 4 x 84 x 84
# Conv2D 32 filtros 8x8 stride 4
# Conv2D 64 filtros 4x4 stride 2
# Conv2D 64 filtros 3x3 stride 1
# Capa densa de 512 características

class CustomCNN84(BaseFeaturesExtractor):
    def __init__(self, observation_space, features_dim=512):
        super().__init__(observation_space, features_dim)

        n_input_channels = observation_space.shape[0]

        self.cnn = nn.Sequential(
            nn.Conv2d(n_input_channels, 32, kernel_size=8, stride=4),
            nn.ReLU(),
            nn.Conv2d(32, 64, kernel_size=4, stride=2),
            nn.ReLU(),
            nn.Conv2d(64, 64, kernel_size=3, stride=1),
            nn.ReLU(),
            nn.Flatten()
        )

```

```

        with th.no_grad():
            sample = th.as_tensor(observation_space.sample()[None]).float()
            n_flatten = self.cnn(sample).shape[1]

        self.linear = nn.Sequential(
            nn.Linear(n_flatten, features_dim),
            nn.ReLU()
        )

    def forward(self, observations):
        return self.linear(self.cnn(observations))

policy_kwargs = dict(
    features_extractor_class=CustomCNN84,
    features_extractor_kwargs=dict(features_dim=512),
    net_arch=[]
)

```

```

In [ ]: # =====
# 9. Función auxiliar de guardado
# =====
# Esta función solo se usa como alternativa si model.save(...) falla por problem
# de serialización en Colab/Python. El logger oficial será el de SB3/TensorBoard

def save_policy_checkpoint(model, path):
    path = Path(path)
    path.parent.mkdir(parents=True, exist_ok=True)

    th.save(
        {
            "policy_state_dict": model.policy.state_dict(),
            "optimizer_state_dict": model.policy.optimizer.state_dict(),
            "num_timesteps": model.num_timesteps,
            "exploration_rate": getattr(model, "exploration_rate", None),
        },
        path
    )

```

```

In [ ]: # =====
# 10. Logger completo y callback personalizado
# =====
# Guarda todas las métricas propias en JSON y CSV.
# Además registra métricas custom en TensorBoard/SB3.

class SB3TrainingLogger:
    def __init__(self, json_path, csv_path):
        self.json_path = Path(json_path)
        self.csv_path = Path(csv_path)
        self.json_path.parent.mkdir(parents=True, exist_ok=True)
        self.csv_path.parent.mkdir(parents=True, exist_ok=True)

        if self.json_path.exists():
            with open(self.json_path, 'r') as f:
                self.data = json.load(f)
            print('Log JSON previo cargado:', self.json_path)
        else:
            self.data = {
                'episode': [],

```

```

        'total_steps': [],
        'episode_reward': [],
        'clipped_episode_reward': [],
        'consecutive_clipped_over_20': [],
        'max_consecutive_clipped_over_20': [],
        'avg_clip50': [],
        'duration': [],
        'exploration_rate': []
    }

def add_episode(self, episode, total_steps, reward, clipped_reward,
               consecutive, max_consecutive, avg_clip50, duration,
               exploration_rate):
    row = {
        'episode': int(episode),
        'total_steps': int(total_steps),
        'episode_reward': float(reward),
        'clipped_episode_reward': float(clipped_reward),
        'consecutive_clipped_over_20': int(consecutive),
        'max_consecutive_clipped_over_20': int(max_consecutive),
        'avg_clip50': float(avg_clip50),
        'duration': float(duration),
        'exploration_rate': float(exploration_rate) if exploration_rate is not None
    }

    for key, value in row.items():
        self.data[key].append(value)

    # CSV incremental: se guarda en cada episodio para no perder información
    pd.DataFrame([row]).to_csv(
        self.csv_path,
        mode='a',
        header=not self.csv_path.exists(),
        index=False
    )

def save_json(self):
    with open(self.json_path, 'w') as f:
        json.dump(self.data, f, indent=2)

def __len__(self):
    return len(self.data.get('episode', []))

class ConsecutiveRewardCallback(BaseCallback):
    def __init__(self, training_logger, verbose=0):
        super().__init__(verbose)
        self.training_logger = training_logger
        self.episode = len(training_logger)
        self.consecutive = 0
        self.max_consecutive = 0
        self.recent_clipped_rewards = deque(maxlen=50)
        self.start_time = time.time()
        self.last_json_save_step = 0

        previous_clipped = training_logger.data.get('clipped_episode_reward', [])
        for value in previous_clipped:
            self.recent_clipped_rewards.append(float(value))

        previous_consec = training_logger.data.get('consecutive_clipped_over_20', [])

```

```

if len(previous_consec) > 0:
    self.consecutive = int(previous_consec[-1])

previous_max = training_logger.data.get('max_consecutive_clipped_over_20')
if len(previous_max) > 0:
    self.max_consecutive = int(max(previous_max))

def _on_step(self) -> bool:
    infos = self.locals.get('infos', [])

    for info in infos:
        if 'episode' not in info:
            continue

        ep_reward = float(info['episode']['r'])

        # Métrica comparable con tu práctica anterior.
        # En esta configuración SB3 informa reward real; para Space Invaders
        clipped_reward = ep_reward / 10.0

        duration = time.time() - self.start_time
        self.start_time = time.time()

        self.recent_clipped_rewards.append(clipped_reward)

        if clipped_reward > CONSECUTIVE_THRESHOLD:
            self.consecutive += 1
        else:
            self.consecutive = 0

        if self.consecutive > self.max_consecutive:
            self.max_consecutive = self.consecutive

        avg_clip50 = (
            float(np.mean(self.recent_clipped_rewards))
            if len(self.recent_clipped_rewards) > 0
            else 0.0
        )

        exploration_rate = getattr(self.model, 'exploration_rate', np.nan)

        self.training_logger.add_episode(
            episode=self.episode,
            total_steps=self.num_timesteps,
            reward=ep_reward,
            clipped_reward=clipped_reward,
            consecutive=self.consecutive,
            max_consecutive=self.max_consecutive,
            avg_clip50=avg_clip50,
            duration=duration,
            exploration_rate=exploration_rate
        )

        # Métricas personalizadas para TensorBoard/SB3.
        self.logger.record('custom/clipped_episode_reward', clipped_reward)
        self.logger.record('custom/consecutive_clipped_over_20', self.consecutive)
        self.logger.record('custom/max_consecutive_clipped_over_20', self.max_consecutive)
        self.logger.record('custom/avg_clip50', avg_clip50)

    if self.episode % PRINT_EVERY_EPISODES == 0:

```

```

        print(
            f'EP={self.episode:,} | '
            f'STEP={self.num_timesteps:,} | '
            f'clip={clipped_reward:.0f} | '
            f'consec={self.consecutive} | '
            f'max_consec={self.max_consecutive} | '
            f'avg_clip50={avg_clip50:.2f}'
        )

        self.episode += 1

        if self.max_consecutive >= TARGET_CONSECUTIVE_EPISODES:
            print('Objetivo conseguido: 100 episodios consecutivos superando')
            self.training_logger.save_json()
            return False

        if self.num_timesteps - self.last_json_save_step >= SAVE_FREQ_STEPS:
            self.training_logger.save_json()
            self.last_json_save_step = self.num_timesteps

        return True

    def _on_training_end(self):
        self.training_logger.save_json()
        print('Entrenamiento finalizado.')
        print('max_consec alcanzado:', self.max_consecutive)
        print('Log JSON:', LOG_FILENAME)
        print('Log CSV:', CSV_LOG_FILENAME)

```

```

In [ ]: # =====
# 11. Crear o cargar modelo DQN
# =====

if LOAD_MODEL and MODEL_LATEST.exists():
    print("Cargando modelo desde:", MODEL_LATEST)

    model = DQN.load(
        str(MODEL_LATEST),
        env=env,
        device="auto",
        tensorboard_log=str(TENSORBOARD_DIR)
    )

    print("Modelo cargado correctamente.")

else:
    print("Creando modelo nuevo.")

    model = DQN(
        policy="CnnPolicy",
        env=env,
        learning_rate=LEARNING_RATE,
        buffer_size=BUFFER_SIZE,
        learning_starts=LEARNING_STARTS,
        batch_size=BATCH_SIZE,
        gamma=GAMMA,
        train_freq=TRAIN_FREQ,
        target_update_interval=TARGET_UPDATE_INTERVAL,
        exploration_initial_eps=EXPLORATION_INITIAL_EPS,
        exploration_final_eps=EXPLORATION_FINAL_EPS,

```

```

        exploration_fraction=EXPLORATION_FRACTION,
        policy_kwargs=policy_kwargs,
        tensorboard_log=str(TENSORBOARD_DIR),
        verbose=1,
        seed=SEED,
        device="auto"
    )

    print("Modelo nuevo creado correctamente.")

print("Política usada:")
print(model.policy)

```

Creando modelo nuevo.

Using cuda device

Wrapping the env in a VecTransposeImage.

Modelo nuevo creado correctamente.

Política usada:

CnnPolicy(

```

  (q_net): QNetwork(
    (features_extractor): CustomCNN84(
      (cnn): Sequential(
        (0): Conv2d(4, 32, kernel_size=(8, 8), stride=(4, 4))
        (1): ReLU()
        (2): Conv2d(32, 64, kernel_size=(4, 4), stride=(2, 2))
        (3): ReLU()
        (4): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1))
        (5): ReLU()
        (6): Flatten(start_dim=1, end_dim=-1)
      )
      (linear): Sequential(
        (0): Linear(in_features=3136, out_features=512, bias=True)
        (1): ReLU()
      )
    )
  )
  (q_net): Sequential(
    (0): Linear(in_features=512, out_features=6, bias=True)
  )
)
(q_net_target): QNetwork(
  (features_extractor): CustomCNN84(
    (cnn): Sequential(
      (0): Conv2d(4, 32, kernel_size=(8, 8), stride=(4, 4))
      (1): ReLU()
      (2): Conv2d(32, 64, kernel_size=(4, 4), stride=(2, 2))
      (3): ReLU()
      (4): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1))
      (5): ReLU()
      (6): Flatten(start_dim=1, end_dim=-1)
    )
    (linear): Sequential(
      (0): Linear(in_features=3136, out_features=512, bias=True)
      (1): ReLU()
    )
  )
  (q_net): Sequential(
    (0): Linear(in_features=512, out_features=6, bias=True)
  )
)
)
)

```

```
In [ ]: # =====
# 12. Entrenamiento
# =====

training_logger = SB3TrainingLogger(
    json_path=LOG_FILENAME,
    csv_path=CSV_LOG_FILENAME
)

callback = ConsecutiveRewardCallback(
    training_logger=training_logger,
    verbose=1
)

model.learn(
    total_timesteps=TOTAL_TIMESTEPS,
    callback=callback,
    reset_num_timesteps=not LOAD_MODEL,
    tb_log_name='DQN_SpaceInvaders_v0',
    log_interval=100
)
```

Se han truncado las últimas 5000 líneas del flujo de salida.

EP=4,590 | STEP=802,238 | clip=22 | consec=3 | max_consec=8 | avg_clip50=22.83

custom/		
avg_clip50	22.8	
clipped_episode_r...	16.5	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	838	
ep_rew_mean	225	
exploration rate	0.732	
time/		
episodes	13700	
fps	262	
time_elapsed	3057	
total timesteps	803356	
train/		
learning_rate	0.0001	
loss	0.0988	
n_updates	188338	

EP=4,600 | STEP=804,242 | clip=3 | consec=0 | max_consec=8 | avg_clip50=21.31

EP=4,610 | STEP=806,407 | clip=12 | consec=0 | max_consec=8 | avg_clip50=21.86

EP=4,620 | STEP=808,665 | clip=34 | consec=5 | max_consec=8 | avg_clip50=24.62

custom/		
avg_clip50	24.9	
clipped_episode_r...	39.5	
consecutive_clipp...	2	
max_consecutive_c...	8	
rollout/		
ep_len_mean	865	
ep_rew_mean	236	
exploration rate	0.73	
time/		
episodes	13800	
fps	262	
time_elapsed	3085	
total timesteps	810578	
train/		
learning_rate	0.0001	
loss	0.157	
n_updates	190144	

EP=4,630 | STEP=810,660 | clip=38 | consec=3 | max_consec=8 | avg_clip50=25.48

EP=4,640 | STEP=812,758 | clip=4 | consec=0 | max_consec=8 | avg_clip50=25.26

EP=4,650 | STEP=814,780 | clip=38 | consec=1 | max_consec=8 | avg_clip50=26.46

EP=4,660 | STEP=816,706 | clip=34 | consec=2 | max_consec=8 | avg_clip50=26.78

custom/		
avg_clip50	27.2	
clipped_episode_r...	11.5	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	866	
ep_rew_mean	247	
exploration rate	0.728	
time/		

episodes	13900	
fps	262	
time_elapsed	3112	
total timesteps	817497	
train/		
learning_rate	0.0001	
loss	0.127	
n_updates	191874	

EP=4,670	STEP=818,693	clip=30	consec=1	max_consec=8	avg_clip50=24.79
EP=4,680	STEP=820,395	clip=24	consec=1	max_consec=8	avg_clip50=24.63
EP=4,690	STEP=822,592	clip=19	consec=0	max_consec=8	avg_clip50=24.28

custom/		
avg_clip50	23.9	
clipped_episode_r...	7.5	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	835	
ep_rew_mean	249	
exploration rate	0.726	
time/		
episodes	14000	
fps	262	
time_elapsed	3138	
total timesteps	823777	
train/		
learning_rate	0.0001	
loss	0.0993	
n_updates	193444	

EP=4,700	STEP=824,250	clip=44	consec=2	max_consec=8	avg_clip50=24.43
EP=4,710	STEP=826,413	clip=8	consec=0	max_consec=8	avg_clip50=24.32
EP=4,720	STEP=828,194	clip=17	consec=0	max_consec=8	avg_clip50=23.50
EP=4,730	STEP=830,118	clip=30	consec=1	max_consec=8	avg_clip50=24.23

custom/		
avg_clip50	23.7	
clipped_episode_r...	11	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	826	
ep_rew_mean	245	
exploration rate	0.723	
time/		
episodes	14100	
fps	262	
time_elapsed	3164	
total timesteps	830476	
train/		
learning_rate	0.0001	
loss	0.0857	
n_updates	195118	

EP=4,740	STEP=832,118	clip=9	consec=0	max_consec=8	avg_clip50=23.95
EP=4,750	STEP=834,449	clip=28	consec=1	max_consec=8	avg_clip50=24.89
EP=4,760	STEP=836,389	clip=8	consec=0	max_consec=8	avg_clip50=23.07

custom/	
avg_clip50	25.4
clipped_episode_r...	20
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	846
ep_rew_mean	247
exploration rate	0.721
time/	
episodes	14200
fps	262
time_elapsed	3193
total timesteps	837790
train/	
learning_rate	0.0001
loss	0.124
n_updates	196947

EP=4,770	STEP=838,677	clip=17	consec=0	max_consec=8	avg_clip50=25.18
EP=4,780	STEP=841,062	clip=88	consec=1	max_consec=8	avg_clip50=27.49
EP=4,790	STEP=843,098	clip=14	consec=0	max_consec=8	avg_clip50=27.62

custom/	
avg_clip50	27.7
clipped_episode_r...	58
consecutive_clipp...	5
max_consecutive_c...	8
rollout/	
ep_len_mean	891
ep_rew_mean	265
exploration rate	0.719
time/	
episodes	14300
fps	262
time_elapsed	3222
total timesteps	845165
train/	
learning_rate	0.0001
loss	0.062
n_updates	198791

EP=4,800	STEP=845,408	clip=33	consec=6	max_consec=8	avg_clip50=27.83
EP=4,810	STEP=847,667	clip=38	consec=2	max_consec=8	avg_clip50=29.81
EP=4,820	STEP=849,856	clip=37	consec=1	max_consec=8	avg_clip50=30.00
EP=4,830	STEP=852,072	clip=23	consec=4	max_consec=8	avg_clip50=27.77

custom/	
avg_clip50	27
clipped_episode_r...	6
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	911
ep_rew_mean	270
exploration rate	0.716
time/	
episodes	14400
fps	262
time_elapsed	3251

total timesteps	852552
train/	
learning_rate	0.0001
loss	0.041
n_updates	200637

EP=4,840	STEP=853,640	clip=10	consec=0	max_consec=8	avg_clip50=25.91
EP=4,850	STEP=855,848	clip=24	consec=4	max_consec=8	avg_clip50=24.87
EP=4,860	STEP=857,771	clip=22	consec=5	max_consec=8	avg_clip50=24.59

custom/	
avg_clip50	24.1
clipped_episode_r...	19
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	894
ep_rew_mean	268
exploration rate	0.714
time/	
episodes	14500
fps	262
time_elapsed	3280
total timesteps	859639
train/	
learning_rate	0.0001
loss	0.0443
n_updates	202409

EP=4,870	STEP=860,189	clip=32	consec=2	max_consec=8	avg_clip50=25.22
EP=4,880	STEP=862,498	clip=34	consec=3	max_consec=8	avg_clip50=25.27
EP=4,890	STEP=864,594	clip=14	consec=0	max_consec=8	avg_clip50=27.98
EP=4,900	STEP=866,662	clip=40	consec=3	max_consec=8	avg_clip50=27.68

custom/	
avg_clip50	27.5
clipped_episode_r...	10
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	889
ep_rew_mean	261
exploration rate	0.711
time/	
episodes	14600
fps	262
time_elapsed	3309
total timesteps	867006
train/	
learning_rate	0.0001
loss	0.297
n_updates	204251

EP=4,910	STEP=868,526	clip=31	consec=1	max_consec=8	avg_clip50=25.99
EP=4,920	STEP=870,659	clip=28	consec=3	max_consec=8	avg_clip50=24.12
EP=4,930	STEP=872,759	clip=17	consec=0	max_consec=8	avg_clip50=23.65

custom/	
avg_clip50	22.5
clipped_episode_r...	25.5

consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	879
ep_rew_mean	248
exploration rate	0.709
time/	
episodes	14700
fps	261
time_elapsed	3337
total timesteps	874106
train/	
learning_rate	0.0001
loss	0.224
n_updates	206026

EP=4,940	STEP=874,739	clip=12	consec=0	max_consec=8	avg_clip50=22.05
EP=4,950	STEP=877,285	clip=47	consec=3	max_consec=8	avg_clip50=25.04
EP=4,960	STEP=879,123	clip=30	consec=1	max_consec=8	avg_clip50=25.65
EP=4,970	STEP=881,229	clip=32	consec=1	max_consec=8	avg_clip50=26.04

custom/	
avg_clip50	26
clipped_episode_r...	32.5
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	887
ep_rew_mean	251
exploration rate	0.707
time/	
episodes	14800
fps	261
time_elapsed	3365
total timesteps	881229
train/	
learning_rate	0.0001
loss	0.0561
n_updates	207807

EP=4,980	STEP=883,305	clip=18	consec=0	max_consec=8	avg_clip50=25.85
EP=4,990	STEP=885,249	clip=15	consec=0	max_consec=8	avg_clip50=26.18
EP=5,000	STEP=887,343	clip=22	consec=2	max_consec=8	avg_clip50=23.16

custom/	
avg_clip50	23
clipped_episode_r...	5.5
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	869
ep_rew_mean	242
exploration rate	0.704
time/	
episodes	14900
fps	261
time_elapsed	3392
total timesteps	887810
train/	
learning_rate	0.0001

loss	0.0852
n_updates	209452

EP=5,010	STEP=889,027	clip=12	consec=0	max_consec=8	avg_clip50=22.52
EP=5,020	STEP=890,756	clip=55	consec=2	max_consec=8	avg_clip50=21.05
EP=5,030	STEP=892,664	clip=50	consec=4	max_consec=8	avg_clip50=21.37

custom/	
avg_clip50	21.2
clipped_episode_r...	21.5
consecutive_clipp...	3
max_consecutive_c...	8
rollout/	
ep_len_mean	846
ep_rew_mean	235
exploration rate	0.702
time/	
episodes	15000
fps	261
time_elapsed	3417
total timesteps	894170
train/	
learning_rate	0.0001
loss	0.324
n_updates	211042

EP=5,040	STEP=894,690	clip=8	consec=0	max_consec=8	avg_clip50=20.84
EP=5,050	STEP=896,740	clip=35	consec=1	max_consec=8	avg_clip50=20.88
EP=5,060	STEP=898,678	clip=14	consec=0	max_consec=8	avg_clip50=21.73
EP=5,070	STEP=900,878	clip=32	consec=1	max_consec=8	avg_clip50=22.69

custom/	
avg_clip50	23.4
clipped_episode_r...	48.5
consecutive_clipp...	2
max_consecutive_c...	8
rollout/	
ep_len_mean	831
ep_rew_mean	218
exploration rate	0.7
time/	
episodes	15100
fps	261
time_elapsed	3445
total timesteps	901192
train/	
learning_rate	0.0001
loss	0.449
n_updates	212797

EP=5,080	STEP=903,263	clip=22	consec=5	max_consec=8	avg_clip50=23.07
EP=5,090	STEP=905,513	clip=41	consec=2	max_consec=8	avg_clip50=25.47
EP=5,100	STEP=907,410	clip=16	consec=0	max_consec=8	avg_clip50=24.39

custom/	
avg_clip50	24.4
clipped_episode_r...	12.5
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	

ep_len_mean	845
ep_rew_mean	224
exploration rate	0.697
time/	
episodes	15200
fps	261
time_elapsed	3474
total timesteps	908410
train/	
learning_rate	0.0001
loss	0.103
n_updates	214602

EP=5,110 | STEP=909,150 | clip=14 | consec=0 | max_consec=8 | avg_clip50=23.56
EP=5,120 | STEP=911,310 | clip=12 | consec=0 | max_consec=8 | avg_clip50=25.26
EP=5,130 | STEP=913,151 | clip=14 | consec=0 | max_consec=8 | avg_clip50=24.58
EP=5,140 | STEP=915,136 | clip=12 | consec=0 | max_consec=8 | avg_clip50=22.49

custom/	
avg_clip50	22.5
clipped_episode_r...	11.5
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	863
ep_rew_mean	240
exploration rate	0.695
time/	
episodes	15300
fps	261
time_elapsed	3501
total timesteps	915346
train/	
learning_rate	0.0001
loss	0.111
n_updates	216336

EP=5,150 | STEP=917,157 | clip=18 | consec=0 | max_consec=8 | avg_clip50=22.64
EP=5,160 | STEP=919,154 | clip=34 | consec=2 | max_consec=8 | avg_clip50=23.29
EP=5,170 | STEP=920,662 | clip=10 | consec=0 | max_consec=8 | avg_clip50=19.90

custom/	
avg_clip50	18.8
clipped_episode_r...	27.5
consecutive_clipp...	2
max_consecutive_c...	8
rollout/	
ep_len_mean	830
ep_rew_mean	221
exploration rate	0.693
time/	
episodes	15400
fps	261
time_elapsed	3526
total timesteps	921468
train/	
learning_rate	0.0001
loss	0.0593
n_updates	217866

EP=5,180	STEP=922,953	clip=35	consec=4	max_consec=8	avg_clip50=21.19
EP=5,190	STEP=925,360	clip=62	consec=1	max_consec=8	avg_clip50=23.97
EP=5,200	STEP=927,596	clip=38	consec=5	max_consec=8	avg_clip50=25.92

custom/	
avg_clip50	25.9
clipped_episode_r...	10.5
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	854
ep_rew_mean	244
exploration rate	0.691
time/	
episodes	15500
fps	261
time_elapsed	3555
total timesteps	928835
train/	
learning_rate	0.0001
loss	0.108
n_updates	219708

EP=5,210	STEP=929,255	clip=3	consec=0	max_consec=8	avg_clip50=25.45
EP=5,220	STEP=931,281	clip=28	consec=2	max_consec=8	avg_clip50=27.22
EP=5,230	STEP=933,098	clip=8	consec=0	max_consec=8	avg_clip50=24.87
EP=5,240	STEP=935,139	clip=6	consec=0	max_consec=8	avg_clip50=23.56

custom/	
avg_clip50	23.7
clipped_episode_r...	17.5
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	843
ep_rew_mean	237
exploration rate	0.688
time/	
episodes	15600
fps	261
time_elapsed	3582
total timesteps	935436
train/	
learning_rate	0.0001
loss	0.353
n_updates	221358

EP=5,250	STEP=937,146	clip=6	consec=0	max_consec=8	avg_clip50=22.18
EP=5,260	STEP=939,553	clip=59	consec=6	max_consec=8	avg_clip50=25.75
EP=5,270	STEP=941,509	clip=18	consec=0	max_consec=8	avg_clip50=26.54

custom/	
avg_clip50	27.7
clipped_episode_r...	65.5
consecutive_clipp...	2
max_consecutive_c...	8
rollout/	
ep_len_mean	880
ep_rew_mean	273
exploration rate	0.686

time/	
episodes	15700
fps	260
time_elapsed	3612
total timesteps	942795
train/	
learning_rate	0.0001
loss	0.06
n_updates	223198

EP=5,280	STEP=943,547	clip=29	consec=1	max_consec=8	avg_clip50=28.13
EP=5,290	STEP=945,503	clip=26	consec=1	max_consec=8	avg_clip50=26.00
EP=5,300	STEP=947,676	clip=5	consec=0	max_consec=8	avg_clip50=27.73
EP=5,310	STEP=949,730	clip=36	consec=3	max_consec=8	avg_clip50=25.06

custom/	
avg_clip50	25.1
clipped_episode_r...	36.5
consecutive_clipp...	3
max_consecutive_c...	8
rollout/	
ep_len_mean	864
ep_rew_mean	254
exploration rate	0.684
time/	
episodes	15800
fps	260
time_elapsed	3640
total timesteps	949730
train/	
learning_rate	0.0001
loss	0.0798
n_updates	224932

EP=5,320	STEP=952,100	clip=10	consec=0	max_consec=9	avg_clip50=27.21
EP=5,330	STEP=954,287	clip=16	consec=0	max_consec=9	avg_clip50=27.54
EP=5,340	STEP=955,931	clip=14	consec=0	max_consec=9	avg_clip50=27.41

custom/	
avg_clip50	27.3
clipped_episode_r...	7
consecutive_clipp...	0
max_consecutive_c...	9
rollout/	
ep_len_mean	874
ep_rew_mean	268
exploration rate	0.681
time/	
episodes	15900
fps	260
time_elapsed	3668
total timesteps	956741
train/	
learning_rate	0.0001
loss	0.249
n_updates	226685

EP=5,350	STEP=958,027	clip=63	consec=1	max_consec=9	avg_clip50=27.70
EP=5,360	STEP=959,747	clip=34	consec=4	max_consec=9	avg_clip50=26.97
EP=5,370	STEP=962,015	clip=53	consec=3	max_consec=9	avg_clip50=24.62

custom/	
avg_clip50	25.3
clipped_episode_r...	42
consecutive_clipp...	2
max_consecutive_c...	9
rollout/	
ep_len_mean	870
ep_rew_mean	256
exploration rate	0.679
time/	
episodes	16000
fps	260
time_elapsed	3695
total timesteps	963568
train/	
learning_rate	0.0001
loss	0.0589
n_updates	228391

EP=5,380 | STEP=964,343 | clip=34 | consec=5 | max_consec=9 | avg_clip50=24.83
EP=5,390 | STEP=966,296 | clip=48 | consec=3 | max_consec=9 | avg_clip50=26.91
EP=5,400 | STEP=968,427 | clip=28 | consec=2 | max_consec=9 | avg_clip50=26.55
EP=5,410 | STEP=970,314 | clip=22 | consec=1 | max_consec=9 | avg_clip50=25.96

custom/	
avg_clip50	26
clipped_episode_r...	21.5
consecutive_clipp...	1
max_consecutive_c...	9
rollout/	
ep_len_mean	869
ep_rew_mean	265
exploration rate	0.677
time/	
episodes	16100
fps	260
time_elapsed	3723
total timesteps	970367
train/	
learning_rate	0.0001
loss	0.0278
n_updates	230091

EP=5,420 | STEP=972,210 | clip=32 | consec=2 | max_consec=9 | avg_clip50=25.15
EP=5,430 | STEP=974,608 | clip=29 | consec=6 | max_consec=9 | avg_clip50=24.90
EP=5,440 | STEP=976,665 | clip=6 | consec=0 | max_consec=9 | avg_clip50=23.80

custom/	
avg_clip50	21.5
clipped_episode_r...	15.5
consecutive_clipp...	0
max_consecutive_c...	9
rollout/	
ep_len_mean	866
ep_rew_mean	247
exploration rate	0.675
time/	
episodes	16200
fps	260

	time_elapsed		3751	
	total timesteps		977346	
	train/			
	learning_rate		0.0001	
	loss		0.0391	
	n_updates		231836	

EP=5,450		STEP=978,412		clip=23		consec=2		max_consec=9		avg_clip50=21.48
EP=5,460		STEP=980,372		clip=20		consec=0		max_consec=9		avg_clip50=22.49
EP=5,470		STEP=982,249		clip=7		consec=0		max_consec=9		avg_clip50=23.42

	custom/			
	avg_clip50		22.9	
	clipped_episode_r...		39	
	consecutive_clipp...		1	
	max_consecutive_c...		9	
	rollout/			
	ep_len_mean		846	
	ep_rew_mean		239	
	exploration rate		0.672	
	time/			
	episodes		16300	
	fps		260	
	time_elapsed		3778	
	total timesteps		984106	
	train/			
	learning_rate		0.0001	
	loss		0.0411	
	n_updates		233526	

EP=5,480		STEP=984,265		clip=14		consec=0		max_consec=9		avg_clip50=22.61
EP=5,490		STEP=986,890		clip=77		consec=1		max_consec=9		avg_clip50=25.43
EP=5,500		STEP=989,275		clip=12		consec=0		max_consec=9		avg_clip50=27.17
EP=5,510		STEP=991,344		clip=2		consec=0		max_consec=9		avg_clip50=27.47

	custom/			
	avg_clip50		27.4	
	clipped_episode_r...		18	
	consecutive_clipp...		0	
	max_consecutive_c...		9	
	rollout/			
	ep_len_mean		887	
	ep_rew_mean		250	
	exploration rate		0.67	
	time/			
	episodes		16400	
	fps		260	
	time_elapsed		3809	
	total timesteps		991685	
	train/			
	learning_rate		0.0001	
	loss		0.371	
	n_updates		235421	

EP=5,520		STEP=993,090		clip=13		consec=0		max_consec=9		avg_clip50=25.79
EP=5,530		STEP=995,488		clip=12		consec=0		max_consec=9		avg_clip50=26.53
EP=5,540		STEP=997,565		clip=18		consec=0		max_consec=9		avg_clip50=24.81

	custom/			
	avg_clip50		25.3	

	clipped_episode_r...	9.5	
	consecutive_clipp...	0	
	max_consecutive_c...	9	
	rollout/		
	ep_len_mean	900	
	ep_rew_mean	263	
	exploration rate	0.667	
	time/		
	episodes	16500	
	fps	260	
	time_elapsed	3837	
	total timesteps	998867	
	train/		
	learning_rate	0.0001	
	loss	0.201	
	n_updates	237216	

EP=5,550 | STEP=999,974 | clip=22 | consec=4 | max_consec=9 | avg_clip50=25.51
EP=5,560 | STEP=1,001,919 | clip=19 | consec=0 | max_consec=9 | avg_clip50=24.94
EP=5,570 | STEP=1,004,484 | clip=74 | consec=4 | max_consec=9 | avg_clip50=28.37
EP=5,580 | STEP=1,006,482 | clip=8 | consec=0 | max_consec=9 | avg_clip50=27.24

	custom/		
	avg_clip50	27.2	
	clipped_episode_r...	8.5	
	consecutive_clipp...	0	
	max_consecutive_c...	9	
	rollout/		
	ep_len_mean	934	
	ep_rew_mean	269	
	exploration rate	0.665	
	time/		
	episodes	16600	
	fps	260	
	time_elapsed	3868	
	total timesteps	1006537	
	train/		
	learning_rate	0.0001	
	loss	0.198	
	n_updates	239134	

EP=5,590 | STEP=1,008,554 | clip=14 | consec=0 | max_consec=9 | avg_clip50=26.11
EP=5,600 | STEP=1,010,530 | clip=22 | consec=1 | max_consec=9 | avg_clip50=22.91
EP=5,610 | STEP=1,012,293 | clip=24 | consec=1 | max_consec=9 | avg_clip50=22.93

	custom/		
	avg_clip50	23.3	
	clipped_episode_r...	17	
	consecutive_clipp...	0	
	max_consecutive_c...	9	
	rollout/		
	ep_len_mean	891	
	ep_rew_mean	244	
	exploration rate	0.663	
	time/		
	episodes	16700	
	fps	260	
	time_elapsed	3895	
	total timesteps	1013181	
	train/		

learning_rate	0.0001
loss	0.152
n_updates	240795

EP=5,620	STEP=1,014,504	clip=12	consec=0	max_consec=9	avg_clip50=20.73
EP=5,630	STEP=1,016,520	clip=17	consec=0	max_consec=9	avg_clip50=21.48
EP=5,640	STEP=1,018,627	clip=24	consec=2	max_consec=9	avg_clip50=22.27

custom/	
avg_clip50	24.1
clipped_episode_r...	18.5
consecutive_clipp...	0
max_consecutive_c...	9
rollout/	
ep_len_mean	894
ep_rew_mean	239
exploration rate	0.66
time/	
episodes	16800
fps	260
time_elapsed	3925
total timesteps	1020552
train/	
learning_rate	0.0001
loss	0.102
n_updates	242637

EP=5,650	STEP=1,021,144	clip=30	consec=2	max_consec=9	avg_clip50=24.51
EP=5,660	STEP=1,023,306	clip=14	consec=0	max_consec=9	avg_clip50=25.40
EP=5,670	STEP=1,025,286	clip=21	consec=1	max_consec=9	avg_clip50=25.61
EP=5,680	STEP=1,027,652	clip=4	consec=0	max_consec=9	avg_clip50=26.25

custom/	
avg_clip50	26.8
clipped_episode_r...	45
consecutive_clipp...	1
max_consecutive_c...	9
rollout/	
ep_len_mean	899
ep_rew_mean	242
exploration rate	0.658
time/	
episodes	16900
fps	259
time_elapsed	3955
total timesteps	1028201
train/	
learning_rate	0.0001
loss	0.413
n_updates	244550

EP=5,690	STEP=1,029,819	clip=16	consec=0	max_consec=9	avg_clip50=26.55
EP=5,700	STEP=1,031,771	clip=48	consec=1	max_consec=9	avg_clip50=25.86
EP=5,710	STEP=1,034,088	clip=38	consec=1	max_consec=9	avg_clip50=26.26

custom/	
avg_clip50	27.3
clipped_episode_r...	26
consecutive_clipp...	6
max_consecutive_c...	9

rollout/	
ep_len_mean	933
ep_rew_mean	263
exploration rate	0.655
time/	
episodes	17000
fps	259
time_elapsed	3986
total timesteps	1035720
train/	
learning_rate	0.0001
loss	0.179
n_updates	246429

EP=5,720		STEP=1,036,973		clip=32		consec=1		max_consec=9		avg_clip50=28.82
EP=5,730		STEP=1,039,308		clip=46		consec=1		max_consec=9		avg_clip50=28.86
EP=5,740		STEP=1,041,363		clip=19		consec=0		max_consec=9		avg_clip50=28.83

custom/	
avg_clip50	29.4
clipped_episode_r...	11.5
consecutive_clipp...	0
max_consecutive_c...	9
rollout/	
ep_len_mean	940
ep_rew_mean	275
exploration rate	0.653
time/	
episodes	17100
fps	259
time_elapsed	4016
total timesteps	1043253
train/	
learning_rate	0.0001
loss	0.438
n_updates	248313

EP=5,750		STEP=1,043,476		clip=17		consec=0		max_consec=9		avg_clip50=28.81
EP=5,760		STEP=1,045,715		clip=28		consec=4		max_consec=9		avg_clip50=28.59
EP=5,770		STEP=1,047,846		clip=35		consec=4		max_consec=9		avg_clip50=27.96
EP=5,780		STEP=1,049,904		clip=13		consec=0		max_consec=9		avg_clip50=28.24

custom/	
avg_clip50	27.9
clipped_episode_r...	13
consecutive_clipp...	0
max_consecutive_c...	9
rollout/	
ep_len_mean	919
ep_rew_mean	276
exploration rate	0.65
time/	
episodes	17200
fps	259
time_elapsed	4045
total timesteps	1050458
train/	
learning_rate	0.0001
loss	0.296
n_updates	250114

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EP=5,790 | STEP=1,051,940 | clip=17 | consec=0 | max_consec=9 | avg_clip50=27.83
EP=5,800 | STEP=1,053,775 | clip=4 | consec=0 | max_consec=9 | avg_clip50=27.90
EP=5,810 | STEP=1,056,497 | clip=50 | consec=7 | max_consec=9 | avg_clip50=30.02
-----

```

```

| custom/
|   avg_clip50           | 29.4
|   clipped_episode_r... | 34.5
|   consecutive_clipp... | 1
|   max_consecutive_c... | 9
| rollout/
|   ep_len_mean          | 906
|   ep_rew_mean          | 281
|   exploration rate     | 0.648
| time/
|   episodes             | 17300
|   fps                  | 259
|   time_elapsed         | 4075
|   total timesteps      | 1057961
| train/
|   learning_rate        | 0.0001
|   loss                 | 0.236
|   n_updates            | 251990
-----

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EP=5,820 | STEP=1,058,384 | clip=4 | consec=0 | max_consec=9 | avg_clip50=27.58
EP=5,830 | STEP=1,060,600 | clip=16 | consec=0 | max_consec=9 | avg_clip50=27.61
EP=5,840 | STEP=1,062,961 | clip=48 | consec=3 | max_consec=9 | avg_clip50=28.68
EP=5,850 | STEP=1,064,732 | clip=20 | consec=0 | max_consec=9 | avg_clip50=28.63
-----

```

```

| custom/
|   avg_clip50           | 28.4
|   clipped_episode_r... | 15.5
|   consecutive_clipp... | 0
|   max_consecutive_c... | 9
| rollout/
|   ep_len_mean          | 898
|   ep_rew_mean          | 285
|   exploration rate     | 0.645
| time/
|   episodes             | 17400
|   fps                  | 259
|   time_elapsed         | 4104
|   total timesteps      | 1065234
| train/
|   learning_rate        | 0.0001
|   loss                 | 0.237
|   n_updates            | 253808
-----

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EP=5,860 | STEP=1,066,996 | clip=6 | consec=0 | max_consec=9 | avg_clip50=28.12
EP=5,870 | STEP=1,069,138 | clip=8 | consec=0 | max_consec=9 | avg_clip50=28.46
EP=5,880 | STEP=1,071,218 | clip=16 | consec=0 | max_consec=9 | avg_clip50=27.68
-----

```

```

| custom/
|   avg_clip50           | 27.9
|   clipped_episode_r... | 15
|   consecutive_clipp... | 0
|   max_consecutive_c... | 9
| rollout/
|   ep_len_mean          | 905
|   ep_rew_mean          | 284
-----

```

exploration rate	0.643
time/	
episodes	17500
fps	259
time_elapsed	4134
total timesteps	1072685
train/	
learning_rate	0.0001
loss	0.548
n_updates	255671

EP=5,890 | STEP=1,073,279 | clip=19 | consec=0 | max_consec=9 | avg_clip50=27.42
EP=5,900 | STEP=1,075,571 | clip=80 | consec=1 | max_consec=9 | avg_clip50=29.13
EP=5,910 | STEP=1,077,503 | clip=12 | consec=0 | max_consec=9 | avg_clip50=26.38
EP=5,920 | STEP=1,079,618 | clip=5 | consec=0 | max_consec=9 | avg_clip50=26.19

custom/	
avg_clip50	26.2
clipped_episode_r...	5
consecutive_clipp...	0
max_consecutive_c...	9
rollout/	
ep_len_mean	894
ep_rew_mean	273
exploration rate	0.64
time/	
episodes	17600
fps	259
time_elapsed	4162
total timesteps	1079618
train/	
learning_rate	0.0001
loss	0.0494
n_updates	257404

EP=5,930 | STEP=1,081,535 | clip=28 | consec=1 | max_consec=9 | avg_clip50=26.05
EP=5,940 | STEP=1,083,903 | clip=50 | consec=2 | max_consec=9 | avg_clip50=26.81
EP=5,950 | STEP=1,086,240 | clip=12 | consec=0 | max_consec=9 | avg_clip50=26.44

custom/	
avg_clip50	27.1
clipped_episode_r...	21
consecutive_clipp...	2
max_consecutive_c...	9
rollout/	
ep_len_mean	897
ep_rew_mean	270
exploration rate	0.638
time/	
episodes	17700
fps	259
time_elapsed	4193
total timesteps	1087346
train/	
learning_rate	0.0001
loss	0.0778
n_updates	259336

EP=5,960 | STEP=1,088,747 | clip=17 | consec=0 | max_consec=9 | avg_clip50=29.42
EP=5,970 | STEP=1,091,041 | clip=84 | consec=2 | max_consec=9 | avg_clip50=30.85

EP=5,980 | STEP=1,093,438 | clip=48 | consec=1 | max_consec=9 | avg_clip50=31.29

custom/	
avg_clip50	30.2
clipped_episode_r...	14.5
consecutive_clipp...	0
max_consecutive_c...	9
rollout/	
ep_len_mean	934
ep_rew_mean	282
exploration rate	0.635
time/	
episodes	17800
fps	259
time_elapsed	4226
total timesteps	1095444
train/	
learning_rate	0.0001
loss	0.126
n_updates	261360

EP=5,990 | STEP=1,095,512 | clip=43 | consec=1 | max_consec=9 | avg_clip50=30.09
EP=6,000 | STEP=1,097,870 | clip=44 | consec=3 | max_consec=9 | avg_clip50=30.38
EP=6,010 | STEP=1,099,501 | clip=14 | consec=0 | max_consec=9 | avg_clip50=26.69
EP=6,020 | STEP=1,101,616 | clip=61 | consec=3 | max_consec=9 | avg_clip50=26.02

custom/	
avg_clip50	25.8
clipped_episode_r...	33
consecutive_clipp...	2
max_consecutive_c...	9
rollout/	
ep_len_mean	926
ep_rew_mean	281
exploration rate	0.633
time/	
episodes	17900
fps	259
time_elapsed	4254
total timesteps	1102522
train/	
learning_rate	0.0001
loss	0.0923
n_updates	263130

EP=6,030 | STEP=1,103,829 | clip=15 | consec=0 | max_consec=9 | avg_clip50=26.59
EP=6,040 | STEP=1,105,755 | clip=50 | consec=1 | max_consec=9 | avg_clip50=25.96
EP=6,050 | STEP=1,108,014 | clip=22 | consec=1 | max_consec=9 | avg_clip50=24.47

custom/	
avg_clip50	26.4
clipped_episode_r...	13
consecutive_clipp...	0
max_consecutive_c...	9
rollout/	
ep_len_mean	904
ep_rew_mean	266
exploration rate	0.63
time/	
episodes	18000

fps	259
time_elapsed	4284
total timesteps	1109921
train/	
learning_rate	0.0001
loss	0.0688
n_updates	264980

EP=6,060	STEP=1,110,257	clip=36	consec=1	max_consec=9	avg_clip50=26.40
EP=6,070	STEP=1,112,557	clip=14	consec=0	max_consec=9	avg_clip50=27.37
EP=6,080	STEP=1,114,470	clip=6	consec=0	max_consec=9	avg_clip50=25.22
EP=6,090	STEP=1,116,599	clip=18	consec=0	max_consec=9	avg_clip50=25.27

custom/	
avg_clip50	26.6
clipped_episode_r...	34
consecutive_clipp...	3
max_consecutive_c...	9
rollout/	
ep_len_mean	901
ep_rew_mean	260
exploration rate	0.628
time/	
episodes	18100
fps	258
time_elapsed	4314
total timesteps	1117482
train/	
learning_rate	0.0001
loss	0.395
n_updates	266870

EP=6,100	STEP=1,118,667	clip=16	consec=0	max_consec=9	avg_clip50=25.04
EP=6,110	STEP=1,120,855	clip=25	consec=3	max_consec=9	avg_clip50=25.19
EP=6,120	STEP=1,122,549	clip=20	consec=0	max_consec=9	avg_clip50=22.86

custom/	
avg_clip50	24
clipped_episode_r...	4.5
consecutive_clipp...	0
max_consecutive_c...	9
rollout/	
ep_len_mean	876
ep_rew_mean	250
exploration rate	0.626
time/	
episodes	18200
fps	258
time_elapsed	4340
total timesteps	1123851
train/	
learning_rate	0.0001
loss	0.343
n_updates	268462

EP=6,130	STEP=1,124,529	clip=18	consec=0	max_consec=9	avg_clip50=24.26
EP=6,140	STEP=1,126,543	clip=16	consec=0	max_consec=9	avg_clip50=24.14
EP=6,150	STEP=1,128,624	clip=25	consec=1	max_consec=9	avg_clip50=24.08
EP=6,160	STEP=1,130,848	clip=60	consec=1	max_consec=9	avg_clip50=24.44

custom/	
avg_clip50	24.4
clipped_episode_r...	60.5
consecutive_clipp...	1
max_consecutive_c...	9
rollout/	
ep_len_mean	870
ep_rew_mean	248
exploration rate	0.623
time/	
episodes	18300
fps	258
time_elapsed	4369
total timesteps	1130923
train/	
learning_rate	0.0001
loss	0.0686
n_updates	270230

EP=6,170	STEP=1,132,933	clip=52	consec=1	max_consec=9	avg_clip50=24.92
EP=6,180	STEP=1,134,966	clip=14	consec=0	max_consec=9	avg_clip50=24.26
EP=6,190	STEP=1,136,995	clip=6	consec=0	max_consec=9	avg_clip50=24.34

custom/	
avg_clip50	24
clipped_episode_r...	2
consecutive_clipp...	0
max_consecutive_c...	9
rollout/	
ep_len_mean	847
ep_rew_mean	238
exploration rate	0.621
time/	
episodes	18400
fps	258
time_elapsed	4397
total timesteps	1137793
train/	
learning_rate	0.0001
loss	0.0686
n_updates	271948

EP=6,200	STEP=1,138,993	clip=9	consec=0	max_consec=9	avg_clip50=25.60
EP=6,210	STEP=1,140,903	clip=14	consec=0	max_consec=9	avg_clip50=24.37
EP=6,220	STEP=1,143,114	clip=11	consec=0	max_consec=9	avg_clip50=25.61

custom/	
avg_clip50	25.1
clipped_episode_r...	36.5
consecutive_clipp...	1
max_consecutive_c...	9
rollout/	
ep_len_mean	869
ep_rew_mean	247
exploration rate	0.619
time/	
episodes	18500
fps	258
time_elapsed	4427
total timesteps	1144909

train/	
learning_rate	0.0001
loss	0.0667
n_updates	273727

EP=6,230	STEP=1,145,120	clip=16	consec=0	max_consec=9	avg_clip50=25.13
EP=6,240	STEP=1,147,058	clip=52	consec=2	max_consec=9	avg_clip50=25.62
EP=6,250	STEP=1,149,154	clip=21	consec=3	max_consec=9	avg_clip50=25.70
EP=6,260	STEP=1,151,190	clip=29	consec=1	max_consec=9	avg_clip50=27.09

custom/	
avg_clip50	26.8
clipped_episode_r...	15.5
consecutive_clipp...	0
max_consecutive_c...	9
rollout/	
ep_len_mean	854
ep_rew_mean	255
exploration rate	0.616
time/	
episodes	18600
fps	258
time_elapsed	4456
total timesteps	1152002
train/	
learning_rate	0.0001
loss	0.405
n_updates	275500

EP=6,270	STEP=1,153,213	clip=50	consec=1	max_consec=9	avg_clip50=26.59
EP=6,280	STEP=1,155,173	clip=28	consec=3	max_consec=9	avg_clip50=26.68
EP=6,290	STEP=1,157,038	clip=24	consec=3	max_consec=9	avg_clip50=26.96

custom/	
avg_clip50	26.5
clipped_episode_r...	14
consecutive_clipp...	0
max_consecutive_c...	9
rollout/	
ep_len_mean	842
ep_rew_mean	257
exploration rate	0.614
time/	
episodes	18700
fps	258
time_elapsed	4481
total timesteps	1158337
train/	
learning_rate	0.0001
loss	0.0593
n_updates	277084

EP=6,300	STEP=1,158,796	clip=27	consec=1	max_consec=9	avg_clip50=25.01
EP=6,310	STEP=1,160,981	clip=8	consec=0	max_consec=9	avg_clip50=23.78
EP=6,320	STEP=1,162,839	clip=19	consec=0	max_consec=9	avg_clip50=23.44
EP=6,330	STEP=1,165,211	clip=42	consec=1	max_consec=9	avg_clip50=24.45

custom/	
avg_clip50	25.1
clipped_episode_r...	35

consecutive_clipp...	2
max_consecutive_c...	9
rollout/	
ep_len_mean	852
ep_rew_mean	259
exploration rate	0.612
time/	
episodes	18800
fps	258
time_elapsed	4510
total timesteps	1165419
train/	
learning_rate	0.0001
loss	0.0867
n_updates	278854

EP=6,340 | STEP=1,167,413 | clip=28 | consec=5 | max_consec=9 | avg_clip50=25.42
EP=6,350 | STEP=1,169,712 | clip=12 | consec=0 | max_consec=9 | avg_clip50=27.39
EP=6,360 | STEP=1,171,928 | clip=20 | consec=0 | max_consec=9 | avg_clip50=29.00

custom/	
avg_clip50	27.6
clipped_episode_r...	29
consecutive_clipp...	1
max_consecutive_c...	9
rollout/	
ep_len_mean	868
ep_rew_mean	260
exploration rate	0.609
time/	
episodes	18900
fps	258
time_elapsed	4540
total timesteps	1172925
train/	
learning_rate	0.0001
loss	0.0958
n_updates	280731

EP=6,370 | STEP=1,174,002 | clip=15 | consec=0 | max_consec=9 | avg_clip50=29.45
EP=6,380 | STEP=1,176,268 | clip=51 | consec=3 | max_consec=9 | avg_clip50=31.98
EP=6,390 | STEP=1,178,362 | clip=34 | consec=2 | max_consec=9 | avg_clip50=31.07
EP=6,400 | STEP=1,180,591 | clip=30 | consec=3 | max_consec=9 | avg_clip50=31.45

custom/	
avg_clip50	31.4
clipped_episode_r...	29.5
consecutive_clipp...	3
max_consecutive_c...	9
rollout/	
ep_len_mean	917
ep_rew_mean	294
exploration rate	0.607
time/	
episodes	19000
fps	258
time_elapsed	4571
total timesteps	1180591
train/	
learning_rate	0.0001

loss	0.0547
n_updates	282647

EP=6,410 | STEP=1,182,726 | clip=28 | consec=4 | max_consec=9 | avg_clip50=30.28
EP=6,420 | STEP=1,184,534 | clip=50 | consec=1 | max_consec=9 | avg_clip50=29.22
EP=6,430 | STEP=1,186,684 | clip=29 | consec=2 | max_consec=9 | avg_clip50=27.15

custom/	
avg_clip50	26.2
clipped_episode_r...	22
consecutive_clipp...	2
max_consecutive_c...	9
rollout/	
ep_len_mean	901
ep_rew_mean	291
exploration rate	0.605
time/	
episodes	19100
fps	258
time_elapsed	4599
total timesteps	1187377
train/	
learning_rate	0.0001
loss	0.276
n_updates	284344

EP=6,440 | STEP=1,188,572 | clip=8 | consec=0 | max_consec=9 | avg_clip50=25.65
EP=6,450 | STEP=1,190,475 | clip=36 | consec=2 | max_consec=9 | avg_clip50=24.72
EP=6,460 | STEP=1,192,696 | clip=70 | consec=5 | max_consec=9 | avg_clip50=25.90

custom/	
avg_clip50	27.6
clipped_episode_r...	71.5
consecutive_clipp...	2
max_consecutive_c...	9
rollout/	
ep_len_mean	892
ep_rew_mean	285
exploration rate	0.602
time/	
episodes	19200
fps	258
time_elapsed	4628
total timesteps	1194513
train/	
learning_rate	0.0001
loss	0.417
n_updates	286128

EP=6,470 | STEP=1,194,942 | clip=30 | consec=1 | max_consec=9 | avg_clip50=27.55
EP=6,480 | STEP=1,197,210 | clip=30 | consec=2 | max_consec=9 | avg_clip50=28.03
EP=6,490 | STEP=1,199,838 | clip=16 | consec=0 | max_consec=9 | avg_clip50=31.44
EP=6,500 | STEP=1,202,359 | clip=20 | consec=0 | max_consec=9 | avg_clip50=33.38

custom/	
avg_clip50	32.9
clipped_episode_r...	32.5
consecutive_clipp...	1
max_consecutive_c...	9
rollout/	

ep_len_mean	913
ep_rew_mean	288
exploration rate	0.599
time/	
episodes	19300
fps	258
time_elapsed	4661
total timesteps	1202706
train/	
learning_rate	0.0001
loss	0.461
n_updates	288176

EP=6,510 | STEP=1,204,272 | clip=17 | consec=0 | max_consec=9 | avg_clip50=31.63
EP=6,520 | STEP=1,206,523 | clip=44 | consec=1 | max_consec=9 | avg_clip50=32.16
EP=6,530 | STEP=1,208,078 | clip=9 | consec=0 | max_consec=9 | avg_clip50=29.33

custom/	
avg_clip50	27.1
clipped_episode_r...	12.5
consecutive_clipp...	0
max_consecutive_c...	9
rollout/	
ep_len_mean	904
ep_rew_mean	289
exploration rate	0.597
time/	
episodes	19400
fps	257
time_elapsed	4688
total timesteps	1209299
train/	
learning_rate	0.0001
loss	0.0251
n_updates	289824

EP=6,540 | STEP=1,210,147 | clip=10 | consec=0 | max_consec=9 | avg_clip50=26.81
EP=6,550 | STEP=1,212,031 | clip=10 | consec=0 | max_consec=9 | avg_clip50=23.99
EP=6,560 | STEP=1,213,916 | clip=29 | consec=3 | max_consec=9 | avg_clip50=23.77
EP=6,570 | STEP=1,216,002 | clip=32 | consec=1 | max_consec=9 | avg_clip50=23.28

custom/	
avg_clip50	23.3
clipped_episode_r...	32.5
consecutive_clipp...	1
max_consecutive_c...	9
rollout/	
ep_len_mean	888
ep_rew_mean	277
exploration rate	0.595
time/	
episodes	19500
fps	257
time_elapsed	4715
total timesteps	1216036
train/	
learning_rate	0.0001
loss	0.0224
n_updates	291508

EP=6,580 | STEP=1,218,530 | clip=36 | consec=11 | max_consec=11 | avg_clip50=27.45

EP=6,590 | STEP=1,220,773 | clip=24 | consec=1 | max_consec=12 | avg_clip50=29.26

EP=6,600 | STEP=1,222,711 | clip=16 | consec=0 | max_consec=12 | avg_clip50=29.78

custom/	
avg_clip50	29.9
clipped_episode_r...	8
consecutive_clipp...	0
max_consecutive_c...	12
rollout/	
ep_len_mean	856
ep_rew_mean	269
exploration rate	0.593
time/	
episodes	19600
fps	257
time_elapsed	4745
total timesteps	1223327
train/	
learning_rate	0.0001
loss	0.172
n_updates	293331

EP=6,610 | STEP=1,224,631 | clip=47 | consec=2 | max_consec=12 | avg_clip50=30.19

EP=6,620 | STEP=1,226,427 | clip=4 | consec=0 | max_consec=12 | avg_clip50=28.98

EP=6,630 | STEP=1,228,539 | clip=18 | consec=0 | max_consec=12 | avg_clip50=26.84

custom/	
avg_clip50	26.4
clipped_episode_r...	30
consecutive_clipp...	2
max_consecutive_c...	12
rollout/	
ep_len_mean	876
ep_rew_mean	280
exploration rate	0.59
time/	
episodes	19700
fps	257
time_elapsed	4774
total timesteps	1230611
train/	
learning_rate	0.0001
loss	0.109
n_updates	295152

EP=6,640 | STEP=1,231,052 | clip=35 | consec=4 | max_consec=12 | avg_clip50=26.81

EP=6,650 | STEP=1,232,985 | clip=26 | consec=2 | max_consec=12 | avg_clip50=26.36

EP=6,660 | STEP=1,235,327 | clip=60 | consec=2 | max_consec=12 | avg_clip50=28.01

EP=6,670 | STEP=1,237,921 | clip=4 | consec=0 | max_consec=12 | avg_clip50=30.58

custom/	
avg_clip50	29.6
clipped_episode_r...	19
consecutive_clipp...	0
max_consecutive_c...	12
rollout/	
ep_len_mean	920
ep_rew_mean	295

exploration rate	0.587
time/	
episodes	19800
fps	257
time_elapsed	4808
total timesteps	1238773
train/	
learning_rate	0.0001
loss	0.0616
n_updates	297193

EP=6,680 | STEP=1,240,269 | clip=35 | consec=1 | max_consec=12 | avg_clip50=31.00
EP=6,690 | STEP=1,242,358 | clip=25 | consec=1 | max_consec=12 | avg_clip50=29.87
EP=6,700 | STEP=1,244,463 | clip=34 | consec=3 | max_consec=12 | avg_clip50=30.68

custom/	
avg_clip50	30.4
clipped_episode_r...	13.5
consecutive_clipp...	0
max_consecutive_c...	12
rollout/	
ep_len_mean	934
ep_rew_mean	292
exploration rate	0.585
time/	
episodes	19900
fps	257
time_elapsed	4838
total timesteps	1246133
train/	
learning_rate	0.0001
loss	0.296
n_updates	299033

EP=6,710 | STEP=1,246,719 | clip=11 | consec=0 | max_consec=12 | avg_clip50=29.73
EP=6,720 | STEP=1,248,746 | clip=17 | consec=0 | max_consec=12 | avg_clip50=27.51
EP=6,730 | STEP=1,251,278 | clip=22 | consec=1 | max_consec=12 | avg_clip50=28.16
EP=6,740 | STEP=1,253,224 | clip=34 | consec=2 | max_consec=12 | avg_clip50=27.51

custom/	
avg_clip50	27.6
clipped_episode_r...	64.5
consecutive_clipp...	3
max_consecutive_c...	12
rollout/	
ep_len_mean	941
ep_rew_mean	292
exploration rate	0.583
time/	
episodes	20000
fps	257
time_elapsed	4868
total timesteps	1253574
train/	
learning_rate	0.0001
loss	0.0406
n_updates	300893

EP=6,750 | STEP=1,255,475 | clip=56 | consec=1 | max_consec=12 | avg_clip50=28.75
EP=6,760 | STEP=1,257,837 | clip=30 | consec=1 | max_consec=12 | avg_clip50=29.71

EP=6,770 | STEP=1,260,467 | clip=10 | consec=0 | max_consec=12 | avg_clip50=32.38

custom/		
avg_clip50	33.5	
clipped_episode_r...	13	
consecutive_clipp...	0	
max_consecutive_c...	12	
rollout/		
ep_len_mean	947	
ep_rew_mean	303	
exploration rate	0.58	
time/		
episodes	20100	
fps	257	
time_elapsed	4900	
total timesteps	1261644	
train/		
learning_rate	0.0001	
loss	0.112	
n_updates	302910	

EP=6,780 | STEP=1,262,904 | clip=42 | consec=3 | max_consec=12 | avg_clip50=33.07

EP=6,790 | STEP=1,265,333 | clip=9 | consec=0 | max_consec=12 | avg_clip50=35.62

EP=6,800 | STEP=1,268,062 | clip=8 | consec=0 | max_consec=12 | avg_clip50=37.96

custom/		
avg_clip50	36.5	
clipped_episode_r...	6	
consecutive_clipp...	0	
max_consecutive_c...	12	
rollout/		
ep_len_mean	990	
ep_rew_mean	331	
exploration rate	0.577	
time/		
episodes	20200	
fps	257	
time_elapsed	4934	
total timesteps	1269835	
train/		
learning_rate	0.0001	
loss	0.164	
n_updates	304958	

EP=6,810 | STEP=1,270,360 | clip=28 | consec=2 | max_consec=12 | avg_clip50=37.04

EP=6,820 | STEP=1,272,660 | clip=44 | consec=2 | max_consec=12 | avg_clip50=35.47

EP=6,830 | STEP=1,274,816 | clip=40 | consec=2 | max_consec=12 | avg_clip50=33.46

EP=6,840 | STEP=1,276,743 | clip=12 | consec=0 | max_consec=12 | avg_clip50=31.02

custom/		
avg_clip50	29	
clipped_episode_r...	5.5	
consecutive_clipp...	0	
max_consecutive_c...	12	
rollout/		
ep_len_mean	979	
ep_rew_mean	329	
exploration rate	0.575	
time/		
episodes	20300	

fps	257				
time_elapsed	4963				
total timesteps	1277040				
train/					
learning_rate	0.0001				
loss	0.0395				
n_updates	306759				

EP=6,850	STEP=1,278,668	clip=22	consec=2	max_consec=12	avg_clip50=26.78
EP=6,860	STEP=1,280,611	clip=35	consec=1	max_consec=12	avg_clip50=26.28
EP=6,870	STEP=1,282,617	clip=11	consec=0	max_consec=12	avg_clip50=25.08

custom/					
avg_clip50	25.2				
clipped_episode_r...	9				
consecutive_clipp...	0				
max_consecutive_c...	12				
rollout/					
ep_len_mean	919				
ep_rew_mean	294				
exploration rate	0.572				
time/					
episodes	20400				
fps	257				
time_elapsed	4991				
total timesteps	1283816				
train/					
learning_rate	0.0001				
loss	0.307				
n_updates	308453				

EP=6,880	STEP=1,284,793	clip=40	consec=2	max_consec=12	avg_clip50=25.58
EP=6,890	STEP=1,287,173	clip=64	consec=2	max_consec=12	avg_clip50=27.42
EP=6,900	STEP=1,289,588	clip=62	consec=2	max_consec=12	avg_clip50=29.52

custom/					
avg_clip50	32.7				
clipped_episode_r...	46.5				
consecutive_clipp...	2				
max_consecutive_c...	12				
rollout/					
ep_len_mean	922				
ep_rew_mean	294				
exploration rate	0.57				
time/					
episodes	20500				
fps	257				
time_elapsed	5025				
total timesteps	1292154				
train/					
learning_rate	0.0001				
loss	0.0732				
n_updates	310538				

EP=6,910	STEP=1,292,350	clip=38	consec=3	max_consec=12	avg_clip50=32.74
EP=6,920	STEP=1,294,403	clip=16	consec=0	max_consec=12	avg_clip50=32.82
EP=6,930	STEP=1,297,102	clip=59	consec=7	max_consec=12	avg_clip50=34.83
EP=6,940	STEP=1,299,508	clip=28	consec=8	max_consec=12	avg_clip50=34.68

custom/					

avg_clip50	34
clipped_episode_r...	12
consecutive_clipp...	0
max_consecutive_c...	12
rollout/	
ep_len_mean	958
ep_rew_mean	311
exploration rate	0.567
time/	
episodes	20600
fps	257
time_elapsed	5057
total timesteps	1300022
train/	
learning_rate	0.0001
loss	0.554
n_updates	312505

EP=6,950	STEP=1,301,273	clip=11	consec=0	max_consec=12	avg_clip50=31.81
EP=6,960	STEP=1,303,388	clip=16	consec=0	max_consec=12	avg_clip50=28.99
EP=6,970	STEP=1,305,616	clip=22	consec=2	max_consec=12	avg_clip50=30.14

custom/	
avg_clip50	30.2
clipped_episode_r...	12
consecutive_clipp...	0
max_consecutive_c...	12
rollout/	
ep_len_mean	967
ep_rew_mean	319
exploration rate	0.565
time/	
episodes	20700
fps	256
time_elapsed	5085
total timesteps	1306866
train/	
learning_rate	0.0001
loss	0.0983
n_updates	314216

EP=6,980	STEP=1,307,732	clip=5	consec=0	max_consec=12	avg_clip50=28.07
EP=6,990	STEP=1,309,653	clip=22	consec=2	max_consec=12	avg_clip50=26.01
EP=7,000	STEP=1,311,658	clip=2	consec=0	max_consec=12	avg_clip50=27.34
EP=7,010	STEP=1,313,841	clip=32	consec=2	max_consec=12	avg_clip50=27.10

custom/	
avg_clip50	27.1
clipped_episode_r...	32
consecutive_clipp...	2
max_consecutive_c...	12
rollout/	
ep_len_mean	906
ep_rew_mean	280
exploration rate	0.562
time/	
episodes	20800
fps	256
time_elapsed	5114
total timesteps	1313904

train/	
learning_rate	0.0001
loss	0.419
n_updates	315975

EP=7,020	STEP=1,316,057	clip=26	consec=5	max_consec=12	avg_clip50=26.57
EP=7,030	STEP=1,318,434	clip=14	consec=0	max_consec=12	avg_clip50=26.16
EP=7,040	STEP=1,320,876	clip=38	consec=1	max_consec=12	avg_clip50=28.76

custom/	
avg_clip50	26.8
clipped_episode_r...	15
consecutive_clipp...	0
max_consecutive_c...	12
rollout/	
ep_len_mean	901
ep_rew_mean	273
exploration rate	0.56
time/	
episodes	20900
fps	256
time_elapsed	5145
total timesteps	1321661
train/	
learning_rate	0.0001
loss	0.0516
n_updates	317915

EP=7,050	STEP=1,323,275	clip=16	consec=0	max_consec=12	avg_clip50=30.20
EP=7,060	STEP=1,325,765	clip=55	consec=2	max_consec=12	avg_clip50=31.64
EP=7,070	STEP=1,328,094	clip=21	consec=3	max_consec=12	avg_clip50=32.05

custom/	
avg_clip50	33.1
clipped_episode_r...	16
consecutive_clipp...	0
max_consecutive_c...	12
rollout/	
ep_len_mean	953
ep_rew_mean	294
exploration rate	0.557
time/	
episodes	21000
fps	256
time_elapsed	5179
total timesteps	1330055
train/	
learning_rate	0.0001
loss	0.0728
n_updates	320013

EP=7,080	STEP=1,330,421	clip=26	consec=2	max_consec=12	avg_clip50=33.02
EP=7,090	STEP=1,332,669	clip=50	consec=2	max_consec=12	avg_clip50=32.46
EP=7,100	STEP=1,335,093	clip=41	consec=1	max_consec=12	avg_clip50=32.84
EP=7,110	STEP=1,337,438	clip=35	consec=1	max_consec=12	avg_clip50=32.15

custom/	
avg_clip50	32
clipped_episode_r...	12.5
consecutive_clipp...	0

max_consecutive_c...	12
rollout/	
ep_len_mean	983
ep_rew_mean	316
exploration rate	0.555
time/	
episodes	21100
fps	256
time_elapsed	5210
total timesteps	1337634
train/	
learning_rate	0.0001
loss	0.488
n_updates	321908

EP=7,120	STEP=1,339,849	clip=49	consec=1	max_consec=12	avg_clip50=32.52
EP=7,130	STEP=1,342,021	clip=30	consec=3	max_consec=12	avg_clip50=32.53
EP=7,140	STEP=1,344,723	clip=34	consec=5	max_consec=12	avg_clip50=33.41

custom/	
avg_clip50	33.5
clipped_episode_r...	42
consecutive_clipp...	1
max_consecutive_c...	12
rollout/	
ep_len_mean	1.01e+03
ep_rew_mean	335
exploration rate	0.552
time/	
episodes	21200
fps	256
time_elapsed	5244
total timesteps	1346006
train/	
learning_rate	0.0001
loss	0.104
n_updates	324001

EP=7,150	STEP=1,347,156	clip=22	consec=4	max_consec=12	avg_clip50=33.78
EP=7,160	STEP=1,348,873	clip=12	consec=0	max_consec=12	avg_clip50=31.60
EP=7,170	STEP=1,350,729	clip=22	consec=4	max_consec=12	avg_clip50=29.37

custom/	
avg_clip50	30.5
clipped_episode_r...	14.5
consecutive_clipp...	0
max_consecutive_c...	12
rollout/	
ep_len_mean	946
ep_rew_mean	310
exploration rate	0.55
time/	
episodes	21300
fps	256
time_elapsed	5272
total timesteps	1352674
train/	
learning_rate	0.0001
loss	0.234
n_updates	325668

```

-----
EP=7,180 | STEP=1,352,832 | clip=8 | consec=0 | max_consec=12 | avg_clip50=28.73
EP=7,190 | STEP=1,355,404 | clip=24 | consec=2 | max_consec=12 | avg_clip50=27.87
EP=7,200 | STEP=1,357,124 | clip=3 | consec=0 | max_consec=12 | avg_clip50=24.52
EP=7,210 | STEP=1,359,030 | clip=4 | consec=0 | max_consec=12 | avg_clip50=25.32

```

```

-----
| custom/
|   avg_clip50          | 25.3
|   clipped_episode_r... | 17.5
|   consecutive_clipp... | 0
|   max_consecutive_c... | 12
| rollout/
|   ep_len_mean         | 912
|   ep_rew_mean         | 286
|   exploration rate     | 0.547
| time/
|   episodes            | 21400
|   fps                 | 256
|   time_elapsed        | 5300
|   total timesteps     | 1359541
| train/
|   learning_rate       | 0.0001
|   loss                | 0.219
|   n_updates           | 327385

```

```

-----
EP=7,220 | STEP=1,361,469 | clip=47 | consec=3 | max_consec=12 | avg_clip50=27.38
EP=7,230 | STEP=1,363,420 | clip=6 | consec=0 | max_consec=12 | avg_clip50=26.95
EP=7,240 | STEP=1,365,632 | clip=70 | consec=4 | max_consec=12 | avg_clip50=26.69

```

```

-----
| custom/
|   avg_clip50          | 27.6
|   clipped_episode_r... | 49
|   consecutive_clipp... | 1
|   max_consecutive_c... | 12
| rollout/
|   ep_len_mean         | 872
|   ep_rew_mean         | 271
|   exploration rate     | 0.545
| time/
|   episodes            | 21500
|   fps                 | 256
|   time_elapsed        | 5330
|   total timesteps     | 1366777
| train/
|   learning_rate       | 0.0001
|   loss                | 0.0694
|   n_updates           | 329194

```

```

-----
EP=7,250 | STEP=1,367,861 | clip=26 | consec=2 | max_consec=12 | avg_clip50=28.35
EP=7,260 | STEP=1,369,979 | clip=13 | consec=0 | max_consec=12 | avg_clip50=29.40
EP=7,270 | STEP=1,372,295 | clip=41 | consec=1 | max_consec=12 | avg_clip50=28.98

```

```

-----
| custom/
|   avg_clip50          | 30.1
|   clipped_episode_r... | 57.5
|   consecutive_clipp... | 1
|   max_consecutive_c... | 12
| rollout/
|   ep_len_mean         | 898
|   ep_rew_mean         | 280

```

exploration rate	0.542
time/	
episodes	21600
fps	256
time_elapsed	5359
total timesteps	1374029
train/	
learning_rate	0.0001
loss	0.0553
n_updates	331007

EP=7,280 | STEP=1,374,429 | clip=32 | consec=3 | max_consec=12 | avg_clip50=29.97
EP=7,290 | STEP=1,377,459 | clip=37 | consec=2 | max_consec=12 | avg_clip50=31.35
EP=7,300 | STEP=1,379,682 | clip=22 | consec=1 | max_consec=12 | avg_clip50=32.71
EP=7,310 | STEP=1,382,128 | clip=70 | consec=8 | max_consec=12 | avg_clip50=34.57

custom/	
avg_clip50	35.1
clipped_episode_r...	18.5
consecutive_clipp...	0
max_consecutive_c...	12
rollout/	
ep_len_mean	958
ep_rew_mean	316
exploration rate	0.539
time/	
episodes	21700
fps	256
time_elapsed	5396
total timesteps	1383008
train/	
learning_rate	0.0001
loss	0.319
n_updates	333251

EP=7,320 | STEP=1,384,575 | clip=32 | consec=6 | max_consec=12 | avg_clip50=36.00
EP=7,330 | STEP=1,386,849 | clip=12 | consec=0 | max_consec=12 | avg_clip50=36.21
EP=7,340 | STEP=1,389,344 | clip=66 | consec=3 | max_consec=12 | avg_clip50=36.51

custom/	
avg_clip50	37.6
clipped_episode_r...	26.5
consecutive_clipp...	11
max_consecutive_c...	12
rollout/	
ep_len_mean	1.02e+03
ep_rew_mean	356
exploration rate	0.537
time/	
episodes	21800
fps	256
time_elapsed	5431
total timesteps	1391651
train/	
learning_rate	0.0001
loss	0.0579
n_updates	335412

EP=7,350 | STEP=1,392,126 | clip=36 | consec=1 | max_consec=12 | avg_clip50=37.87
EP=7,360 | STEP=1,394,384 | clip=47 | consec=3 | max_consec=12 | avg_clip50=36.80

EP=7,370	STEP=1,396,422	clip=16	consec=0	max_consec=12	avg_clip50=35.21
EP=7,380	STEP=1,398,358	clip=22	consec=1	max_consec=12	avg_clip50=34.52

custom/	
avg_clip50	34.2
clipped_episode_r...	31
consecutive_clipp...	2
max_consecutive_c...	12
rollout/	
ep_len_mean	998
ep_rew_mean	351
exploration rate	0.534
time/	
episodes	21900
fps	256
time_elapsed	5460
total timesteps	1398683
train/	
learning_rate	0.0001
loss	0.503
n_updates	337170

EP=7,390	STEP=1,400,967	clip=68	consec=7	max_consec=12	avg_clip50=34.48
EP=7,400	STEP=1,403,107	clip=48	consec=1	max_consec=12	avg_clip50=32.00
EP=7,410	STEP=1,405,496	clip=43	consec=4	max_consec=12	avg_clip50=33.03

custom/	
avg_clip50	33.2
clipped_episode_r...	51.5
consecutive_clipp...	1
max_consecutive_c...	12
rollout/	
ep_len_mean	983
ep_rew_mean	349
exploration rate	0.532
time/	
episodes	22000
fps	256
time_elapsed	5492
total timesteps	1406541
train/	
learning_rate	0.0001
loss	0.326
n_updates	339135

EP=7,420	STEP=1,408,176	clip=44	consec=3	max_consec=12	avg_clip50=35.07
EP=7,430	STEP=1,410,197	clip=14	consec=0	max_consec=12	avg_clip50=34.14
EP=7,440	STEP=1,411,833	clip=23	consec=1	max_consec=12	avg_clip50=29.92

custom/	
avg_clip50	29.6
clipped_episode_r...	17
consecutive_clipp...	0
max_consecutive_c...	12
rollout/	
ep_len_mean	923
ep_rew_mean	307
exploration rate	0.529
time/	
episodes	22100

fps	255				
time_elapsed	5522				
total timesteps	1413565				
train/					
learning_rate	0.0001				
loss	0.102				
n_updates	340891				

EP=7,450	STEP=1,413,942	clip=17	consec=0	max_consec=12	avg_clip50=29.09
EP=7,460	STEP=1,415,791	clip=25	consec=2	max_consec=12	avg_clip50=26.30
EP=7,470	STEP=1,418,494	clip=4	consec=0	max_consec=12	avg_clip50=27.47
EP=7,480	STEP=1,420,448	clip=28	consec=3	max_consec=12	avg_clip50=27.90

custom/					
avg_clip50	28.1				
clipped_episode_r...	25.5				
consecutive_clipp...	4				
max_consecutive_c...	12				
rollout/					
ep_len_mean	929				
ep_rew_mean	310				
exploration rate	0.527				
time/					
episodes	22200				
fps	255				
time_elapsed	5552				
total timesteps	1420830				
train/					
learning_rate	0.0001				
loss	0.0735				
n_updates	342707				

EP=7,490	STEP=1,422,757	clip=36	consec=1	max_consec=12	avg_clip50=30.10
EP=7,500	STEP=1,424,832	clip=42	consec=1	max_consec=12	avg_clip50=30.15
EP=7,510	STEP=1,426,411	clip=36	consec=4	max_consec=12	avg_clip50=29.09

custom/					
avg_clip50	28.7				
clipped_episode_r...	92.5				
consecutive_clipp...	1				
max_consecutive_c...	12				
rollout/					
ep_len_mean	881				
ep_rew_mean	283				
exploration rate	0.525				
time/					
episodes	22300				
fps	255				
time_elapsed	5579				
total timesteps	1427325				
train/					
learning_rate	0.0001				
loss	0.0337				
n_updates	344331				

EP=7,520	STEP=1,428,836	clip=49	consec=2	max_consec=12	avg_clip50=28.39
EP=7,530	STEP=1,431,371	clip=8	consec=0	max_consec=12	avg_clip50=29.99
EP=7,540	STEP=1,433,378	clip=8	consec=0	max_consec=12	avg_clip50=28.53

custom/					

avg_clip50	29.2
clipped_episode_r...	32.5
consecutive_clipp...	3
max_consecutive_c...	12
rollout/	
ep_len_mean	913
ep_rew_mean	295
exploration rate	0.522
time/	
episodes	22400
fps	255
time_elapsed	5611
total timesteps	1435287
train/	
learning_rate	0.0001
loss	0.276
n_updates	346321

EP=7,550	STEP=1,435,779	clip=70	consec=5	max_consec=12	avg_clip50=30.69
EP=7,560	STEP=1,437,935	clip=6	consec=0	max_consec=12	avg_clip50=34.06
EP=7,570	STEP=1,439,969	clip=14	consec=0	max_consec=12	avg_clip50=30.62
EP=7,580	STEP=1,441,978	clip=3	consec=0	max_consec=12	avg_clip50=29.86

custom/	
avg_clip50	29.5
clipped_episode_r...	33
consecutive_clipp...	1
max_consecutive_c...	12
rollout/	
ep_len_mean	904
ep_rew_mean	295
exploration rate	0.52
time/	
episodes	22500
fps	255
time_elapsed	5641
total timesteps	1442551
train/	
learning_rate	0.0001
loss	0.0684
n_updates	348137

EP=7,590	STEP=1,443,969	clip=16	consec=0	max_consec=12	avg_clip50=30.50
EP=7,600	STEP=1,446,564	clip=19	consec=0	max_consec=12	avg_clip50=31.47
EP=7,610	STEP=1,448,455	clip=10	consec=0	max_consec=12	avg_clip50=29.43

custom/	
avg_clip50	30.5
clipped_episode_r...	2
consecutive_clipp...	0
max_consecutive_c...	12
rollout/	
ep_len_mean	928
ep_rew_mean	310
exploration rate	0.517
time/	
episodes	22600
fps	255
time_elapsed	5672
total timesteps	1449889

train/						
learning_rate		0.0001				
loss		0.73				
n_updates		349972				

EP=7,620	STEP=1,450,792	clip=14	consec=0	max_consec=12	avg_clip50=31.38	
EP=7,630	STEP=1,452,909	clip=11	consec=0	max_consec=12	avg_clip50=30.83	
EP=7,640	STEP=1,455,572	clip=50	consec=2	max_consec=12	avg_clip50=33.85	
EP=7,650	STEP=1,457,839	clip=20	consec=0	max_consec=12	avg_clip50=31.91	

custom/						
avg_clip50		31.9				
clipped_episode_r...		19.5				
consecutive_clipp...		0				
max_consecutive_c...		12				
rollout/						
ep_len_mean		929				
ep_rew_mean		317				
exploration rate		0.515				
time/						
episodes		22700				
fps		255				
time_elapsed		5704				
total timesteps		1457839				
train/						
learning_rate		0.0001				
loss		0.0446				
n_updates		351959				

EP=7,660	STEP=1,460,284	clip=9	consec=0	max_consec=12	avg_clip50=33.58	
EP=7,670	STEP=1,462,609	clip=26	consec=6	max_consec=12	avg_clip50=33.37	
EP=7,680	STEP=1,465,013	clip=14	consec=0	max_consec=12	avg_clip50=34.75	

custom/						
avg_clip50		33.4				
clipped_episode_r...		33.5				
consecutive_clipp...		1				
max_consecutive_c...		12				
rollout/						
ep_len_mean		970				
ep_rew_mean		329				
exploration rate		0.512				
time/						
episodes		22800				
fps		255				
time_elapsed		5737				
total timesteps		1465880				
train/						
learning_rate		0.0001				
loss		0.0728				
n_updates		353969				

EP=7,690	STEP=1,467,732	clip=39	consec=3	max_consec=12	avg_clip50=35.04	
EP=7,700	STEP=1,469,834	clip=10	consec=0	max_consec=12	avg_clip50=34.57	
EP=7,710	STEP=1,472,282	clip=19	consec=0	max_consec=12	avg_clip50=35.25	

custom/						
avg_clip50		35.5				
clipped_episode_r...		39.5				
consecutive_clipp...		1				

max_consecutive_c...	12
rollout/	
ep_len_mean	994
ep_rew_mean	342
exploration rate	0.509
time/	
episodes	22900
fps	255
time_elapsed	5771
total timesteps	1474131
train/	
learning_rate	0.0001
loss	0.0411
n_updates	356032

EP=7,720	STEP=1,474,493	clip=10	consec=0	max_consec=12	avg_clip50=34.85
EP=7,730	STEP=1,476,863	clip=60	consec=1	max_consec=12	avg_clip50=35.45
EP=7,740	STEP=1,479,144	clip=13	consec=0	max_consec=12	avg_clip50=35.30
EP=7,750	STEP=1,481,296	clip=44	consec=2	max_consec=12	avg_clip50=35.07

custom/	
avg_clip50	35.5
clipped_episode_r...	39.5
consecutive_clipp...	3
max_consecutive_c...	12
rollout/	
ep_len_mean	975
ep_rew_mean	347
exploration rate	0.507
time/	
episodes	23000
fps	255
time_elapsed	5802
total timesteps	1481678
train/	
learning_rate	0.0001
loss	0.278
n_updates	357919

EP=7,760	STEP=1,483,445	clip=47	consec=2	max_consec=12	avg_clip50=34.12
EP=7,770	STEP=1,485,695	clip=45	consec=2	max_consec=12	avg_clip50=34.91
EP=7,780	STEP=1,487,931	clip=25	consec=5	max_consec=12	avg_clip50=34.52

custom/	
avg_clip50	34.9
clipped_episode_r...	29.5
consecutive_clipp...	3
max_consecutive_c...	12
rollout/	
ep_len_mean	960
ep_rew_mean	346
exploration rate	0.504
time/	
episodes	23100
fps	255
time_elapsed	5834
total timesteps	1489420
train/	
learning_rate	0.0001
loss	0.0366

n_updates		359854			
EP=7,790	STEP=1,490,257	clip=26	consec=7	max_consec=12	avg_clip50=33.09
EP=7,800	STEP=1,492,802	clip=20	consec=0	max_consec=12	avg_clip50=33.53
EP=7,810	STEP=1,495,576	clip=53	consec=4	max_consec=12	avg_clip50=35.68
EP=7,820	STEP=1,497,876	clip=28	consec=1	max_consec=12	avg_clip50=35.79
custom/					
avg_clip50		35.8			
clipped_episode_r...		38.5			
consecutive_clipp...		2			
max_consecutive_c...		12			
rollout/					
ep_len_mean		981			
ep_rew_mean		354			
exploration rate		0.501			
time/					
episodes		23200			
fps		255			
time_elapsed		5870			
total timesteps		1498190			
train/					
learning_rate		0.0001			
loss		0.0648			
n_updates		362047			
EP=7,830	STEP=1,500,176	clip=37	consec=2	max_consec=12	avg_clip50=35.71
EP=7,840	STEP=1,502,343	clip=11	consec=0	max_consec=12	avg_clip50=35.38
EP=7,850	STEP=1,505,345	clip=49	consec=1	max_consec=12	avg_clip50=39.26
custom/					
avg_clip50		38.6			
clipped_episode_r...		35.5			
consecutive_clipp...		2			
max_consecutive_c...		12			
rollout/					
ep_len_mean		1.01e+03			
ep_rew_mean		364			
exploration rate		0.498			
time/					
episodes		23300			
fps		255			
time_elapsed		5905			
total timesteps		1506561			
train/					
learning_rate		0.0001			
loss		0.434			
n_updates		364140			
EP=7,860	STEP=1,507,663	clip=35	consec=3	max_consec=12	avg_clip50=37.06
EP=7,870	STEP=1,510,122	clip=48	consec=3	max_consec=12	avg_clip50=36.96
EP=7,880	STEP=1,512,505	clip=39	consec=1	max_consec=12	avg_clip50=36.43
custom/					
avg_clip50		37.6			
clipped_episode_r...		77			
consecutive_clipp...		2			
max_consecutive_c...		12			
rollout/					
ep_len_mean		1.03e+03			

ep_rew_mean	366
exploration rate	0.496
time/	
episodes	23400
fps	255
time_elapsed	5939
total timesteps	1514830
train/	
learning_rate	0.0001
loss	0.358
n_updates	366207

EP=7,890 | STEP=1,514,990 | clip=32 | consec=3 | max_consec=12 | avg_clip50=37.99
EP=7,900 | STEP=1,517,787 | clip=44 | consec=5 | max_consec=12 | avg_clip50=36.68
EP=7,910 | STEP=1,520,436 | clip=50 | consec=3 | max_consec=12 | avg_clip50=37.86
EP=7,920 | STEP=1,522,831 | clip=12 | consec=0 | max_consec=12 | avg_clip50=38.07

custom/	
avg_clip50	36.5
clipped_episode_r...	9.5
consecutive_clipp...	0
max_consecutive_c...	12
rollout/	
ep_len_mean	1.05e+03
ep_rew_mean	371
exploration rate	0.493
time/	
episodes	23500
fps	254
time_elapsed	5976
total timesteps	1523985
train/	
learning_rate	0.0001
loss	0.0515
n_updates	368496

EP=7,930 | STEP=1,525,147 | clip=69 | consec=2 | max_consec=12 | avg_clip50=37.89
EP=7,940 | STEP=1,527,758 | clip=34 | consec=2 | max_consec=12 | avg_clip50=38.08
EP=7,950 | STEP=1,529,968 | clip=13 | consec=0 | max_consec=12 | avg_clip50=35.62

custom/	
avg_clip50	33.4
clipped_episode_r...	22
consecutive_clipp...	1
max_consecutive_c...	12
rollout/	
ep_len_mean	1.02e+03
ep_rew_mean	356
exploration rate	0.49
time/	
episodes	23600
fps	254
time_elapsed	6007
total timesteps	1531532
train/	
learning_rate	0.0001
loss	0.0629
n_updates	370382

EP=7,960 | STEP=1,532,084 | clip=50 | consec=3 | max_consec=12 | avg_clip50=33.46

EP=7,970 | STEP=1,534,498 | clip=66 | consec=7 | max_consec=12 | avg_clip50=33.53
EP=7,980 | STEP=1,536,970 | clip=32 | consec=5 | max_consec=12 | avg_clip50=34.15
EP=7,990 | STEP=1,539,482 | clip=48 | consec=15 | max_consec=15 | avg_clip50=34.45

custom/	
avg_clip50	33.8
clipped_episode_r...	49
consecutive_clipp...	1
max_consecutive_c...	15
rollout/	
ep_len_mean	1.03e+03
ep_rew_mean	363
exploration rate	0.487
time/	
episodes	23700
fps	254
time_elapsed	6044
total timesteps	1540264
train/	
learning_rate	0.0001
loss	0.326
n_updates	372565

EP=8,000 | STEP=1,541,786 | clip=38 | consec=1 | max_consec=15 | avg_clip50=34.34
EP=8,010 | STEP=1,544,212 | clip=37 | consec=2 | max_consec=15 | avg_clip50=35.70
EP=8,020 | STEP=1,546,362 | clip=68 | consec=4 | max_consec=15 | avg_clip50=34.74

custom/	
avg_clip50	36
clipped_episode_r...	32
consecutive_clipp...	10
max_consecutive_c...	15
rollout/	
ep_len_mean	982
ep_rew_mean	350
exploration rate	0.485
time/	
episodes	23800
fps	254
time_elapsed	6075
total timesteps	1547827
train/	
learning_rate	0.0001
loss	0.0594
n_updates	374456

EP=8,030 | STEP=1,548,908 | clip=42 | consec=14 | max_consec=15 | avg_clip50=36.88
EP=8,040 | STEP=1,551,768 | clip=56 | consec=8 | max_consec=15 | avg_clip50=37.58
EP=8,050 | STEP=1,554,126 | clip=55 | consec=2 | max_consec=15 | avg_clip50=38.00
EP=8,060 | STEP=1,556,626 | clip=26 | consec=1 | max_consec=15 | avg_clip50=39.18

custom/	
avg_clip50	39.2
clipped_episode_r...	26
consecutive_clipp...	1
max_consecutive_c...	15
rollout/	
ep_len_mean	1.03e+03

ep_rew_mean	374
exploration rate	0.482
time/	
episodes	23900
fps	254
time_elapsed	6111
total timesteps	1556671
train/	
learning_rate	0.0001
loss	0.028
n_updates	376667

EP=8,070 | STEP=1,559,615 | clip=100 | consec=3 | max_consec=15 | avg_clip50=42.78

EP=8,080 | STEP=1,562,051 | clip=8 | consec=0 | max_consec=15 | avg_clip50=41.09

EP=8,090 | STEP=1,564,600 | clip=13 | consec=0 | max_consec=15 | avg_clip50=39.16

custom/	
avg_clip50	39.1
clipped_episode_r...	26.5
consecutive_clipp...	1
max_consecutive_c...	15
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	377
exploration rate	0.479
time/	
episodes	24000
fps	254
time_elapsed	6148
total timesteps	1565599
train/	
learning_rate	0.0001
loss	0.0602
n_updates	378899

EP=8,100 | STEP=1,566,920 | clip=46 | consec=3 | max_consec=15 | avg_clip50=39.49

EP=8,110 | STEP=1,569,706 | clip=68 | consec=9 | max_consec=15 | avg_clip50=42.19

EP=8,120 | STEP=1,572,181 | clip=57 | consec=4 | max_consec=15 | avg_clip50=40.16

EP=8,130 | STEP=1,574,750 | clip=38 | consec=5 | max_consec=15 | avg_clip50=39.95

custom/	
avg_clip50	40
clipped_episode_r...	38
consecutive_clipp...	5
max_consecutive_c...	15
rollout/	
ep_len_mean	1.08e+03
ep_rew_mean	405
exploration rate	0.476
time/	
episodes	24100
fps	254
time_elapsed	6186
total timesteps	1574750
train/	
learning_rate	0.0001
loss	0.283
n_updates	381187

EP=8,140	STEP=1,577,142	clip=6	consec=0	max_consec=15	avg_clip50=39.87
EP=8,150	STEP=1,580,048	clip=24	consec=3	max_consec=15	avg_clip50=42.33
EP=8,160	STEP=1,582,626	clip=34	consec=4	max_consec=15	avg_clip50=39.29

custom/	
avg_clip50	37.9
clipped_episode_r...	23
consecutive_clipp...	2
max_consecutive_c...	15
rollout/	
ep_len_mean	1.06e+03
ep_rew_mean	399
exploration rate	0.473
time/	
episodes	24200
fps	254
time_elapsed	6221
total timesteps	1583429
train/	
learning_rate	0.0001
loss	0.207
n_updates	383357

EP=8,170	STEP=1,584,972	clip=52	consec=2	max_consec=15	avg_clip50=37.83
EP=8,180	STEP=1,587,351	clip=43	consec=1	max_consec=15	avg_clip50=37.79
EP=8,190	STEP=1,590,303	clip=44	consec=11	max_consec=15	avg_clip50=39.1

2

custom/	
avg_clip50	37.3
clipped_episode_r...	12
consecutive_clipp...	0
max_consecutive_c...	15
rollout/	
ep_len_mean	1.08e+03
ep_rew_mean	400
exploration rate	0.47
time/	
episodes	24300
fps	254
time_elapsed	6256
total timesteps	1592023
train/	
learning_rate	0.0001
loss	0.149
n_updates	385505

EP=8,200	STEP=1,592,748	clip=52	consec=3	max_consec=15	avg_clip50=37.32
EP=8,210	STEP=1,595,025	clip=21	consec=1	max_consec=15	avg_clip50=37.65
EP=8,220	STEP=1,597,290	clip=27	consec=1	max_consec=15	avg_clip50=37.16
EP=8,230	STEP=1,600,045	clip=28	consec=8	max_consec=15	avg_clip50=38.23

custom/	
avg_clip50	38.5
clipped_episode_r...	50
consecutive_clipp...	9
max_consecutive_c...	15
rollout/	
ep_len_mean	1.06e+03
ep_rew_mean	384

exploration rate	0.467
time/	
episodes	24400
fps	254
time_elapsed	6291
total timesteps	1600481
train/	
learning_rate	0.0001
loss	0.404
n_updates	387620

EP=8,240 | STEP=1,602,287 | clip=30 | consec=4 | max_consec=15 | avg_clip50=37.14
EP=8,250 | STEP=1,605,219 | clip=24 | consec=14 | max_consec=15 | avg_clip50=39.0
2
EP=8,260 | STEP=1,607,894 | clip=28 | consec=3 | max_consec=20 | avg_clip50=39.02

custom/	
avg_clip50	37.1
clipped_episode_r...	19.5
consecutive_clipp...	0
max_consecutive_c...	20
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	377
exploration rate	0.464
time/	
episodes	24500
fps	254
time_elapsed	6325
total timesteps	1608706
train/	
learning_rate	0.0001
loss	0.0379
n_updates	389676

EP=8,270 | STEP=1,610,171 | clip=50 | consec=5 | max_consec=20 | avg_clip50=39.48
EP=8,280 | STEP=1,612,824 | clip=74 | consec=1 | max_consec=20 | avg_clip50=40.64
EP=8,290 | STEP=1,615,562 | clip=47 | consec=11 | max_consec=20 | avg_clip50=42.1
5

custom/	
avg_clip50	39.5
clipped_episode_r...	29.5
consecutive_clipp...	3
max_consecutive_c...	20
rollout/	
ep_len_mean	1.06e+03
ep_rew_mean	396
exploration rate	0.461
time/	
episodes	24600
fps	254
time_elapsed	6362
total timesteps	1617791
train/	
learning_rate	0.0001
loss	0.114
n_updates	391947

EP=8,300 | STEP=1,617,872 | clip=12 | consec=0 | max_consec=20 | avg_clip50=39.28

EP=8,310 | STEP=1,620,338 | clip=20 | consec=0 | max_consec=20 | avg_clip50=38.69
EP=8,320 | STEP=1,622,476 | clip=6 | consec=0 | max_consec=20 | avg_clip50=38.51
EP=8,330 | STEP=1,625,304 | clip=60 | consec=7 | max_consec=20 | avg_clip50=38.64

custom/	
avg_clip50	39
clipped_episode_r...	54.5
consecutive_clipp...	10
max_consecutive_c...	20
rollout/	
ep_len_mean	1.07e+03
ep_rew_mean	398
exploration rate	0.458
time/	
episodes	24700
fps	254
time_elapsed	6398
total timesteps	1626402
train/	
learning_rate	0.0001
loss	0.0231
n_updates	394100

EP=8,340 | STEP=1,628,170 | clip=44 | consec=17 | max_consec=20 | avg_clip50=39.45
EP=8,350 | STEP=1,631,058 | clip=48 | consec=5 | max_consec=20 | avg_clip50=41.58
EP=8,360 | STEP=1,633,599 | clip=44 | consec=7 | max_consec=20 | avg_clip50=41.89

custom/	
avg_clip50	42.7
clipped_episode_r...	8
consecutive_clipp...	0
max_consecutive_c...	20
rollout/	
ep_len_mean	1.08e+03
ep_rew_mean	409
exploration rate	0.455
time/	
episodes	24800
fps	254
time_elapsed	6436
total timesteps	1635472
train/	
learning_rate	0.0001
loss	0.0785
n_updates	396367

EP=8,370 | STEP=1,636,093 | clip=35 | consec=3 | max_consec=20 | avg_clip50=43.78
EP=8,380 | STEP=1,638,446 | clip=26 | consec=1 | max_consec=20 | avg_clip50=40.60
EP=8,390 | STEP=1,641,197 | clip=60 | consec=8 | max_consec=20 | avg_clip50=39.29
EP=8,400 | STEP=1,643,565 | clip=46 | consec=2 | max_consec=20 | avg_clip50=37.71

custom/	
avg_clip50	36.2
clipped_episode_r...	12
consecutive_clipp...	0
max_consecutive_c...	20
rollout/	
ep_len_mean	1.06e+03
ep_rew_mean	389

exploration rate	0.453
time/	
episodes	24900
fps	254
time_elapsed	6470
total timesteps	1643778
train/	
learning_rate	0.0001
loss	0.0509
n_updates	398444

EP=8,410	STEP=1,645,856	clip=10	consec=0	max_consec=20	avg_clip50=36.29
EP=8,420	STEP=1,648,442	clip=26	consec=8	max_consec=20	avg_clip50=36.99
EP=8,430	STEP=1,651,180	clip=58	consec=8	max_consec=20	avg_clip50=40.34

custom/	
avg_clip50	39.5
clipped_episode_r...	25
consecutive_clipp...	13
max_consecutive_c...	20
rollout/	
ep_len_mean	1.07e+03
ep_rew_mean	398
exploration rate	0.45
time/	
episodes	25000
fps	253
time_elapsed	6506
total timesteps	1652404
train/	
learning_rate	0.0001
loss	0.457
n_updates	400600

EP=8,440	STEP=1,653,809	clip=42	consec=4	max_consec=20	avg_clip50=39.64
EP=8,450	STEP=1,656,119	clip=6	consec=0	max_consec=20	avg_clip50=38.56
EP=8,460	STEP=1,658,581	clip=27	consec=3	max_consec=20	avg_clip50=39.63

custom/	
avg_clip50	37.3
clipped_episode_r...	32.5
consecutive_clipp...	2
max_consecutive_c...	20
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	378
exploration rate	0.447
time/	
episodes	25100
fps	253
time_elapsed	6540
total timesteps	1660585
train/	
learning_rate	0.0001
loss	0.0542
n_updates	402646

EP=8,470	STEP=1,661,015	clip=76	consec=4	max_consec=20	avg_clip50=38.18
EP=8,480	STEP=1,663,851	clip=52	consec=6	max_consec=20	avg_clip50=36.54
EP=8,490	STEP=1,666,378	clip=44	consec=2	max_consec=20	avg_clip50=35.93

EP=8,500 | STEP=1,669,256 | clip=54 | consec=7 | max_consec=20 | avg_clip50=38.82

custom/	
avg_clip50	37
clipped_episode_r...	46.5
consecutive_clipp...	1
max_consecutive_c...	20
rollout/	
ep_len_mean	1.09e+03
ep_rew_mean	391
exploration rate	0.444
time/	
episodes	25200
fps	253
time_elapsed	6578
total timesteps	1669943
train/	
learning_rate	0.0001
loss	0.0571
n_updates	404985

EP=8,510 | STEP=1,671,700 | clip=52 | consec=4 | max_consec=20 | avg_clip50=38.40

EP=8,520 | STEP=1,673,836 | clip=4 | consec=0 | max_consec=20 | avg_clip50=36.73

EP=8,530 | STEP=1,676,275 | clip=20 | consec=0 | max_consec=20 | avg_clip50=35.75

custom/	
avg_clip50	36.1
clipped_episode_r...	21.5
consecutive_clipp...	1
max_consecutive_c...	20
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	360
exploration rate	0.441
time/	
episodes	25300
fps	253
time_elapsed	6613
total timesteps	1678192
train/	
learning_rate	0.0001
loss	0.0993
n_updates	407047

EP=8,540 | STEP=1,678,762 | clip=57 | consec=3 | max_consec=20 | avg_clip50=36.15

EP=8,550 | STEP=1,680,835 | clip=12 | consec=0 | max_consec=20 | avg_clip50=33.28

EP=8,560 | STEP=1,683,745 | clip=30 | consec=7 | max_consec=20 | avg_clip50=35.94

EP=8,570 | STEP=1,686,260 | clip=44 | consec=1 | max_consec=20 | avg_clip50=36.98

custom/	
avg_clip50	37
clipped_episode_r...	33.5
consecutive_clipp...	2
max_consecutive_c...	20
rollout/	
ep_len_mean	1.06e+03
ep_rew_mean	370
exploration rate	0.438
time/	
episodes	25400

fps	253
time_elapsed	6648
total timesteps	1686607
train/	
learning_rate	0.0001
loss	0.0979
n_updates	409151

EP=8,580 | STEP=1,689,132 | clip=74 | consec=11 | max_consec=20 | avg_clip50=40.05

EP=8,590 | STEP=1,691,581 | clip=46 | consec=1 | max_consec=20 | avg_clip50=39.08

EP=8,600 | STEP=1,694,207 | clip=18 | consec=0 | max_consec=20 | avg_clip50=41.50

custom/	
avg_clip50	41.2
clipped_episode_r...	60
consecutive_clipp...	6
max_consecutive_c...	20
rollout/	
ep_len_mean	1.06e+03
ep_rew_mean	386
exploration rate	0.435
time/	
episodes	25500
fps	253
time_elapsed	6687
total timesteps	1696141
train/	
learning_rate	0.0001
loss	0.185
n_updates	411535

EP=8,610 | STEP=1,697,038 | clip=46 | consec=1 | max_consec=20 | avg_clip50=40.16

EP=8,620 | STEP=1,699,277 | clip=55 | consec=2 | max_consec=20 | avg_clip50=40.17

EP=8,630 | STEP=1,701,640 | clip=46 | consec=7 | max_consec=20 | avg_clip50=36.81

custom/	
avg_clip50	36.8
clipped_episode_r...	8
consecutive_clipp...	0
max_consecutive_c...	20
rollout/	
ep_len_mean	1.05e+03
ep_rew_mean	380
exploration rate	0.433
time/	
episodes	25600
fps	253
time_elapsed	6720
total timesteps	1703968
train/	
learning_rate	0.0001
loss	0.0904
n_updates	413491

EP=8,640 | STEP=1,704,086 | clip=85 | consec=1 | max_consec=20 | avg_clip50=37.55

EP=8,650 | STEP=1,706,975 | clip=72 | consec=11 | max_consec=20 | avg_clip50=38.72

EP=8,660 | STEP=1,709,387 | clip=54 | consec=2 | max_consec=20 | avg_clip50=38.84

EP=8,670 | STEP=1,711,376 | clip=14 | consec=0 | max_consec=20 | avg_clip50=38.69

custom/	
avg_clip50	39
clipped_episode_r...	12
consecutive_clipp...	0
max_consecutive_c...	20
rollout/	
ep_len_mean	1.05e+03
ep_rew_mean	387
exploration rate	0.43
time/	
episodes	25700
fps	253
time_elapsed	6755
total timesteps	1712437
train/	
learning_rate	0.0001
loss	0.0711
n_updates	415609

EP=8,680 | STEP=1,714,001 | clip=52 | consec=6 | max_consec=20 | avg_clip50=39.49
EP=8,690 | STEP=1,716,441 | clip=37 | consec=1 | max_consec=20 | avg_clip50=38.73
EP=8,700 | STEP=1,718,623 | clip=42 | consec=1 | max_consec=20 | avg_clip50=36.45

custom/	
avg_clip50	36.9
clipped_episode_r...	8.5
consecutive_clipp...	0
max_consecutive_c...	20
rollout/	
ep_len_mean	1.02e+03
ep_rew_mean	374
exploration rate	0.427
time/	
episodes	25800
fps	253
time_elapsed	6790
total timesteps	1720807
train/	
learning_rate	0.0001
loss	0.0449
n_updates	417701

EP=8,710 | STEP=1,721,199 | clip=15 | consec=0 | max_consec=20 | avg_clip50=35.66
EP=8,720 | STEP=1,723,739 | clip=72 | consec=6 | max_consec=20 | avg_clip50=36.23
EP=8,730 | STEP=1,726,309 | clip=60 | consec=2 | max_consec=20 | avg_clip50=36.06
EP=8,740 | STEP=1,728,706 | clip=38 | consec=6 | max_consec=20 | avg_clip50=36.62

custom/	
avg_clip50	36
clipped_episode_r...	11
consecutive_clipp...	0
max_consecutive_c...	20
rollout/	
ep_len_mean	1.03e+03
ep_rew_mean	374
exploration rate	0.424
time/	
episodes	25900
fps	253

	time_elapsed		6825	
	total timesteps		1729179	
	train/			
	learning_rate		0.0001	
	loss		0.0307	
	n_updates		419794	

EP=8,750		STEP=1,730,943		clip=18		consec=0		max_consec=20		avg_clip50=35.92
EP=8,760		STEP=1,733,216		clip=78		consec=3		max_consec=20		avg_clip50=35.31
EP=8,770		STEP=1,735,608		clip=32		consec=2		max_consec=20		avg_clip50=35.38

	custom/			
	avg_clip50		34.2	
	clipped_episode_r...		21.5	
	consecutive_clipp...		8	
	max_consecutive_c...		20	
	rollout/			
	ep_len_mean		1.01e+03	
	ep_rew_mean		357	
	exploration rate		0.422	
	time/			
	episodes		26000	
	fps		253	
	time_elapsed		6858	
	total timesteps		1737061	
	train/			
	learning_rate		0.0001	
	loss		0.0705	
	n_updates		421765	

EP=8,780		STEP=1,738,139		clip=28		consec=12		max_consec=20		avg_clip50=36.08
EP=8,790		STEP=1,740,676		clip=36		consec=7		max_consec=20		avg_clip50=37.43
EP=8,800		STEP=1,743,214		clip=26		consec=7		max_consec=20		avg_clip50=38.26

	custom/			
	avg_clip50		39.3	
	clipped_episode_r...		41.5	
	consecutive_clipp...		4	
	max_consecutive_c...		20	
	rollout/			
	ep_len_mean		1.02e+03	
	ep_rew_mean		367	
	exploration rate		0.419	
	time/			
	episodes		26100	
	fps		253	
	time_elapsed		6894	
	total timesteps		1745681	
	train/			
	learning_rate		0.0001	
	loss		0.155	
	n_updates		423920	

EP=8,810		STEP=1,745,773		clip=36		consec=5		max_consec=20		avg_clip50=38.45
EP=8,820		STEP=1,748,333		clip=58		consec=1		max_consec=20		avg_clip50=38.87
EP=8,830		STEP=1,750,847		clip=52		consec=2		max_consec=20		avg_clip50=39.19
EP=8,840		STEP=1,753,233		clip=12		consec=0		max_consec=20		avg_clip50=37.79

	custom/			
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avg_clip50	39
clipped_episode_r...	50
consecutive_clipp...	3
max_consecutive_c...	20
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	386
exploration rate	0.416
time/	
episodes	26200
fps	253
time_elapsed	6930
total timesteps	1754229
train/	
learning_rate	0.0001
loss	0.0829
n_updates	426057

EP=8,850 | STEP=1,755,833 | clip=16 | consec=0 | max_consec=20 | avg_clip50=38.80
EP=8,860 | STEP=1,758,356 | clip=43 | consec=1 | max_consec=20 | avg_clip50=39.59
EP=8,870 | STEP=1,760,841 | clip=26 | consec=11 | max_consec=20 | avg_clip50=40.00

custom/	
avg_clip50	38.7
clipped_episode_r...	38
consecutive_clipp...	3
max_consecutive_c...	20
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	392
exploration rate	0.413
time/	
episodes	26300
fps	253
time_elapsed	6964
total timesteps	1762362
train/	
learning_rate	0.0001
loss	0.1
n_updates	428090

EP=8,880 | STEP=1,763,093 | clip=42 | consec=6 | max_consec=20 | avg_clip50=39.08
EP=8,890 | STEP=1,765,954 | clip=50 | consec=5 | max_consec=20 | avg_clip50=40.93
EP=8,900 | STEP=1,768,177 | clip=55 | consec=2 | max_consec=20 | avg_clip50=38.80
EP=8,910 | STEP=1,770,407 | clip=44 | consec=3 | max_consec=20 | avg_clip50=37.85

custom/	
avg_clip50	38.1
clipped_episode_r...	32
consecutive_clipp...	5
max_consecutive_c...	20
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	389
exploration rate	0.41
time/	
episodes	26400
fps	252
time_elapsed	7000

total timesteps	1770958
train/	
learning_rate	0.0001
loss	0.559
n_updates	430239

EP=8,920 | STEP=1,773,203 | clip=28 | consec=13 | max_consec=20 | avg_clip50=39.84

EP=8,930 | STEP=1,775,854 | clip=56 | consec=2 | max_consec=20 | avg_clip50=40.63

EP=8,940 | STEP=1,777,995 | clip=12 | consec=0 | max_consec=20 | avg_clip50=37.85

custom/	
avg_clip50	39.3
clipped_episode_r...	54
consecutive_clipp...	1
max_consecutive_c...	20
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	392
exploration rate	0.407
time/	
episodes	26500
fps	252
time_elapsed	7037
total timesteps	1779867
train/	
learning_rate	0.0001
loss	0.275
n_updates	432466

EP=8,950 | STEP=1,780,637 | clip=42 | consec=4 | max_consec=20 | avg_clip50=39.78

EP=8,960 | STEP=1,782,870 | clip=9 | consec=0 | max_consec=20 | avg_clip50=39.05

EP=8,970 | STEP=1,785,629 | clip=4 | consec=0 | max_consec=20 | avg_clip50=37.13

EP=8,980 | STEP=1,788,488 | clip=46 | consec=10 | max_consec=20 | avg_clip50=38.55

custom/	
avg_clip50	38.3
clipped_episode_r...	18
consecutive_clipp...	0
max_consecutive_c...	20
rollout/	
ep_len_mean	1.05e+03
ep_rew_mean	392
exploration rate	0.404
time/	
episodes	26600
fps	252
time_elapsed	7075
total timesteps	1788877
train/	
learning_rate	0.0001
loss	0.0194
n_updates	434719

EP=8,990 | STEP=1,790,873 | clip=17 | consec=0 | max_consec=20 | avg_clip50=39.25

EP=9,000 | STEP=1,793,679 | clip=38 | consec=10 | max_consec=20 | avg_clip50=39.98

EP=9,010 | STEP=1,795,973 | clip=38 | consec=8 | max_consec=20 | avg_clip50=40.81

custom/	
avg_clip50	41.3
clipped_episode_r...	49
consecutive_clipp...	14
max_consecutive_c...	20
rollout/	
ep_len_mean	1.07e+03
ep_rew_mean	404
exploration rate	0.401
time/	
episodes	26700
fps	252
time_elapsed	7112
total timesteps	1797759
train/	
learning_rate	0.0001
loss	0.136
n_updates	436939

EP=9,020 | STEP=1,799,006 | clip=40 | consec=1 | max_consec=20 | avg_clip50=41.98
EP=9,030 | STEP=1,801,148 | clip=4 | consec=0 | max_consec=20 | avg_clip50=38.97
EP=9,040 | STEP=1,802,913 | clip=52 | consec=2 | max_consec=20 | avg_clip50=37.87
EP=9,050 | STEP=1,805,606 | clip=34 | consec=12 | max_consec=20 | avg_clip50=38.3
1

custom/	
avg_clip50	38.9
clipped_episode_r...	41.5
consecutive_clipp...	13
max_consecutive_c...	20
rollout/	
ep_len_mean	1.05e+03
ep_rew_mean	393
exploration rate	0.399
time/	
episodes	26800
fps	252
time_elapsed	7146
total timesteps	1805884
train/	
learning_rate	0.0001
loss	0.0669
n_updates	438970

EP=9,060 | STEP=1,807,864 | clip=32 | consec=3 | max_consec=20 | avg_clip50=38.70
EP=9,070 | STEP=1,810,460 | clip=13 | consec=0 | max_consec=20 | avg_clip50=37.34
EP=9,080 | STEP=1,812,520 | clip=16 | consec=0 | max_consec=20 | avg_clip50=36.57

custom/	
avg_clip50	36.6
clipped_episode_r...	19
consecutive_clipp...	0
max_consecutive_c...	20
rollout/	
ep_len_mean	996
ep_rew_mean	370
exploration rate	0.396
time/	
episodes	26900
fps	252

	time_elapsed	7178	
	total timesteps	1813457	
	train/		
	learning_rate	0.0001	
	loss	0.0958	
	n_updates	440864	

EP=9,090		STEP=1,814,651		clip=20		consec=2		max_consec=20		avg_clip50=36.32
EP=9,100		STEP=1,817,024		clip=48		consec=4		max_consec=20		avg_clip50=35.31
EP=9,110		STEP=1,819,456		clip=58		consec=3		max_consec=20		avg_clip50=35.70

	custom/		
	avg_clip50	35.6	
	clipped_episode_r...	52	
	consecutive_clipp...	3	
	max_consecutive_c...	20	
	rollout/		
	ep_len_mean	965	
	ep_rew_mean	367	
	exploration rate	0.393	
	time/		
	episodes	27000	
	fps	252	
	time_elapsed	7213	
	total timesteps	1821797	
	train/		
	learning_rate	0.0001	
	loss	0.0523	
	n_updates	442949	

EP=9,120		STEP=1,821,916		clip=24		consec=4		max_consec=20		avg_clip50=35.78
EP=9,130		STEP=1,824,364		clip=54		consec=4		max_consec=20		avg_clip50=38.11
EP=9,140		STEP=1,827,124		clip=48		consec=14		max_consec=20		avg_clip50=42.34
EP=9,150		STEP=1,829,582		clip=38		consec=7		max_consec=20		avg_clip50=41.32

	custom/		
	avg_clip50	43.8	
	clipped_episode_r...	47	
	consecutive_clipp...	10	
	max_consecutive_c...	20	
	rollout/		
	ep_len_mean	1.03e+03	
	ep_rew_mean	390	
	exploration rate	0.39	
	time/		
	episodes	27100	
	fps	252	
	time_elapsed	7252	
	total timesteps	1831000	
	train/		
	learning_rate	0.0001	
	loss	0.0993	
	n_updates	445249	

EP=9,160		STEP=1,833,300		clip=4		consec=0		max_consec=20		avg_clip50=45.31
EP=9,170		STEP=1,835,916		clip=38		consec=2		max_consec=20		avg_clip50=45.52
EP=9,180		STEP=1,838,568		clip=28		consec=9		max_consec=20		avg_clip50=45.87

	custom/		
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avg_clip50	45.4
clipped_episode_r...	26.5
consecutive_clipp...	15
max_consecutive_c...	20
rollout/	
ep_len_mean	1.1e+03
ep_rew_mean	429
exploration rate	0.387
time/	
episodes	27200
fps	252
time_elapsed	7291
total timesteps	1840404
train/	
learning_rate	0.0001
loss	0.0535
n_updates	447600

EP=9,190	STEP=1,841,092	clip=27	consec=2	max_consec=20	avg_clip50=43.77
EP=9,200	STEP=1,843,954	clip=38	consec=7	max_consec=20	avg_clip50=45.09
EP=9,210	STEP=1,846,854	clip=54	consec=4	max_consec=20	avg_clip50=42.37
EP=9,220	STEP=1,849,386	clip=40	consec=2	max_consec=20	avg_clip50=42.20

custom/	
avg_clip50	42.5
clipped_episode_r...	36.5
consecutive_clipp...	3
max_consecutive_c...	20
rollout/	
ep_len_mean	1.14e+03
ep_rew_mean	436
exploration rate	0.384
time/	
episodes	27300
fps	252
time_elapsed	7330
total timesteps	1849810
train/	
learning_rate	0.0001
loss	0.0731
n_updates	449952

EP=9,230	STEP=1,852,038	clip=41	consec=7	max_consec=20	avg_clip50=43.89
EP=9,240	STEP=1,854,756	clip=48	consec=6	max_consec=20	avg_clip50=43.90
EP=9,250	STEP=1,857,329	clip=44	consec=16	max_consec=20	avg_clip50=43.30

custom/	
avg_clip50	44.1
clipped_episode_r...	30.5
consecutive_clipp...	5
max_consecutive_c...	20
rollout/	
ep_len_mean	1.11e+03
ep_rew_mean	429
exploration rate	0.381
time/	
episodes	27400
fps	252
time_elapsed	7369

total timesteps	1859014
train/	
learning_rate	0.0001
loss	0.0559
n_updates	452253

EP=9,260 | STEP=1,859,893 | clip=64 | consec=8 | max_consec=20 | avg_clip50=43.79
EP=9,270 | STEP=1,862,967 | clip=52 | consec=9 | max_consec=20 | avg_clip50=46.09
EP=9,280 | STEP=1,866,074 | clip=34 | consec=19 | max_consec=20 | avg_clip50=46.21

EP=9,290 | STEP=1,868,644 | clip=14 | consec=0 | max_consec=20 | avg_clip50=46.10

custom/	
avg_clip50	46.5
clipped_episode_r...	46
consecutive_clipp...	1
max_consecutive_c...	20
rollout/	
ep_len_mean	1.14e+03
ep_rew_mean	452
exploration rate	0.378
time/	
episodes	27500
fps	252
time_elapsed	7410
total timesteps	1868993
train/	
learning_rate	0.0001
loss	0.068
n_updates	454748

EP=9,300 | STEP=1,871,285 | clip=24 | consec=10 | max_consec=20 | avg_clip50=47.17

EP=9,310 | STEP=1,874,135 | clip=58 | consec=6 | max_consec=20 | avg_clip50=46.93

EP=9,320 | STEP=1,877,172 | clip=46 | consec=16 | max_consec=20 | avg_clip50=47.10

custom/	
avg_clip50	45.2
clipped_episode_r...	32.5
consecutive_clipp...	4
max_consecutive_c...	20
rollout/	
ep_len_mean	1.16e+03
ep_rew_mean	468
exploration rate	0.374
time/	
episodes	27600
fps	252
time_elapsed	7450
total timesteps	1878653
train/	
learning_rate	0.0001
loss	0.0549
n_updates	457163

EP=9,330 | STEP=1,879,564 | clip=6 | consec=0 | max_consec=20 | avg_clip50=43.89

EP=9,340 | STEP=1,882,296 | clip=49 | consec=4 | max_consec=20 | avg_clip50=45.16

EP=9,350 | STEP=1,885,260 | clip=30 | consec=14 | max_consec=20 | avg_clip50=46.56

custom/	
avg_clip50	47.8
clipped_episode_r...	25.5
consecutive_clipp...	23
max_consecutive_c...	23
rollout/	
ep_len_mean	1.18e+03
ep_rew_mean	474
exploration rate	0.371
time/	
episodes	27700
fps	252
time_elapsed	7490
total timesteps	1888189
train/	
learning_rate	0.0001
loss	0.556
n_updates	459547

EP=9,360 | STEP=1,888,202 | clip=49 | consec=24 | max_consec=24 | avg_clip50=47.6
1

EP=9,370 | STEP=1,891,050 | clip=46 | consec=4 | max_consec=29 | avg_clip50=46.47

EP=9,380 | STEP=1,894,027 | clip=20 | consec=0 | max_consec=29 | avg_clip50=48.87

EP=9,390 | STEP=1,896,330 | clip=21 | consec=1 | max_consec=29 | avg_clip50=47.47

custom/	
avg_clip50	47
clipped_episode_r...	60
consecutive_clipp...	1
max_consecutive_c...	29
rollout/	
ep_len_mean	1.15e+03
ep_rew_mean	462
exploration rate	0.368
time/	
episodes	27800
fps	252
time_elapsed	7527
total timesteps	1897104
train/	
learning_rate	0.0001
loss	0.455
n_updates	461775

EP=9,400 | STEP=1,898,853 | clip=34 | consec=6 | max_consec=29 | avg_clip50=44.95

EP=9,410 | STEP=1,901,488 | clip=80 | consec=16 | max_consec=29 | avg_clip50=44.6
6

EP=9,420 | STEP=1,903,800 | clip=18 | consec=0 | max_consec=29 | avg_clip50=41.54

custom/	
avg_clip50	40.2
clipped_episode_r...	2
consecutive_clipp...	0
max_consecutive_c...	29
rollout/	
ep_len_mean	1.11e+03
ep_rew_mean	441
exploration rate	0.365
time/	

episodes	27900
fps	251
time_elapsed	7563
total timesteps	1905675
train/	
learning_rate	0.0001
loss	0.0465
n_updates	463918

EP=9,430 STEP=1,906,323 clip=64 consec=2 max_consec=29 avg_clip50=40.25	
EP=9,440 STEP=1,909,118 clip=29 consec=3 max_consec=29 avg_clip50=41.56	
EP=9,450 STEP=1,911,847 clip=94 consec=1 max_consec=29 avg_clip50=42.06	
EP=9,460 STEP=1,914,423 clip=28 consec=8 max_consec=29 avg_clip50=41.53	

custom/	
avg_clip50	42.1
clipped_episode_r...	55
consecutive_clipp...	9
max_consecutive_c...	29
rollout/	
ep_len_mean	1.1e+03
ep_rew_mean	430
exploration rate	0.362
time/	
episodes	28000
fps	251
time_elapsed	7602
total timesteps	1914870
train/	
learning_rate	0.0001
loss	0.0543
n_updates	466217

EP=9,470 STEP=1,917,532 clip=92 consec=18 max_consec=29 avg_clip50=46.00	
EP=9,480 STEP=1,919,978 clip=27 consec=3 max_consec=29 avg_clip50=45.08	
EP=9,490 STEP=1,923,098 clip=74 consec=13 max_consec=29 avg_clip50=47.86	

custom/	
avg_clip50	48.8
clipped_episode_r...	47.5
consecutive_clipp...	20
max_consecutive_c...	29
rollout/	
ep_len_mean	1.12e+03
ep_rew_mean	451
exploration rate	0.359
time/	
episodes	28100
fps	251
time_elapsed	7644
total timesteps	1925042
train/	
learning_rate	0.0001
loss	0.474
n_updates	468760

EP=9,500 STEP=1,925,927 clip=60 consec=23 max_consec=29 avg_clip50=48.58	

EP=9,510 | STEP=1,928,519 | clip=58 | consec=33 | max_consec=33 | avg_clip50=47.2
4

EP=9,520 | STEP=1,930,852 | clip=80 | consec=1 | max_consec=40 | avg_clip50=43.98

EP=9,530 | STEP=1,933,336 | clip=48 | consec=3 | max_consec=40 | avg_clip50=44.88

custom/	
avg_clip50	44.9
clipped_episode_r...	48
consecutive_clipp...	3
max_consecutive_c...	40
rollout/	
ep_len_mean	1.12e+03
ep_rew_mean	450
exploration rate	0.356
time/	
episodes	28200
fps	251
time_elapsed	7680
total timesteps	1933525
train/	
learning_rate	0.0001
loss	0.043
n_updates	470881

EP=9,540 | STEP=1,935,787 | clip=47 | consec=2 | max_consec=40 | avg_clip50=41.64

EP=9,550 | STEP=1,938,535 | clip=48 | consec=12 | max_consec=40 | avg_clip50=41.0

1

EP=9,560 | STEP=1,941,153 | clip=54 | consec=4 | max_consec=40 | avg_clip50=41.34

custom/	
avg_clip50	41.5
clipped_episode_r...	38
consecutive_clipp...	3
max_consecutive_c...	40
rollout/	
ep_len_mean	1.1e+03
ep_rew_mean	438
exploration rate	0.353
time/	
episodes	28300
fps	251
time_elapsed	7719
total timesteps	1942763
train/	
learning_rate	0.0001
loss	0.0652
n_updates	473190

EP=9,570 | STEP=1,943,917 | clip=47 | consec=1 | max_consec=40 | avg_clip50=42.35

EP=9,580 | STEP=1,946,526 | clip=60 | consec=11 | max_consec=40 | avg_clip50=42.5

2

EP=9,590 | STEP=1,949,202 | clip=49 | consec=9 | max_consec=40 | avg_clip50=42.91

EP=9,600 | STEP=1,951,472 | clip=18 | consec=0 | max_consec=40 | avg_clip50=41.26

custom/	
avg_clip50	41.3
clipped_episode_r...	18
consecutive_clipp...	0
max_consecutive_c...	40
rollout/	

ep_len_mean	1.07e+03
ep_rew_mean	411
exploration rate	0.35
time/	
episodes	28400
fps	251
time_elapsed	7757
total timesteps	1951653
train/	
learning_rate	0.0001
loss	0.0761
n_updates	475413

EP=9,610 | STEP=1,953,995 | clip=58 | consec=10 | max_consec=40 | avg_clip50=40.97

EP=9,620 | STEP=1,956,182 | clip=12 | consec=0 | max_consec=40 | avg_clip50=39.55

EP=9,630 | STEP=1,958,840 | clip=64 | consec=10 | max_consec=40 | avg_clip50=40.43

custom/	
avg_clip50	40.3
clipped_episode_r...	24
consecutive_clipp...	14
max_consecutive_c...	40
rollout/	
ep_len_mean	1.06e+03
ep_rew_mean	414
exploration rate	0.347
time/	
episodes	28500
fps	251
time_elapsed	7792
total timesteps	1960127
train/	
learning_rate	0.0001
loss	0.043
n_updates	477531

EP=9,640 | STEP=1,961,787 | clip=60 | consec=20 | max_consec=40 | avg_clip50=41.86

EP=9,650 | STEP=1,964,527 | clip=22 | consec=30 | max_consec=40 | avg_clip50=43.49

EP=9,660 | STEP=1,967,312 | clip=50 | consec=2 | max_consec=40 | avg_clip50=43.65

EP=9,670 | STEP=1,970,017 | clip=75 | consec=5 | max_consec=40 | avg_clip50=47.12

custom/	
avg_clip50	47.7
clipped_episode_r...	54.5
consecutive_clipp...	6
max_consecutive_c...	40
rollout/	
ep_len_mean	1.09e+03
ep_rew_mean	433
exploration rate	0.344
time/	
episodes	28600
fps	251
time_elapsed	7836
total timesteps	1970573
train/	

learning_rate	0.0001
loss	0.464
n_updates	480143

EP=9,680	STEP=1,972,810	clip=54	consec=3	max_consec=40	avg_clip50=47.79
EP=9,690	STEP=1,975,480	clip=13	consec=0	max_consec=40	avg_clip50=45.65
EP=9,700	STEP=1,978,199	clip=58	consec=4	max_consec=40	avg_clip50=45.09

custom/	
avg_clip50	46.5
clipped_episode_r...	30
consecutive_clipp...	1
max_consecutive_c...	40
rollout/	
ep_len_mean	1.12e+03
ep_rew_mean	452
exploration rate	0.341
time/	
episodes	28700
fps	251
time_elapsed	7876
total timesteps	1980169
train/	
learning_rate	0.0001
loss	0.044
n_updates	482542

EP=9,710	STEP=1,980,921	clip=52	consec=4	max_consec=40	avg_clip50=46.39
EP=9,720	STEP=1,983,081	clip=22	consec=3	max_consec=40	avg_clip50=43.43
EP=9,730	STEP=1,985,895	clip=72	consec=13	max_consec=40	avg_clip50=42.63
EP=9,740	STEP=1,988,280	clip=48	consec=23	max_consec=40	avg_clip50=42.44

custom/	
avg_clip50	42
clipped_episode_r...	23.5
consecutive_clipp...	24
max_consecutive_c...	40
rollout/	
ep_len_mean	1.1e+03
ep_rew_mean	439
exploration rate	0.338
time/	
episodes	28800
fps	251
time_elapsed	7912
total timesteps	1988590
train/	
learning_rate	0.0001
loss	0.0295
n_updates	484647

EP=9,750	STEP=1,990,867	clip=30	consec=6	max_consec=40	avg_clip50=42.89
EP=9,760	STEP=1,993,214	clip=46	consec=2	max_consec=40	avg_clip50=40.49
EP=9,770	STEP=1,995,903	clip=66	consec=2	max_consec=40	avg_clip50=42.99

custom/	
avg_clip50	42.3
clipped_episode_r...	24.5

	consecutive_clipp...	10	
	max_consecutive_c...	40	
	rollout/		
	ep_len_mean	1.08e+03	
	ep_rew_mean	424	
	exploration rate	0.335	
	time/		
	episodes	28900	
	fps	251	
	time_elapsed	7951	
	total timesteps	1997909	
	train/		
	learning_rate	0.0001	
	loss	0.0408	
	n_updates	486977	

EP=9,780 | STEP=1,998,652 | clip=56 | consec=12 | max_consec=40 | avg_clip50=41.92

EP=9,790 | STEP=2,001,186 | clip=21 | consec=1 | max_consec=40 | avg_clip50=42.24

EP=9,800 | STEP=2,004,573 | clip=46 | consec=11 | max_consec=40 | avg_clip50=44.07

EP=9,810 | STEP=2,007,216 | clip=54 | consec=6 | max_consec=40 | avg_clip50=45.15

	custom/		
	avg_clip50	45.1	
	clipped_episode_r...	51.5	
	consecutive_clipp...	8	
	max_consecutive_c...	40	
	rollout/		
	ep_len_mean	1.1e+03	
	ep_rew_mean	434	
	exploration rate	0.331	
	time/		
	episodes	29000	
	fps	251	
	time_elapsed	7994	
	total timesteps	2008034	
	train/		
	learning_rate	0.0001	
	loss	0.0493	
	n_updates	489508	

EP=9,820 | STEP=2,010,015 | clip=58 | consec=1 | max_consec=40 | avg_clip50=44.86

EP=9,830 | STEP=2,012,465 | clip=55 | consec=11 | max_consec=40 | avg_clip50=45.47

EP=9,840 | STEP=2,015,511 | clip=77 | consec=21 | max_consec=40 | avg_clip50=47.19

	custom/		
	avg_clip50	46	
	clipped_episode_r...	20	
	consecutive_clipp...	0	
	max_consecutive_c...	40	
	rollout/		
	ep_len_mean	1.15e+03	
	ep_rew_mean	450	
	exploration rate	0.328	
	time/		
	episodes	29100	
	fps	251	

	time_elapsed		8034	
	total timesteps		2017819	
	train/			
	learning_rate		0.0001	
	loss		0.0402	
	n_updates		491954	

EP=9,850		STEP=2,018,378		clip=46		consec=2		max_consec=40		avg_clip50=46.36
EP=9,860		STEP=2,020,816		clip=50		consec=5		max_consec=40		avg_clip50=46.59
EP=9,870		STEP=2,023,380		clip=60		consec=4		max_consec=40		avg_clip50=44.50
EP=9,880		STEP=2,026,479		clip=58		consec=3		max_consec=40		avg_clip50=45.83

	custom/			
	avg_clip50		45	
	clipped_episode_r...		28.5	
	consecutive_clipp...		5	
	max_consecutive_c...		40	
	rollout/			
	ep_len_mean		1.15e+03	
	ep_rew_mean		453	
	exploration rate		0.325	
	time/			
	episodes		29200	
	fps		251	
	time_elapsed		8073	
	total timesteps		2027046	
	train/			
	learning_rate		0.0001	
	loss		0.0441	
	n_updates		494261	

EP=9,890		STEP=2,029,432		clip=24		consec=13		max_consec=40		avg_clip50=47.94
EP=9,900		STEP=2,032,429		clip=48		consec=23		max_consec=40		avg_clip50=49.10
EP=9,910		STEP=2,034,598		clip=52		consec=3		max_consec=40		avg_clip50=47.57

	custom/			
	avg_clip50		51.1	
	clipped_episode_r...		73	
	consecutive_clipp...		12	
	max_consecutive_c...		40	
	rollout/			
	ep_len_mean		1.15e+03	
	ep_rew_mean		478	
	exploration rate		0.321	
	time/			
	episodes		29300	
	fps		251	
	time_elapsed		8118	
	total timesteps		2037721	
	train/			
	learning_rate		0.0001	
	loss		0.0478	
	n_updates		496930	

EP=9,920		STEP=2,037,765		clip=53		consec=13		max_consec=40		avg_clip50=50.97
EP=9,930		STEP=2,040,632		clip=48		consec=23		max_consec=40		avg_clip50=49.89

EP=9,940 | STEP=2,043,409 | clip=81 | consec=33 | max_consec=40 | avg_clip50=46.6
9

EP=9,950 | STEP=2,045,841 | clip=8 | consec=0 | max_consec=40 | avg_clip50=43.08

custom/		
avg_clip50	43.7	
clipped_episode_r...	19	
consecutive_clipp...	0	
max_consecutive_c...	40	
rollout/		
ep_len_mean	1.13e+03	
ep_rew_mean	452	
exploration rate	0.318	
time/		
episodes	29400	
fps	250	
time_elapsed	8156	
total timesteps	2046832	
train/		
learning_rate	0.0001	
loss	0.451	
n_updates	499207	

EP=9,960 | STEP=2,048,396 | clip=31 | consec=7 | max_consec=40 | avg_clip50=44.53

EP=9,970 | STEP=2,050,628 | clip=42 | consec=6 | max_consec=40 | avg_clip50=40.87

EP=9,980 | STEP=2,053,717 | clip=45 | consec=16 | max_consec=40 | avg_clip50=43.0

3

custom/		
avg_clip50	43.8	
clipped_episode_r...	26	
consecutive_clipp...	3	
max_consecutive_c...	40	
rollout/		
ep_len_mean	1.13e+03	
ep_rew_mean	454	
exploration rate	0.315	
time/		
episodes	29500	
fps	250	
time_elapsed	8194	
total timesteps	2055913	
train/		
learning_rate	0.0001	
loss	0.0347	
n_updates	501478	

EP=9,990 | STEP=2,056,544 | clip=54 | consec=6 | max_consec=40 | avg_clip50=43.61

EP=10,000 | STEP=2,059,461 | clip=58 | consec=16 | max_consec=40 | avg_clip50=48.
05

EP=10,010 | STEP=2,062,012 | clip=21 | consec=7 | max_consec=40 | avg_clip50=47.9

2

EP=10,020 | STEP=2,064,815 | clip=48 | consec=17 | max_consec=40 | avg_clip50=50.

75

custom/		
avg_clip50	50.3	
clipped_episode_r...	10	
consecutive_clipp...	0	
max_consecutive_c...	40	

rollout/		
ep_len_mean	1.11e+03	
ep_rew_mean	454	
exploration rate	0.312	
time/		
episodes	29600	
fps	250	
time_elapsed	8233	
total timesteps	2065095	
train/		
learning_rate	0.0001	
loss	0.0976	
n_updates	503773	

EP=10,030 | STEP=2,067,094 | clip=68 | consec=6 | max_consec=40 | avg_clip50=47.43

EP=10,040 | STEP=2,069,240 | clip=47 | consec=2 | max_consec=40 | avg_clip50=44.95

EP=10,050 | STEP=2,071,556 | clip=73 | consec=1 | max_consec=40 | avg_clip50=42.18

custom/		
avg_clip50	42.4	
clipped_episode_r...	31.5	
consecutive_clipp...	2	
max_consecutive_c...	40	
rollout/		
ep_len_mean	1.08e+03	
ep_rew_mean	452	
exploration rate	0.309	
time/		
episodes	29700	
fps	250	
time_elapsed	8269	
total timesteps	2073588	
train/		
learning_rate	0.0001	
loss	0.516	
n_updates	505896	

EP=10,060 | STEP=2,074,376 | clip=34 | consec=5 | max_consec=40 | avg_clip50=43.75

EP=10,070 | STEP=2,077,255 | clip=14 | consec=0 | max_consec=40 | avg_clip50=42.94

EP=10,080 | STEP=2,080,283 | clip=70 | consec=10 | max_consec=40 | avg_clip50=45.71

EP=10,090 | STEP=2,083,030 | clip=54 | consec=20 | max_consec=40 | avg_clip50=46.49

custom/		
avg_clip50	46.9	
clipped_episode_r...	46.5	
consecutive_clipp...	21	
max_consecutive_c...	40	
rollout/		
ep_len_mean	1.11e+03	
ep_rew_mean	454	
exploration rate	0.306	
time/		
episodes	29800	

fps	250
time_elapsed	8310
total timesteps	2083388
train/	
learning_rate	0.0001
loss	0.519
n_updates	508346

EP=10,100 | STEP=2,085,746 | clip=12 | consec=0 | max_consec=40 | avg_clip50=45.96

EP=10,110 | STEP=2,089,226 | clip=64 | consec=10 | max_consec=40 | avg_clip50=48.37

EP=10,120 | STEP=2,092,087 | clip=26 | consec=1 | max_consec=40 | avg_clip50=47.81

custom/	
avg_clip50	46.7
clipped_episode_r...	60
consecutive_clipp...	1
max_consecutive_c...	40
rollout/	
ep_len_mean	1.15e+03
ep_rew_mean	458
exploration rate	0.303
time/	
episodes	29900
fps	250
time_elapsed	8352
total timesteps	2093347
train/	
learning_rate	0.0001
loss	0.0906
n_updates	510836

EP=10,130 | STEP=2,094,918 | clip=74 | consec=6 | max_consec=40 | avg_clip50=47.48

EP=10,140 | STEP=2,097,798 | clip=41 | consec=16 | max_consec=40 | avg_clip50=49.56

EP=10,150 | STEP=2,100,435 | clip=24 | consec=4 | max_consec=40 | avg_clip50=49.15

custom/	
avg_clip50	48.6
clipped_episode_r...	39.5
consecutive_clipp...	6
max_consecutive_c...	40
rollout/	
ep_len_mean	1.2e+03
ep_rew_mean	482
exploration rate	0.3
time/	
episodes	30000
fps	250
time_elapsed	8393
total timesteps	2103084
train/	
learning_rate	0.0001
loss	0.663
n_updates	513270

EP=10,160 | STEP=2,103,168 | clip=4 | consec=0 | max_consec=40 | avg_clip50=47.37
 EP=10,170 | STEP=2,105,801 | clip=52 | consec=10 | max_consec=40 | avg_clip50=48.02
 EP=10,180 | STEP=2,108,686 | clip=36 | consec=20 | max_consec=40 | avg_clip50=48.07
 EP=10,190 | STEP=2,111,718 | clip=54 | consec=30 | max_consec=40 | avg_clip50=50.31

custom/		
avg_clip50	51.3	
clipped_episode_r...	44.5	
consecutive_clipp...	34	
max_consecutive_c...	40	
rollout/		
ep_len_mean	1.2e+03	
ep_rew_mean	504	
exploration rate	0.296	
time/		
episodes	30100	
fps	250	
time_elapsed	8435	
total timesteps	2112945	
train/		
learning_rate	0.0001	
loss	0.133	
n_updates	515736	

EP=10,200 | STEP=2,114,596 | clip=34 | consec=40 | max_consec=40 | avg_clip50=52.51
 EP=10,210 | STEP=2,117,566 | clip=36 | consec=9 | max_consec=40 | avg_clip50=52.04
 EP=10,220 | STEP=2,120,344 | clip=58 | consec=19 | max_consec=40 | avg_clip50=52.31

custom/		
avg_clip50	50.1	
clipped_episode_r...	55.5	
consecutive_clipp...	6	
max_consecutive_c...	40	
rollout/		
ep_len_mean	1.17e+03	
ep_rew_mean	494	
exploration rate	0.293	
time/		
episodes	30200	
fps	250	
time_elapsed	8475	
total timesteps	2122528	
train/		
learning_rate	0.0001	
loss	0.0564	
n_updates	518131	

EP=10,230 | STEP=2,123,061 | clip=54 | consec=8 | max_consec=40 | avg_clip50=50.32
 EP=10,240 | STEP=2,125,678 | clip=38 | consec=4 | max_consec=40 | avg_clip50=49.15
 EP=10,250 | STEP=2,128,342 | clip=58 | consec=6 | max_consec=40 | avg_clip50=47.32
 EP=10,260 | STEP=2,131,483 | clip=52 | consec=16 | max_consec=40 | avg_clip50=47.

custom/		
avg_clip50	48.9	
clipped_episode_r...	86	
consecutive_clipp...	18	
max_consecutive_c...	40	
rollout/		
ep_len_mean	1.18e+03	
ep_rew_mean	500	
exploration rate	0.29	
time/		
episodes	30300	
fps	250	
time_elapsed	8517	
total timesteps	2132454	
train/		
learning_rate	0.0001	
loss	0.0847	
n_updates	520613	

EP=10,270 | STEP=2,134,266 | clip=77 | consec=1 | max_consec=40 | avg_clip50=47.5
4
EP=10,280 | STEP=2,137,192 | clip=28 | consec=8 | max_consec=40 | avg_clip50=49.6
3
EP=10,290 | STEP=2,140,332 | clip=43 | consec=18 | max_consec=40 | avg_clip50=48.
78

custom/		
avg_clip50	50.7	
clipped_episode_r...	77.5	
consecutive_clipp...	25	
max_consecutive_c...	40	
rollout/		
ep_len_mean	1.2e+03	
ep_rew_mean	491	
exploration rate	0.286	
time/		
episodes	30400	
fps	250	
time_elapsed	8561	
total timesteps	2142870	
train/		
learning_rate	0.0001	
loss	0.0642	
n_updates	523217	

EP=10,300 | STEP=2,143,467 | clip=74 | consec=28 | max_consec=40 | avg_clip50=51.
86
EP=10,310 | STEP=2,146,267 | clip=72 | consec=1 | max_consec=40 | avg_clip50=52.7
2
EP=10,320 | STEP=2,149,472 | clip=56 | consec=11 | max_consec=40 | avg_clip50=53.
26
EP=10,330 | STEP=2,152,584 | clip=46 | consec=21 | max_consec=40 | avg_clip50=53.
86

custom/		
avg_clip50	54.5	
clipped_episode_r...	68.5	
consecutive_clipp...	23	

	max_consecutive_c...	40	
	rollout/		
	ep_len_mean	1.22e+03	
	ep_rew_mean	513	
	exploration rate	0.283	
	time/		
	episodes	30500	
	fps	250	
	time_elapsed	8605	
	total timesteps	2153271	
	train/		
	learning_rate	0.0001	
	loss	0.085	
	n_updates	525817	

EP=10,340 | STEP=2,155,779 | clip=68 | consec=31 | max_consec=40 | avg_clip50=54.43

EP=10,350 | STEP=2,158,518 | clip=12 | consec=0 | max_consec=40 | avg_clip50=52.99

EP=10,360 | STEP=2,161,570 | clip=23 | consec=5 | max_consec=40 | avg_clip50=51.22

	custom/		
	avg_clip50	50.1	
	clipped_episode_r...	33.5	
	consecutive_clipp...	4	
	max_consecutive_c...	40	
	rollout/		
	ep_len_mean	1.24e+03	
	ep_rew_mean	512	
	exploration rate	0.28	
	time/		
	episodes	30600	
	fps	250	
	time_elapsed	8648	
	total timesteps	2163244	
	train/		
	learning_rate	0.0001	
	loss	0.266	
	n_updates	528310	

EP=10,370 | STEP=2,164,164 | clip=7 | consec=0 | max_consec=40 | avg_clip50=48.85

EP=10,380 | STEP=2,167,101 | clip=54 | consec=6 | max_consec=40 | avg_clip50=47.31

EP=10,390 | STEP=2,169,897 | clip=48 | consec=16 | max_consec=40 | avg_clip50=46.17

EP=10,400 | STEP=2,172,718 | clip=54 | consec=26 | max_consec=40 | avg_clip50=46.55

	custom/		
	avg_clip50	46.5	
	clipped_episode_r...	53.5	
	consecutive_clipp...	26	
	max_consecutive_c...	40	
	rollout/		
	ep_len_mean	1.22e+03	
	ep_rew_mean	498	
	exploration rate	0.276	
	time/		
	episodes	30700	

fps	250
time_elapsed	8689
total timesteps	2173059
train/	
learning_rate	0.0001
loss	0.0687
n_updates	530764

EP=10,410 | STEP=2,175,889 | clip=42 | consec=36 | max_consec=40 | avg_clip50=47.87

EP=10,420 | STEP=2,178,346 | clip=58 | consec=3 | max_consec=42 | avg_clip50=49.08

EP=10,430 | STEP=2,181,726 | clip=60 | consec=13 | max_consec=42 | avg_clip50=50.62

custom/	
avg_clip50	50
clipped_episode_r...	45.5
consecutive_clipp...	2
max_consecutive_c...	42
rollout/	
ep_len_mean	1.21e+03
ep_rew_mean	490
exploration rate	0.273
time/	
episodes	30800
fps	250
time_elapsed	8732
total timesteps	2183184
train/	
learning_rate	0.0001
loss	0.516
n_updates	533295

EP=10,440 | STEP=2,184,897 | clip=60 | consec=7 | max_consec=42 | avg_clip50=51.76

EP=10,450 | STEP=2,187,596 | clip=44 | consec=2 | max_consec=42 | avg_clip50=51.44

EP=10,460 | STEP=2,190,289 | clip=24 | consec=12 | max_consec=42 | avg_clip50=49.30

EP=10,470 | STEP=2,193,241 | clip=72 | consec=22 | max_consec=42 | avg_clip50=50.62

custom/	
avg_clip50	50.6
clipped_episode_r...	46.5
consecutive_clipp...	23
max_consecutive_c...	42
rollout/	
ep_len_mean	1.21e+03
ep_rew_mean	501
exploration rate	0.27
time/	
episodes	30900
fps	249
time_elapsed	8776
total timesteps	2193510
train/	
learning_rate	0.0001
loss	0.358

n_updates	535877
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EP=10,480	STEP=2,196,359	clip=54	consec=2	max_consec=42	avg_clip50=49.98
EP=10,490	STEP=2,199,283	clip=14	consec=0	max_consec=42	avg_clip50=50.83
EP=10,500	STEP=2,201,977	clip=31	consec=10	max_consec=42	avg_clip50=49.31

custom/	
avg_clip50	49.2
clipped_episode_r...	60
consecutive_clipp...	16
max_consecutive_c...	42
rollout/	
ep_len_mean	1.21e+03
ep_rew_mean	500
exploration rate	0.266
time/	
episodes	31000
fps	249
time_elapsed	8820
total timesteps	2203969
train/	
learning_rate	0.0001
loss	0.0568
n_updates	538492

EP=10,510	STEP=2,205,095	clip=52	consec=20	max_consec=42	avg_clip50=50.68
EP=10,520	STEP=2,207,860	clip=47	consec=30	max_consec=42	avg_clip50=49.46
EP=10,530	STEP=2,211,099	clip=54	consec=40	max_consec=42	avg_clip50=50.48
EP=10,540	STEP=2,213,937	clip=44	consec=5	max_consec=44	avg_clip50=48.10

custom/	
avg_clip50	48.3
clipped_episode_r...	60.5
consecutive_clipp...	6
max_consecutive_c...	44
rollout/	
ep_len_mean	1.21e+03
ep_rew_mean	498
exploration rate	0.263
time/	
episodes	31100
fps	249
time_elapsed	8865
total timesteps	2214513
train/	
learning_rate	0.0001
loss	0.231
n_updates	541128

EP=10,550	STEP=2,217,009	clip=48	consec=15	max_consec=44	avg_clip50=50.72
EP=10,560	STEP=2,219,562	clip=12	consec=0	max_consec=44	avg_clip50=50.05

EP=10,570 | STEP=2,222,699 | clip=58 | consec=10 | max_consec=44 | avg_clip50=51.02

custom/		
avg_clip50	52.1	
clipped_episode_r...	47	
consecutive_clipp...	17	
max_consecutive_c...	44	
rollout/		
ep_len_mean	1.21e+03	
ep_rew_mean	504	
exploration rate	0.259	
time/		
episodes	31200	
fps	249	
time_elapsed	8908	
total timesteps	2224862	
train/		
learning_rate	0.0001	
loss	0.0435	
n_updates	543715	

EP=10,580 | STEP=2,225,591 | clip=66 | consec=20 | max_consec=44 | avg_clip50=50.43

EP=10,590 | STEP=2,228,616 | clip=12 | consec=0 | max_consec=44 | avg_clip50=51.03

EP=10,600 | STEP=2,231,647 | clip=16 | consec=0 | max_consec=44 | avg_clip50=50.19

EP=10,610 | STEP=2,234,701 | clip=64 | consec=9 | max_consec=44 | avg_clip50=51.16

custom/		
avg_clip50	51.7	
clipped_episode_r...	48.5	
consecutive_clipp...	13	
max_consecutive_c...	44	
rollout/		
ep_len_mean	1.23e+03	
ep_rew_mean	509	
exploration rate	0.255	
time/		
episodes	31300	
fps	249	
time_elapsed	8955	
total timesteps	2235955	
train/		
learning_rate	0.0001	
loss	0.0808	
n_updates	546488	

EP=10,620 | STEP=2,237,625 | clip=65 | consec=19 | max_consec=44 | avg_clip50=51.24

EP=10,630 | STEP=2,240,147 | clip=60 | consec=7 | max_consec=44 | avg_clip50=48.94

EP=10,640 | STEP=2,242,703 | clip=60 | consec=6 | max_consec=44 | avg_clip50=48.32

custom/		
avg_clip50	46.2	
clipped_episode_r...	31	

	consecutive_clipp...	3	
	max_consecutive_c...	44	
	rollout/		
	ep_len_mean	1.18e+03	
	ep_rew_mean	485	
	exploration rate	0.252	
	time/		
	episodes	31400	
	fps	249	
	time_elapsed	8995	
	total timesteps	2245185	
	train/		
	learning_rate	0.0001	
	loss	0.0398	
	n_updates	548796	

EP=10,650 | STEP=2,245,521 | clip=54 | consec=4 | max_consec=44 | avg_clip50=46.94

EP=10,660 | STEP=2,248,669 | clip=60 | consec=3 | max_consec=44 | avg_clip50=47.50

EP=10,670 | STEP=2,251,693 | clip=30 | consec=13 | max_consec=44 | avg_clip50=48.31

EP=10,680 | STEP=2,254,616 | clip=54 | consec=2 | max_consec=44 | avg_clip50=49.49

	custom/		
	avg_clip50	50.4	
	clipped_episode_r...	28	
	consecutive_clipp...	7	
	max_consecutive_c...	44	
	rollout/		
	ep_len_mean	1.2e+03	
	ep_rew_mean	490	
	exploration rate	0.249	
	time/		
	episodes	31500	
	fps	249	
	time_elapsed	9041	
	total timesteps	2256097	
	train/		
	learning_rate	0.0001	
	loss	0.309	
	n_updates	551524	

EP=10,690 | STEP=2,257,427 | clip=94 | consec=12 | max_consec=44 | avg_clip50=50.80

EP=10,700 | STEP=2,260,175 | clip=80 | consec=22 | max_consec=44 | avg_clip50=52.61

EP=10,710 | STEP=2,262,984 | clip=39 | consec=32 | max_consec=44 | avg_clip50=51.42

EP=10,720 | STEP=2,265,738 | clip=52 | consec=42 | max_consec=44 | avg_clip50=50.17

	custom/		
	avg_clip50	50.2	
	clipped_episode_r...	77.5	
	consecutive_clipp...	44	
	max_consecutive_c...	44	
	rollout/		
	ep_len_mean	1.17e+03	

	ep_rew_mean	497	
	exploration rate	0.245	
	time/		
	episodes	31600	
	fps	249	
	time_elapsed	9084	
	total timesteps	2266273	
	train/		
	learning_rate	0.0001	
	loss	0.0571	
	n_updates	554068	

EP=10,730 | STEP=2,268,353 | clip=54 | consec=52 | max_consec=52 | avg_clip50=50.33

EP=10,740 | STEP=2,270,941 | clip=12 | consec=0 | max_consec=53 | avg_clip50=49.11

EP=10,750 | STEP=2,274,166 | clip=44 | consec=10 | max_consec=53 | avg_clip50=50.36

	custom/		
	avg_clip50	52.4	
	clipped_episode_r...	60	
	consecutive_clipp...	18	
	max_consecutive_c...	53	
	rollout/		
	ep_len_mean	1.19e+03	
	ep_rew_mean	519	
	exploration rate	0.242	
	time/		
	episodes	31700	
	fps	249	
	time_elapsed	9130	
	total timesteps	2277151	
	train/		
	learning_rate	0.0001	
	loss	0.278	
	n_updates	556787	

EP=10,760 | STEP=2,277,496 | clip=46 | consec=20 | max_consec=53 | avg_clip50=52.28

EP=10,770 | STEP=2,280,440 | clip=43 | consec=30 | max_consec=53 | avg_clip50=51.39

EP=10,780 | STEP=2,283,327 | clip=54 | consec=40 | max_consec=53 | avg_clip50=50.32

EP=10,790 | STEP=2,286,448 | clip=98 | consec=50 | max_consec=53 | avg_clip50=53.00

	custom/		
	avg_clip50	52.4	
	clipped_episode_r...	22.5	
	consecutive_clipp...	53	
	max_consecutive_c...	53	
	rollout/		
	ep_len_mean	1.21e+03	
	ep_rew_mean	515	
	exploration rate	0.238	
	time/		
	episodes	31800	
	fps	249	
	time_elapsed	9174	

total timesteps	2287476
train/	
learning_rate	0.0001
loss	0.515
n_updates	559368

EP=10,800 | STEP=2,288,804 | clip=42 | consec=1 | max_consec=53 | avg_clip50=48.42

EP=10,810 | STEP=2,291,828 | clip=60 | consec=11 | max_consec=53 | avg_clip50=47.20

EP=10,820 | STEP=2,294,463 | clip=36 | consec=21 | max_consec=53 | avg_clip50=46.92

custom/	
avg_clip50	48.8
clipped_episode_r...	45.5
consecutive_clipp...	29
max_consecutive_c...	53
rollout/	
ep_len_mean	1.21e+03
ep_rew_mean	495
exploration rate	0.235
time/	
episodes	31900
fps	249
time_elapsed	9215
total timesteps	2296956
train/	
learning_rate	0.0001
loss	0.413
n_updates	561738

EP=10,830 | STEP=2,297,472 | clip=17 | consec=0 | max_consec=53 | avg_clip50=48.46

EP=10,840 | STEP=2,300,364 | clip=42 | consec=8 | max_consec=53 | avg_clip50=46.74

EP=10,850 | STEP=2,303,282 | clip=94 | consec=1 | max_consec=53 | avg_clip50=49.47

EP=10,860 | STEP=2,306,913 | clip=59 | consec=4 | max_consec=53 | avg_clip50=51.46

custom/	
avg_clip50	52.3
clipped_episode_r...	69
consecutive_clipp...	6
max_consecutive_c...	53
rollout/	
ep_len_mean	1.23e+03
ep_rew_mean	500
exploration rate	0.231
time/	
episodes	32000
fps	249
time_elapsed	9262
total timesteps	2307868
train/	
learning_rate	0.0001
loss	0.0904
n_updates	564466

EP=10,870 | STEP=2,310,251 | clip=50 | consec=14 | max_consec=53 | avg_clip50=53.84
EP=10,880 | STEP=2,312,985 | clip=52 | consec=24 | max_consec=53 | avg_clip50=53.25
EP=10,890 | STEP=2,315,938 | clip=60 | consec=2 | max_consec=53 | avg_clip50=52.88

custom/		
avg_clip50	55.1	
clipped_episode_r...	91.5	
consecutive_clipp...	10	
max_consecutive_c...	53	
rollout/		
ep_len_mean	1.24e+03	
ep_rew_mean	518	
exploration rate	0.228	
time/		
episodes	32100	
fps	249	
time_elapsed	9309	
total timesteps	2318554	
train/		
learning_rate	0.0001	
loss	0.0501	
n_updates	567138	

EP=10,900 | STEP=2,319,043 | clip=60 | consec=12 | max_consec=53 | avg_clip50=55.19
EP=10,910 | STEP=2,322,430 | clip=46 | consec=22 | max_consec=53 | avg_clip50=55.04
EP=10,920 | STEP=2,325,154 | clip=60 | consec=1 | max_consec=53 | avg_clip50=52.27
EP=10,930 | STEP=2,327,960 | clip=35 | consec=11 | max_consec=53 | avg_clip50=51.65

custom/		
avg_clip50	52.8	
clipped_episode_r...	37.5	
consecutive_clipp...	14	
max_consecutive_c...	53	
rollout/		
ep_len_mean	1.27e+03	
ep_rew_mean	534	
exploration rate	0.224	
time/		
episodes	32200	
fps	248	
time_elapsed	9354	
total timesteps	2329032	
train/		
learning_rate	0.0001	
loss	0.0458	
n_updates	569757	

EP=10,940 | STEP=2,331,262 | clip=100 | consec=3 | max_consec=53 | avg_clip50=53.63
EP=10,950 | STEP=2,334,213 | clip=57 | consec=13 | max_consec=53 | avg_clip50=51.76
EP=10,960 | STEP=2,337,248 | clip=74 | consec=3 | max_consec=53 | avg_clip50=50.54

custom/		
avg_clip50	53.6	
clipped_episode_r...	54.5	
consecutive_clipp...	11	
max_consecutive_c...	53	
rollout/		
ep_len_mean	1.25e+03	
ep_rew_mean	529	
exploration rate	0.221	
time/		
episodes	32300	
fps	248	
time_elapsed	9401	
total timesteps	2339964	
train/		
learning_rate	0.0001	
loss	0.049	
n_updates	572490	

EP=10,970 | STEP=2,340,421 | clip=40 | consec=13 | max_consec=53 | avg_clip50=53.84
EP=10,980 | STEP=2,342,902 | clip=45 | consec=4 | max_consec=53 | avg_clip50=52.41
EP=10,990 | STEP=2,346,215 | clip=27 | consec=14 | max_consec=53 | avg_clip50=51.80
EP=11,000 | STEP=2,349,282 | clip=74 | consec=24 | max_consec=53 | avg_clip50=51.96

custom/		
avg_clip50	52.3	
clipped_episode_r...	80	
consecutive_clipp...	26	
max_consecutive_c...	53	
rollout/		
ep_len_mean	1.25e+03	
ep_rew_mean	520	
exploration rate	0.217	
time/		
episodes	32400	
fps	248	
time_elapsed	9444	
total timesteps	2350035	
train/		
learning_rate	0.0001	
loss	0.0543	
n_updates	575008	

EP=11,010 | STEP=2,351,895 | clip=38 | consec=6 | max_consec=53 | avg_clip50=51.67
EP=11,020 | STEP=2,354,704 | clip=50 | consec=16 | max_consec=53 | avg_clip50=48.89
EP=11,030 | STEP=2,358,141 | clip=54 | consec=26 | max_consec=53 | avg_clip50=52.66

custom/		
avg_clip50	53.5	
clipped_episode_r...	49	
consecutive_clipp...	33	
max_consecutive_c...	53	

rollout/	
ep_len_mean	1.28e+03
ep_rew_mean	535
exploration rate	0.214
time/	
episodes	32500
fps	248
time_elapsed	9491
total timesteps	2360896
train/	
learning_rate	0.0001
loss	0.0446
n_updates	577723

EP=11,040 | STEP=2,361,654 | clip=52 | consec=2 | max_consec=53 | avg_clip50=53.60

EP=11,050 | STEP=2,364,700 | clip=50 | consec=12 | max_consec=53 | avg_clip50=53.76

EP=11,060 | STEP=2,367,837 | clip=54 | consec=22 | max_consec=53 | avg_clip50=55.65

EP=11,070 | STEP=2,370,683 | clip=38 | consec=4 | max_consec=53 | avg_clip50=57.13

custom/	
avg_clip50	57.6
clipped_episode_r...	54
consecutive_clipp...	6
max_consecutive_c...	53
rollout/	
ep_len_mean	1.26e+03
ep_rew_mean	534
exploration rate	0.21
time/	
episodes	32600
fps	248
time_elapsed	9536
total timesteps	2371613
train/	
learning_rate	0.0001
loss	0.0419
n_updates	580403

EP=11,080 | STEP=2,373,438 | clip=52 | consec=14 | max_consec=53 | avg_clip50=56.59

EP=11,090 | STEP=2,376,385 | clip=96 | consec=24 | max_consec=53 | avg_clip50=56.20

EP=11,100 | STEP=2,379,194 | clip=22 | consec=34 | max_consec=53 | avg_clip50=55.62

EP=11,110 | STEP=2,382,481 | clip=74 | consec=44 | max_consec=53 | avg_clip50=56.69

custom/	
avg_clip50	56.7
clipped_episode_r...	74.5
consecutive_clipp...	44
max_consecutive_c...	53
rollout/	
ep_len_mean	1.27e+03
ep_rew_mean	562
exploration rate	0.207

time/	
episodes	32700
fps	248
time_elapsed	9583
total timesteps	2382481
train/	
learning_rate	0.0001
loss	0.489
n_updates	583120

EP=11,120 | STEP=2,385,544 | clip=60 | consec=54 | max_consec=54 | avg_clip50=57.44

EP=11,130 | STEP=2,388,211 | clip=28 | consec=64 | max_consec=64 | avg_clip50=55.39

EP=11,140 | STEP=2,390,993 | clip=37 | consec=4 | max_consec=69 | avg_clip50=53.16

custom/	
avg_clip50	52.4
clipped_episode_r...	45
consecutive_clipp...	9
max_consecutive_c...	69
rollout/	
ep_len_mean	1.22e+03
ep_rew_mean	546
exploration rate	0.203
time/	
episodes	32800
fps	248
time_elapsed	9627
total timesteps	2392602
train/	
learning_rate	0.0001
loss	0.099
n_updates	585650

EP=11,150 | STEP=2,394,250 | clip=112 | consec=14 | max_consec=69 | avg_clip50=54.49

EP=11,160 | STEP=2,397,203 | clip=77 | consec=24 | max_consec=69 | avg_clip50=52.11

EP=11,170 | STEP=2,400,464 | clip=60 | consec=34 | max_consec=69 | avg_clip50=52.56

EP=11,180 | STEP=2,403,273 | clip=41 | consec=44 | max_consec=69 | avg_clip50=54.63

custom/	
avg_clip50	54.6
clipped_episode_r...	41
consecutive_clipp...	44
max_consecutive_c...	69
rollout/	
ep_len_mean	1.24e+03
ep_rew_mean	550
exploration rate	0.2
time/	
episodes	32900
fps	248
time_elapsed	9673
total timesteps	2403548
train/	

learning_rate	0.0001
loss	0.0606
n_updates	588386

EP=11,190 | STEP=2,406,016 | clip=74 | consec=1 | max_consec=69 | avg_clip50=55.12

EP=11,200 | STEP=2,409,283 | clip=96 | consec=11 | max_consec=69 | avg_clip50=55.20

EP=11,210 | STEP=2,412,548 | clip=37 | consec=21 | max_consec=69 | avg_clip50=56.07

custom/	
avg_clip50	54.9
clipped_episode_r...	52
consecutive_clipp...	30
max_consecutive_c...	69
rollout/	
ep_len_mean	1.24e+03
ep_rew_mean	538
exploration rate	0.196
time/	
episodes	33000
fps	248
time_elapsed	9723
total timesteps	2415048
train/	
learning_rate	0.0001
loss	0.0554
n_updates	591261

EP=11,220 | STEP=2,415,174 | clip=18 | consec=0 | max_consec=69 | avg_clip50=54.11

EP=11,230 | STEP=2,417,811 | clip=52 | consec=5 | max_consec=69 | avg_clip50=52.22

EP=11,240 | STEP=2,420,848 | clip=38 | consec=15 | max_consec=69 | avg_clip50=52.47

EP=11,250 | STEP=2,423,936 | clip=57 | consec=25 | max_consec=69 | avg_clip50=52.55

custom/	
avg_clip50	52.2
clipped_episode_r...	45
consecutive_clipp...	30
max_consecutive_c...	69
rollout/	
ep_len_mean	1.22e+03
ep_rew_mean	537
exploration rate	0.192
time/	
episodes	33100
fps	248
time_elapsed	9768
total timesteps	2425413
train/	
learning_rate	0.0001
loss	0.0682
n_updates	593853

EP=11,260 | STEP=2,427,272 | clip=58 | consec=35 | max_consec=69 | avg_clip50=53.44

EP=11,270 | STEP=2,430,045 | clip=54 | consec=3 | max_consec=69 | avg_clip50=53.67

EP=11,280 | STEP=2,433,139 | clip=47 | consec=13 | max_consec=69 | avg_clip50=54.95

custom/		
avg_clip50	54.3	
clipped_episode_r...	22	
consecutive_clipp...	22	
max_consecutive_c...	69	
rollout/		
ep_len_mean	1.24e+03	
ep_rew_mean	537	
exploration rate	0.189	
time/		
episodes	33200	
fps	248	
time_elapsed	9812	
total timesteps	2435745	
train/		
learning_rate	0.0001	
loss	0.0688	
n_updates	596436	

EP=11,290 | STEP=2,435,963 | clip=60 | consec=23 | max_consec=69 | avg_clip50=54.73

EP=11,300 | STEP=2,438,715 | clip=55 | consec=1 | max_consec=69 | avg_clip50=53.27

EP=11,310 | STEP=2,442,448 | clip=54 | consec=11 | max_consec=69 | avg_clip50=54.30

EP=11,320 | STEP=2,445,785 | clip=55 | consec=21 | max_consec=69 | avg_clip50=55.55

custom/		
avg_clip50	56.2	
clipped_episode_r...	57.5	
consecutive_clipp...	25	
max_consecutive_c...	69	
rollout/		
ep_len_mean	1.28e+03	
ep_rew_mean	553	
exploration rate	0.185	
time/		
episodes	33300	
fps	248	
time_elapsed	9861	
total timesteps	2447190	
train/		
learning_rate	0.0001	
loss	0.0345	
n_updates	599297	

EP=11,330 | STEP=2,449,007 | clip=36 | consec=31 | max_consec=69 | avg_clip50=56.70

EP=11,340 | STEP=2,452,119 | clip=60 | consec=41 | max_consec=69 | avg_clip50=58.01

EP=11,350 | STEP=2,456,000 | clip=52 | consec=51 | max_consec=69 | avg_clip50=61.06

custom/		

	avg_clip50	58.5	
	clipped_episode_r...	75	
	consecutive_clipp...	7	
	max_consecutive_c...	69	
	rollout/		
	ep_len_mean	1.31e+03	
	ep_rew_mean	564	
	exploration rate	0.181	
	time/		
	episodes	33400	
	fps	248	
	time_elapsed	9911	
	total timesteps	2458909	
	train/		
	learning_rate	0.0001	
	loss	0.0422	
	n_updates	602227	

EP=11,360 | STEP=2,458,983 | clip=48 | consec=8 | max_consec=69 | avg_clip50=58.35

EP=11,370 | STEP=2,461,515 | clip=80 | consec=2 | max_consec=69 | avg_clip50=56.33

EP=11,380 | STEP=2,464,749 | clip=60 | consec=12 | max_consec=69 | avg_clip50=56.58

EP=11,390 | STEP=2,468,043 | clip=38 | consec=22 | max_consec=69 | avg_clip50=56.91

	custom/		
	avg_clip50	53.8	
	clipped_episode_r...	20.5	
	consecutive_clipp...	27	
	max_consecutive_c...	69	
	rollout/		
	ep_len_mean	1.31e+03	
	ep_rew_mean	568	
	exploration rate	0.178	
	time/		
	episodes	33500	
	fps	248	
	time_elapsed	9955	
	total timesteps	2469143	
	train/		
	learning_rate	0.0001	
	loss	0.442	
	n_updates	604785	

EP=11,400 | STEP=2,470,766 | clip=38 | consec=32 | max_consec=69 | avg_clip50=52.63

EP=11,410 | STEP=2,473,851 | clip=39 | consec=42 | max_consec=69 | avg_clip50=50.97

EP=11,420 | STEP=2,477,198 | clip=50 | consec=52 | max_consec=69 | avg_clip50=54.55

EP=11,430 | STEP=2,479,997 | clip=57 | consec=7 | max_consec=69 | avg_clip50=52.37

	custom/		
	avg_clip50	52.4	
	clipped_episode_r...	57	
	consecutive_clipp...	7	
	max_consecutive_c...	69	

rollout/		
ep_len_mean	1.28e+03	
ep_rew_mean	545	
exploration rate	0.174	
time/		
episodes	33600	
fps	247	
time_elapsed	10002	
total timesteps	2479997	
train/		
learning_rate	0.0001	
loss	0.614	
n_updates	607499	

EP=11,440 | STEP=2,483,181 | clip=49 | consec=17 | max_consec=69 | avg_clip50=52.49

EP=11,450 | STEP=2,486,221 | clip=70 | consec=27 | max_consec=69 | avg_clip50=56.39

EP=11,460 | STEP=2,489,837 | clip=54 | consec=37 | max_consec=69 | avg_clip50=59.91

custom/		
avg_clip50	58.2	
clipped_episode_r...	66.5	
consecutive_clipp...	44	
max_consecutive_c...	69	
rollout/		
ep_len_mean	1.28e+03	
ep_rew_mean	553	
exploration rate	0.17	
time/		
episodes	33700	
fps	247	
time_elapsed	10053	
total timesteps	2491787	
train/		
learning_rate	0.0001	
loss	0.222	
n_updates	610446	

EP=11,470 | STEP=2,492,721 | clip=54 | consec=47 | max_consec=69 | avg_clip50=56.95

EP=11,480 | STEP=2,495,099 | clip=78 | consec=57 | max_consec=69 | avg_clip50=56.26

EP=11,490 | STEP=2,498,197 | clip=38 | consec=2 | max_consec=69 | avg_clip50=55.66

EP=11,500 | STEP=2,502,252 | clip=54 | consec=12 | max_consec=69 | avg_clip50=56.70

custom/		
avg_clip50	55.5	
clipped_episode_r...	63	
consecutive_clipp...	14	
max_consecutive_c...	69	
rollout/		
ep_len_mean	1.31e+03	
ep_rew_mean	566	
exploration rate	0.167	
time/		
episodes	33800	

fps	247
time_elapsed	10100
total timesteps	2502853
train/	
learning_rate	0.0001
loss	0.325
n_updates	613213

EP=11,510 | STEP=2,505,379 | clip=19 | consec=0 | max_consec=69 | avg_clip50=55.59

EP=11,520 | STEP=2,508,419 | clip=10 | consec=0 | max_consec=69 | avg_clip50=57.94

EP=11,530 | STEP=2,511,509 | clip=80 | consec=10 | max_consec=69 | avg_clip50=60.16

custom/	
avg_clip50	61
clipped_episode_r...	110
consecutive_clipp...	16
max_consecutive_c...	69
rollout/	
ep_len_mean	1.32e+03
ep_rew_mean	584
exploration rate	0.163
time/	
episodes	33900
fps	247
time_elapsed	10148
total timesteps	2513956
train/	
learning_rate	0.0001
loss	0.0604
n_updates	615988

EP=11,540 | STEP=2,514,890 | clip=49 | consec=20 | max_consec=69 | avg_clip50=61.18

EP=11,550 | STEP=2,517,844 | clip=42 | consec=3 | max_consec=69 | avg_clip50=57.70

EP=11,560 | STEP=2,520,916 | clip=58 | consec=13 | max_consec=69 | avg_clip50=56.00

EP=11,570 | STEP=2,524,269 | clip=60 | consec=23 | max_consec=69 | avg_clip50=57.65

custom/	
avg_clip50	57.8
clipped_episode_r...	50
consecutive_clipp...	25
max_consecutive_c...	69
rollout/	
ep_len_mean	1.31e+03
ep_rew_mean	579
exploration rate	0.159
time/	
episodes	34000
fps	247
time_elapsed	10195
total timesteps	2524902
train/	
learning_rate	0.0001
loss	0.0842

n_updates	618725
-----------	--------

EP=11,580 | STEP=2,527,212 | clip=34 | consec=33 | max_consec=69 | avg_clip50=57.87

EP=11,590 | STEP=2,530,171 | clip=44 | consec=2 | max_consec=69 | avg_clip50=56.26

EP=11,600 | STEP=2,533,460 | clip=54 | consec=12 | max_consec=69 | avg_clip50=56.85

custom/	
avg_clip50	56.8
clipped_episode_r...	38.5
consecutive_clipp...	18
max_consecutive_c...	69
rollout/	
ep_len_mean	1.28e+03
ep_rew_mean	567
exploration rate	0.156
time/	
episodes	34100
fps	247
time_elapsed	10240
total timesteps	2535267
train/	
learning_rate	0.0001
loss	0.0522
n_updates	621316

EP=11,610 | STEP=2,536,221 | clip=54 | consec=22 | max_consec=69 | avg_clip50=57.59

EP=11,620 | STEP=2,539,228 | clip=57 | consec=32 | max_consec=69 | avg_clip50=56.24

EP=11,630 | STEP=2,542,212 | clip=6 | consec=0 | max_consec=69 | avg_clip50=55.83

EP=11,640 | STEP=2,544,943 | clip=58 | consec=1 | max_consec=69 | avg_clip50=55.49

custom/	
avg_clip50	55.3
clipped_episode_r...	35
consecutive_clipp...	3
max_consecutive_c...	69
rollout/	
ep_len_mean	1.24e+03
ep_rew_mean	553
exploration rate	0.152
time/	
episodes	34200
fps	247
time_elapsed	10285
total timesteps	2545698
train/	
learning_rate	0.0001
loss	0.232
n_updates	623924

EP=11,650 | STEP=2,547,640 | clip=74 | consec=2 | max_consec=69 | avg_clip50=54.11

EP=11,660 | STEP=2,550,778 | clip=44 | consec=12 | max_consec=69 | avg_clip50=55.90

EP=11,670 | STEP=2,553,539 | clip=74 | consec=22 | max_consec=69 | avg_clip50=55.

custom/		
avg_clip50	52.9	
clipped_episode_r...	54.5	
consecutive_clipp...	5	
max_consecutive_c...	69	
rollout/		
ep_len_mean	1.19e+03	
ep_rew_mean	544	
exploration rate	0.149	
time/		
episodes	34300	
fps	247	
time_elapsed	10328	
total timesteps	2555616	
train/		
learning_rate	0.0001	
loss	0.106	
n_updates	626403	

EP=11,680 | STEP=2,555,875 | clip=24 | consec=7 | max_consec=69 | avg_clip50=52.9
5
EP=11,690 | STEP=2,559,368 | clip=52 | consec=6 | max_consec=69 | avg_clip50=55.0
0
EP=11,700 | STEP=2,562,086 | clip=60 | consec=3 | max_consec=69 | avg_clip50=55.5
0
EP=11,710 | STEP=2,565,235 | clip=103 | consec=4 | max_consec=69 | avg_clip50=54.
41

custom/		
avg_clip50	54.3	
clipped_episode_r...	35.5	
consecutive_clipp...	7	
max_consecutive_c...	69	
rollout/		
ep_len_mean	1.21e+03	
ep_rew_mean	552	
exploration rate	0.145	
time/		
episodes	34400	
fps	247	
time_elapsed	10374	
total timesteps	2566170	
train/		
learning_rate	0.0001	
loss	0.0418	
n_updates	629042	

EP=11,720 | STEP=2,568,130 | clip=57 | consec=14 | max_consec=69 | avg_clip50=53.
02
EP=11,730 | STEP=2,571,459 | clip=60 | consec=24 | max_consec=69 | avg_clip50=59.
03
EP=11,740 | STEP=2,574,582 | clip=60 | consec=34 | max_consec=69 | avg_clip50=59.
13
EP=11,750 | STEP=2,577,609 | clip=60 | consec=44 | max_consec=69 | avg_clip50=60.
11

custom/		
avg_clip50	60.1	

	clipped_episode_r...	60.5	
	consecutive_clipp...	44	
	max_consecutive_c...	69	
	rollout/		
	ep_len_mean	1.24e+03	
	ep_rew_mean	578	
	exploration rate	0.142	
	time/		
	episodes	34500	
	fps	247	
	time_elapsed	10423	
	total timesteps	2577635	
	train/		
	learning_rate	0.0001	
	loss	0.0536	
	n_updates	631908	

EP=11,760 | STEP=2,580,153 | clip=77 | consec=3 | max_consec=69 | avg_clip50=57.2
2

EP=11,770 | STEP=2,583,127 | clip=60 | consec=5 | max_consec=69 | avg_clip50=57.5
2

EP=11,780 | STEP=2,586,115 | clip=61 | consec=15 | max_consec=69 | avg_clip50=53.
67

	custom/		
	avg_clip50	49.7	
	clipped_episode_r...	39.5	
	consecutive_clipp...	2	
	max_consecutive_c...	69	
	rollout/		
	ep_len_mean	1.23e+03	
	ep_rew_mean	549	
	exploration rate	0.138	
	time/		
	episodes	34600	
	fps	247	
	time_elapsed	10467	
	total timesteps	2587582	
	train/		
	learning_rate	0.0001	
	loss	0.0394	
	n_updates	634395	

EP=11,790 | STEP=2,588,849 | clip=54 | consec=6 | max_consec=69 | avg_clip50=51.1
3

EP=11,800 | STEP=2,591,672 | clip=61 | consec=7 | max_consec=69 | avg_clip50=50.1
3

EP=11,810 | STEP=2,595,334 | clip=96 | consec=17 | max_consec=69 | avg_clip50=54.
97

EP=11,820 | STEP=2,598,298 | clip=36 | consec=27 | max_consec=69 | avg_clip50=54.
56

	custom/		
	avg_clip50	55.5	
	clipped_episode_r...	54.5	
	consecutive_clipp...	29	
	max_consecutive_c...	69	
	rollout/		
	ep_len_mean	1.25e+03	
	ep_rew_mean	562	

exploration rate	0.135	
time/		
episodes	34700	
fps	247	
time_elapsed	10515	
total timesteps	2598850	
train/		
learning_rate	0.0001	
loss	0.0442	
n_updates	637212	

EP=11,830 | STEP=2,601,071 | clip=18 | consec=0 | max_consec=69 | avg_clip50=56.17

EP=11,840 | STEP=2,604,005 | clip=31 | consec=10 | max_consec=69 | avg_clip50=57.27

EP=11,850 | STEP=2,607,372 | clip=95 | consec=20 | max_consec=69 | avg_clip50=60.09

custom/		
avg_clip50	60.1	
clipped_episode_r...	80.5	
consecutive_clipp...	28	
max_consecutive_c...	69	
rollout/		
ep_len_mean	1.27e+03	
ep_rew_mean	572	
exploration rate	0.131	
time/		
episodes	34800	
fps	247	
time_elapsed	10565	
total timesteps	2610276	
train/		
learning_rate	0.0001	
loss	0.0539	
n_updates	640068	

EP=11,860 | STEP=2,610,658 | clip=66 | consec=30 | max_consec=69 | avg_clip50=59.45

EP=11,870 | STEP=2,613,953 | clip=52 | consec=40 | max_consec=69 | avg_clip50=60.41

EP=11,880 | STEP=2,617,612 | clip=54 | consec=50 | max_consec=69 | avg_clip50=61.55

EP=11,890 | STEP=2,621,058 | clip=60 | consec=60 | max_consec=69 | avg_clip50=63.25

custom/		
avg_clip50	63.3	
clipped_episode_r...	75	
consecutive_clipp...	65	
max_consecutive_c...	69	
rollout/		
ep_len_mean	1.33e+03	
ep_rew_mean	607	
exploration rate	0.127	
time/		
episodes	34900	
fps	246	
time_elapsed	10618	
total timesteps	2622524	

train/	
learning_rate	0.0001
loss	0.0272
n_updates	643130

```
-----
EP=11,900 | STEP=2,624,159 | clip=52 | consec=70 | max_consec=70 | avg_clip50=61.99
EP=11,910 | STEP=2,627,060 | clip=39 | consec=80 | max_consec=80 | avg_clip50=59.96
EP=11,920 | STEP=2,630,106 | clip=72 | consec=90 | max_consec=90 | avg_clip50=60.35
EP=11,930 | STEP=2,633,424 | clip=98 | consec=100 | max_consec=100 | avg_clip50=60.29
```

Objetivo conseguido: 100 episodios consecutivos superando el umbral.

Entrenamiento finalizado.

max_consec alcanzado: 100

Log JSON: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.json

Log CSV: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.csv

Out[]: <stable_baselines3.dqn.DQN at 0x7d661b344290>

```
In [ ]: # =====
# 13. Guardado manual del modelo
# =====
# Primero se intenta guardar con SB3. Si falla por serialización ,
# se guarda únicamente la política y el optimizador con PyTorch.

try:
    model.save(str(MODEL_LATEST))
    print('Modelo completo SB3 guardado en:', MODEL_LATEST)

except Exception as e:
    print('No se pudo guardar el modelo completo con SB3.')
    print('Error:', repr(e))
    save_policy_checkpoint(model, POLICY_LATEST)
    print('Política/optimizador guardados con PyTorch en:', POLICY_LATEST)
```

No se pudo guardar el modelo completo con SB3.

Error: IndexError('tuple index out of range')

Política/optimizador guardados con PyTorch en: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_spaceinvaders_v0_policy_latest.pt

```
In [ ]: # =====
# 14. Tablas y gráficas del entrenamiento
# =====

if not CSV_LOG_FILENAME.exists():
    raise FileNotFoundError(f'No existe el log CSV: {CSV_LOG_FILENAME}')

df = pd.read_csv(CSV_LOG_FILENAME)
print('Filas registradas:', len(df))
print('Columnas:', list(df.columns))

display(df.tail(10))

summary_table = pd.DataFrame({
    'metrica': [
        'episodios_registrados',
        'ultimo_step',
```

```

        'mejor_clip',
        'mejor_avg_clip50',
        'mejor_max_consec',
        'ultimo_avg_clip50',
        'ultimo_exploration_rate'
    ],
    'valor': [
        len(df),
        int(df['total_steps'].iloc[-1]) if len(df) else 0,
        float(df['clipped_episode_reward'].max()) if len(df) else 0.0,
        float(df['avg_clip50'].max()) if len(df) else 0.0,
        int(df['max_consecutive_clipped_over_20'].max()) if len(df) else 0,
        float(df['avg_clip50'].iloc[-1]) if len(df) else 0.0,
        float(df['exploration_rate'].iloc[-1]) if 'exploration_rate' in df and 1
    ]
})

display(summary_table)

steps = df['total_steps'].to_numpy()
clipped = df['clipped_episode_reward'].to_numpy()
consec = df['consecutive_clipped_over_20'].to_numpy()
max_consec = df['max_consecutive_clipped_over_20'].to_numpy()
avg50 = df['avg_clip50'].to_numpy()

plt.figure(figsize=(10, 4))
plt.plot(steps, clipped)
plt.title('Recompensa clipped por episodio')
plt.xlabel('Steps')
plt.ylabel('Clipped reward')
plt.grid(True)
plt.show()

plt.figure(figsize=(10, 4))
plt.plot(steps, avg50)
plt.title('Media de recompensa clipped en los últimos 50 episodios')
plt.xlabel('Steps')
plt.ylabel('avg_clip50')
plt.grid(True)
plt.show()

plt.figure(figsize=(10, 4))
plt.plot(steps, consec, label='consec')
plt.plot(steps, max_consec, label='max_consec')
plt.axhline(TARGET_CONSECUTIVE_EPISODES, linestyle='--')
plt.title('Episodios consecutivos con recompensa clipped > 20')
plt.xlabel('Steps')
plt.ylabel('Consecutivos')
plt.legend()
plt.grid(True)
plt.show()

if 'exploration_rate' in df.columns:
    plt.figure(figsize=(10, 4))
    plt.plot(steps, df['exploration_rate'])
    plt.title('Evolución de epsilon / exploration_rate')
    plt.xlabel('Steps')
    plt.ylabel('Exploration rate')
    plt.grid(True)
    plt.show()

```



```
print('Mejor max_consec:', int(max(max_consec)) if len(max_consec) else 0)
print('Mejor avg_clip50:', float(max(avg50)) if len(avg50) else 0.0)
```

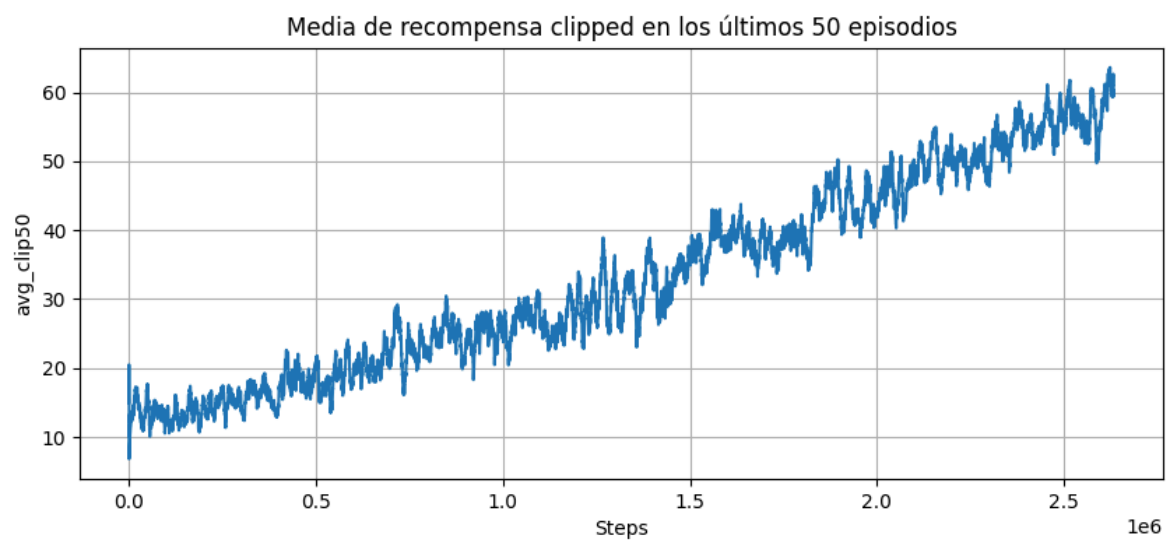
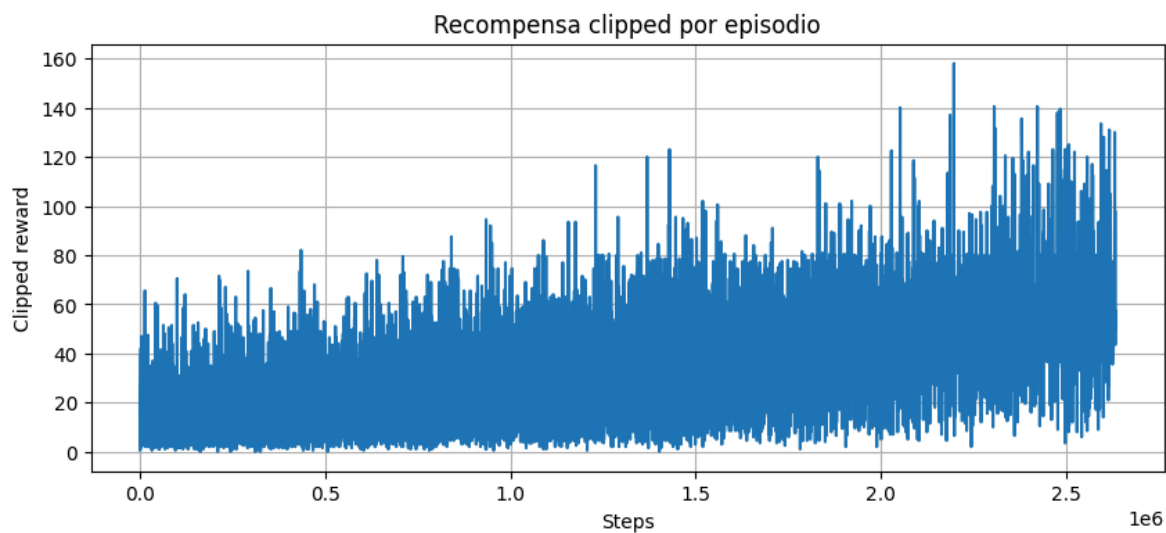
Filas registradas: 11961

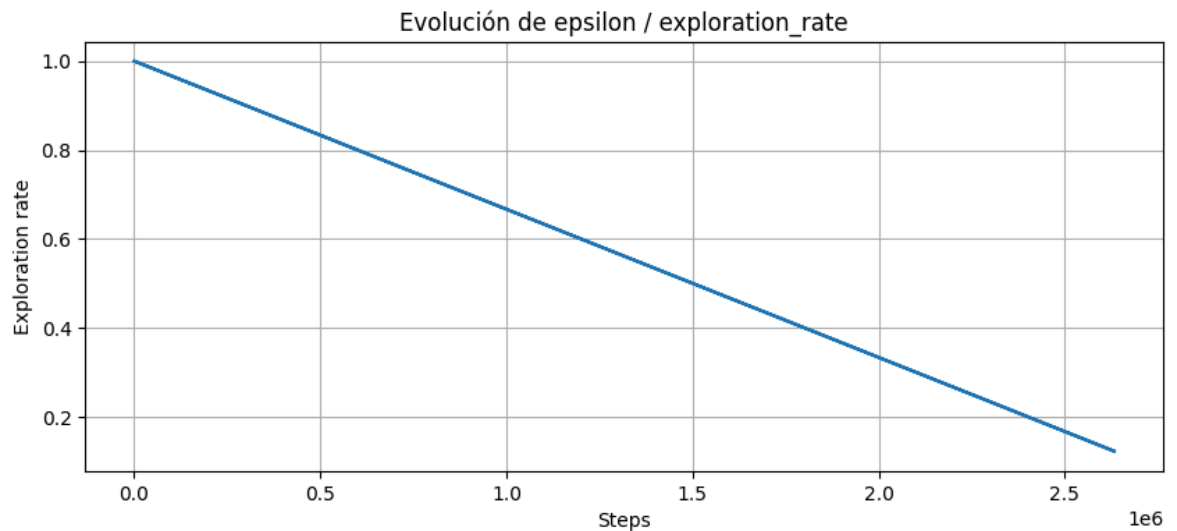
Columnas: ['episode', 'total_steps', 'episode_reward', 'clipped_episode_reward', 'consecutive_clipped_over_20', 'max_consecutive_clipped_over_20', 'avg_clip50', 'duration', 'exploration_rate']

	episode	total_steps	episode_reward	clipped_episode_reward	consecutive_clipped_
11951	11921	2630487	525.0	52.5	
11952	11922	2630696	595.0	59.5	
11953	11923	2631025	520.0	52.0	
11954	11924	2631553	1300.0	130.0	
11955	11925	2631852	530.0	53.0	
11956	11926	2632170	550.0	55.0	
11957	11927	2632426	500.0	50.0	
11958	11928	2632752	575.0	57.5	
11959	11929	2632938	435.0	43.5	
11960	11930	2633424	975.0	97.5	



	metrica	valor
0	episodios_registrados	1.196100e+04
1	ultimo_step	2.633424e+06
2	mejor_clip	1.580000e+02
3	mejor_avg_clip50	6.361000e+01
4	mejor_max_consec	1.000000e+02
5	ultimo_avg_clip50	6.029000e+01
6	ultimo_exploration_rate	1.230701e-01





Mejor max_consec: 100
 Mejor avg_clip50: 63.61

```
In [ ]: # =====
# 15. Evaluación visual rápida
# =====
# Si existe el modelo completo de SB3, se carga.
# Si no, se usa el modelo actual de memoria.

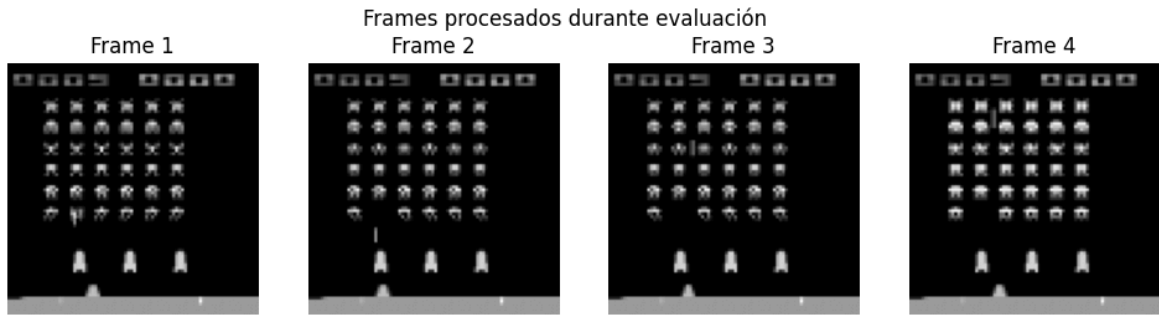
if MODEL_LATEST.exists():
    try:
        model = DQN.load(str(MODEL_LATEST), env=env, device='auto')
        print('Modelo cargado para evaluación:', MODEL_LATEST)
    except Exception as e:
        print('No se pudo cargar MODEL_LATEST. Se usa el modelo actual en memoria.')
        print('Error:', repr(e))
else:
    print('No existe MODEL_LATEST. Se usa el modelo actual en memoria.')

obs = env.reset()
for _ in range(10):
    action, _ = model.predict(obs, deterministic=True)
    obs, reward, done, info = env.step(action)

obs0 = obs[0]
if obs0.shape[0] in [1, 3, 4, 12]:
    frames = obs0
else:
    frames = np.transpose(obs0, (2, 0, 1))

num_frames = min(4, frames.shape[0])
plt.figure(figsize=(12, 3))
for i in range(num_frames):
    plt.subplot(1, num_frames, i + 1)
    plt.imshow(frames[i], cmap='gray')
    plt.title(f'Frame {i + 1}')
    plt.axis('off')
plt.suptitle('Frames procesados durante evaluación')
plt.show()
```

No se pudo cargar MODEL_LATEST. Se usa el modelo actual en memoria.
 Error: ValueError("Error: the file <_io.BufferedReader name='/content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_spaceinvaders_v0_sb3_la test.zip'> wasn't a zip-file")



```
In [ ]: # =====
# 16. Cerrar entorno
# =====

# env.close()
```

```
In [ ]: # =====
# 17. Comprobación final de ficheros y resultados
# =====

print('OUTPUT_DIR:', OUTPUT_DIR)
print('Existe OUTPUT_DIR:', OUTPUT_DIR.exists())
print('MODEL_LATEST:', MODEL_LATEST, '| existe:', MODEL_LATEST.exists())
print('POLICY_LATEST:', POLICY_LATEST, '| existe:', POLICY_LATEST.exists())
print('LOG_FILENAME:', LOG_FILENAME, '| existe:', LOG_FILENAME.exists())
print('CSV_LOG_FILENAME:', CSV_LOG_FILENAME, '| existe:', CSV_LOG_FILENAME.exists())
print('TENSORBOARD_DIR:', TENSORBOARD_DIR, '| existe:', TENSORBOARD_DIR.exists())
print('MONITOR_DIR:', MONITOR_DIR, '| existe:', MONITOR_DIR.exists())

if CSV_LOG_FILENAME.exists():
    df_check = pd.read_csv(CSV_LOG_FILENAME)
    print('\nResumen del log CSV')
    print('Episodios registrados:', len(df_check))

    if len(df_check) > 0:
        print('Último episodio:', int(df_check['episode'].iloc[-1]))
        print('Último step:', int(df_check['total_steps'].iloc[-1]))
        print('Mejor clipped_episode_reward:', float(df_check['clipped_episode_r']))
        print('Mejor avg_clip50:', float(df_check['avg_clip50'].max()))
        print('Mejor max_consec:', int(df_check['max_consecutive_clipped_over_20']))
        print('Último avg_clip50:', float(df_check['avg_clip50'].iloc[-1]))

        display(df_check.tail(5))
    else:
        print('\nNo se encontró el CSV de entrenamiento. Ejecuta el bloque de entren

if LOG_FILENAME.exists():
    with open(LOG_FILENAME, 'r') as f:
        data_check = json.load(f)
        print('\nResumen del log JSON')
        print('Claves:', list(data_check.keys()))
        print('Episodios en JSON:', len(data_check.get('episode', [])))
    else:
        print('\nNo se encontró el JSON de entrenamiento.')

# Listado breve de ficheros creados en la carpeta del proyecto.
print('\nFicheros principales en OUTPUT_DIR:')
for p in sorted(OUTPUT_DIR.glob('*')):
    if p.is_file():

```

```

        print(f'- {p.name} ({p.stat().st_size / 1024:.1f} KB)')
    else:
        print(f'- {p.name}/')

```

OUTPUT_DIR: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0

Existe OUTPUT_DIR: True

MODEL_LATEST: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_spaceinvaders_v0_sb3_latest.zip | existe: True

POLICY_LATEST: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_spaceinvaders_v0_policy_latest.pt | existe: True

LOG_FILENAME: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.json | existe: True

CSV_LOG_FILENAME: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.csv | existe: True

TENSORBOARD_DIR: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/tensorboard | existe: True

MONITOR_DIR: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/monitor | existe: True

Resumen del log CSV

Episodios registrados: 11961

Último episodio: 11930

Último step: 2633424

Mejor clipped_episode_reward: 158.0

Mejor avg_clip50: 63.61

Mejor max_consec: 100

Último avg_clip50: 60.29

	episode	total_steps	episode_reward	clipped_episode_reward	consecutive_clipped
11956	11926	2632170	550.0	55.0	
11957	11927	2632426	500.0	50.0	
11958	11928	2632752	575.0	57.5	
11959	11929	2632938	435.0	43.5	
11960	11930	2633424	975.0	97.5	

Resumen del log JSON

Claves: ['episode', 'total_steps', 'episode_reward', 'clipped_episode_reward', 'consecutive_clipped_over_20', 'max_consecutive_clipped_over_20', 'avg_clip50', 'duration', 'exploration_rate']

Episodios en JSON: 11931

Ficheros principales en OUTPUT_DIR:

- .DS_Store (6.0 KB)
- dqn_SpaceInvaders-v0_sb3_log.csv (819.1 KB)
- dqn_SpaceInvaders-v0_sb3_log.json (1341.4 KB)
- dqn_spaceinvaders_v0_policy_latest.pt (26381.2 KB)
- dqn_spaceinvaders_v0_sb3_latest.zip (0.0 KB)
- monitor/
- tensorboard/

```

In [ ]: # =====
        # Entorno de evaluación real
        # =====

```

```

import gym
import numpy as np
from stable_baselines3.common.atari_wrappers import (
    MaxAndSkipEnv,
    FireResetEnv,
    WarpFrame
)
from stable_baselines3.common.vec_env import DummyVecEnv, VecFrameStack, VecTran

def create_eval_env():
    def make_env():
        env = gym.make(ENV_ID)

        env.seed(SEED)
        env = MaxAndSkipEnv(env, skip=4)

        if "FIRE" in env.unwrapped.get_action_meanings():
            env = FireResetEnv(env)

        env = WarpFrame(env)

        return env

    env = DummyVecEnv([make_env])
    env = VecFrameStack(env, n_stack=4)
    env = VecTransposeImage(env)

    return env

```

```

In [ ]: # =====
# Cargar policy aprendida
# =====

import torch
from stable_baselines3 import DQN

POLICY_LATEST = OUTPUT_DIR / "dqn_spaceinvaders_v0_policy_latest.pt"

eval_env = create_eval_env()

model = DQN(
    policy="CnnPolicy",
    env=eval_env,
    learning_rate=LEARNING_RATE,
    buffer_size=BUFFER_SIZE,
    learning_starts=LEARNING_STARTS,
    batch_size=BATCH_SIZE,
    gamma=GAMMA,
    train_freq=TRAIN_FREQ,
    target_update_interval=TARGET_UPDATE_INTERVAL,
    exploration_initial_eps=EXPLORATION_INITIAL_EPS,
    exploration_final_eps=EXPLORATION_FINAL_EPS,
    exploration_fraction=EXPLORATION_FRACTION,
    policy_kwargs=policy_kwargs,
    verbose=1,
    seed=SEED,
    device="auto"
)

checkpoint = torch.load(POLICY_LATEST, map_location=model.device)

```

```

model.policy.load_state_dict(checkpoint["policy_state_dict"])
model.num_timesteps = checkpoint["num_timesteps"]
model.exploration_rate = checkpoint["exploration_rate"]

print("Policy cargada correctamente")
print("Timesteps:", model.num_timesteps)
print("Exploration rate:", model.exploration_rate)

```

Using cuda device

Policy cargada correctamente

Timesteps: 2633424

Exploration rate: 0.12307014100000035

```

In [ ]: # =====
# Testing sin entrenamiento
# =====

TEST_EPISODES = 120

rewards = []
clipped_rewards = []

for ep in range(TEST_EPISODES):
    obs = eval_env.reset()
    done = False
    total_reward = 0.0

    while not done:
        action, _ = model.predict(obs, deterministic=True)
        obs, reward, done, infos = eval_env.step(action)
        total_reward += float(reward[0])
        done = bool(done[0])

    clipped_reward = total_reward / 10.0
    rewards.append(total_reward)
    clipped_rewards.append(clipped_reward)

    print(
        f"TEST EP={ep+1:03d} | "
        f"reward={total_reward:.0f} | "
        f"clip={clipped_reward:.0f}"
    )

print("Reward medio:", np.mean(rewards))
print("Clip medio:", np.mean(clipped_rewards))
print("Reward máximo:", np.max(rewards))
print("Clip máximo:", np.max(clipped_rewards))

eval_env.close()

```

TEST	EP=001		reward=900		clip=90
TEST	EP=002		reward=600		clip=60
TEST	EP=003		reward=575		clip=58
TEST	EP=004		reward=570		clip=57
TEST	EP=005		reward=800		clip=80
TEST	EP=006		reward=600		clip=60
TEST	EP=007		reward=545		clip=54
TEST	EP=008		reward=555		clip=56
TEST	EP=009		reward=360		clip=36
TEST	EP=010		reward=575		clip=58
TEST	EP=011		reward=1180		clip=118
TEST	EP=012		reward=600		clip=60
TEST	EP=013		reward=575		clip=58
TEST	EP=014		reward=545		clip=54
TEST	EP=015		reward=600		clip=60
TEST	EP=016		reward=915		clip=92
TEST	EP=017		reward=800		clip=80
TEST	EP=018		reward=575		clip=58
TEST	EP=019		reward=600		clip=60
TEST	EP=020		reward=605		clip=60
TEST	EP=021		reward=775		clip=78
TEST	EP=022		reward=575		clip=58
TEST	EP=023		reward=1295		clip=130
TEST	EP=024		reward=600		clip=60
TEST	EP=025		reward=425		clip=42
TEST	EP=026		reward=1000		clip=100
TEST	EP=027		reward=605		clip=60
TEST	EP=028		reward=600		clip=60
TEST	EP=029		reward=800		clip=80
TEST	EP=030		reward=600		clip=60
TEST	EP=031		reward=980		clip=98
TEST	EP=032		reward=605		clip=60
TEST	EP=033		reward=545		clip=54
TEST	EP=034		reward=630		clip=63
TEST	EP=035		reward=755		clip=76
TEST	EP=036		reward=545		clip=54
TEST	EP=037		reward=570		clip=57
TEST	EP=038		reward=545		clip=54
TEST	EP=039		reward=485		clip=48
TEST	EP=040		reward=800		clip=80
TEST	EP=041		reward=575		clip=58
TEST	EP=042		reward=600		clip=60
TEST	EP=043		reward=650		clip=65
TEST	EP=044		reward=750		clip=75
TEST	EP=045		reward=600		clip=60
TEST	EP=046		reward=470		clip=47
TEST	EP=047		reward=770		clip=77
TEST	EP=048		reward=800		clip=80
TEST	EP=049		reward=895		clip=90
TEST	EP=050		reward=785		clip=78
TEST	EP=051		reward=380		clip=38
TEST	EP=052		reward=545		clip=54
TEST	EP=053		reward=545		clip=54
TEST	EP=054		reward=775		clip=78
TEST	EP=055		reward=570		clip=57
TEST	EP=056		reward=750		clip=75
TEST	EP=057		reward=520		clip=52
TEST	EP=058		reward=1155		clip=116
TEST	EP=059		reward=570		clip=57
TEST	EP=060		reward=745		clip=74

TEST	EP=061		reward=580		clip=58
TEST	EP=062		reward=545		clip=54
TEST	EP=063		reward=570		clip=57
TEST	EP=064		reward=415		clip=42
TEST	EP=065		reward=550		clip=55
TEST	EP=066		reward=815		clip=82
TEST	EP=067		reward=445		clip=44
TEST	EP=068		reward=600		clip=60
TEST	EP=069		reward=1175		clip=118
TEST	EP=070		reward=575		clip=58
TEST	EP=071		reward=815		clip=82
TEST	EP=072		reward=515		clip=52
TEST	EP=073		reward=575		clip=58
TEST	EP=074		reward=575		clip=58
TEST	EP=075		reward=605		clip=60
TEST	EP=076		reward=540		clip=54
TEST	EP=077		reward=600		clip=60
TEST	EP=078		reward=510		clip=51
TEST	EP=079		reward=500		clip=50
TEST	EP=080		reward=465		clip=46
TEST	EP=081		reward=545		clip=54
TEST	EP=082		reward=605		clip=60
TEST	EP=083		reward=555		clip=56
TEST	EP=084		reward=770		clip=77
TEST	EP=085		reward=1005		clip=100
TEST	EP=086		reward=600		clip=60
TEST	EP=087		reward=555		clip=56
TEST	EP=088		reward=600		clip=60
TEST	EP=089		reward=540		clip=54
TEST	EP=090		reward=600		clip=60
TEST	EP=091		reward=600		clip=60
TEST	EP=092		reward=1055		clip=106
TEST	EP=093		reward=570		clip=57
TEST	EP=094		reward=600		clip=60
TEST	EP=095		reward=515		clip=52
TEST	EP=096		reward=580		clip=58
TEST	EP=097		reward=785		clip=78
TEST	EP=098		reward=575		clip=58
TEST	EP=099		reward=575		clip=58
TEST	EP=100		reward=1030		clip=103
TEST	EP=101		reward=550		clip=55
TEST	EP=102		reward=1150		clip=115
TEST	EP=103		reward=630		clip=63
TEST	EP=104		reward=575		clip=58
TEST	EP=105		reward=515		clip=52
TEST	EP=106		reward=725		clip=72
TEST	EP=107		reward=575		clip=58
TEST	EP=108		reward=545		clip=54
TEST	EP=109		reward=1405		clip=140
TEST	EP=110		reward=600		clip=60
TEST	EP=111		reward=600		clip=60
TEST	EP=112		reward=415		clip=42
TEST	EP=113		reward=470		clip=47
TEST	EP=114		reward=1170		clip=117
TEST	EP=115		reward=575		clip=58
TEST	EP=116		reward=1205		clip=120
TEST	EP=117		reward=585		clip=58
TEST	EP=118		reward=570		clip=57
TEST	EP=119		reward=600		clip=60
TEST	EP=120		reward=520		clip=52

Reward medio: 662.25
Clip medio: 66.225
Reward máximo: 1405.0
Clip máximo: 140.5

Vídeo del juego

Se graban dos vídeos con la política **del modelo base ya entrenado** (determinista, cargada previamente desde su checkpoint con la celda "Cargar policy aprendida"). Esto sustituye a la renderización en vivo de gym, que no es compatible con Colab:

1. **Una partida individual** (el wrapper de Atari corta el episodio al perder una vida).
2. **10 episodios seguidos** en un único mp4, para observar el comportamiento del agente de forma más representativa.

Los vídeos quedan embebidos en el notebook. Las variantes de la Parte 2 se comparan de forma **cuantitativa** mediante las gráficas y el test de 120 episodios, por lo que no aporta información adicional grabar una partida por cada variante.

```
In [ ]: # =====  
# Grabar una partida en vídeo  
# =====  
  
from stable_baselines3.common.vec_env import VecFrameStack, VecVideoRecorder  
from IPython.display import display  
import shutil  
  
VIDEO_DIR = OUTPUT_DIR / 'video_1'  
if VIDEO_DIR.exists():  
    shutil.rmtree(VIDEO_DIR)  
VIDEO_DIR.mkdir(parents=True, exist_ok=True)  
  
video_env = make_atari_env(  
    ENV_ID,  
    n_envs=1,  
    seed=SEED,  
    wrapper_kwargs=dict(clip_reward=True)  
)  
video_env = VecFrameStack(video_env, n_stack=4)  
video_env = VecVideoRecorder(  
    video_env,  
    str(VIDEO_DIR),  
    record_video_trigger=lambda step: step == 0,  
    video_length=5_000,  
    name_prefix='sb3-spaceinvaders'  
)  
  
obs = video_env.reset()  
for _ in range(5_000):  
    action, _ = model.predict(obs, deterministic=True)  
    obs, rewards, dones, infos = video_env.step(action)  
    if bool(dones[0]):  
        break  
  
video_env.close()
```

```

video_files = sorted(VIDEO_DIR.glob('*.mp4'))
print('Videos generados:', [video_file.name for video_file in video_files])
display(video_files[0] if video_files else 'No se ha generado ningún vídeo.')

```

Videos generados: ['sb3-spaceinvaders-step-0-to-step-5000.mp4']
 PosixPath('/content/gdrive/My Drive/08_MIAR/sb3/grupo/A_Refuerzo/datos/sb3_1_spaceinvaders_v0/video_1/sb3-spaceinvaders-step-0-to-step-5000.mp4')

```

In [ ]: # Reproducir el MP4 generado dentro del notebook
import io
import base64
from IPython.display import HTML

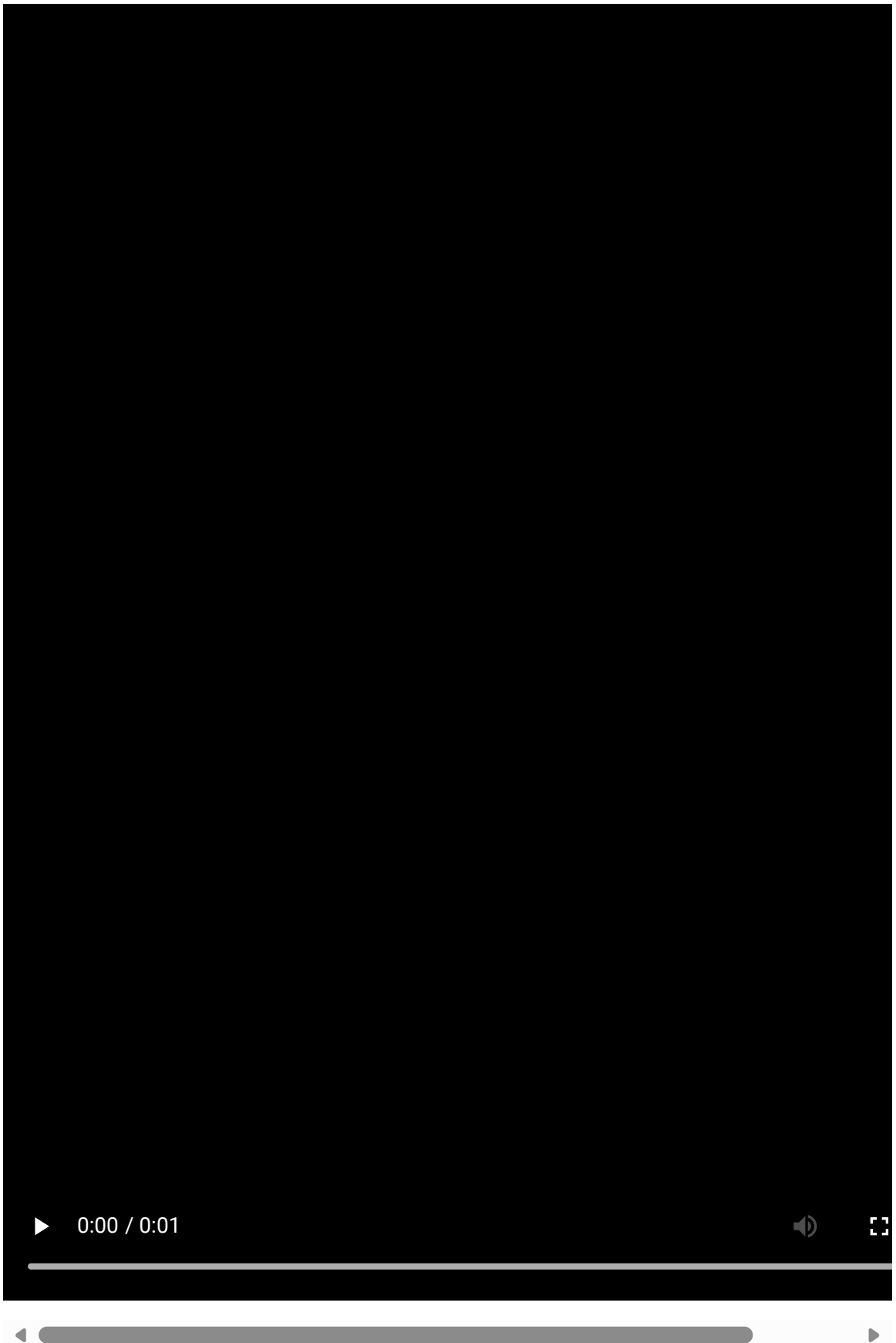
video_files = sorted(VIDEO_DIR.glob('*.mp4'))
if not video_files:
    raise FileNotFoundError(f'No se encontró ningún MP4 en {VIDEO_DIR}')

video_path = video_files[0]
with io.open(video_path, 'r+b') as video_file:
    video = video_file.read()

encoded = base64.b64encode(video).decode('ascii')
HTML(
    data=f'''
    <video width="640" height="auto" alt="space-invaders" controls>
      <source src="data:video/mp4;base64,{encoded}" type="video/mp4" />
    </video>'''
)

```

Out[]:



```
In [ ]: # =====  
# Grabar 10 partidas en un único mp4  
# =====  
  
from stable_baselines3.common.vec_env import VecFrameStack, VecVideoRecorder  
import shutil  
  
N_EPISODES = 10
```

```

MAX_STEPS_PER_EP = 5_000 # margen por episodio, ajusta si Space Invaders dura m

VIDEO_DIR = OUTPUT_DIR / 'video_10'
if VIDEO_DIR.exists():
    shutil.rmtree(VIDEO_DIR)
VIDEO_DIR.mkdir(parents=True, exist_ok=True)

video_env = make_atari_env(
    ENV_ID,
    n_envs=1,
    seed=SEED,
    wrapper_kwargs=dict(clip_reward=False)
)
video_env = VecFrameStack(video_env, n_stack=4)
video_env = VecVideoRecorder(
    video_env,
    str(VIDEO_DIR),
    record_video_trigger=lambda step: step == 0,
    video_length=N_EPISODES * MAX_STEPS_PER_EP, # cubre las 10 partidas seguidas
    name_prefix='sb3-spaceinvaders-10eps'
)

obs = video_env.reset()
episodes_done = 0
step = 0
while episodes_done < N_EPISODES:
    action, _ = model.predict(obs, deterministic=True)
    obs, rewards, dones, infos = video_env.step(action)
    step += 1
    if bool(dones[0]):
        episodes_done += 1 # el VecEnv ya resetea internamente, obs ya es del n
video_env.close()

```

```

In [ ]: # Reproducir el MP4 generado dentro del notebook
import io
import base64
from IPython.display import HTML

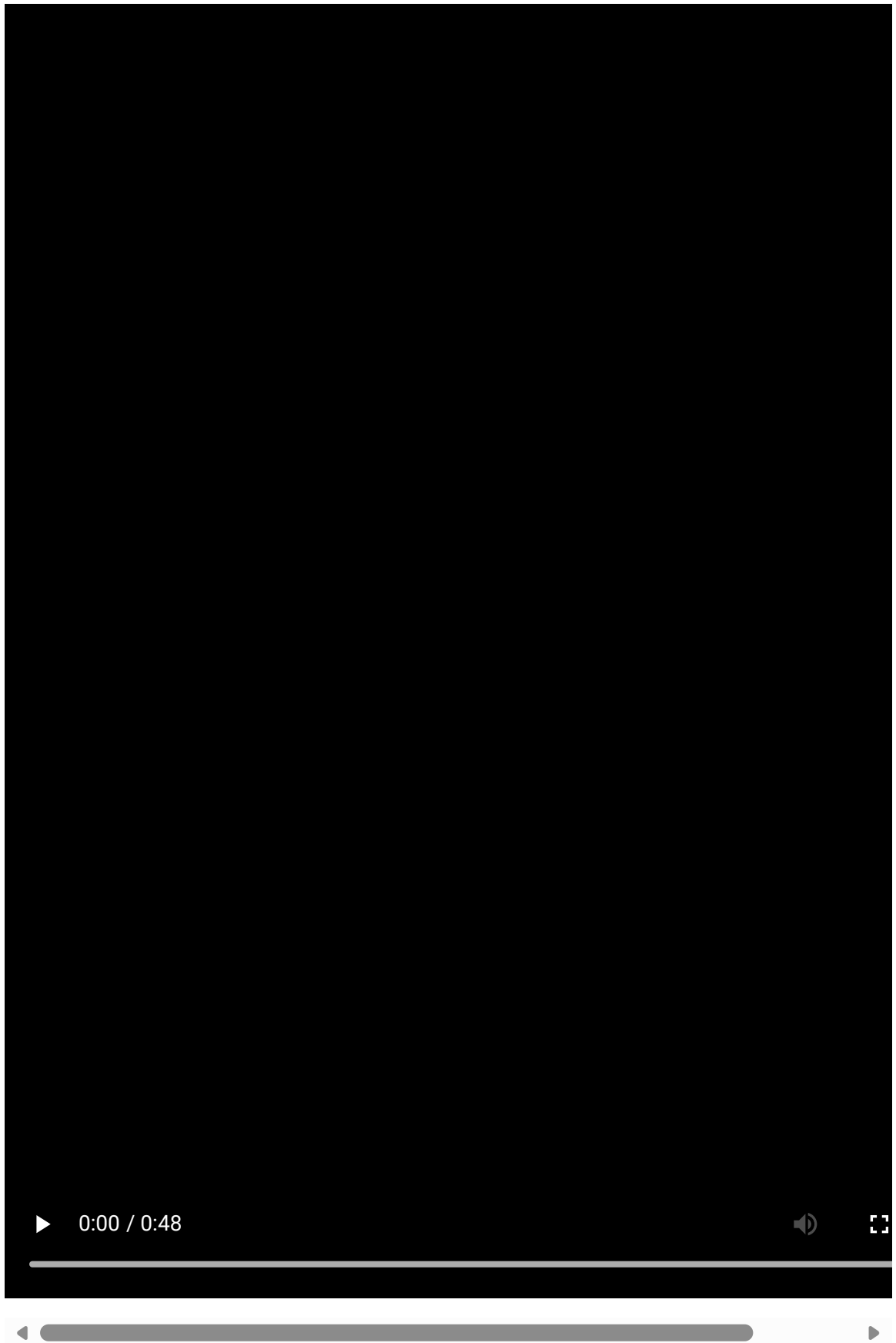
video_files = sorted(VIDEO_DIR.glob('*.mp4'))
if not video_files:
    raise FileNotFoundError(f'No se encontró ningún MP4 en {VIDEO_DIR}')

video_path = video_files[0]
with io.open(video_path, 'r+b') as video_file:
    video = video_file.read()

encoded = base64.b64encode(video).decode('ascii')
HTML(
    data=f'''
    <video width="640" height="auto" alt="space-invaders" controls>
      <source src="data:video/mp4;base64,{encoded}" type="video/mp4" />
    </video>'''
)

```

Out[]:



PARTE 2 (ampliación). Propuestas de mejora sobre el DQN base

El DQN entrenado en las secciones anteriores actúa como **referencia (baseline)**. Sobre él se implementan y comparan **tres propuestas de mejora** independientes. No se combinan entre sí a propósito: así se puede aislar y discutir el efecto de cada una.

#	Propuesta	Qué cambia	Motivación
1	Double DQN (van Hasselt et al., 2016)	El target pasa de <code>max_a Q_target(s',a)</code> a <code>Q_target(s', argmax_a Q_online(s',a))</code>	El operador <code>max</code> sobre estimaciones ruidosas sobreestima sistemáticamente los valores Q; separar la <i>selección</i> de la acción (red online) de su <i>evaluación</i> (red target) reduce ese sesgo
2	Dueling DQN (Wang et al., 2016)	La cabeza de la red se divide en dos streams: <code>V(s)</code> y <code>A(s,a)</code> , con <code>Q = V + A - mean(A)</code>	En muchos estados de Space Invaders la acción elegida apenas cambia el valor del estado (p. ej. cuando no hay proyectiles cerca); estimar <code>V(s)</code> por separado propaga aprendizaje a todas las acciones a la vez
3	Exploración ajustada	<code>exploration_fraction</code> 0.30 → 0.10, <code>ε_final</code> 0.001 → 0.01, <code>target_update_interval</code> 10k → 5k, <code>learning_starts</code> 50k → 25k	El run base terminó con $\epsilon \approx 0.12$: el 12 % de las acciones seguían siendo aleatorias incluso al final del entrenamiento. Un annealing más corto, adaptado al presupuesto real, permite explotar antes lo aprendido

Protocolo de comparación justa: misma semilla (42), mismo preprocesado y misma red convolucional base, mismo presupuesto máximo de steps, mismo criterio de parada (100 episodios consecutivos con recompensa clipped > 20) y mismo test final (120 episodios con política determinista).

```
In [ ]: # =====
# M1. Infraestructura multi-run: rutas por variante y helpers
# =====
# Cada variante escribe sus logs y checkpoints en su propia carpeta de Drive,
# reutilizando SB3TrainingLogger y ConsecutiveRewardCallback sin cambios.
# La variante 'base' apunta a la carpeta del entrenamiento ya realizado.

RUN_NAMES = ['base', 'double', 'dueling', 'tuned']

RUN_LABELS = {
    'base': 'DQN base',
    'double': 'Double DQN',
    'dueling': 'Dueling DQN',
    'tuned': 'DQN exploración ajustada',
}

def setup_run_paths(run_name):
    if run_name == 'base':
        run_dir = OUTPUT_DIR
    else:
        run_dir = Path(drive_root) / 'datos' / (PROJECT_NAME + '_' + run_name)
    run_dir.mkdir(parents=True, exist_ok=True)
```

```

(run_dir / 'monitor').mkdir(parents=True, exist_ok=True)
(run_dir / 'tensorboard').mkdir(parents=True, exist_ok=True)
return {
    'dir': run_dir,
    'policy': run_dir / 'dqn_spaceinvaders_v0_policy_latest.pt',
    'model_zip': run_dir / 'dqn_spaceinvaders_v0_sb3_latest.zip',
    'json': run_dir / 'dqn_SpaceInvaders-v0_sb3_log.json',
    'csv': run_dir / 'dqn_SpaceInvaders-v0_sb3_log.csv',
    'monitor': run_dir / 'monitor',
    'tensorboard': run_dir / 'tensorboard',
    'test_json': run_dir / 'test_results.json',
}

RUN_PATHS = {name: setup_run_paths(name) for name in RUN_NAMES}

def create_train_env(monitor_dir):
    env = make_atari_env(
        ENV_ID,
        n_envs=1,
        seed=SEED,
        monitor_dir=str(monitor_dir),
        wrapper_kwargs=dict(clip_reward=True)
    )
    return VecFrameStack(env, n_stack=4)

def train_variant(run_name, model, total_timesteps=TOTAL_TIMESTEPS):
    paths = RUN_PATHS[run_name]
    training_logger = SB3TrainingLogger(json_path=paths['json'], csv_path=paths['csv'],
    callback = ConsecutiveRewardCallback(training_logger=training_logger, verbose=True)

    model.learn(
        total_timesteps=total_timesteps,
        callback=callback,
        reset_num_timesteps=True,
        tb_log_name='DQN_' + run_name,
        log_interval=100
    )

    # Siempre se guarda el checkpoint PyTorch (el formato que sabemos cargar);
    # el .zip de SB3 se intenta como extra (en el run base fallaba en Colab).
    save_policy_checkpoint(model, paths['policy'])
    print('Checkpoint PyTorch guardado en:', paths['policy'])
    try:
        model.save(str(paths['model_zip']))
        print('Modelo SB3 guardado en:', paths['model_zip'])
    except Exception as e:
        print('model.save() falló (no es crítico, se usa el .pt):', repr(e))

    return model

print('Rutas por variante creadas:')
for name in RUN_NAMES:
    print(' -', RUN_LABELS[name], '->', RUN_PATHS[name]['dir'])

```


Rutas por variante creadas:

- DQN base -> /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0
- Double DQN -> /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_double
- Dueling DQN -> /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_dueling
- DQN exploración ajustada -> /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_tuned

```
In [ ]: # =====
# M2. Propuesta 1: Double DQN
# =====
# DQN estándar usa como target:  $r + \gamma * \max_a Q_{\text{target}}(s', a)$ .
# El max sobre estimaciones con ruido produce sobreestimación sistemática
# de los valores Q. Double DQN (van Hasselt et al., 2016) desacopla:
#   - SELECCIÓN de la acción con la red online:  $a^* = \operatorname{argmax}_a Q_{\text{online}}(s', a)$ 
#   - EVALUACIÓN de esa acción con la red target:  $Q_{\text{target}}(s', a^*)$ 
# Solo cambia el cálculo del target; el resto (replay buffer, epsilon-greedy,
# red target) es idéntico al DQN base, por lo que basta sobrescribir train().

import torch.nn.functional as F

try:
    from stable_baselines3.common import logger as sb3_logger
except Exception:
    sb3_logger = None

class DoubleDQN(DQN):

    def train(self, gradient_steps, batch_size=100):
        self._update_learning_rate(self.policy.optimizer)

        losses = []
        for _ in range(gradient_steps):
            replay_data = self.replay_buffer.sample(batch_size, env=self._vec_no

            with th.no_grad():
                # Selección con la red online (única diferencia con DQN)
                next_actions = self.q_net(replay_data.next_observations).argmax(
                # Evaluación con la red target
                next_q = th.gather(self.q_net_target(replay_data.next_observatio
                target_q = replay_data.rewards + (1 - replay_data.dones) * self.

            current_q = th.gather(self.q_net(replay_data.observations), dim=1, i

            loss = F.smooth_l1_loss(current_q, target_q)
            losses.append(loss.item())

            self.policy.optimizer.zero_grad()
            loss.backward()
            th.nn.utils.clip_grad_norm_(self.policy.parameters(), self.max_grad_
            self.policy.optimizer.step()

        self._n_updates += gradient_steps

        log = getattr(self, 'logger', None) or sb3_logger
        if log is not None:
            log.record('train/n_updates', self._n_updates, exclude='tensorboard')
```

```
log.record('train/loss', float(np.mean(losses)))

print('DoubleDQN definido.')
```

DoubleDQN definido.

```
In [ ]: # =====
# M3. Propuesta 2: Dueling DQN
# =====
# Q(s,a) se descompone en dos streams sobre las mismas features convolucionales:
#   V(s): valor del estado
#   A(s,a): ventaja de cada acción respecto a la media
#   Q(s,a) = V(s) + A(s,a) - mean_a' A(s,a') (Wang et al., 2016)
# Restar la media de las ventajas hace la descomposición identificable.
# Se implementa sustituyendo la cabeza lineal de QNetwork por los dos streams;
# el extractor convolucional (CustomCNN84) no cambia.

from stable_baselines3.dqn.policies import CnnPolicy as DQNCnnPolicy
from stable_baselines3.dqn.policies import QNetwork

class DuelingQNetwork(QNetwork):

    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        action_dim = self.action_space.n

        # Se anula la cabeza lineal original y se crean los dos streams
        self.q_net = nn.Identity()
        self.value_stream = nn.Sequential(
            nn.Linear(self.features_dim, 256),
            nn.ReLU(),
            nn.Linear(256, 1)
        )
        self.advantage_stream = nn.Sequential(
            nn.Linear(self.features_dim, 256),
            nn.ReLU(),
            nn.Linear(256, action_dim)
        )

    def forward(self, obs):
        features = self.extract_features(obs)
        value = self.value_stream(features)
        advantage = self.advantage_stream(features)
        return value + advantage - advantage.mean(dim=1, keepdim=True)

class DuelingCnnPolicy(DQNCnnPolicy):

    def make_q_net(self):
        net_args = self._update_features_extractor(self.net_args, features_extra)
        return DuelingQNetwork(**net_args).to(self.device)

print('DuelingCnnPolicy definida.')
```

DuelingCnnPolicy definida.

```
In [ ]: # =====
# M4. Propuesta 3 (exploración ajustada) y constructor de modelos
```

```

# =====
# El run base se detuvo (criterio de 100 consecutivos) en ~2.6M steps con
# epsilon ~ 0.12: el schedule (30% de 10M steps) era demasiado largo para el
# presupuesto real. La propuesta 3 usa el mismo algoritmo DQN base pero:
# - annealing de epsilon en el 10% del presupuesto y epsilon final 0.01
# - red target actualizada cada 5k steps (en lugar de 10k)
# - learning_starts en 25k (empieza a aprender antes)

TUNED_PARAMS = dict(
    exploration_fraction=0.10,
    exploration_final_eps=0.01,
    target_update_interval=5_000,
    learning_starts=25_000,
)

def build_model(run_name, env, tensorboard_dir=None, for_eval=False):
    # for_eval=True reduce el replay buffer a 1k transiciones para no
    # reservar ~2.8 GB de RAM cuando solo se quiere evaluar la política.
    common = dict(
        env=env,
        learning_rate=LEARNING_RATE,
        buffer_size=1_000 if for_eval else BUFFER_SIZE,
        learning_starts=LEARNING_STARTS,
        batch_size=BATCH_SIZE,
        gamma=GAMMA,
        train_freq=TRAIN_FREQ,
        target_update_interval=TARGET_UPDATE_INTERVAL,
        exploration_initial_eps=EXPLORATION_INITIAL_EPS,
        exploration_final_eps=EXPLORATION_FINAL_EPS,
        exploration_fraction=EXPLORATION_FRACTION,
        policy_kwargs=policy_kwargs,
        tensorboard_log=str(tensorboard_dir) if tensorboard_dir is not None else
        verbose=1,
        seed=SEED,
        device='auto',
    )

    if run_name == 'base':
        return DQN(policy='CnnPolicy', **common)
    if run_name == 'double':
        return DoubleDQN(policy='CnnPolicy', **common)
    if run_name == 'dueling':
        return DQN(policy=DuelingCnnPolicy, **common)
    if run_name == 'tuned':
        common.update(TUNED_PARAMS)
        return DQN(policy='CnnPolicy', **common)

    raise ValueError('Variante desconocida: ' + run_name)

print('build_model listo. Variantes disponibles:', RUN_NAMES)

```

build_model listo. Variantes disponibles: ['base', 'double', 'dueling', 'tuned']

Notas de ejecución. Con la GPU estándar de Colab el run base avanzó a ~260 fps, es decir, el criterio de parada (~2.6M steps) se alcanzó en unas 3 horas. Cada variante se entrena en una celda independiente para poder repartirlas entre sesiones de Colab. Si una sesión se corta, el checkpoint

`.pt` y los logs CSV/JSON quedan guardados en Drive: para reanudar, cargar el `policy_state_dict` en un modelo nuevo y ajustar `exploration_initial_eps` al último `exploration_rate` registrado en el CSV (tal y como indica el enunciado, al retomar un entrenamiento hay que adaptar los parámetros de exploración).

```
In [ ]: # =====  
# M5. Entrenamiento Propuesta 1: Double DQN  
# =====  
  
env_double = create_train_env(RUN_PATHS['double']['monitor'])  
model_double = build_model('double', env_double, RUN_PATHS['double']['tensorboard'])  
model_double = train_variant('double', model_double)  
env_double.close()
```

Se han truncado las últimas 5000 líneas del flujo de salida.

```
-----
EP=3,490 | STEP=595,871 | clip=55 | consec=4 | max_consec=6 | avg_clip50=20.10
EP=3,500 | STEP=597,374 | clip=10 | consec=0 | max_consec=6 | avg_clip50=18.17
EP=3,510 | STEP=599,564 | clip=15 | consec=0 | max_consec=6 | avg_clip50=20.56
-----
```

```
| custom/
|   avg_clip50          | 19.9
|   clipped_episode_r... | 19.5
|   consecutive_clipp... | 0
|   max_consecutive_c... | 6
| rollout/
|   ep_len_mean         | 773
|   ep_rew_mean         | 187
|   exploration rate     | 0.8
| time/
|   episodes            | 10500
|   fps                 | 214
|   time_elapsed        | 2794
|   total timesteps     | 600641
| train/
|   learning_rate       | 0.0001
|   loss                | 0.202
|   n_updates           | 137660
-----
```

```
EP=3,520 | STEP=601,661 | clip=14 | consec=0 | max_consec=6 | avg_clip50=21.30
EP=3,530 | STEP=603,282 | clip=28 | consec=1 | max_consec=6 | avg_clip50=20.21
EP=3,540 | STEP=605,012 | clip=30 | consec=1 | max_consec=6 | avg_clip50=17.38
-----
```

```
| custom/
|   avg_clip50          | 18.4
|   clipped_episode_r... | 27.5
|   consecutive_clipp... | 2
|   max_consecutive_c... | 6
| rollout/
|   ep_len_mean         | 781
|   ep_rew_mean         | 183
|   exploration rate     | 0.798
| time/
|   episodes            | 10600
|   fps                 | 214
|   time_elapsed        | 2822
|   total timesteps     | 606646
| train/
|   learning_rate       | 0.0001
|   loss                | 0.0389
|   n_updates           | 139161
-----
```

```
EP=3,550 | STEP=606,793 | clip=14 | consec=0 | max_consec=6 | avg_clip50=18.43
EP=3,560 | STEP=608,889 | clip=6  | consec=0 | max_consec=6 | avg_clip50=17.94
EP=3,570 | STEP=610,569 | clip=8  | consec=0 | max_consec=6 | avg_clip50=16.32
EP=3,580 | STEP=612,611 | clip=32 | consec=2 | max_consec=6 | avg_clip50=18.03
-----
```

```
| custom/
|   avg_clip50          | 18.9
|   clipped_episode_r... | 48
|   consecutive_clipp... | 1
|   max_consecutive_c... | 6
| rollout/
|   ep_len_mean         | 812
-----
```

ep_rew_mean	195
exploration rate	0.796
time/	
episodes	10700
fps	214
time_elapsed	2852
total timesteps	613074
train/	
learning_rate	0.0001
loss	0.0804
n_updates	140768

EP=3,590 | STEP=614,818 | clip=21 | consec=2 | max_consec=6 | avg_clip50=19.96
EP=3,600 | STEP=616,361 | clip=5 | consec=0 | max_consec=6 | avg_clip50=19.80
EP=3,610 | STEP=618,588 | clip=50 | consec=1 | max_consec=6 | avg_clip50=21.13

custom/	
avg_clip50	22.1
clipped_episode_r...	9
consecutive_clipp...	0
max_consecutive_c...	6
rollout/	
ep_len_mean	815
ep_rew_mean	198
exploration rate	0.794
time/	
episodes	10800
fps	214
time_elapsed	2885
total timesteps	619928
train/	
learning_rate	0.0001
loss	0.0351
n_updates	142481

EP=3,620 | STEP=620,650 | clip=8 | consec=0 | max_consec=6 | avg_clip50=22.54
EP=3,630 | STEP=622,779 | clip=32 | consec=2 | max_consec=6 | avg_clip50=22.51
EP=3,640 | STEP=624,393 | clip=10 | consec=0 | max_consec=6 | avg_clip50=20.26

custom/	
avg_clip50	21.1
clipped_episode_r...	12
consecutive_clipp...	0
max_consecutive_c...	6
rollout/	
ep_len_mean	825
ep_rew_mean	206
exploration rate	0.791
time/	
episodes	10900
fps	214
time_elapsed	2915
total timesteps	626195
train/	
learning_rate	0.0001
loss	0.0409
n_updates	144048

EP=3,650 | STEP=626,293 | clip=14 | consec=0 | max_consec=6 | avg_clip50=21.31
EP=3,660 | STEP=627,829 | clip=18 | consec=0 | max_consec=6 | avg_clip50=18.99

EP=3,670 | STEP=629,721 | clip=15 | consec=0 | max_consec=6 | avg_clip50=18.79
EP=3,680 | STEP=631,664 | clip=20 | consec=0 | max_consec=6 | avg_clip50=18.70

custom/		
avg_clip50	18.4	
clipped_episode_r...	5	
consecutive_clipp...	0	
max_consecutive_c...	6	
rollout/		
ep_len_mean	799	
ep_rew_mean	201	
exploration rate	0.79	
time/		
episodes	11000	
fps	214	
time_elapsed	2943	
total timesteps	632008	
train/		
learning_rate	0.0001	
loss	0.164	
n_updates	145501	

EP=3,690 | STEP=633,231 | clip=20 | consec=0 | max_consec=6 | avg_clip50=18.55
EP=3,700 | STEP=635,099 | clip=18 | consec=0 | max_consec=6 | avg_clip50=17.77
EP=3,710 | STEP=636,858 | clip=6 | consec=0 | max_consec=6 | avg_clip50=18.82

custom/		
avg_clip50	19.2	
clipped_episode_r...	33	
consecutive_clipp...	1	
max_consecutive_c...	6	
rollout/		
ep_len_mean	770	
ep_rew_mean	187	
exploration rate	0.787	
time/		
episodes	11100	
fps	214	
time_elapsed	2973	
total timesteps	638268	
train/		
learning_rate	0.0001	
loss	0.0665	
n_updates	147066	

EP=3,720 | STEP=638,805 | clip=22 | consec=1 | max_consec=6 | avg_clip50=19.36
EP=3,730 | STEP=640,742 | clip=17 | consec=0 | max_consec=6 | avg_clip50=19.84
EP=3,740 | STEP=642,628 | clip=24 | consec=3 | max_consec=6 | avg_clip50=20.71
EP=3,750 | STEP=644,771 | clip=44 | consec=1 | max_consec=6 | avg_clip50=22.11

custom/		
avg_clip50	22.1	
clipped_episode_r...	44	
consecutive_clipp...	1	
max_consecutive_c...	6	
rollout/		
ep_len_mean	783	
ep_rew_mean	199	
exploration rate	0.785	
time/		

	episodes		11200	
	fps		214	
	time_elapsed		3004	
	total timesteps		644771	
	train/			
	learning_rate		0.0001	
	loss		0.0694	
	n_updates		148692	

EP=3,760 | STEP=646,505 | clip=17 | consec=0 | max_consec=6 | avg_clip50=21.80
EP=3,770 | STEP=648,325 | clip=2 | consec=0 | max_consec=6 | avg_clip50=20.47
EP=3,780 | STEP=650,215 | clip=16 | consec=0 | max_consec=6 | avg_clip50=20.15

	custom/			
	avg_clip50		19.8	
	clipped_episode_r...		10.5	
	consecutive_clipp...		0	
	max_consecutive_c...		6	
	rollout/			
	ep_len_mean		789	
	ep_rew_mean		199	
	exploration rate		0.783	
	time/			
	episodes		11300	
	fps		214	
	time_elapsed		3036	
	total timesteps		651165	
	train/			
	learning_rate		0.0001	
	loss		0.0814	
	n_updates		150291	

EP=3,790 | STEP=652,178 | clip=15 | consec=0 | max_consec=6 | avg_clip50=20.32
EP=3,800 | STEP=654,023 | clip=19 | consec=0 | max_consec=6 | avg_clip50=20.16
EP=3,810 | STEP=655,711 | clip=10 | consec=0 | max_consec=6 | avg_clip50=19.97

	custom/			
	avg_clip50		21.9	
	clipped_episode_r...		52	
	consecutive_clipp...		2	
	max_consecutive_c...		6	
	rollout/			
	ep_len_mean		807	
	ep_rew_mean		215	
	exploration rate		0.781	
	time/			
	episodes		11400	
	fps		214	
	time_elapsed		3066	
	total timesteps		657536	
	train/			
	learning_rate		0.0001	
	loss		0.0409	
	n_updates		151883	

EP=3,820 | STEP=657,737 | clip=15 | consec=0 | max_consec=6 | avg_clip50=22.22
EP=3,830 | STEP=660,132 | clip=12 | consec=0 | max_consec=6 | avg_clip50=22.35
EP=3,840 | STEP=662,242 | clip=6 | consec=0 | max_consec=6 | avg_clip50=23.27
EP=3,850 | STEP=664,398 | clip=30 | consec=2 | max_consec=6 | avg_clip50=23.07

custom/	
avg_clip50	23.7
clipped_episode_r...	26.5
consecutive_clipp...	4
max_consecutive_c...	6
rollout/	
ep_len_mean	836
ep_rew_mean	220
exploration rate	0.779
time/	
episodes	11500
fps	214
time_elapsed	3100
total timesteps	664871
train/	
learning_rate	0.0001
loss	0.0954
n_updates	153717

EP=3,860 | STEP=666,164 | clip=6 | consec=0 | max_consec=6 | avg_clip50=22.92
EP=3,870 | STEP=668,198 | clip=32 | consec=1 | max_consec=6 | avg_clip50=21.74
EP=3,880 | STEP=670,206 | clip=26 | consec=1 | max_consec=6 | avg_clip50=21.88

custom/	
avg_clip50	21.1
clipped_episode_r...	15
consecutive_clipp...	0
max_consecutive_c...	6
rollout/	
ep_len_mean	844
ep_rew_mean	225
exploration rate	0.776
time/	
episodes	11600
fps	214
time_elapsed	3131
total timesteps	671248
train/	
learning_rate	0.0001
loss	0.0822
n_updates	155311

EP=3,890 | STEP=672,119 | clip=19 | consec=0 | max_consec=6 | avg_clip50=21.04
EP=3,900 | STEP=674,492 | clip=11 | consec=0 | max_consec=6 | avg_clip50=22.87
EP=3,910 | STEP=676,715 | clip=10 | consec=0 | max_consec=6 | avg_clip50=23.94

custom/	
avg_clip50	24.4
clipped_episode_r...	41
consecutive_clipp...	1
max_consecutive_c...	6
rollout/	
ep_len_mean	879
ep_rew_mean	229
exploration rate	0.774
time/	
episodes	11700
fps	214
time_elapsed	3165
total timesteps	678456

train/	
learning_rate	0.0001
loss	0.0487
n_updates	157113

EP=3,920	STEP=678,578	clip=10	consec=0	max_consec=6	avg_clip50=24.01
EP=3,930	STEP=680,579	clip=18	consec=0	max_consec=6	avg_clip50=23.95
EP=3,940	STEP=682,665	clip=26	consec=1	max_consec=6	avg_clip50=24.12
EP=3,950	STEP=684,586	clip=20	consec=0	max_consec=6	avg_clip50=22.17

custom/	
avg_clip50	22.2
clipped_episode_r...	14
consecutive_clipp...	0
max_consecutive_c...	6
rollout/	
ep_len_mean	852
ep_rew_mean	223
exploration rate	0.772
time/	
episodes	11800
fps	214
time_elapsed	3199
total timesteps	685210
train/	
learning_rate	0.0001
loss	0.0219
n_updates	158802

EP=3,960	STEP=686,540	clip=2	consec=0	max_consec=6	avg_clip50=21.61
EP=3,970	STEP=688,599	clip=8	consec=0	max_consec=6	avg_clip50=21.25
EP=3,980	STEP=690,446	clip=4	consec=0	max_consec=6	avg_clip50=20.33

custom/	
avg_clip50	20.4
clipped_episode_r...	32
consecutive_clipp...	1
max_consecutive_c...	6
rollout/	
ep_len_mean	858
ep_rew_mean	222
exploration rate	0.77
time/	
episodes	11900
fps	214
time_elapsed	3230
total timesteps	691698
train/	
learning_rate	0.0001
loss	0.122
n_updates	160424

EP=3,990	STEP=692,464	clip=44	consec=1	max_consec=6	avg_clip50=20.87
EP=4,000	STEP=694,245	clip=8	consec=0	max_consec=6	avg_clip50=20.00
EP=4,010	STEP=696,334	clip=19	consec=0	max_consec=6	avg_clip50=20.16

custom/	
avg_clip50	19.4
clipped_episode_r...	12
consecutive_clipp...	0

max_consecutive_c...	6
rollout/	
ep_len_mean	829
ep_rew_mean	204
exploration rate	0.768
time/	
episodes	12000
fps	214
time_elapsed	3261
total timesteps	698085
train/	
learning_rate	0.0001
loss	0.0799
n_updates	162021

EP=4,020	STEP=698,113	clip=12	consec=0	max_consec=6	avg_clip50=19.50
EP=4,030	STEP=700,213	clip=30	consec=2	max_consec=6	avg_clip50=19.29
EP=4,040	STEP=702,096	clip=19	consec=0	max_consec=6	avg_clip50=18.29
EP=4,050	STEP=703,730	clip=12	consec=0	max_consec=6	avg_clip50=17.67

custom/	
avg_clip50	17.6
clipped_episode_r...	8.5
consecutive_clipp...	0
max_consecutive_c...	6
rollout/	
ep_len_mean	812
ep_rew_mean	188
exploration rate	0.765
time/	
episodes	12100
fps	213
time_elapsed	3292
total timesteps	704370
train/	
learning_rate	0.0001
loss	0.104
n_updates	163592

EP=4,060	STEP=705,439	clip=11	consec=0	max_consec=6	avg_clip50=17.55
EP=4,070	STEP=707,633	clip=8	consec=0	max_consec=6	avg_clip50=19.88
EP=4,080	STEP=709,821	clip=40	consec=2	max_consec=6	avg_clip50=21.80

custom/	
avg_clip50	23.9
clipped_episode_r...	31.5
consecutive_clipp...	5
max_consecutive_c...	6
rollout/	
ep_len_mean	837
ep_rew_mean	213
exploration rate	0.763
time/	
episodes	12200
fps	213
time_elapsed	3328
total timesteps	711632
train/	
learning_rate	0.0001
loss	0.268

n_updates	165407
-----------	--------

EP=4,090	STEP=712,084	clip=7	consec=0	max_consec=6	avg_clip50=23.45
EP=4,100	STEP=713,835	clip=59	consec=1	max_consec=6	avg_clip50=24.60
EP=4,110	STEP=716,034	clip=14	consec=0	max_consec=6	avg_clip50=25.85
EP=4,120	STEP=718,040	clip=22	consec=1	max_consec=6	avg_clip50=24.46

custom/	
avg_clip50	23.2
clipped_episode_r...	18.5
consecutive_clipp...	0
max_consecutive_c...	6
rollout/	
ep_len_mean	843
ep_rew_mean	222
exploration rate	0.761
time/	
episodes	12300
fps	213
time_elapsed	3359
total timesteps	718231
train/	
learning_rate	0.0001
loss	0.0669
n_updates	167057

EP=4,130	STEP=719,875	clip=36	consec=1	max_consec=6	avg_clip50=22.50
EP=4,140	STEP=721,411	clip=9	consec=0	max_consec=6	avg_clip50=20.25
EP=4,150	STEP=723,186	clip=22	consec=1	max_consec=6	avg_clip50=20.14

custom/	
avg_clip50	20.1
clipped_episode_r...	21
consecutive_clipp...	1
max_consecutive_c...	6
rollout/	
ep_len_mean	824
ep_rew_mean	225
exploration rate	0.759
time/	
episodes	12400
fps	213
time_elapsed	3388
total timesteps	724026
train/	
learning_rate	0.0001
loss	0.0235
n_updates	168506

EP=4,160	STEP=724,957	clip=10	consec=0	max_consec=6	avg_clip50=18.57
EP=4,170	STEP=726,929	clip=43	consec=1	max_consec=6	avg_clip50=18.91
EP=4,180	STEP=729,069	clip=29	consec=3	max_consec=6	avg_clip50=19.36

custom/	
avg_clip50	21.1
clipped_episode_r...	30
consecutive_clipp...	2
max_consecutive_c...	7
rollout/	
ep_len_mean	810

ep_rew_mean	206
exploration rate	0.757
time/	
episodes	12500
fps	213
time_elapsed	3420
total timesteps	730879
train/	
learning_rate	0.0001
loss	0.114
n_updates	170219

EP=4,190	STEP=731,131	clip=4	consec=0	max_consec=7	avg_clip50=21.02
EP=4,200	STEP=733,116	clip=20	consec=0	max_consec=7	avg_clip50=20.79
EP=4,210	STEP=734,970	clip=26	consec=2	max_consec=7	avg_clip50=21.44
EP=4,220	STEP=737,246	clip=16	consec=0	max_consec=7	avg_clip50=23.56

custom/	
avg_clip50	23.9
clipped_episode_r...	33
consecutive_clipp...	1
max_consecutive_c...	7
rollout/	
ep_len_mean	819
ep_rew_mean	214
exploration rate	0.754
time/	
episodes	12600
fps	213
time_elapsed	3452
total timesteps	737607
train/	
learning_rate	0.0001
loss	0.078
n_updates	171901

EP=4,230	STEP=738,993	clip=11	consec=0	max_consec=7	avg_clip50=22.94
EP=4,240	STEP=740,959	clip=74	consec=1	max_consec=7	avg_clip50=23.64
EP=4,250	STEP=742,723	clip=35	consec=1	max_consec=7	avg_clip50=23.88

custom/	
avg_clip50	24.5
clipped_episode_r...	32.5
consecutive_clipp...	2
max_consecutive_c...	7
rollout/	
ep_len_mean	834
ep_rew_mean	225
exploration rate	0.752
time/	
episodes	12700
fps	213
time_elapsed	3482
total timesteps	744010
train/	
learning_rate	0.0001
loss	0.173
n_updates	173502

EP=4,260	STEP=744,864	clip=4	consec=0	max_consec=7	avg_clip50=24.05
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EP=4,270 | STEP=747,147 | clip=4 | consec=0 | max_consec=7 | avg_clip50=23.43
EP=4,280 | STEP=749,092 | clip=22 | consec=1 | max_consec=7 | avg_clip50=23.21

custom/		
avg_clip50	23.1	
clipped_episode_r...	62.5	
consecutive_clipp...	1	
max_consecutive_c...	7	
rollout/		
ep_len_mean	827	
ep_rew_mean	227	
exploration rate	0.75	
time/		
episodes	12800	
fps	213	
time_elapsed	3515	
total timesteps	750810	
train/		
learning_rate	0.0001	
loss	0.214	
n_updates	175202	

EP=4,290 | STEP=750,842 | clip=45 | consec=2 | max_consec=7 | avg_clip50=22.51
EP=4,300 | STEP=753,144 | clip=34 | consec=5 | max_consec=7 | avg_clip50=24.49
EP=4,310 | STEP=755,038 | clip=14 | consec=0 | max_consec=7 | avg_clip50=24.36
EP=4,320 | STEP=756,873 | clip=50 | consec=2 | max_consec=7 | avg_clip50=22.85

custom/		
avg_clip50	22.6	
clipped_episode_r...	12	
consecutive_clipp...	0	
max_consecutive_c...	7	
rollout/		
ep_len_mean	836	
ep_rew_mean	233	
exploration rate	0.748	
time/		
episodes	12900	
fps	213	
time_elapsed	3548	
total timesteps	757556	
train/		
learning_rate	0.0001	
loss	0.166	
n_updates	176888	

EP=4,330 | STEP=758,582 | clip=15 | consec=0 | max_consec=7 | avg_clip50=22.23
EP=4,340 | STEP=760,547 | clip=14 | consec=0 | max_consec=7 | avg_clip50=22.43
EP=4,350 | STEP=762,687 | clip=10 | consec=0 | max_consec=7 | avg_clip50=22.47

custom/		
avg_clip50	21.3	
clipped_episode_r...	8	
consecutive_clipp...	0	
max_consecutive_c...	7	
rollout/		
ep_len_mean	835	
ep_rew_mean	229	
exploration rate	0.746	
time/		

	episodes		13000	
	fps		213	
	time_elapsed		3578	
	total timesteps		763876	
	train/			
	learning_rate		0.0001	
	loss		0.0568	
	n_updates		178468	

EP=4,360		STEP=764,598		clip=52		consec=3		max_consec=7		avg_clip50=22.04
EP=4,370		STEP=766,523		clip=22		consec=1		max_consec=7		avg_clip50=22.15
EP=4,380		STEP=768,318		clip=10		consec=0		max_consec=7		avg_clip50=23.25
EP=4,390		STEP=770,312		clip=12		consec=0		max_consec=7		avg_clip50=22.66

	custom/			
	avg_clip50		22.7	
	clipped_episode_r...		12.5	
	consecutive_clipp...		0	
	max_consecutive_c...		7	
	rollout/			
	ep_len_mean		825	
	ep_rew_mean		225	
	exploration rate		0.743	
	time/			
	episodes		13100	
	fps		213	
	time_elapsed		3610	
	total timesteps		770510	
	train/			
	learning_rate		0.0001	
	loss		0.0915	
	n_updates		180127	

EP=4,400		STEP=772,000		clip=16		consec=0		max_consec=7		avg_clip50=19.56
EP=4,410		STEP=774,258		clip=10		consec=0		max_consec=7		avg_clip50=22.22
EP=4,420		STEP=776,422		clip=62		consec=3		max_consec=7		avg_clip50=23.09

	custom/			
	avg_clip50		23.5	
	clipped_episode_r...		47	
	consecutive_clipp...		1	
	max_consecutive_c...		7	
	rollout/			
	ep_len_mean		828	
	ep_rew_mean		228	
	exploration rate		0.741	
	time/			
	episodes		13200	
	fps		213	
	time_elapsed		3643	
	total timesteps		777250	
	train/			
	learning_rate		0.0001	
	loss		0.155	
	n_updates		181812	

EP=4,430		STEP=778,252		clip=24		consec=1		max_consec=7		avg_clip50=22.68
EP=4,440		STEP=780,212		clip=50		consec=1		max_consec=7		avg_clip50=22.96
EP=4,450		STEP=781,873		clip=6		consec=0		max_consec=7		avg_clip50=23.76

custom/	
avg_clip50	23.6
clipped_episode_r...	28.5
consecutive_clipp...	1
max_consecutive_c...	7
rollout/	
ep_len_mean	836
ep_rew_mean	230
exploration rate	0.739
time/	
episodes	13300
fps	213
time_elapsed	3675
total timesteps	783777
train/	
learning_rate	0.0001
loss	0.0468
n_updates	183444

EP=4,460	STEP=784,173	clip=22	consec=1	max_consec=7	avg_clip50=22.96
EP=4,470	STEP=785,931	clip=31	consec=3	max_consec=7	avg_clip50=21.81
EP=4,480	STEP=787,830	clip=3	consec=0	max_consec=7	avg_clip50=22.02
EP=4,490	STEP=789,746	clip=28	consec=2	max_consec=7	avg_clip50=21.78

custom/	
avg_clip50	21.9
clipped_episode_r...	9
consecutive_clipp...	0
max_consecutive_c...	7
rollout/	
ep_len_mean	819
ep_rew_mean	222
exploration rate	0.737
time/	
episodes	13400
fps	213
time_elapsed	3705
total timesteps	790014
train/	
learning_rate	0.0001
loss	0.0901
n_updates	185003

EP=4,500	STEP=791,300	clip=20	consec=0	max_consec=7	avg_clip50=21.09
EP=4,510	STEP=793,250	clip=9	consec=0	max_consec=7	avg_clip50=19.74
EP=4,520	STEP=794,623	clip=18	consec=0	max_consec=7	avg_clip50=17.53

custom/	
avg_clip50	15.8
clipped_episode_r...	3.5
consecutive_clipp...	0
max_consecutive_c...	7
rollout/	
ep_len_mean	762
ep_rew_mean	190
exploration rate	0.735
time/	
episodes	13500
fps	213
time_elapsed	3729

total timesteps	795183
train/	
learning_rate	0.0001
loss	0.035
n_updates	186295

EP=4,530	STEP=796,365	clip=42	consec=2	max_consec=7	avg_clip50=18.35
EP=4,540	STEP=798,680	clip=33	consec=3	max_consec=7	avg_clip50=20.25
EP=4,550	STEP=800,401	clip=20	consec=1	max_consec=7	avg_clip50=20.49

custom/	
avg_clip50	21.1
clipped_episode_r...	38
consecutive_clipp...	2
max_consecutive_c...	7
rollout/	
ep_len_mean	772
ep_rew_mean	201
exploration rate	0.733
time/	
episodes	13600
fps	213
time_elapsed	3763
total timesteps	801929
train/	
learning_rate	0.0001
loss	0.19
n_updates	187982

EP=4,560	STEP=802,179	clip=11	consec=0	max_consec=7	avg_clip50=20.07
EP=4,570	STEP=804,269	clip=76	consec=2	max_consec=7	avg_clip50=23.72
EP=4,580	STEP=806,194	clip=9	consec=0	max_consec=7	avg_clip50=23.98
EP=4,590	STEP=808,382	clip=14	consec=0	max_consec=7	avg_clip50=24.03

custom/	
avg_clip50	23.9
clipped_episode_r...	9.5
consecutive_clipp...	0
max_consecutive_c...	7
rollout/	
ep_len_mean	788
ep_rew_mean	221
exploration rate	0.731
time/	
episodes	13700
fps	213
time_elapsed	3796
total timesteps	808667
train/	
learning_rate	0.0001
loss	0.328
n_updates	189666

EP=4,600	STEP=810,017	clip=4	consec=0	max_consec=7	avg_clip50=24.16
EP=4,610	STEP=811,848	clip=16	consec=0	max_consec=7	avg_clip50=23.95
EP=4,620	STEP=813,835	clip=46	consec=1	max_consec=7	avg_clip50=22.92

custom/	
avg_clip50	22
clipped_episode_r...	13.5

consecutive_clipp...	0
max_consecutive_c...	7
rollout/	
ep_len_mean	818
ep_rew_mean	233
exploration rate	0.729
time/	
episodes	13800
fps	212
time_elapsed	3826
total timesteps	814906
train/	
learning_rate	0.0001
loss	0.0275
n_updates	191226

EP=4,630 | STEP=815,663 | clip=43 | consec=1 | max_consec=7 | avg_clip50=21.82
EP=4,640 | STEP=818,180 | clip=64 | consec=1 | max_consec=7 | avg_clip50=22.66
EP=4,650 | STEP=819,921 | clip=10 | consec=0 | max_consec=7 | avg_clip50=23.64

custom/	
avg_clip50	25.9
clipped_episode_r...	30.5
consecutive_clipp...	1
max_consecutive_c...	7
rollout/	
ep_len_mean	858
ep_rew_mean	249
exploration rate	0.726
time/	
episodes	13900
fps	212
time_elapsed	3861
total timesteps	822358
train/	
learning_rate	0.0001
loss	0.0499
n_updates	193089

EP=4,660 | STEP=822,379 | clip=12 | consec=0 | max_consec=7 | avg_clip50=25.78
EP=4,670 | STEP=824,937 | clip=64 | consec=1 | max_consec=7 | avg_clip50=28.05
EP=4,680 | STEP=826,767 | clip=10 | consec=0 | max_consec=7 | avg_clip50=27.58
EP=4,690 | STEP=828,857 | clip=7 | consec=0 | max_consec=7 | avg_clip50=24.83

custom/	
avg_clip50	25.5
clipped_episode_r...	40.5
consecutive_clipp...	3
max_consecutive_c...	7
rollout/	
ep_len_mean	884
ep_rew_mean	243
exploration rate	0.724
time/	
episodes	14000
fps	212
time_elapsed	3898
total timesteps	829815
train/	
learning_rate	0.0001

loss	0.0601
n_updates	194953

EP=4,700	STEP=830,994	clip=5	consec=0	max_consec=7	avg_clip50=25.95
EP=4,710	STEP=832,966	clip=32	consec=1	max_consec=7	avg_clip50=24.09
EP=4,720	STEP=835,147	clip=14	consec=0	max_consec=7	avg_clip50=21.96

custom/	
avg_clip50	23.6
clipped_episode_r...	31
consecutive_clipp...	1
max_consecutive_c...	7
rollout/	
ep_len_mean	912
ep_rew_mean	256
exploration rate	0.721
time/	
episodes	14100
fps	212
time_elapsed	3930
total timesteps	836668
train/	
learning_rate	0.0001
loss	0.042
n_updates	196666

EP=4,730	STEP=837,235	clip=8	consec=0	max_consec=7	avg_clip50=23.84
EP=4,740	STEP=839,301	clip=70	consec=1	max_consec=7	avg_clip50=25.74
EP=4,750	STEP=841,646	clip=58	consec=2	max_consec=7	avg_clip50=26.36
EP=4,760	STEP=843,754	clip=22	consec=3	max_consec=7	avg_clip50=26.39

custom/	
avg_clip50	25.9
clipped_episode_r...	3.5
consecutive_clipp...	0
max_consecutive_c...	7
rollout/	
ep_len_mean	894
ep_rew_mean	250
exploration rate	0.719
time/	
episodes	14200
fps	212
time_elapsed	3965
total timesteps	843838
train/	
learning_rate	0.0001
loss	0.218
n_updates	198459

EP=4,770	STEP=845,307	clip=12	consec=0	max_consec=7	avg_clip50=24.42
EP=4,780	STEP=847,271	clip=54	consec=2	max_consec=7	avg_clip50=24.30
EP=4,790	STEP=849,182	clip=58	consec=4	max_consec=7	avg_clip50=22.87

custom/	
avg_clip50	22.4
clipped_episode_r...	15.5
consecutive_clipp...	0
max_consecutive_c...	7
rollout/	

ep_len_mean	850
ep_rew_mean	239
exploration rate	0.717
time/	
episodes	14300
fps	212
time_elapsed	3997
total timesteps	850177
train/	
learning_rate	0.0001
loss	0.112
n_updates	200044

EP=4,800	STEP=851,283	clip=30	consec=1	max_consec=7	avg_clip50=22.31
EP=4,810	STEP=852,953	clip=24	consec=1	max_consec=7	avg_clip50=21.35
EP=4,820	STEP=854,933	clip=4	consec=0	max_consec=7	avg_clip50=23.86

custom/	
avg_clip50	24.5
clipped_episode_r...	22.5
consecutive_clipp...	2
max_consecutive_c...	7
rollout/	
ep_len_mean	832
ep_rew_mean	238
exploration rate	0.715
time/	
episodes	14400
fps	212
time_elapsed	4029
total timesteps	856825
train/	
learning_rate	0.0001
loss	0.0284
n_updates	201706

EP=4,830	STEP=857,207	clip=28	consec=4	max_consec=7	avg_clip50=24.41
EP=4,840	STEP=859,740	clip=34	consec=1	max_consec=7	avg_clip50=25.59
EP=4,850	STEP=861,201	clip=2	consec=0	max_consec=7	avg_clip50=22.99
EP=4,860	STEP=863,016	clip=28	consec=2	max_consec=7	avg_clip50=23.53

custom/	
avg_clip50	23.1
clipped_episode_r...	24.5
consecutive_clipp...	4
max_consecutive_c...	7
rollout/	
ep_len_mean	824
ep_rew_mean	227
exploration rate	0.712
time/	
episodes	14500
fps	212
time_elapsed	4061
total timesteps	863547
train/	
learning_rate	0.0001
loss	0.0606
n_updates	203386

EP=4,870 | STEP=864,973 | clip=4 | consec=0 | max_consec=8 | avg_clip50=23.04
EP=4,880 | STEP=866,991 | clip=29 | consec=1 | max_consec=8 | avg_clip50=22.25
EP=4,890 | STEP=868,781 | clip=6 | consec=0 | max_consec=8 | avg_clip50=19.96

custom/		
avg_clip50	19	
clipped_episode_r...	5.5	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	808	
ep_rew_mean	217	
exploration rate	0.71	
time/		
episodes	14600	
fps	212	
time_elapsed	4091	
total timesteps	869534	
train/		
learning_rate	0.0001	
loss	0.0687	
n_updates	204883	

EP=4,900 | STEP=870,199 | clip=16 | consec=0 | max_consec=8 | avg_clip50=19.64
EP=4,910 | STEP=872,467 | clip=43 | consec=1 | max_consec=8 | avg_clip50=21.79
EP=4,920 | STEP=874,657 | clip=40 | consec=2 | max_consec=8 | avg_clip50=22.95
EP=4,930 | STEP=876,822 | clip=46 | consec=1 | max_consec=8 | avg_clip50=24.48

custom/		
avg_clip50	24.5	
clipped_episode_r...	46.5	
consecutive_clipp...	1	
max_consecutive_c...	8	
rollout/		
ep_len_mean	829	
ep_rew_mean	234	
exploration rate	0.708	
time/		
episodes	14700	
fps	212	
time_elapsed	4128	
total timesteps	877085	
train/		
learning_rate	0.0001	
loss	0.0543	
n_updates	206771	

EP=4,940 | STEP=878,827 | clip=30 | consec=1 | max_consec=8 | avg_clip50=25.08
EP=4,950 | STEP=880,735 | clip=14 | consec=0 | max_consec=8 | avg_clip50=27.10
EP=4,960 | STEP=882,598 | clip=29 | consec=1 | max_consec=8 | avg_clip50=25.94

custom/		
avg_clip50	25.7	
clipped_episode_r...	44	
consecutive_clipp...	1	
max_consecutive_c...	8	
rollout/		
ep_len_mean	823	
ep_rew_mean	235	
exploration rate	0.706	

time/	
episodes	14800
fps	212
time_elapsed	4159
total timesteps	883413
train/	
learning_rate	0.0001
loss	0.142
n_updates	208353

EP=4,970	STEP=884,766	clip=16	consec=0	max_consec=8	avg_clip50=25.20
EP=4,980	STEP=886,893	clip=12	consec=0	max_consec=8	avg_clip50=23.43
EP=4,990	STEP=888,784	clip=12	consec=0	max_consec=8	avg_clip50=24.13

custom/	
avg_clip50	24.8
clipped_episode_r...	19.5
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	876
ep_rew_mean	262
exploration rate	0.703
time/	
episodes	14900
fps	212
time_elapsed	4192
total timesteps	890468
train/	
learning_rate	0.0001
loss	0.0475
n_updates	210116

EP=5,000	STEP=890,863	clip=30	consec=1	max_consec=8	avg_clip50=24.86
EP=5,010	STEP=893,050	clip=30	consec=6	max_consec=8	avg_clip50=25.30
EP=5,020	STEP=895,353	clip=45	consec=4	max_consec=8	avg_clip50=24.97
EP=5,030	STEP=897,318	clip=21	consec=1	max_consec=8	avg_clip50=24.95

custom/	
avg_clip50	24.9
clipped_episode_r...	6.5
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	860
ep_rew_mean	240
exploration rate	0.701
time/	
episodes	15000
fps	212
time_elapsed	4227
total timesteps	897508
train/	
learning_rate	0.0001
loss	0.0679
n_updates	211876

EP=5,040	STEP=899,007	clip=44	consec=2	max_consec=8	avg_clip50=23.48
EP=5,050	STEP=901,205	clip=36	consec=4	max_consec=8	avg_clip50=23.17
EP=5,060	STEP=903,305	clip=14	consec=0	max_consec=8	avg_clip50=23.07

custom/		
avg_clip50	22.4	
clipped_episode_r...	27.5	
consecutive_clipp...	1	
max_consecutive_c...	8	
rollout/		
ep_len_mean	864	
ep_rew_mean	233	
exploration rate	0.699	
time/		
episodes	15100	
fps	212	
time_elapsed	4261	
total timesteps	904398	
train/		
learning_rate	0.0001	
loss	0.127	
n_updates	213599	

EP=5,070 | STEP=905,338 | clip=67 | consec=3 | max_consec=8 | avg_clip50=22.93
EP=5,080 | STEP=907,260 | clip=42 | consec=2 | max_consec=8 | avg_clip50=22.28
EP=5,090 | STEP=909,243 | clip=14 | consec=0 | max_consec=8 | avg_clip50=22.18

custom/		
avg_clip50	22.5	
clipped_episode_r...	32	
consecutive_clipp...	3	
max_consecutive_c...	8	
rollout/		
ep_len_mean	861	
ep_rew_mean	228	
exploration rate	0.697	
time/		
episodes	15200	
fps	212	
time_elapsed	4294	
total timesteps	911209	
train/		
learning_rate	0.0001	
loss	0.0839	
n_updates	215302	

EP=5,100 | STEP=911,227 | clip=24 | consec=4 | max_consec=8 | avg_clip50=22.30
EP=5,110 | STEP=913,164 | clip=6 | consec=0 | max_consec=8 | avg_clip50=22.28
EP=5,120 | STEP=915,050 | clip=23 | consec=1 | max_consec=8 | avg_clip50=21.07
EP=5,130 | STEP=917,303 | clip=14 | consec=0 | max_consec=9 | avg_clip50=23.58

custom/		
avg_clip50	26.4	
clipped_episode_r...	34.5	
consecutive_clipp...	1	
max_consecutive_c...	9	
rollout/		
ep_len_mean	867	
ep_rew_mean	248	
exploration rate	0.694	
time/		
episodes	15300	
fps	212	

	time_elapsed		4329	
	total timesteps		918558	
	train/			
	learning_rate		0.0001	
	loss		0.0941	
	n_updates		217139	

EP=5,140		STEP=919,531		clip=12		consec=0		max_consec=9		avg_clip50=27.64
EP=5,150		STEP=921,799		clip=16		consec=0		max_consec=9		avg_clip50=28.91
EP=5,160		STEP=923,458		clip=14		consec=0		max_consec=9		avg_clip50=27.51

	custom/			
	avg_clip50		30.5	
	clipped_episode_r...		14.5	
	consecutive_clipp...		0	
	max_consecutive_c...		9	
	rollout/			
	ep_len_mean		871	
	ep_rew_mean		262	
	exploration rate		0.692	
	time/			
	episodes		15400	
	fps		212	
	time_elapsed		4365	
	total timesteps		925691	
	train/			
	learning_rate		0.0001	
	loss		0.0574	
	n_updates		218922	

EP=5,170		STEP=925,920		clip=28		consec=1		max_consec=9		avg_clip50=30.64
EP=5,180		STEP=928,175		clip=7		consec=0		max_consec=9		avg_clip50=30.79
EP=5,190		STEP=929,708		clip=6		consec=0		max_consec=9		avg_clip50=26.41
EP=5,200		STEP=931,774		clip=39		consec=1		max_consec=9		avg_clip50=25.28

	custom/			
	avg_clip50		26.2	
	clipped_episode_r...		24.5	
	consecutive_clipp...		4	
	max_consecutive_c...		9	
	rollout/			
	ep_len_mean		869	
	ep_rew_mean		271	
	exploration rate		0.689	
	time/			
	episodes		15500	
	fps		212	
	time_elapsed		4398	
	total timesteps		932494	
	train/			
	learning_rate		0.0001	
	loss		0.0419	
	n_updates		220623	

EP=5,210		STEP=933,521		clip=15		consec=0		max_consec=9		avg_clip50=25.50
EP=5,220		STEP=935,390		clip=30		consec=1		max_consec=9		avg_clip50=22.80
EP=5,230		STEP=937,468		clip=62		consec=1		max_consec=9		avg_clip50=21.01

	custom/			
	avg_clip50		22.5	

	clipped_episode_r...	65	
	consecutive_clipp...	4	
	max_consecutive_c...	9	
	rollout/		
	ep_len_mean	851	
	ep_rew_mean	251	
	exploration rate	0.687	
	time/		
	episodes	15600	
	fps	211	
	time_elapsed	4431	
	total timesteps	939039	
	train/		
	learning_rate	0.0001	
	loss	0.0652	
	n_updates	222259	

EP=5,240		STEP=939,773		clip=48		consec=1		max_consec=9		avg_clip50=23.96
EP=5,250		STEP=941,346		clip=11		consec=0		max_consec=9		avg_clip50=22.51
EP=5,260		STEP=943,535		clip=8		consec=0		max_consec=9		avg_clip50=24.71
EP=5,270		STEP=945,749		clip=54		consec=1		max_consec=9		avg_clip50=26.40

	custom/		
	avg_clip50	26.6	
	clipped_episode_r...	25.5	
	consecutive_clipp...	2	
	max_consecutive_c...	9	
	rollout/		
	ep_len_mean	839	
	ep_rew_mean	247	
	exploration rate	0.685	
	time/		
	episodes	15700	
	fps	211	
	time_elapsed	4467	
	total timesteps	946048	
	train/		
	learning_rate	0.0001	
	loss	0.0503	
	n_updates	224011	

EP=5,280		STEP=947,777		clip=7		consec=0		max_consec=9		avg_clip50=27.50
EP=5,290		STEP=949,822		clip=12		consec=0		max_consec=9		avg_clip50=26.35
EP=5,300		STEP=951,753		clip=14		consec=0		max_consec=9		avg_clip50=26.52

	custom/		
	avg_clip50	25.8	
	clipped_episode_r...	33.5	
	consecutive_clipp...	1	
	max_consecutive_c...	9	
	rollout/		
	ep_len_mean	838	
	ep_rew_mean	243	
	exploration rate	0.683	
	time/		
	episodes	15800	
	fps	211	
	time_elapsed	4499	
	total timesteps	952638	
	train/		

learning_rate	0.0001
loss	0.0457
n_updates	225659

EP=5,310	STEP=953,659	clip=26	consec=2	max_consec=9	avg_clip50=25.48
EP=5,320	STEP=956,046	clip=42	consec=4	max_consec=9	avg_clip50=26.34
EP=5,330	STEP=958,323	clip=55	consec=1	max_consec=9	avg_clip50=26.99
EP=5,340	STEP=960,692	clip=34	consec=1	max_consec=9	avg_clip50=27.92

custom/	
avg_clip50	27.9
clipped_episode_r...	33.5
consecutive_clipp...	1
max_consecutive_c...	9
rollout/	
ep_len_mean	881
ep_rew_mean	271
exploration rate	0.68
time/	
episodes	15900
fps	211
time_elapsed	4540
total timesteps	960999
train/	
learning_rate	0.0001
loss	0.115
n_updates	227749

EP=5,350	STEP=962,766	clip=10	consec=0	max_consec=9	avg_clip50=29.52
EP=5,360	STEP=965,059	clip=24	consec=2	max_consec=9	avg_clip50=30.31
EP=5,370	STEP=967,795	clip=82	consec=8	max_consec=9	avg_clip50=30.68

custom/	
avg_clip50	31.8
clipped_episode_r...	23
consecutive_clipp...	12
max_consecutive_c...	12
rollout/	
ep_len_mean	928
ep_rew_mean	290
exploration rate	0.677
time/	
episodes	16000
fps	211
time_elapsed	4577
total timesteps	968635
train/	
learning_rate	0.0001
loss	0.127
n_updates	229658

EP=5,380	STEP=969,994	clip=36	consec=3	max_consec=14	avg_clip50=30.68
EP=5,390	STEP=972,261	clip=53	consec=2	max_consec=14	avg_clip50=31.42
EP=5,400	STEP=974,513	clip=10	consec=0	max_consec=14	avg_clip50=30.81

custom/	
avg_clip50	30.5
clipped_episode_r...	29
consecutive_clipp...	3
max_consecutive_c...	14

rollout/	
ep_len_mean	973
ep_rew_mean	304
exploration rate	0.675
time/	
episodes	16100
fps	211
time_elapsed	4617
total timesteps	976726
train/	
learning_rate	0.0001
loss	0.141
n_updates	231681

EP=5,410	STEP=976,758	clip=10	consec=0	max_consec=14	avg_clip50=30.26
EP=5,420	STEP=978,603	clip=12	consec=0	max_consec=14	avg_clip50=28.05
EP=5,430	STEP=980,573	clip=12	consec=0	max_consec=14	avg_clip50=26.61
EP=5,440	STEP=982,833	clip=26	consec=2	max_consec=14	avg_clip50=25.49

custom/	
avg_clip50	25.3
clipped_episode_r...	14.5
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	922
ep_rew_mean	279
exploration rate	0.673
time/	
episodes	16200
fps	211
time_elapsed	4651
total timesteps	983387
train/	
learning_rate	0.0001
loss	0.15
n_updates	233346

EP=5,450	STEP=984,935	clip=10	consec=0	max_consec=14	avg_clip50=26.09
EP=5,460	STEP=987,158	clip=36	consec=1	max_consec=14	avg_clip50=27.23
EP=5,470	STEP=989,275	clip=46	consec=3	max_consec=14	avg_clip50=28.08

custom/	
avg_clip50	29.5
clipped_episode_r...	16
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	922
ep_rew_mean	284
exploration rate	0.67
time/	
episodes	16300
fps	211
time_elapsed	4689
total timesteps	991226
train/	
learning_rate	0.0001
loss	0.123
n_updates	235306

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-----
EP=5,480 | STEP=991,877 | clip=31 | consec=1 | max_consec=14 | avg_clip50=29.46
EP=5,490 | STEP=994,258 | clip=45 | consec=5 | max_consec=14 | avg_clip50=30.65
EP=5,500 | STEP=996,143 | clip=21 | consec=2 | max_consec=14 | avg_clip50=30.54
EP=5,510 | STEP=998,373 | clip=26 | consec=1 | max_consec=14 | avg_clip50=30.76

```

```

-----
| custom/
|   avg_clip50           | 30.4
|   clipped_episode_r... | 26.5
|   consecutive_clipp... | 2
|   max_consecutive_c... | 14
| rollout/
|   ep_len_mean          | 909
|   ep_rew_mean           | 290
|   exploration rate      | 0.667
| time/
|   episodes              | 16400
|   fps                   | 211
|   time_elapsed          | 4726
|   total timesteps       | 998610
| train/
|   learning_rate         | 0.0001
|   loss                  | 0.657
|   n_updates             | 237152

```

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-----
EP=5,520 | STEP=1,001,016 | clip=56 | consec=1 | max_consec=14 | avg_clip50=32.09
EP=5,530 | STEP=1,003,131 | clip=10 | consec=0 | max_consec=14 | avg_clip50=30.70
EP=5,540 | STEP=1,005,238 | clip=58 | consec=2 | max_consec=14 | avg_clip50=30.40

```

```

-----
| custom/
|   avg_clip50           | 30.3
|   clipped_episode_r... | 10.5
|   consecutive_clipp... | 0
|   max_consecutive_c... | 14
| rollout/
|   ep_len_mean          | 938
|   ep_rew_mean           | 302
|   exploration rate      | 0.665
| time/
|   episodes              | 16500
|   fps                   | 211
|   time_elapsed          | 4761
|   total timesteps       | 1006036
| train/
|   learning_rate         | 0.0001
|   loss                  | 0.042
|   n_updates             | 239008

```

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-----
EP=5,550 | STEP=1,007,154 | clip=32 | consec=2 | max_consec=14 | avg_clip50=30.18
EP=5,560 | STEP=1,009,503 | clip=23 | consec=3 | max_consec=14 | avg_clip50=29.78
EP=5,570 | STEP=1,011,928 | clip=9  | consec=0 | max_consec=14 | avg_clip50=28.24

```

```

-----
| custom/
|   avg_clip50           | 26.5
|   clipped_episode_r... | 18.5
|   consecutive_clipp... | 0
|   max_consecutive_c... | 14
| rollout/
|   ep_len_mean          | 917
|   ep_rew_mean           | 288

```

exploration rate	0.663
time/	
episodes	16600
fps	211
time_elapsed	4797
total timesteps	1013284
train/	
learning_rate	0.0001
loss	0.135
n_updates	240820

EP=5,580 | STEP=1,013,657 | clip=24 | consec=2 | max_consec=14 | avg_clip50=27.11
EP=5,590 | STEP=1,015,602 | clip=25 | consec=1 | max_consec=14 | avg_clip50=26.36
EP=5,600 | STEP=1,017,364 | clip=16 | consec=0 | max_consec=14 | avg_clip50=25.15
EP=5,610 | STEP=1,019,595 | clip=32 | consec=2 | max_consec=14 | avg_clip50=24.61

custom/	
avg_clip50	24.6
clipped_episode_r...	33.5
consecutive_clipp...	3
max_consecutive_c...	14
rollout/	
ep_len_mean	895
ep_rew_mean	273
exploration rate	0.66
time/	
episodes	16700
fps	211
time_elapsed	4830
total timesteps	1019890
train/	
learning_rate	0.0001
loss	0.287
n_updates	242472

EP=5,620 | STEP=1,021,910 | clip=38 | consec=2 | max_consec=14 | avg_clip50=25.44
EP=5,630 | STEP=1,023,723 | clip=11 | consec=0 | max_consec=14 | avg_clip50=26.27
EP=5,640 | STEP=1,025,835 | clip=38 | consec=2 | max_consec=14 | avg_clip50=26.12

custom/	
avg_clip50	26.3
clipped_episode_r...	18.5
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	884
ep_rew_mean	266
exploration rate	0.658
time/	
episodes	16800
fps	211
time_elapsed	4865
total timesteps	1027078
train/	
learning_rate	0.0001
loss	0.0602
n_updates	244269

EP=5,650 | STEP=1,028,017 | clip=30 | consec=1 | max_consec=14 | avg_clip50=27.40
EP=5,660 | STEP=1,029,595 | clip=14 | consec=0 | max_consec=14 | avg_clip50=24.89

EP=5,670 | STEP=1,031,295 | clip=18 | consec=0 | max_consec=14 | avg_clip50=22.28

custom/	
avg_clip50	22.3
clipped_episode_r...	27.5
consecutive_clipp...	2
max_consecutive_c...	14
rollout/	
ep_len_mean	836
ep_rew_mean	244
exploration rate	0.656
time/	
episodes	16900
fps	211
time_elapsed	4896
total timesteps	1033137
train/	
learning_rate	0.0001
loss	0.0616
n_updates	245784

EP=5,680 | STEP=1,033,525 | clip=65 | consec=1 | max_consec=14 | avg_clip50=23.18
EP=5,690 | STEP=1,035,178 | clip=46 | consec=1 | max_consec=14 | avg_clip50=21.75
EP=5,700 | STEP=1,037,537 | clip=10 | consec=0 | max_consec=14 | avg_clip50=22.30
EP=5,710 | STEP=1,039,457 | clip=58 | consec=3 | max_consec=14 | avg_clip50=24.74

custom/	
avg_clip50	25.7
clipped_episode_r...	47.5
consecutive_clipp...	1
max_consecutive_c...	14
rollout/	
ep_len_mean	840
ep_rew_mean	246
exploration rate	0.654
time/	
episodes	17000
fps	210
time_elapsed	4931
total timesteps	1039925
train/	
learning_rate	0.0001
loss	0.0684
n_updates	247481

EP=5,720 | STEP=1,041,366 | clip=26 | consec=3 | max_consec=14 | avg_clip50=26.07
EP=5,730 | STEP=1,043,358 | clip=16 | consec=0 | max_consec=14 | avg_clip50=25.11
EP=5,740 | STEP=1,045,267 | clip=28 | consec=1 | max_consec=14 | avg_clip50=25.99

custom/	
avg_clip50	25.2
clipped_episode_r...	14
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	820
ep_rew_mean	242
exploration rate	0.651
time/	
episodes	17100

	fps		210	
	time_elapsed		4964	
	total timesteps		1046563	
	train/			
	learning_rate		0.0001	
	loss		0.464	
	n_updates		249140	

EP=5,750		STEP=1,047,360		clip=54		consec=1		max_consec=14		avg_clip50=25.70
EP=5,760		STEP=1,050,080		clip=63		consec=1		max_consec=14		avg_clip50=28.00
EP=5,770		STEP=1,051,894		clip=41		consec=2		max_consec=14		avg_clip50=27.18
EP=5,780		STEP=1,054,057		clip=8		consec=0		max_consec=14		avg_clip50=27.92

	custom/			
	avg_clip50		27.9	
	clipped_episode_r...		8	
	consecutive_clipp...		0	
	max_consecutive_c...		14	
	rollout/			
	ep_len_mean		867	
	ep_rew_mean		265	
	exploration rate		0.649	
	time/			
	episodes		17200	
	fps		210	
	time_elapsed		5003	
	total timesteps		1054057	
	train/			
	learning_rate		0.0001	
	loss		0.286	
	n_updates		251014	

EP=5,790		STEP=1,056,233		clip=52		consec=1		max_consec=14		avg_clip50=30.28
EP=5,800		STEP=1,057,997		clip=14		consec=0		max_consec=14		avg_clip50=28.62
EP=5,810		STEP=1,059,891		clip=4		consec=0		max_consec=14		avg_clip50=25.08

	custom/			
	avg_clip50		24.9	
	clipped_episode_r...		19	
	consecutive_clipp...		0	
	max_consecutive_c...		14	
	rollout/			
	ep_len_mean		853	
	ep_rew_mean		261	
	exploration rate		0.647	
	time/			
	episodes		17300	
	fps		210	
	time_elapsed		5035	
	total timesteps		1060500	
	train/			
	learning_rate		0.0001	
	loss		0.0435	
	n_updates		252624	

EP=5,820		STEP=1,061,942		clip=4		consec=0		max_consec=14		avg_clip50=25.22
EP=5,830		STEP=1,063,617		clip=10		consec=0		max_consec=14		avg_clip50=23.41
EP=5,840		STEP=1,065,639		clip=26		consec=4		max_consec=14		avg_clip50=21.05

	custom/			
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avg_clip50	20.8
clipped_episode_r...	10
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	848
ep_rew_mean	248
exploration rate	0.645
time/	
episodes	17400
fps	210
time_elapsed	5066
total timesteps	1066933
train/	
learning_rate	0.0001
loss	0.0746
n_updates	254233

EP=5,850	STEP=1,067,629	clip=67	consec=1	max_consec=14	avg_clip50=21.76
EP=5,860	STEP=1,070,118	clip=12	consec=0	max_consec=14	avg_clip50=24.66
EP=5,870	STEP=1,072,094	clip=34	consec=1	max_consec=14	avg_clip50=24.96
EP=5,880	STEP=1,073,841	clip=6	consec=0	max_consec=14	avg_clip50=25.82

custom/	
avg_clip50	26
clipped_episode_r...	25
consecutive_clipp...	2
max_consecutive_c...	14
rollout/	
ep_len_mean	845
ep_rew_mean	249
exploration rate	0.642
time/	
episodes	17500
fps	210
time_elapsed	5102
total timesteps	1074383
train/	
learning_rate	0.0001
loss	0.0496
n_updates	256095

EP=5,890	STEP=1,076,054	clip=2	consec=0	max_consec=14	avg_clip50=27.20
EP=5,900	STEP=1,078,171	clip=12	consec=0	max_consec=14	avg_clip50=28.05
EP=5,910	STEP=1,080,549	clip=24	consec=6	max_consec=14	avg_clip50=26.98

custom/	
avg_clip50	28.1
clipped_episode_r...	42.5
consecutive_clipp...	2
max_consecutive_c...	14
rollout/	
ep_len_mean	887
ep_rew_mean	265
exploration rate	0.64
time/	
episodes	17600
fps	210
time_elapsed	5140
total timesteps	1082015

train/						
learning_rate		0.0001				
loss		0.0511				
n_updates		258003				

EP=5,920	STEP=1,082,974	clip=18	consec=0	max_consec=14	avg_clip50=28.26	
EP=5,930	STEP=1,085,028	clip=16	consec=0	max_consec=14	avg_clip50=28.19	
EP=5,940	STEP=1,086,861	clip=10	consec=0	max_consec=14	avg_clip50=25.65	
EP=5,950	STEP=1,088,948	clip=43	consec=1	max_consec=14	avg_clip50=26.48	

custom/						
avg_clip50		26.5				
clipped_episode_r...		43				
consecutive_clipp...		1				
max_consecutive_c...		14				
rollout/						
ep_len_mean		898				
ep_rew_mean		273				
exploration rate		0.637				
time/						
episodes		17700				
fps		210				
time_elapsed		5175				
total timesteps		1088972				
train/						
learning_rate		0.0001				
loss		0.291				
n_updates		259742				

EP=5,960	STEP=1,090,756	clip=50	consec=1	max_consec=14	avg_clip50=24.65	
EP=5,970	STEP=1,093,094	clip=54	consec=1	max_consec=14	avg_clip50=24.33	
EP=5,980	STEP=1,095,450	clip=43	consec=2	max_consec=14	avg_clip50=26.07	

custom/						
avg_clip50		25.8				
clipped_episode_r...		4.5				
consecutive_clipp...		0				
max_consecutive_c...		14				
rollout/						
ep_len_mean		892				
ep_rew_mean		264				
exploration rate		0.635				
time/						
episodes		17800				
fps		210				
time_elapsed		5211				
total timesteps		1096257				
train/						
learning_rate		0.0001				
loss		0.0377				
n_updates		261564				

EP=5,990	STEP=1,097,362	clip=20	consec=0	max_consec=14	avg_clip50=27.68	
EP=6,000	STEP=1,099,428	clip=25	consec=4	max_consec=14	avg_clip50=27.08	
EP=6,010	STEP=1,101,546	clip=60	consec=2	max_consec=14	avg_clip50=28.45	

custom/						
avg_clip50		27.8				
clipped_episode_r...		17.5				
consecutive_clipp...		0				

max_consecutive_c...	14
rollout/	
ep_len_mean	864
ep_rew_mean	257
exploration rate	0.633
time/	
episodes	17900
fps	210
time_elapsed	5247
total timesteps	1103301
train/	
learning_rate	0.0001
loss	0.0912
n_updates	263325

EP=6,020	STEP=1,103,569	clip=30	consec=1	max_consec=14	avg_clip50=27.34
EP=6,030	STEP=1,105,815	clip=8	consec=0	max_consec=14	avg_clip50=27.08
EP=6,040	STEP=1,107,934	clip=28	consec=6	max_consec=14	avg_clip50=28.51
EP=6,050	STEP=1,110,375	clip=34	consec=3	max_consec=14	avg_clip50=29.79

custom/	
avg_clip50	29
clipped_episode_r...	12.5
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	901
ep_rew_mean	285
exploration rate	0.63
time/	
episodes	18000
fps	210
time_elapsed	5289
total timesteps	1110930
train/	
learning_rate	0.0001
loss	0.0877
n_updates	265232

EP=6,060	STEP=1,112,596	clip=44	consec=1	max_consec=14	avg_clip50=29.95
EP=6,070	STEP=1,114,629	clip=34	consec=1	max_consec=14	avg_clip50=30.94
EP=6,080	STEP=1,117,065	clip=42	consec=1	max_consec=14	avg_clip50=30.42

custom/	
avg_clip50	31.1
clipped_episode_r...	6.5
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	923
ep_rew_mean	296
exploration rate	0.628
time/	
episodes	18100
fps	209
time_elapsed	5328
total timesteps	1118570
train/	
learning_rate	0.0001
loss	0.159

n_updates		267142			

EP=6,090	STEP=1,119,178	clip=32	consec=2	max_consec=14	avg_clip50=29.82
EP=6,100	STEP=1,121,010	clip=20	consec=2	max_consec=14	avg_clip50=27.32
EP=6,110	STEP=1,123,028	clip=10	consec=0	max_consec=14	avg_clip50=26.30
EP=6,120	STEP=1,125,072	clip=36	consec=1	max_consec=14	avg_clip50=25.32

custom/					
avg_clip50		25			
clipped_episode_r...		14			
consecutive_clipp...		0			
max_consecutive_c...		14			
rollout/					
ep_len_mean		906			
ep_rew_mean		282			
exploration rate		0.625			
time/					
episodes		18200			
fps		209			
time_elapsed		5362			
total timesteps		1125429			
train/					
learning_rate		0.0001			
loss		0.306			
n_updates		268857			

EP=6,130	STEP=1,126,926	clip=14	consec=0	max_consec=14	avg_clip50=23.74
EP=6,140	STEP=1,129,256	clip=20	consec=0	max_consec=14	avg_clip50=22.42
EP=6,150	STEP=1,131,252	clip=26	consec=3	max_consec=14	avg_clip50=23.29

custom/					
avg_clip50		24.7			
clipped_episode_r...		34.5			
consecutive_clipp...		2			
max_consecutive_c...		14			
rollout/					
ep_len_mean		880			
ep_rew_mean		256			
exploration rate		0.623			
time/					
episodes		18300			
fps		209			
time_elapsed		5396			
total timesteps		1132115			
train/					
learning_rate		0.0001			
loss		0.133			
n_updates		270528			

EP=6,160	STEP=1,133,395	clip=42	consec=4	max_consec=14	avg_clip50=24.32
EP=6,170	STEP=1,135,461	clip=18	consec=0	max_consec=14	avg_clip50=25.28
EP=6,180	STEP=1,137,485	clip=18	consec=0	max_consec=14	avg_clip50=26.31

custom/					
avg_clip50		27.1			
clipped_episode_r...		3.5			
consecutive_clipp...		0			
max_consecutive_c...		14			
rollout/					
ep_len_mean		871			

ep_rew_mean	252
exploration rate	0.621
time/	
episodes	18400
fps	209
time_elapsed	5433
total timesteps	1139489
train/	
learning_rate	0.0001
loss	0.055
n_updates	272372

EP=6,190	STEP=1,139,950	clip=39	consec=2	max_consec=14	avg_clip50=28.39
EP=6,200	STEP=1,141,951	clip=23	consec=2	max_consec=14	avg_clip50=28.68
EP=6,210	STEP=1,144,047	clip=12	consec=0	max_consec=14	avg_clip50=27.47
EP=6,220	STEP=1,146,296	clip=14	consec=0	max_consec=14	avg_clip50=28.49

custom/	
avg_clip50	27.9
clipped_episode_r...	1
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	887
ep_rew_mean	263
exploration rate	0.618
time/	
episodes	18500
fps	209
time_elapsed	5470
total timesteps	1146745
train/	
learning_rate	0.0001
loss	0.0369
n_updates	274186

EP=6,230	STEP=1,148,260	clip=72	consec=2	max_consec=14	avg_clip50=28.60
EP=6,240	STEP=1,150,574	clip=92	consec=2	max_consec=14	avg_clip50=28.62
EP=6,250	STEP=1,152,640	clip=55	consec=1	max_consec=14	avg_clip50=29.09

custom/	
avg_clip50	29.7
clipped_episode_r...	31.5
consecutive_clipp...	3
max_consecutive_c...	14
rollout/	
ep_len_mean	911
ep_rew_mean	290
exploration rate	0.616
time/	
episodes	18600
fps	209
time_elapsed	5507
total timesteps	1154050
train/	
learning_rate	0.0001
loss	0.0497
n_updates	276012

EP=6,260	STEP=1,154,871	clip=49	consec=1	max_consec=14	avg_clip50=30.58
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EP=6,270 | STEP=1,156,825 | clip=8 | consec=0 | max_consec=14 | avg_clip50=29.11
EP=6,280 | STEP=1,159,125 | clip=71 | consec=5 | max_consec=14 | avg_clip50=30.65

custom/	
avg_clip50	31
clipped_episode_r...	23.5
consecutive_clipp...	1
max_consecutive_c...	14
rollout/	
ep_len_mean	905
ep_rew_mean	293
exploration rate	0.613
time/	
episodes	18700
fps	209
time_elapsed	5545
total timesteps	1161344
train/	
learning_rate	0.0001
loss	0.0647
n_updates	277835

EP=6,290 | STEP=1,161,417 | clip=40 | consec=2 | max_consec=14 | avg_clip50=29.98
EP=6,300 | STEP=1,163,390 | clip=36 | consec=1 | max_consec=14 | avg_clip50=28.80
EP=6,310 | STEP=1,165,187 | clip=10 | consec=0 | max_consec=14 | avg_clip50=26.61
EP=6,320 | STEP=1,167,608 | clip=16 | consec=0 | max_consec=14 | avg_clip50=28.78

custom/	
avg_clip50	29
clipped_episode_r...	11.5
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	902
ep_rew_mean	295
exploration rate	0.611
time/	
episodes	18800
fps	209
time_elapsed	5579
total timesteps	1168206
train/	
learning_rate	0.0001
loss	0.0605
n_updates	279551

EP=6,330 | STEP=1,169,571 | clip=28 | consec=3 | max_consec=14 | avg_clip50=26.98
EP=6,340 | STEP=1,171,628 | clip=22 | consec=1 | max_consec=14 | avg_clip50=25.94
EP=6,350 | STEP=1,174,045 | clip=54 | consec=6 | max_consec=14 | avg_clip50=28.48

custom/	
avg_clip50	29.3
clipped_episode_r...	27
consecutive_clipp...	2
max_consecutive_c...	14
rollout/	
ep_len_mean	911
ep_rew_mean	284
exploration rate	0.608
time/	

	episodes	18900	
	fps	209	
	time_elapsed	5619	
	total timesteps	1175920	
	train/		
	learning_rate	0.0001	
	loss	0.0605	
	n_updates	281479	

EP=6,360		STEP=1,176,584		clip=50		consec=5		max_consec=14		avg_clip50=30.43
EP=6,370		STEP=1,178,584		clip=21		consec=1		max_consec=14		avg_clip50=28.80
EP=6,380		STEP=1,180,575		clip=44		consec=1		max_consec=14		avg_clip50=27.95
EP=6,390		STEP=1,183,024		clip=37		consec=1		max_consec=14		avg_clip50=29.62

	custom/		
	avg_clip50	29.5	
	clipped_episode_r...	16	
	consecutive_clipp...	0	
	max_consecutive_c...	14	
	rollout/		
	ep_len_mean	913	
	ep_rew_mean	277	
	exploration rate	0.606	
	time/		
	episodes	19000	
	fps	209	
	time_elapsed	5656	
	total timesteps	1183489	
	train/		
	learning_rate	0.0001	
	loss	0.392	
	n_updates	283372	

EP=6,400		STEP=1,185,388		clip=35		consec=1		max_consec=14		avg_clip50=27.71
EP=6,410		STEP=1,187,162		clip=8		consec=0		max_consec=14		avg_clip50=25.76
EP=6,420		STEP=1,189,100		clip=25		consec=2		max_consec=14		avg_clip50=24.30

	custom/		
	avg_clip50	27.4	
	clipped_episode_r...	65.5	
	consecutive_clipp...	3	
	max_consecutive_c...	14	
	rollout/		
	ep_len_mean	922	
	ep_rew_mean	279	
	exploration rate	0.603	
	time/		
	episodes	19100	
	fps	209	
	time_elapsed	5695	
	total timesteps	1190868	
	train/		
	learning_rate	0.0001	
	loss	0.0414	
	n_updates	285216	

EP=6,430		STEP=1,191,669		clip=29		consec=7		max_consec=14		avg_clip50=28.39
EP=6,440		STEP=1,193,933		clip=32		consec=1		max_consec=14		avg_clip50=28.31
EP=6,450		STEP=1,196,731		clip=52		consec=3		max_consec=14		avg_clip50=31.83
EP=6,460		STEP=1,199,121		clip=43		consec=2		max_consec=14		avg_clip50=34.43

custom/	
avg_clip50	34.4
clipped_episode_r...	43
consecutive_clipp...	2
max_consecutive_c...	14
rollout/	
ep_len_mean	946
ep_rew_mean	301
exploration rate	0.601
time/	
episodes	19200
fps	208
time_elapsed	5737
total timesteps	1199146
train/	
learning_rate	0.0001
loss	0.0659
n_updates	287286

EP=6,470 | STEP=1,201,744 | clip=62 | consec=1 | max_consec=14 | avg_clip50=38.46
EP=6,480 | STEP=1,203,711 | clip=46 | consec=1 | max_consec=14 | avg_clip50=35.46
EP=6,490 | STEP=1,205,702 | clip=16 | consec=0 | max_consec=14 | avg_clip50=34.35

custom/	
avg_clip50	34
clipped_episode_r...	37.5
consecutive_clipp...	1
max_consecutive_c...	14
rollout/	
ep_len_mean	951
ep_rew_mean	317
exploration rate	0.598
time/	
episodes	19300
fps	208
time_elapsed	5774
total timesteps	1206571
train/	
learning_rate	0.0001
loss	0.0536
n_updates	289142

EP=6,500 | STEP=1,207,968 | clip=22 | consec=1 | max_consec=14 | avg_clip50=32.89
EP=6,510 | STEP=1,210,141 | clip=48 | consec=1 | max_consec=14 | avg_clip50=31.97
EP=6,520 | STEP=1,212,456 | clip=20 | consec=0 | max_consec=14 | avg_clip50=30.57

custom/	
avg_clip50	31
clipped_episode_r...	13.5
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	972
ep_rew_mean	332
exploration rate	0.596
time/	
episodes	19400
fps	208
time_elapsed	5813

total timesteps	1214408
train/	
learning_rate	0.0001
loss	0.114
n_updates	291101

EP=6,530 | STEP=1,214,853 | clip=28 | consec=2 | max_consec=14 | avg_clip50=31.28
EP=6,540 | STEP=1,216,948 | clip=12 | consec=0 | max_consec=14 | avg_clip50=29.80
EP=6,550 | STEP=1,218,922 | clip=45 | consec=2 | max_consec=14 | avg_clip50=28.24
EP=6,560 | STEP=1,221,239 | clip=56 | consec=2 | max_consec=14 | avg_clip50=29.25

custom/	
avg_clip50	28.5
clipped_episode_r...	6.5
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	924
ep_rew_mean	301
exploration rate	0.593
time/	
episodes	19500
fps	208
time_elapsed	5849
total timesteps	1221710
train/	
learning_rate	0.0001
loss	0.0637
n_updates	292927

EP=6,570 | STEP=1,223,357 | clip=22 | consec=2 | max_consec=14 | avg_clip50=27.57
EP=6,580 | STEP=1,225,260 | clip=2 | consec=0 | max_consec=14 | avg_clip50=26.39
EP=6,590 | STEP=1,227,625 | clip=56 | consec=2 | max_consec=14 | avg_clip50=28.97

custom/	
avg_clip50	28.4
clipped_episode_r...	15.5
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	922
ep_rew_mean	288
exploration rate	0.591
time/	
episodes	19600
fps	208
time_elapsed	5887
total timesteps	1229089
train/	
learning_rate	0.0001
loss	0.0393
n_updates	294772

EP=6,600 | STEP=1,229,688 | clip=5 | consec=0 | max_consec=14 | avg_clip50=27.79
EP=6,610 | STEP=1,232,280 | clip=12 | consec=0 | max_consec=14 | avg_clip50=28.42
EP=6,620 | STEP=1,234,123 | clip=27 | consec=1 | max_consec=14 | avg_clip50=27.75
EP=6,630 | STEP=1,236,025 | clip=8 | consec=0 | max_consec=14 | avg_clip50=28.91

custom/	
avg_clip50	28.9

	clipped_episode_r...	8	
	consecutive_clipp...	0	
	max_consecutive_c...	14	
	rollout/		
	ep_len_mean	892	
	ep_rew_mean	276	
	exploration rate	0.588	
	time/		
	episodes	19700	
	fps	208	
	time_elapsed	5923	
	total timesteps	1236106	
	train/		
	learning_rate	0.0001	
	loss	0.254	
	n_updates	296526	

EP=6,640		STEP=1,238,356		clip=28		consec=5		max_consec=14		avg_clip50=27.97
EP=6,650		STEP=1,240,197		clip=30		consec=2		max_consec=14		avg_clip50=27.44
EP=6,660		STEP=1,242,680		clip=32		consec=4		max_consec=14		avg_clip50=26.82

	custom/		
	avg_clip50	27.6	
	clipped_episode_r...	9.5	
	consecutive_clipp...	0	
	max_consecutive_c...	14	
	rollout/		
	ep_len_mean	902	
	ep_rew_mean	278	
	exploration rate	0.586	
	time/		
	episodes	19800	
	fps	208	
	time_elapsed	5960	
	total timesteps	1243523	
	train/		
	learning_rate	0.0001	
	loss	0.162	
	n_updates	298380	

EP=6,670		STEP=1,244,878		clip=36		consec=5		max_consec=14		avg_clip50=28.47
EP=6,680		STEP=1,247,054		clip=23		consec=2		max_consec=14		avg_clip50=27.93
EP=6,690		STEP=1,249,769		clip=69		consec=3		max_consec=14		avg_clip50=29.94

	custom/		
	avg_clip50	31.4	
	clipped_episode_r...	5	
	consecutive_clipp...	0	
	max_consecutive_c...	14	
	rollout/		
	ep_len_mean	928	
	ep_rew_mean	292	
	exploration rate	0.583	
	time/		
	episodes	19900	
	fps	208	
	time_elapsed	6001	
	total timesteps	1251670	
	train/		
	learning_rate	0.0001	

	loss	0.0669				
	n_updates	300417				

EP=6,700	STEP=1,251,889	clip=34	consec=1	max_consec=14	avg_clip50=31.50	
EP=6,710	STEP=1,253,751	clip=39	consec=1	max_consec=14	avg_clip50=30.01	
EP=6,720	STEP=1,256,126	clip=49	consec=6	max_consec=14	avg_clip50=30.33	
EP=6,730	STEP=1,258,367	clip=30	consec=1	max_consec=14	avg_clip50=31.71	

	custom/					
	avg_clip50	32.4				
	clipped_episode_r...	36				
	consecutive_clipp...	2				
	max_consecutive_c...	14				
	rollout/					
	ep_len_mean	937				
	ep_rew_mean	296				
	exploration rate	0.581				
	time/					
	episodes	20000				
	fps	208				
	time_elapsed	6038				
	total timesteps	1259092				
	train/					
	learning_rate	0.0001				
	loss	0.395				
	n_updates	302272				

EP=6,740	STEP=1,260,396	clip=16	consec=0	max_consec=14	avg_clip50=29.39	
EP=6,750	STEP=1,262,323	clip=14	consec=0	max_consec=14	avg_clip50=28.46	
EP=6,760	STEP=1,264,727	clip=64	consec=2	max_consec=14	avg_clip50=31.74	

	custom/					
	avg_clip50	31				
	clipped_episode_r...	23				
	consecutive_clipp...	1				
	max_consecutive_c...	14				
	rollout/					
	ep_len_mean	921				
	ep_rew_mean	303				
	exploration rate	0.578				
	time/					
	episodes	20100				
	fps	208				
	time_elapsed	6073				
	total timesteps	1266070				
	train/					
	learning_rate	0.0001				
	loss	0.139				
	n_updates	304017				

EP=6,770	STEP=1,266,764	clip=51	consec=4	max_consec=14	avg_clip50=30.82	
EP=6,780	STEP=1,268,908	clip=6	consec=0	max_consec=14	avg_clip50=29.55	
EP=6,790	STEP=1,270,740	clip=14	consec=0	max_consec=14	avg_clip50=28.77	
EP=6,800	STEP=1,273,524	clip=58	consec=5	max_consec=14	avg_clip50=31.83	

	custom/					
	avg_clip50	31.8				
	clipped_episode_r...	57.5				
	consecutive_clipp...	5				
	max_consecutive_c...	14				

rollout/	
ep_len_mean	911
ep_rew_mean	301
exploration rate	0.576
time/	
episodes	20200
fps	208
time_elapsed	6112
total timesteps	1273784
train/	
learning_rate	0.0001
loss	0.0761
n_updates	305945

EP=6,810	STEP=1,275,402	clip=11	consec=0	max_consec=14	avg_clip50=28.18
EP=6,820	STEP=1,277,329	clip=22	consec=3	max_consec=14	avg_clip50=27.46
EP=6,830	STEP=1,279,668	clip=18	consec=0	max_consec=14	avg_clip50=29.65

custom/	
avg_clip50	30.7
clipped_episode_r...	52.5
consecutive_clipp...	2
max_consecutive_c...	14
rollout/	
ep_len_mean	901
ep_rew_mean	296
exploration rate	0.573
time/	
episodes	20300
fps	208
time_elapsed	6148
total timesteps	1280807
train/	
learning_rate	0.0001
loss	0.369
n_updates	307701

EP=6,840	STEP=1,281,889	clip=18	consec=0	max_consec=14	avg_clip50=30.57
EP=6,850	STEP=1,284,448	clip=30	consec=6	max_consec=14	avg_clip50=30.81
EP=6,860	STEP=1,287,187	clip=44	consec=2	max_consec=14	avg_clip50=33.99

custom/	
avg_clip50	37.3
clipped_episode_r...	51.5
consecutive_clipp...	11
max_consecutive_c...	14
rollout/	
ep_len_mean	968
ep_rew_mean	327
exploration rate	0.571
time/	
episodes	20400
fps	208
time_elapsed	6192
total timesteps	1289599
train/	
learning_rate	0.0001
loss	0.182
n_updates	309899

EP=6,870	STEP=1,289,794	clip=32	consec=12	max_consec=14	avg_clip50=37.49
EP=6,880	STEP=1,292,482	clip=58	consec=9	max_consec=14	avg_clip50=37.96
EP=6,890	STEP=1,295,128	clip=14	consec=0	max_consec=14	avg_clip50=40.32
EP=6,900	STEP=1,297,365	clip=25	consec=1	max_consec=14	avg_clip50=38.35

custom/	
avg_clip50	38.2
clipped_episode_r...	50.5
consecutive_clipp...	1
max_consecutive_c...	14
rollout/	
ep_len_mean	1e+03
ep_rew_mean	345
exploration rate	0.568
time/	
episodes	20500
fps	208
time_elapsed	6235
total timesteps	1298027
train/	
learning_rate	0.0001
loss	0.309
n_updates	312006

EP=6,910	STEP=1,299,659	clip=10	consec=0	max_consec=14	avg_clip50=35.87
EP=6,920	STEP=1,302,056	clip=15	consec=0	max_consec=14	avg_clip50=35.01
EP=6,930	STEP=1,304,261	clip=26	consec=1	max_consec=14	avg_clip50=33.20

custom/	
avg_clip50	29.8
clipped_episode_r...	3
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	1.02e+03
ep_rew_mean	349
exploration rate	0.565
time/	
episodes	20600
fps	208
time_elapsed	6275
total timesteps	1305794
train/	
learning_rate	0.0001
loss	0.0466
n_updates	313948

EP=6,940	STEP=1,306,316	clip=50	consec=2	max_consec=14	avg_clip50=30.20
EP=6,950	STEP=1,308,586	clip=17	consec=0	max_consec=14	avg_clip50=28.85
EP=6,960	STEP=1,310,854	clip=24	consec=4	max_consec=14	avg_clip50=28.82
EP=6,970	STEP=1,312,712	clip=20	consec=1	max_consec=14	avg_clip50=26.13

custom/	
avg_clip50	25.9
clipped_episode_r...	16.5
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	959

ep_rew_mean	303
exploration rate	0.563
time/	
episodes	20700
fps	207
time_elapsed	6313
total timesteps	1313095
train/	
learning_rate	0.0001
loss	0.0694
n_updates	315773

EP=6,980 | STEP=1,314,854 | clip=24 | consec=1 | max_consec=14 | avg_clip50=26.07
EP=6,990 | STEP=1,317,355 | clip=14 | consec=0 | max_consec=14 | avg_clip50=28.43
EP=7,000 | STEP=1,319,806 | clip=42 | consec=7 | max_consec=14 | avg_clip50=31.25

custom/	
avg_clip50	32.7
clipped_episode_r...	80
consecutive_clipp...	12
max_consecutive_c...	14
rollout/	
ep_len_mean	951
ep_rew_mean	309
exploration rate	0.56
time/	
episodes	20800
fps	207
time_elapsed	6354
total timesteps	1321242
train/	
learning_rate	0.0001
loss	0.0641
n_updates	317810

EP=7,010 | STEP=1,322,153 | clip=4 | consec=0 | max_consec=14 | avg_clip50=32.26
EP=7,020 | STEP=1,324,450 | clip=23 | consec=10 | max_consec=14 | avg_clip50=35.0
2
EP=7,030 | STEP=1,327,018 | clip=46 | consec=5 | max_consec=14 | avg_clip50=36.11

custom/	
avg_clip50	33.6
clipped_episode_r...	48.5
consecutive_clipp...	2
max_consecutive_c...	14
rollout/	
ep_len_mean	962
ep_rew_mean	314
exploration rate	0.557
time/	
episodes	20900
fps	207
time_elapsed	6393
total timesteps	1329109
train/	
learning_rate	0.0001
loss	0.0861
n_updates	319777

EP=7,040 | STEP=1,329,133 | clip=40 | consec=3 | max_consec=14 | avg_clip50=34.19

EP=7,050	STEP=1,331,565	clip=34	consec=2	max_consec=14	avg_clip50=33.06
EP=7,060	STEP=1,333,956	clip=46	consec=1	max_consec=14	avg_clip50=33.49
EP=7,070	STEP=1,335,977	clip=3	consec=0	max_consec=14	avg_clip50=30.90

custom/	
avg_clip50	32.3
clipped_episode_r...	47.5
consecutive_clipp...	3
max_consecutive_c...	14
rollout/	
ep_len_mean	988
ep_rew_mean	339
exploration rate	0.555
time/	
episodes	21000
fps	207
time_elapsed	6432
total timesteps	1337115
train/	
learning_rate	0.0001
loss	0.0497
n_updates	321778

EP=7,080	STEP=1,338,385	clip=24	consec=1	max_consec=14	avg_clip50=30.49
EP=7,090	STEP=1,340,814	clip=40	consec=1	max_consec=14	avg_clip50=31.81
EP=7,100	STEP=1,342,759	clip=52	consec=1	max_consec=14	avg_clip50=31.20

custom/	
avg_clip50	29.5
clipped_episode_r...	29.5
consecutive_clipp...	3
max_consecutive_c...	14
rollout/	
ep_len_mean	944
ep_rew_mean	311
exploration rate	0.552
time/	
episodes	21100
fps	207
time_elapsed	6470
total timesteps	1344526
train/	
learning_rate	0.0001
loss	0.148
n_updates	323631

EP=7,110	STEP=1,344,828	clip=44	consec=4	max_consec=14	avg_clip50=29.43
EP=7,120	STEP=1,346,787	clip=11	consec=0	max_consec=14	avg_clip50=29.42
EP=7,130	STEP=1,348,816	clip=9	consec=0	max_consec=14	avg_clip50=27.40
EP=7,140	STEP=1,350,811	clip=9	consec=0	max_consec=14	avg_clip50=26.31

custom/	
avg_clip50	25.2
clipped_episode_r...	15
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	903
ep_rew_mean	284
exploration rate	0.55

time/	
episodes	21200
fps	207
time_elapsed	6504
total timesteps	1351258
train/	
learning_rate	0.0001
loss	0.0507
n_updates	325314

EP=7,150 | STEP=1,352,630 | clip=20 | consec=0 | max_consec=14 | avg_clip50=24.12
EP=7,160 | STEP=1,355,166 | clip=6 | consec=0 | max_consec=14 | avg_clip50=25.54
EP=7,170 | STEP=1,357,078 | clip=14 | consec=0 | max_consec=14 | avg_clip50=25.74

custom/	
avg_clip50	26
clipped_episode_r...	16.5
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	888
ep_rew_mean	272
exploration rate	0.548
time/	
episodes	21300
fps	207
time_elapsed	6543
total timesteps	1358609
train/	
learning_rate	0.0001
loss	0.0639
n_updates	327152

EP=7,180 | STEP=1,359,580 | clip=36 | consec=1 | max_consec=14 | avg_clip50=27.67
EP=7,190 | STEP=1,361,898 | clip=35 | consec=3 | max_consec=14 | avg_clip50=28.43
EP=7,200 | STEP=1,364,053 | clip=39 | consec=1 | max_consec=14 | avg_clip50=30.88

custom/	
avg_clip50	29.3
clipped_episode_r...	23.5
consecutive_clipp...	1
max_consecutive_c...	14
rollout/	
ep_len_mean	901
ep_rew_mean	278
exploration rate	0.545
time/	
episodes	21400
fps	207
time_elapsed	6582
total timesteps	1366082
train/	
learning_rate	0.0001
loss	0.376
n_updates	329020

EP=7,210 | STEP=1,366,205 | clip=52 | consec=2 | max_consec=14 | avg_clip50=30.24
EP=7,220 | STEP=1,368,210 | clip=8 | consec=0 | max_consec=14 | avg_clip50=30.28
EP=7,230 | STEP=1,370,516 | clip=16 | consec=0 | max_consec=14 | avg_clip50=29.66
EP=7,240 | STEP=1,372,581 | clip=70 | consec=1 | max_consec=14 | avg_clip50=28.57

custom/	
avg_clip50	29.4
clipped_episode_r...	29
consecutive_clipp...	4
max_consecutive_c...	14
rollout/	
ep_len_mean	923
ep_rew_mean	287
exploration rate	0.543
time/	
episodes	21500
fps	207
time_elapsed	6619
total timesteps	1373420
train/	
learning_rate	0.0001
loss	0.0626
n_updates	330854

EP=7,250 | STEP=1,374,681 | clip=30 | consec=3 | max_consec=14 | avg_clip50=27.53
EP=7,260 | STEP=1,377,459 | clip=18 | consec=0 | max_consec=14 | avg_clip50=30.40
EP=7,270 | STEP=1,379,487 | clip=56 | consec=3 | max_consec=14 | avg_clip50=30.12

custom/	
avg_clip50	30.4
clipped_episode_r...	16
consecutive_clipp...	0
max_consecutive_c...	14
rollout/	
ep_len_mean	945
ep_rew_mean	303
exploration rate	0.54
time/	
episodes	21600
fps	207
time_elapsed	6661
total timesteps	1381560
train/	
learning_rate	0.0001
loss	0.104
n_updates	332889

EP=7,280 | STEP=1,382,199 | clip=68 | consec=2 | max_consec=14 | avg_clip50=31.75
EP=7,290 | STEP=1,384,404 | clip=56 | consec=2 | max_consec=14 | avg_clip50=31.92
EP=7,300 | STEP=1,387,020 | clip=46 | consec=7 | max_consec=14 | avg_clip50=34.53
EP=7,310 | STEP=1,389,386 | clip=30 | consec=3 | max_consec=14 | avg_clip50=32.60

custom/	
avg_clip50	33.1
clipped_episode_r...	35
consecutive_clipp...	4
max_consecutive_c...	14
rollout/	
ep_len_mean	980
ep_rew_mean	318
exploration rate	0.537
time/	
episodes	21700
fps	207

	time_elapsed		6703	
	total timesteps		1389674	
	train/			
	learning_rate		0.0001	
	loss		0.123	
	n_updates		334918	

EP=7,320		STEP=1,391,640		clip=24		consec=1		max_consec=14		avg_clip50=33.65
EP=7,330		STEP=1,394,294		clip=29		consec=6		max_consec=14		avg_clip50=34.15
EP=7,340		STEP=1,396,528		clip=32		consec=3		max_consec=14		avg_clip50=34.59

	custom/			
	avg_clip50		35.5	
	clipped_episode_r...		26.5	
	consecutive_clipp...		7	
	max_consecutive_c...		14	
	rollout/			
	ep_len_mean		1.01e+03	
	ep_rew_mean		338	
	exploration rate		0.535	
	time/			
	episodes		21800	
	fps		207	
	time_elapsed		6744	
	total timesteps		1397609	
	train/			
	learning_rate		0.0001	
	loss		0.302	
	n_updates		336902	

EP=7,350		STEP=1,398,836		clip=66		consec=4		max_consec=14		avg_clip50=33.65
EP=7,360		STEP=1,401,040		clip=24		consec=5		max_consec=14		avg_clip50=32.90
EP=7,370		STEP=1,403,300		clip=28		consec=2		max_consec=14		avg_clip50=33.29

	custom/			
	avg_clip50		30.6	
	clipped_episode_r...		15.5	
	consecutive_clipp...		0	
	max_consecutive_c...		14	
	rollout/			
	ep_len_mean		986	
	ep_rew_mean		329	
	exploration rate		0.532	
	time/			
	episodes		21900	
	fps		207	
	time_elapsed		6781	
	total timesteps		1405147	
	train/			
	learning_rate		0.0001	
	loss		0.105	
	n_updates		338786	

EP=7,380		STEP=1,405,659		clip=42		consec=2		max_consec=14		avg_clip50=31.01
EP=7,390		STEP=1,408,651		clip=24		consec=12		max_consec=14		avg_clip50=35.1
2										
EP=7,400		STEP=1,410,872		clip=39		consec=1		max_consec=18		avg_clip50=34.14
EP=7,410		STEP=1,412,737		clip=34		consec=1		max_consec=18		avg_clip50=33.82

	custom/			
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avg_clip50	34.8
clipped_episode_r...	47.5
consecutive_clipp...	3
max_consecutive_c...	18
rollout/	
ep_len_mean	984
ep_rew_mean	336
exploration rate	0.529
time/	
episodes	22000
fps	207
time_elapsed	6823
total timesteps	1413419
train/	
learning_rate	0.0001
loss	0.0758
n_updates	340854

EP=7,420	STEP=1,415,172	clip=46	consec=3	max_consec=18	avg_clip50=34.36
EP=7,430	STEP=1,417,354	clip=22	consec=4	max_consec=18	avg_clip50=35.48
EP=7,440	STEP=1,419,543	clip=38	consec=4	max_consec=18	avg_clip50=31.69

custom/	
avg_clip50	32.1
clipped_episode_r...	15.5
consecutive_clipp...	0
max_consecutive_c...	18
rollout/	
ep_len_mean	978
ep_rew_mean	341
exploration rate	0.527
time/	
episodes	22100
fps	207
time_elapsed	6864
total timesteps	1421392
train/	
learning_rate	0.0001
loss	0.0523
n_updates	342847

EP=7,450	STEP=1,421,966	clip=38	consec=1	max_consec=18	avg_clip50=33.30
EP=7,460	STEP=1,424,180	clip=5	consec=0	max_consec=18	avg_clip50=33.13
EP=7,470	STEP=1,426,562	clip=18	consec=0	max_consec=18	avg_clip50=34.04
EP=7,480	STEP=1,428,854	clip=13	consec=0	max_consec=18	avg_clip50=34.36

custom/	
avg_clip50	34.6
clipped_episode_r...	20.5
consecutive_clipp...	1
max_consecutive_c...	18
rollout/	
ep_len_mean	968
ep_rew_mean	347
exploration rate	0.524
time/	
episodes	22200
fps	206
time_elapsed	6904
total timesteps	1429030

train/	
learning_rate	0.0001
loss	0.11
n_updates	344757

EP=7,490	STEP=1,430,934	clip=5	consec=0	max_consec=18	avg_clip50=33.18
EP=7,500	STEP=1,433,490	clip=24	consec=4	max_consec=18	avg_clip50=34.08
EP=7,510	STEP=1,435,745	clip=5	consec=0	max_consec=18	avg_clip50=35.18

custom/	
avg_clip50	35.5
clipped_episode_r...	40
consecutive_clipp...	4
max_consecutive_c...	18
rollout/	
ep_len_mean	952
ep_rew_mean	338
exploration rate	0.522
time/	
episodes	22300
fps	206
time_elapsed	6943
total timesteps	1436904
train/	
learning_rate	0.0001
loss	0.146
n_updates	346725

EP=7,520	STEP=1,437,891	clip=18	consec=0	max_consec=18	avg_clip50=34.35
EP=7,530	STEP=1,440,365	clip=40	consec=4	max_consec=18	avg_clip50=35.26
EP=7,540	STEP=1,443,029	clip=54	consec=1	max_consec=18	avg_clip50=39.47
EP=7,550	STEP=1,445,378	clip=27	consec=5	max_consec=18	avg_clip50=37.72

custom/	
avg_clip50	37.7
clipped_episode_r...	27
consecutive_clipp...	5
max_consecutive_c...	18
rollout/	
ep_len_mean	982
ep_rew_mean	359
exploration rate	0.519
time/	
episodes	22400
fps	206
time_elapsed	6987
total timesteps	1445567
train/	
learning_rate	0.0001
loss	0.148
n_updates	348891

EP=7,560	STEP=1,447,334	clip=48	consec=1	max_consec=18	avg_clip50=35.26
EP=7,570	STEP=1,449,459	clip=40	consec=5	max_consec=18	avg_clip50=34.09
EP=7,580	STEP=1,451,651	clip=6	consec=0	max_consec=18	avg_clip50=31.02

custom/	
avg_clip50	30.3
clipped_episode_r...	12.5
consecutive_clipp...	0

max_consecutive_c...	18
rollout/	
ep_len_mean	953
ep_rew_mean	332
exploration rate	0.516
time/	
episodes	22500
fps	206
time_elapsed	7024
total timesteps	1452775
train/	
learning_rate	0.0001
loss	0.162
n_updates	350693

EP=7,590	STEP=1,453,964	clip=47	consec=2	max_consec=18	avg_clip50=28.23
EP=7,600	STEP=1,456,278	clip=32	consec=6	max_consec=18	avg_clip50=28.33
EP=7,610	STEP=1,458,152	clip=12	consec=0	max_consec=18	avg_clip50=28.65

custom/	
avg_clip50	30.6
clipped_episode_r...	21
consecutive_clipp...	6
max_consecutive_c...	18
rollout/	
ep_len_mean	954
ep_rew_mean	319
exploration rate	0.514
time/	
episodes	22600
fps	206
time_elapsed	7064
total timesteps	1460426
train/	
learning_rate	0.0001
loss	0.0729
n_updates	352606

EP=7,620	STEP=1,460,642	clip=51	consec=8	max_consec=18	avg_clip50=30.48
EP=7,630	STEP=1,463,008	clip=48	consec=3	max_consec=18	avg_clip50=32.08
EP=7,640	STEP=1,465,606	clip=13	consec=0	max_consec=18	avg_clip50=33.04
EP=7,650	STEP=1,468,094	clip=52	consec=1	max_consec=18	avg_clip50=34.31

custom/	
avg_clip50	34.2
clipped_episode_r...	6.5
consecutive_clipp...	0
max_consecutive_c...	18
rollout/	
ep_len_mean	955
ep_rew_mean	315
exploration rate	0.511
time/	
episodes	22700
fps	206
time_elapsed	7106
total timesteps	1468834
train/	
learning_rate	0.0001
loss	0.0423

n_updates		354708				

EP=7,660	STEP=1,469,996	clip=14	consec=0	max_consec=18	avg_clip50=35.31	
EP=7,670	STEP=1,472,062	clip=30	consec=2	max_consec=18	avg_clip50=33.67	
EP=7,680	STEP=1,474,501	clip=50	consec=2	max_consec=18	avg_clip50=33.78	

custom/						
avg_clip50		33.3				
clipped_episode_r...		54.5				
consecutive_clipp...		6				
max_consecutive_c...		18				
rollout/						
ep_len_mean		952				
ep_rew_mean		328				
exploration rate		0.508				
time/						
episodes		22800				
fps		206				
time_elapsed		7143				
total timesteps		1476123				
train/						
learning_rate		0.0001				
loss		0.0417				
n_updates		356530				

EP=7,690	STEP=1,476,634	clip=58	consec=1	max_consec=18	avg_clip50=32.47	
EP=7,700	STEP=1,479,140	clip=18	consec=0	max_consec=18	avg_clip50=32.65	
EP=7,710	STEP=1,481,623	clip=66	consec=10	max_consec=18	avg_clip50=34.6	
0						
EP=7,720	STEP=1,483,937	clip=32	consec=7	max_consec=18	avg_clip50=35.73	

custom/						
avg_clip50		35.3				
clipped_episode_r...		12.5				
consecutive_clipp...		0				
max_consecutive_c...		18				
rollout/						
ep_len_mean		972				
ep_rew_mean		343				
exploration rate		0.506				
time/						
episodes		22900				
fps		206				
time_elapsed		7184				
total timesteps		1484094				
train/						
learning_rate		0.0001				
loss		0.233				
n_updates		358523				

EP=7,730	STEP=1,486,128	clip=24	consec=1	max_consec=18	avg_clip50=34.16	
EP=7,740	STEP=1,488,143	clip=43	consec=1	max_consec=18	avg_clip50=31.91	
EP=7,750	STEP=1,490,860	clip=42	consec=3	max_consec=18	avg_clip50=31.48	

custom/						
avg_clip50		31.9				
clipped_episode_r...		69				
consecutive_clipp...		1				
max_consecutive_c...		18				
rollout/						

ep_len_mean	964
ep_rew_mean	328
exploration rate	0.503
time/	
episodes	23000
fps	206
time_elapsed	7225
total timesteps	1492100
train/	
learning_rate	0.0001
loss	0.288
n_updates	360524

EP=7,760 | STEP=1,493,489 | clip=48 | consec=7 | max_consec=18 | avg_clip50=32.45
EP=7,770 | STEP=1,495,642 | clip=32 | consec=3 | max_consec=18 | avg_clip50=32.32
EP=7,780 | STEP=1,497,963 | clip=20 | consec=3 | max_consec=18 | avg_clip50=33.39

custom/	
avg_clip50	36.1
clipped_episode_r...	42.5
consecutive_clipp...	2
max_consecutive_c...	18
rollout/	
ep_len_mean	1e+03
ep_rew_mean	341
exploration rate	0.5
time/	
episodes	23100
fps	206
time_elapsed	7265
total timesteps	1500022
train/	
learning_rate	0.0001
loss	0.179
n_updates	362505

EP=7,790 | STEP=1,500,446 | clip=23 | consec=4 | max_consec=18 | avg_clip50=35.88
EP=7,800 | STEP=1,502,878 | clip=54 | consec=3 | max_consec=18 | avg_clip50=35.51
EP=7,810 | STEP=1,504,988 | clip=45 | consec=1 | max_consec=18 | avg_clip50=33.07
EP=7,820 | STEP=1,506,847 | clip=3 | consec=0 | max_consec=18 | avg_clip50=30.98

custom/	
avg_clip50	29.9
clipped_episode_r...	27.5
consecutive_clipp...	3
max_consecutive_c...	18
rollout/	
ep_len_mean	963
ep_rew_mean	319
exploration rate	0.498
time/	
episodes	23200
fps	206
time_elapsed	7304
total timesteps	1507762
train/	
learning_rate	0.0001
loss	0.0508
n_updates	364440

EP=7,830	STEP=1,509,197	clip=34	consec=2	max_consec=18	avg_clip50=32.52
EP=7,840	STEP=1,511,623	clip=19	consec=0	max_consec=18	avg_clip50=33.11
EP=7,850	STEP=1,513,943	clip=34	consec=1	max_consec=18	avg_clip50=32.88

custom/	
avg_clip50	34.4
clipped_episode_r...	26
consecutive_clipp...	2
max_consecutive_c...	18
rollout/	
ep_len_mean	978
ep_rew_mean	344
exploration rate	0.495
time/	
episodes	23300
fps	206
time_elapsed	7347
total timesteps	1516164
train/	
learning_rate	0.0001
loss	0.307
n_updates	366540

EP=7,860	STEP=1,516,700	clip=29	consec=5	max_consec=18	avg_clip50=35.01
EP=7,870	STEP=1,519,237	clip=54	consec=8	max_consec=18	avg_clip50=37.75
EP=7,880	STEP=1,521,275	clip=43	consec=3	max_consec=18	avg_clip50=35.52
EP=7,890	STEP=1,523,623	clip=64	consec=3	max_consec=18	avg_clip50=35.71

custom/	
avg_clip50	35.6
clipped_episode_r...	18.5
consecutive_clipp...	0
max_consecutive_c...	18
rollout/	
ep_len_mean	971
ep_rew_mean	341
exploration rate	0.492
time/	
episodes	23400
fps	206
time_elapsed	7387
total timesteps	1524046
train/	
learning_rate	0.0001
loss	0.0471
n_updates	368511

EP=7,900	STEP=1,525,784	clip=48	consec=1	max_consec=18	avg_clip50=34.32
EP=7,910	STEP=1,528,013	clip=12	consec=0	max_consec=18	avg_clip50=33.01
EP=7,920	STEP=1,530,409	clip=9	consec=0	max_consec=18	avg_clip50=32.42

custom/	
avg_clip50	35
clipped_episode_r...	28
consecutive_clipp...	5
max_consecutive_c...	18
rollout/	
ep_len_mean	999
ep_rew_mean	353
exploration rate	0.49

time/	
episodes	23500
fps	206
time_elapsed	7429
total timesteps	1532198
train/	
learning_rate	0.0001
loss	0.0875
n_updates	370549

EP=7,930 | STEP=1,533,275 | clip=20 | consec=10 | max_consec=18 | avg_clip50=34.73

EP=7,940 | STEP=1,536,257 | clip=58 | consec=5 | max_consec=18 | avg_clip50=36.95

EP=7,950 | STEP=1,538,649 | clip=9 | consec=0 | max_consec=18 | avg_clip50=39.54

EP=7,960 | STEP=1,540,960 | clip=46 | consec=2 | max_consec=18 | avg_clip50=39.76

custom/	
avg_clip50	39.8
clipped_episode_r...	45.5
consecutive_clipp...	2
max_consecutive_c...	18
rollout/	
ep_len_mean	1.01e+03
ep_rew_mean	364
exploration rate	0.487
time/	
episodes	23600
fps	206
time_elapsed	7475
total timesteps	1541190
train/	
learning_rate	0.0001
loss	0.328
n_updates	372797

EP=7,970 | STEP=1,543,140 | clip=72 | consec=2 | max_consec=18 | avg_clip50=38.94

EP=7,980 | STEP=1,545,735 | clip=16 | consec=0 | max_consec=18 | avg_clip50=39.28

EP=7,990 | STEP=1,548,211 | clip=76 | consec=3 | max_consec=18 | avg_clip50=36.54

custom/	
avg_clip50	36
clipped_episode_r...	51
consecutive_clipp...	3
max_consecutive_c...	18
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	371
exploration rate	0.484
time/	
episodes	23700
fps	206
time_elapsed	7519
total timesteps	1549352
train/	
learning_rate	0.0001
loss	0.0508
n_updates	374837

EP=8,000 | STEP=1,550,508 | clip=26 | consec=1 | max_consec=18 | avg_clip50=35.06

EP=8,010 | STEP=1,552,987 | clip=60 | consec=2 | max_consec=18 | avg_clip50=34.60

EP=8,020 | STEP=1,555,392 | clip=40 | consec=1 | max_consec=18 | avg_clip50=35.43

custom/		
avg_clip50	32.1	
clipped_episode_r...	51	
consecutive_clipp...	2	
max_consecutive_c...	18	
rollout/		
ep_len_mean	1.02e+03	
ep_rew_mean	360	
exploration rate	0.481	
time/		
episodes	23800	
fps	205	
time_elapsed	7560	
total timesteps	1557221	
train/		
learning_rate	0.0001	
loss	0.0317	
n_updates	376805	

EP=8,030 | STEP=1,557,486 | clip=15 | consec=0 | max_consec=18 | avg_clip50=32.01

EP=8,040 | STEP=1,559,774 | clip=44 | consec=3 | max_consec=18 | avg_clip50=32.44

EP=8,050 | STEP=1,562,340 | clip=39 | consec=13 | max_consec=18 | avg_clip50=33.9
2

EP=8,060 | STEP=1,564,691 | clip=28 | consec=1 | max_consec=18 | avg_clip50=34.12

custom/		
avg_clip50	34.5	
clipped_episode_r...	34.5	
consecutive_clipp...	2	
max_consecutive_c...	18	
rollout/		
ep_len_mean	995	
ep_rew_mean	344	
exploration rate	0.479	
time/		
episodes	23900	
fps	205	
time_elapsed	7603	
total timesteps	1565322	
train/		
learning_rate	0.0001	
loss	0.0425	
n_updates	378830	

EP=8,070 | STEP=1,567,577 | clip=78 | consec=7 | max_consec=18 | avg_clip50=36.58

EP=8,080 | STEP=1,569,679 | clip=4 | consec=0 | max_consec=18 | avg_clip50=37.24

EP=8,090 | STEP=1,572,307 | clip=16 | consec=0 | max_consec=18 | avg_clip50=37.95

custom/		
avg_clip50	35.8	
clipped_episode_r...	7	
consecutive_clipp...	0	
max_consecutive_c...	18	
rollout/		
ep_len_mean	985	
ep_rew_mean	339	
exploration rate	0.476	
time/		

	episodes		24000	
	fps		205	
	time_elapsed		7646	
	total timesteps		1573396	
	train/			
	learning_rate		0.0001	
	loss		0.115	
	n_updates		380848	

EP=8,100		STEP=1,574,473		clip=65		consec=4		max_consec=18		avg_clip50=35.42
EP=8,110		STEP=1,576,773		clip=17		consec=0		max_consec=18		avg_clip50=35.07
EP=8,120		STEP=1,579,632		clip=76		consec=2		max_consec=18		avg_clip50=34.76
EP=8,130		STEP=1,581,807		clip=6		consec=0		max_consec=18		avg_clip50=34.77

	custom/			
	avg_clip50		34.8	
	clipped_episode_r...		46	
	consecutive_clipp...		1	
	max_consecutive_c...		18	
	rollout/			
	ep_len_mean		1.02e+03	
	ep_rew_mean		360	
	exploration rate		0.473	
	time/			
	episodes		24100	
	fps		205	
	time_elapsed		7694	
	total timesteps		1582167	
	train/			
	learning_rate		0.0001	
	loss		0.106	
	n_updates		383041	

EP=8,140		STEP=1,584,384		clip=7		consec=0		max_consec=18		avg_clip50=34.25
EP=8,150		STEP=1,586,825		clip=28		consec=1		max_consec=18		avg_clip50=36.92
EP=8,160		STEP=1,589,152		clip=28		consec=3		max_consec=18		avg_clip50=36.57

	custom/			
	avg_clip50		36.4	
	clipped_episode_r...		38.5	
	consecutive_clipp...		4	
	max_consecutive_c...		18	
	rollout/			
	ep_len_mean		1.02e+03	
	ep_rew_mean		359	
	exploration rate		0.47	
	time/			
	episodes		24200	
	fps		205	
	time_elapsed		7740	
	total timesteps		1590526	
	train/			
	learning_rate		0.0001	
	loss		0.0388	
	n_updates		385131	

EP=8,170		STEP=1,591,658		clip=24		consec=4		max_consec=18		avg_clip50=35.38
EP=8,180		STEP=1,594,128		clip=38		consec=2		max_consec=18		avg_clip50=37.69
EP=8,190		STEP=1,596,775		clip=60		consec=12		max_consec=18		avg_clip50=38.7

EP=8,200 | STEP=1,599,243 | clip=44 | consec=22 | max_consec=22 | avg_clip50=37.6
0

custom/		
avg_clip50	37.6	
clipped_episode_r...	44.5	
consecutive_clipp...	22	
max_consecutive_c...	22	
rollout/		
ep_len_mean	1.04e+03	
ep_rew_mean	373	
exploration rate	0.467	
time/		
episodes	24300	
fps	205	
time_elapsed	7785	
total timesteps	1599243	
train/		
learning_rate	0.0001	
loss	0.329	
n_updates	387310	

EP=8,210 | STEP=1,601,618 | clip=72 | consec=2 | max_consec=23 | avg_clip50=39.74
EP=8,220 | STEP=1,604,153 | clip=65 | consec=6 | max_consec=23 | avg_clip50=39.99
EP=8,230 | STEP=1,606,260 | clip=16 | consec=0 | max_consec=23 | avg_clip50=38.58

custom/		
avg_clip50	37.5	
clipped_episode_r...	42.5	
consecutive_clipp...	4	
max_consecutive_c...	23	
rollout/		
ep_len_mean	1.01e+03	
ep_rew_mean	378	
exploration rate	0.465	
time/		
episodes	24400	
fps	205	
time_elapsed	7826	
total timesteps	1607175	
train/		
learning_rate	0.0001	
loss	0.0332	
n_updates	389293	

EP=8,240 | STEP=1,608,775 | clip=26 | consec=10 | max_consec=23 | avg_clip50=37.6
5
EP=8,250 | STEP=1,611,052 | clip=52 | consec=3 | max_consec=23 | avg_clip50=37.05
EP=8,260 | STEP=1,612,849 | clip=14 | consec=0 | max_consec=23 | avg_clip50=33.02

custom/		
avg_clip50	33.4	
clipped_episode_r...	34.5	
consecutive_clipp...	3	
max_consecutive_c...	23	
rollout/		
ep_len_mean	982	
ep_rew_mean	363	
exploration rate	0.462	
time/		

episodes	24500
fps	205
time_elapsed	7864
total timesteps	1614935
train/	
learning_rate	0.0001
loss	0.0647
n_updates	391233

EP=8,270	STEP=1,615,156	clip=36	consec=4	max_consec=23	avg_clip50=32.85
EP=8,280	STEP=1,617,304	clip=14	consec=0	max_consec=23	avg_clip50=32.80
EP=8,290	STEP=1,620,425	clip=69	consec=7	max_consec=23	avg_clip50=35.53
EP=8,300	STEP=1,623,061	clip=49	consec=4	max_consec=23	avg_clip50=36.75

custom/	
avg_clip50	38.7
clipped_episode_r...	49.5
consecutive_clipp...	2
max_consecutive_c...	23
rollout/	
ep_len_mean	991
ep_rew_mean	367
exploration rate	0.459
time/	
episodes	24600
fps	205
time_elapsed	7914
total timesteps	1623946
train/	
learning_rate	0.0001
loss	0.0685
n_updates	393486

EP=8,310	STEP=1,625,275	clip=50	consec=4	max_consec=23	avg_clip50=40.11
EP=8,320	STEP=1,627,294	clip=34	consec=3	max_consec=23	avg_clip50=37.89
EP=8,330	STEP=1,629,236	clip=86	consec=1	max_consec=23	avg_clip50=36.99

custom/	
avg_clip50	37.1
clipped_episode_r...	65.5
consecutive_clipp...	9
max_consecutive_c...	23
rollout/	
ep_len_mean	983
ep_rew_mean	357
exploration rate	0.457
time/	
episodes	24700
fps	205
time_elapsed	7955
total timesteps	1631746
train/	
learning_rate	0.0001
loss	0.0365
n_updates	395436

EP=8,340	STEP=1,632,058	clip=16	consec=0	max_consec=23	avg_clip50=35.15
EP=8,350	STEP=1,634,320	clip=12	consec=0	max_consec=23	avg_clip50=33.58
EP=8,360	STEP=1,636,887	clip=19	consec=0	max_consec=23	avg_clip50=32.36
EP=8,370	STEP=1,638,903	clip=38	consec=2	max_consec=23	avg_clip50=31.88

custom/	
avg_clip50	32.5
clipped_episode_r...	28.5
consecutive_clipp...	1
max_consecutive_c...	23
rollout/	
ep_len_mean	992
ep_rew_mean	344
exploration rate	0.454
time/	
episodes	24800
fps	204
time_elapsed	7996
total timesteps	1639285
train/	
learning_rate	0.0001
loss	0.0453
n_updates	397321

EP=8,380 | STEP=1,641,130 | clip=26 | consec=3 | max_consec=23 | avg_clip50=31.91
EP=8,390 | STEP=1,643,476 | clip=28 | consec=4 | max_consec=23 | avg_clip50=30.46
EP=8,400 | STEP=1,645,756 | clip=32 | consec=1 | max_consec=23 | avg_clip50=30.36

custom/	
avg_clip50	30.7
clipped_episode_r...	38.5
consecutive_clipp...	7
max_consecutive_c...	23
rollout/	
ep_len_mean	978
ep_rew_mean	327
exploration rate	0.451
time/	
episodes	24900
fps	204
time_elapsed	8042
total timesteps	1647694
train/	
learning_rate	0.0001
loss	0.0613
n_updates	399423

EP=8,410 | STEP=1,648,381 | clip=28 | consec=11 | max_consec=23 | avg_clip50=31.80
EP=8,420 | STEP=1,650,688 | clip=13 | consec=0 | max_consec=23 | avg_clip50=33.22
EP=8,430 | STEP=1,653,408 | clip=12 | consec=0 | max_consec=23 | avg_clip50=35.24
EP=8,440 | STEP=1,656,119 | clip=74 | consec=5 | max_consec=23 | avg_clip50=38.32

custom/	
avg_clip50	38.8
clipped_episode_r...	41
consecutive_clipp...	6
max_consecutive_c...	23
rollout/	
ep_len_mean	1.01e+03
ep_rew_mean	342
exploration rate	0.448
time/	
episodes	25000

fps	204
time_elapsed	8088
total timesteps	1656430
train/	
learning_rate	0.0001
loss	0.332
n_updates	401607

EP=8,450 | STEP=1,658,701 | clip=40 | consec=2 | max_consec=23 | avg_clip50=39.62
EP=8,460 | STEP=1,661,381 | clip=18 | consec=0 | max_consec=23 | avg_clip50=39.94
EP=8,470 | STEP=1,664,074 | clip=62 | consec=4 | max_consec=23 | avg_clip50=41.88

custom/	
avg_clip50	42.9
clipped_episode_r...	26.5
consecutive_clipp...	10
max_consecutive_c...	23
rollout/	
ep_len_mean	1.06e+03
ep_rew_mean	386
exploration rate	0.445
time/	
episodes	25100
fps	204
time_elapsed	8138
total timesteps	1665904
train/	
learning_rate	0.0001
loss	0.411
n_updates	403975

EP=8,480 | STEP=1,666,619 | clip=72 | consec=2 | max_consec=23 | avg_clip50=43.33
EP=8,490 | STEP=1,668,950 | clip=34 | consec=1 | max_consec=23 | avg_clip50=40.46
EP=8,500 | STEP=1,671,686 | clip=54 | consec=4 | max_consec=23 | avg_clip50=43.97
EP=8,510 | STEP=1,674,550 | clip=31 | consec=4 | max_consec=23 | avg_clip50=45.10

custom/	
avg_clip50	45.1
clipped_episode_r...	54.5
consecutive_clipp...	5
max_consecutive_c...	23
rollout/	
ep_len_mean	1.09e+03
ep_rew_mean	425
exploration rate	0.442
time/	
episodes	25200
fps	204
time_elapsed	8186
total timesteps	1675037
train/	
learning_rate	0.0001
loss	0.0442
n_updates	406259

EP=8,520 | STEP=1,676,905 | clip=39 | consec=14 | max_consec=23 | avg_clip50=44.18
EP=8,530 | STEP=1,679,324 | clip=51 | consec=6 | max_consec=23 | avg_clip50=43.37
EP=8,540 | STEP=1,682,157 | clip=24 | consec=3 | max_consec=23 | avg_clip50=44.81

custom/	
avg_clip50	42.7
clipped_episode_r...	41
consecutive_clipp...	8
max_consecutive_c...	23
rollout/	
ep_len_mean	1.09e+03
ep_rew_mean	428
exploration rate	0.439
time/	
episodes	25300
fps	204
time_elapsed	8232
total timesteps	1683705
train/	
learning_rate	0.0001
loss	0.172
n_updates	408426

EP=8,550	STEP=1,684,834	clip=36	consec=1	max_consec=23	avg_clip50=41.74
EP=8,560	STEP=1,687,688	clip=51	consec=2	max_consec=23	avg_clip50=41.68
EP=8,570	STEP=1,689,904	clip=32	consec=1	max_consec=23	avg_clip50=40.91
EP=8,580	STEP=1,692,491	clip=50	consec=3	max_consec=23	avg_clip50=40.07

custom/	
avg_clip50	40.1
clipped_episode_r...	49.5
consecutive_clipp...	3
max_consecutive_c...	23
rollout/	
ep_len_mean	1.08e+03
ep_rew_mean	417
exploration rate	0.436
time/	
episodes	25400
fps	204
time_elapsed	8278
total timesteps	1692491
train/	
learning_rate	0.0001
loss	0.0329
n_updates	410622

EP=8,590	STEP=1,695,116	clip=45	consec=6	max_consec=23	avg_clip50=39.73
EP=8,600	STEP=1,697,594	clip=79	consec=1	max_consec=23	avg_clip50=38.91
EP=8,610	STEP=1,700,228	clip=24	consec=11	max_consec=23	avg_clip50=37.8

5

custom/	
avg_clip50	37.2
clipped_episode_r...	17
consecutive_clipp...	0
max_consecutive_c...	23
rollout/	
ep_len_mean	1.06e+03
ep_rew_mean	392
exploration rate	0.434
time/	
episodes	25500
fps	204

	time_elapsed	8324	
	total timesteps	1701085	
	train/		
	learning_rate	0.0001	
	loss	0.122	
	n_updates	412771	

EP=8,620		STEP=1,702,496		clip=35		consec=4		max_consec=23		avg_clip50=37.80
EP=8,630		STEP=1,704,657		clip=14		consec=0		max_consec=23		avg_clip50=36.94
EP=8,640		STEP=1,707,852		clip=54		consec=10		max_consec=23		avg_clip50=39.1

1

	custom/		
	avg_clip50	38.5	
	clipped_episode_r...	27	
	consecutive_clipp...	1	
	max_consecutive_c...	23	
	rollout/		
	ep_len_mean	1.05e+03	
	ep_rew_mean	382	
	exploration rate	0.431	
	time/		
	episodes	25600	
	fps	204	
	time_elapsed	8371	
	total timesteps	1709668	
	train/		
	learning_rate	0.0001	
	loss	0.0606	
	n_updates	414916	

EP=8,650		STEP=1,709,932		clip=41		consec=3		max_consec=23		avg_clip50=38.18
EP=8,660		STEP=1,712,629		clip=36		consec=5		max_consec=23		avg_clip50=38.34
EP=8,670		STEP=1,715,126		clip=72		consec=2		max_consec=23		avg_clip50=39.67
EP=8,680		STEP=1,717,863		clip=50		consec=3		max_consec=23		avg_clip50=41.56

	custom/		
	avg_clip50	41.1	
	clipped_episode_r...	44	
	consecutive_clipp...	6	
	max_consecutive_c...	23	
	rollout/		
	ep_len_mean	1.06e+03	
	ep_rew_mean	394	
	exploration rate	0.428	
	time/		
	episodes	25700	
	fps	204	
	time_elapsed	8418	
	total timesteps	1718679	
	train/		
	learning_rate	0.0001	
	loss	0.0766	
	n_updates	417169	

EP=8,690		STEP=1,720,598		clip=60		consec=4		max_consec=23		avg_clip50=40.03
EP=8,700		STEP=1,723,245		clip=34		consec=14		max_consec=23		avg_clip50=41.5

1

EP=8,710		STEP=1,725,441		clip=13		consec=0		max_consec=23		avg_clip50=39.78
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custom/	
avg_clip50	39.7
clipped_episode_r...	40
consecutive_clipp...	2
max_consecutive_c...	23
rollout/	
ep_len_mean	1.07e+03
ep_rew_mean	398
exploration rate	0.425
time/	
episodes	25800
fps	204
time_elapsed	8463
total timesteps	1727304
train/	
learning_rate	0.0001
loss	0.0582
n_updates	419325

EP=8,720	STEP=1,728,118	clip=21	consec=5	max_consec=23	avg_clip50=39.82
EP=8,730	STEP=1,730,625	clip=45	consec=1	max_consec=23	avg_clip50=39.36
EP=8,740	STEP=1,733,102	clip=28	consec=8	max_consec=23	avg_clip50=38.41
EP=8,750	STEP=1,735,061	clip=32	consec=2	max_consec=23	avg_clip50=36.42

custom/	
avg_clip50	36.9
clipped_episode_r...	38.5
consecutive_clipp...	3
max_consecutive_c...	23
rollout/	
ep_len_mean	1.05e+03
ep_rew_mean	387
exploration rate	0.422
time/	
episodes	25900
fps	204
time_elapsed	8506
total timesteps	1735355
train/	
learning_rate	0.0001
loss	0.483
n_updates	421338

EP=8,760	STEP=1,737,569	clip=50	consec=1	max_consec=23	avg_clip50=36.64
EP=8,770	STEP=1,740,239	clip=56	consec=1	max_consec=23	avg_clip50=36.22
EP=8,780	STEP=1,742,491	clip=12	consec=0	max_consec=23	avg_clip50=34.58

custom/	
avg_clip50	34.4
clipped_episode_r...	59
consecutive_clipp...	5
max_consecutive_c...	23
rollout/	
ep_len_mean	1.03e+03
ep_rew_mean	370
exploration rate	0.419
time/	
episodes	26000
fps	203
time_elapsed	8551

total timesteps	1743928
train/	
learning_rate	0.0001
loss	0.396
n_updates	423481

EP=8,790	STEP=1,744,973	clip=28	consec=1	max_consec=23	avg_clip50=34.37
EP=8,800	STEP=1,747,134	clip=30	consec=1	max_consec=23	avg_clip50=35.54
EP=8,810	STEP=1,749,903	clip=50	consec=4	max_consec=23	avg_clip50=36.81

custom/	
avg_clip50	37.7
clipped_episode_r...	46
consecutive_clipp...	6
max_consecutive_c...	23
rollout/	
ep_len_mean	1.01e+03
ep_rew_mean	366
exploration rate	0.416
time/	
episodes	26100
fps	203
time_elapsed	8595
total timesteps	1752268
train/	
learning_rate	0.0001
loss	0.0924
n_updates	425566

EP=8,820	STEP=1,752,322	clip=67	consec=7	max_consec=23	avg_clip50=37.90
EP=8,830	STEP=1,754,515	clip=27	consec=7	max_consec=23	avg_clip50=38.17
EP=8,840	STEP=1,757,306	clip=68	consec=17	max_consec=23	avg_clip50=41.28
EP=8,850	STEP=1,759,822	clip=20	consec=4	max_consec=23	avg_clip50=42.00

custom/	
avg_clip50	41.7
clipped_episode_r...	16
consecutive_clipp...	0
max_consecutive_c...	23
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	390
exploration rate	0.414
time/	
episodes	26200
fps	203
time_elapsed	8640
total timesteps	1760757
train/	
learning_rate	0.0001
loss	0.0387
n_updates	427689

EP=8,860	STEP=1,762,277	clip=34	consec=1	max_consec=23	avg_clip50=41.72
EP=8,870	STEP=1,764,886	clip=74	consec=6	max_consec=23	avg_clip50=40.78
EP=8,880	STEP=1,767,270	clip=31	consec=2	max_consec=23	avg_clip50=42.70

custom/	
avg_clip50	41.8

	clipped_episode_r...	72.5	
	consecutive_clipp...	2	
	max_consecutive_c...	23	
	rollout/		
	ep_len_mean	1.04e+03	
	ep_rew_mean	408	
	exploration rate	0.411	
	time/		
	episodes	26300	
	fps	203	
	time_elapsed	8686	
	total timesteps	1769578	
	train/		
	learning_rate	0.0001	
	loss	0.07	
	n_updates	429894	

EP=8,890		STEP=1,769,956		clip=27		consec=4		max_consec=23		avg_clip50=40.70
EP=8,900		STEP=1,772,233		clip=36		consec=4		max_consec=23		avg_clip50=40.67
EP=8,910		STEP=1,774,927		clip=50		consec=7		max_consec=23		avg_clip50=42.32
EP=8,920		STEP=1,777,423		clip=28		consec=2		max_consec=23		avg_clip50=43.40

	custom/		
	avg_clip50	42.8	
	clipped_episode_r...	48	
	consecutive_clipp...	4	
	max_consecutive_c...	23	
	rollout/		
	ep_len_mean	1.05e+03	
	ep_rew_mean	422	
	exploration rate	0.408	
	time/		
	episodes	26400	
	fps	203	
	time_elapsed	8730	
	total timesteps	1777964	
	train/		
	learning_rate	0.0001	
	loss	0.052	
	n_updates	431990	

EP=8,930		STEP=1,779,885		clip=50		consec=7		max_consec=23		avg_clip50=43.03
EP=8,940		STEP=1,782,350		clip=63		consec=1		max_consec=23		avg_clip50=43.14
EP=8,950		STEP=1,785,048		clip=43		consec=11		max_consec=23		avg_clip50=44.5

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	custom/		
	avg_clip50	45.4	
	clipped_episode_r...	69	
	consecutive_clipp...	17	
	max_consecutive_c...	23	
	rollout/		
	ep_len_mean	1.06e+03	
	ep_rew_mean	429	
	exploration rate	0.405	
	time/		
	episodes	26500	
	fps	203	
	time_elapsed	8777	
	total timesteps	1786973	

train/	
learning_rate	0.0001
loss	0.297
n_updates	434243

EP=8,960	STEP=1,787,513	clip=14	consec=0	max_consec=23	avg_clip50=42.47
EP=8,970	STEP=1,789,786	clip=16	consec=0	max_consec=23	avg_clip50=39.73
EP=8,980	STEP=1,792,929	clip=35	consec=1	max_consec=23	avg_clip50=41.49

custom/	
avg_clip50	41.3
clipped_episode_r...	23.5
consecutive_clipp...	3
max_consecutive_c...	23
rollout/	
ep_len_mean	1.07e+03
ep_rew_mean	418
exploration rate	0.402
time/	
episodes	26600
fps	203
time_elapsed	8822
total timesteps	1795429
train/	
learning_rate	0.0001
loss	0.0566
n_updates	436357

EP=8,990	STEP=1,795,509	clip=25	consec=4	max_consec=23	avg_clip50=40.51
EP=9,000	STEP=1,797,972	clip=23	consec=3	max_consec=23	avg_clip50=39.72
EP=9,010	STEP=1,800,834	clip=10	consec=0	max_consec=23	avg_clip50=41.57
EP=9,020	STEP=1,803,856	clip=26	consec=10	max_consec=23	avg_clip50=45.7

6

custom/	
avg_clip50	44.1
clipped_episode_r...	54
consecutive_clipp...	3
max_consecutive_c...	23
rollout/	
ep_len_mean	1.11e+03
ep_rew_mean	432
exploration rate	0.399
time/	
episodes	26700
fps	203
time_elapsed	8872
total timesteps	1805044
train/	
learning_rate	0.0001
loss	0.0839
n_updates	438760

EP=9,030	STEP=1,806,802	clip=54	consec=9	max_consec=23	avg_clip50=44.76
EP=9,040	STEP=1,809,249	clip=49	consec=3	max_consec=23	avg_clip50=44.88
EP=9,050	STEP=1,811,909	clip=39	consec=1	max_consec=23	avg_clip50=44.78

custom/	
avg_clip50	41.4
clipped_episode_r...	22

consecutive_clipp...	3
max_consecutive_c...	23
rollout/	
ep_len_mean	1.11e+03
ep_rew_mean	415
exploration rate	0.396
time/	
episodes	26800
fps	203
time_elapsed	8919
total timesteps	1813892
train/	
learning_rate	0.0001
loss	0.0542
n_updates	440972

```

EP=9,060 | STEP=1,814,081 | clip=40 | consec=4 | max_consec=23 | avg_clip50=42.00
EP=9,070 | STEP=1,816,392 | clip=58 | consec=2 | max_consec=23 | avg_clip50=38.67
EP=9,080 | STEP=1,818,839 | clip=54 | consec=5 | max_consec=23 | avg_clip50=38.08
EP=9,090 | STEP=1,821,049 | clip=38 | consec=6 | max_consec=23 | avg_clip50=37.64

```

custom/	
avg_clip50	36.6
clipped_episode_r...	38.5
consecutive_clipp...	10
max_consecutive_c...	23
rollout/	
ep_len_mean	1.07e+03
ep_rew_mean	411
exploration rate	0.393
time/	
episodes	26900
fps	203
time_elapsed	8963
total timesteps	1822009
train/	
learning_rate	0.0001
loss	0.0652
n_updates	443002

```

EP=9,100 | STEP=1,823,561 | clip=26 | consec=16 | max_consec=23 | avg_clip50=36.4
3
EP=9,110 | STEP=1,826,203 | clip=44 | consec=5 | max_consec=23 | avg_clip50=38.58
EP=9,120 | STEP=1,828,720 | clip=38 | consec=1 | max_consec=23 | avg_clip50=39.62

```

custom/	
avg_clip50	39.3
clipped_episode_r...	60.5
consecutive_clipp...	1
max_consecutive_c...	23
rollout/	
ep_len_mean	1.02e+03
ep_rew_mean	387
exploration rate	0.39
time/	
episodes	27000
fps	203
time_elapsed	9010
total timesteps	1830850
train/	

	learning_rate	0.0001				
	loss	0.0618				
	n_updates	445212				
EP=9,130	STEP=1,831,110	clip=52	consec=2	max_consec=23	avg_clip50=39.26	
EP=9,140	STEP=1,833,661	clip=60	consec=7	max_consec=23	avg_clip50=39.50	
EP=9,150	STEP=1,836,207	clip=15	consec=0	max_consec=23	avg_clip50=39.84	
EP=9,160	STEP=1,839,064	clip=64	consec=1	max_consec=23	avg_clip50=41.01	
custom/						
	avg_clip50	40.9				
	clipped_episode_r...	45				
	consecutive_clipp...	4				
	max_consecutive_c...	23				
rollout/						
	ep_len_mean	1.04e+03				
	ep_rew_mean	396				
	exploration rate	0.387				
time/						
	episodes	27100				
	fps	203				
	time_elapsed	9060				
	total timesteps	1840189				
train/						
	learning_rate	0.0001				
	loss	0.0296				
	n_updates	447547				
EP=9,170	STEP=1,841,793	clip=28	consec=11	max_consec=23	avg_clip50=40.4	
7						
EP=9,180	STEP=1,844,578	clip=72	consec=21	max_consec=23	avg_clip50=41.6	
9						
EP=9,190	STEP=1,846,912	clip=34	consec=6	max_consec=23	avg_clip50=41.81	
custom/						
	avg_clip50	42.4				
	clipped_episode_r...	49				
	consecutive_clipp...	13				
	max_consecutive_c...	23				
rollout/						
	ep_len_mean	1.09e+03				
	ep_rew_mean	413				
	exploration rate	0.384				
time/						
	episodes	27200				
	fps	203				
	time_elapsed	9104				
	total timesteps	1848779				
train/						
	learning_rate	0.0001				
	loss	0.0482				
	n_updates	449694				
EP=9,200	STEP=1,849,660	clip=75	consec=16	max_consec=23	avg_clip50=43.6	
1						
EP=9,210	STEP=1,851,855	clip=44	consec=2	max_consec=23	avg_clip50=40.50	
EP=9,220	STEP=1,854,868	clip=48	consec=4	max_consec=23	avg_clip50=42.07	
EP=9,230	STEP=1,857,599	clip=46	consec=9	max_consec=23	avg_clip50=41.91	
custom/						

	avg_clip50	41.9	
	clipped_episode_r...	46	
	consecutive_clipp...	9	
	max_consecutive_c...	23	
	rollout/		
	ep_len_mean	1.11e+03	
	ep_rew_mean	418	
	exploration rate	0.381	
	time/		
	episodes	27300	
	fps	203	
	time_elapsed	9151	
	total timesteps	1857833	
	train/		
	learning_rate	0.0001	
	loss	0.0662	
	n_updates	451958	

EP=9,240		STEP=1,860,440		clip=72		consec=3		max_consec=23		avg_clip50=42.41
EP=9,250		STEP=1,862,950		clip=41		consec=9		max_consec=23		avg_clip50=43.15
EP=9,260		STEP=1,865,308		clip=26		consec=6		max_consec=23		avg_clip50=44.57

	custom/		
	avg_clip50	45.3	
	clipped_episode_r...	29.5	
	consecutive_clipp...	11	
	max_consecutive_c...	23	
	rollout/		
	ep_len_mean	1.09e+03	
	ep_rew_mean	432	
	exploration rate	0.378	
	time/		
	episodes	27400	
	fps	202	
	time_elapsed	9196	
	total timesteps	1866791	
	train/		
	learning_rate	0.0001	
	loss	0.0602	
	n_updates	454197	

EP=9,270		STEP=1,868,011		clip=46		consec=16		max_consec=23		avg_clip50=44.08
EP=9,280		STEP=1,870,681		clip=46		consec=26		max_consec=26		avg_clip50=43.57
EP=9,290		STEP=1,873,261		clip=45		consec=1		max_consec=26		avg_clip50=43.44
EP=9,300		STEP=1,875,934		clip=40		consec=11		max_consec=26		avg_clip50=43.16

	custom/		
	avg_clip50	43.2	
	clipped_episode_r...	66	
	consecutive_clipp...	13	
	max_consecutive_c...	26	
	rollout/		
	ep_len_mean	1.1e+03	
	ep_rew_mean	437	
	exploration rate	0.375	
	time/		
	episodes	27500	

fps	202
time_elapsed	9248
total timesteps	1876646
train/	
learning_rate	0.0001
loss	0.151
n_updates	456661

EP=9,310 | STEP=1,878,377 | clip=56 | consec=1 | max_consec=26 | avg_clip50=42.92
EP=9,320 | STEP=1,880,977 | clip=32 | consec=2 | max_consec=26 | avg_clip50=43.83
EP=9,330 | STEP=1,883,539 | clip=86 | consec=7 | max_consec=26 | avg_clip50=44.47

custom/	
avg_clip50	44.6
clipped_episode_r...	15.5
consecutive_clipp...	0
max_consecutive_c...	26
rollout/	
ep_len_mean	1.09e+03
ep_rew_mean	443
exploration rate	0.372
time/	
episodes	27600
fps	202
time_elapsed	9293
total timesteps	1885180
train/	
learning_rate	0.0001
loss	0.0798
n_updates	458794

EP=9,340 | STEP=1,885,987 | clip=41 | consec=2 | max_consec=26 | avg_clip50=43.80
EP=9,350 | STEP=1,888,497 | clip=7 | consec=0 | max_consec=26 | avg_clip50=41.22
EP=9,360 | STEP=1,891,481 | clip=20 | consec=0 | max_consec=26 | avg_clip50=43.46
EP=9,370 | STEP=1,894,543 | clip=52 | consec=10 | max_consec=26 | avg_clip50=45.07

custom/	
avg_clip50	44.6
clipped_episode_r...	60
consecutive_clipp...	12
max_consecutive_c...	26
rollout/	
ep_len_mean	1.11e+03
ep_rew_mean	450
exploration rate	0.369
time/	
episodes	27700
fps	202
time_elapsed	9344
total timesteps	1895122
train/	
learning_rate	0.0001
loss	0.0645
n_updates	461280

EP=9,380 | STEP=1,897,585 | clip=92 | consec=1 | max_consec=26 | avg_clip50=46.60
EP=9,390 | STEP=1,900,167 | clip=46 | consec=4 | max_consec=26 | avg_clip50=47.68
EP=9,400 | STEP=1,902,637 | clip=54 | consec=4 | max_consec=26 | avg_clip50=50.60

custom/	
avg_clip50	47.2
clipped_episode_r...	38.5
consecutive_clipp...	3
max_consecutive_c...	26
rollout/	
ep_len_mean	1.11e+03
ep_rew_mean	455
exploration rate	0.366
time/	
episodes	27800
fps	202
time_elapsed	9396
total timesteps	1904866
train/	
learning_rate	0.0001
loss	0.0706
n_updates	463716

EP=9,410	STEP=1,905,289	clip=68	consec=5	max_consec=26	avg_clip50=48.58
EP=9,420	STEP=1,907,930	clip=73	consec=2	max_consec=26	avg_clip50=46.14
EP=9,430	STEP=1,910,603	clip=60	consec=12	max_consec=26	avg_clip50=44.4
4					
EP=9,440	STEP=1,913,090	clip=52	consec=9	max_consec=26	avg_clip50=43.68

custom/	
avg_clip50	42.7
clipped_episode_r...	49.5
consecutive_clipp...	12
max_consecutive_c...	26
rollout/	
ep_len_mean	1.13e+03
ep_rew_mean	458
exploration rate	0.363
time/	
episodes	27900
fps	202
time_elapsed	9444
total timesteps	1914009
train/	
learning_rate	0.0001
loss	0.0536
n_updates	466002

EP=9,450	STEP=1,915,883	clip=60	consec=1	max_consec=26	avg_clip50=42.50
EP=9,460	STEP=1,918,743	clip=24	consec=8	max_consec=26	avg_clip50=43.66
EP=9,470	STEP=1,921,430	clip=80	consec=5	max_consec=26	avg_clip50=43.35

custom/	
avg_clip50	41.1
clipped_episode_r...	54
consecutive_clipp...	2
max_consecutive_c...	26
rollout/	
ep_len_mean	1.1e+03
ep_rew_mean	433
exploration rate	0.36
time/	
episodes	28000
fps	202

	time_elapsed		9492	
	total timesteps		1923311	
	train/			
	learning_rate		0.0001	
	loss		0.404	
	n_updates		468327	

EP=9,480 | STEP=1,923,917 | clip=41 | consec=5 | max_consec=26 | avg_clip50=41.92
EP=9,490 | STEP=1,926,960 | clip=50 | consec=15 | max_consec=26 | avg_clip50=44.1
0
EP=9,500 | STEP=1,929,699 | clip=60 | consec=9 | max_consec=26 | avg_clip50=45.83
EP=9,510 | STEP=1,932,365 | clip=72 | consec=19 | max_consec=26 | avg_clip50=45.7
7

	custom/			
	avg_clip50		46.2	
	clipped_episode_r...		40.5	
	consecutive_clipp...		22	
	max_consecutive_c...		26	
	rollout/			
	ep_len_mean		1.13e+03	
	ep_rew_mean		450	
	exploration rate		0.356	
	time/			
	episodes		28100	
	fps		202	
	time_elapsed		9547	
	total timesteps		1933423	
	train/			
	learning_rate		0.0001	
	loss		0.0565	
	n_updates		470855	

EP=9,520 | STEP=1,934,807 | clip=14 | consec=0 | max_consec=27 | avg_clip50=45.64
EP=9,530 | STEP=1,937,566 | clip=58 | consec=10 | max_consec=27 | avg_clip50=47.2
5
EP=9,540 | STEP=1,940,223 | clip=50 | consec=2 | max_consec=27 | avg_clip50=45.44

	custom/			
	avg_clip50		46.3	
	clipped_episode_r...		52	
	consecutive_clipp...		10	
	max_consecutive_c...		27	
	rollout/			
	ep_len_mean		1.13e+03	
	ep_rew_mean		456	
	exploration rate		0.353	
	time/			
	episodes		28200	
	fps		202	
	time_elapsed		9596	
	total timesteps		1942731	
	train/			
	learning_rate		0.0001	
	loss		0.0428	
	n_updates		473182	

EP=9,550 | STEP=1,943,285 | clip=32 | consec=12 | max_consec=27 | avg_clip50=45.5
8
EP=9,560 | STEP=1,946,148 | clip=38 | consec=9 | max_consec=27 | avg_clip50=46.15

EP=9,570 | STEP=1,949,013 | clip=52 | consec=19 | max_consec=27 | avg_clip50=47.14

EP=9,580 | STEP=1,952,179 | clip=51 | consec=29 | max_consec=29 | avg_clip50=48.93

custom/		
avg_clip50	51	
clipped_episode_r...	92.5	
consecutive_clipp...	32	
max_consecutive_c...	32	
rollout/		
ep_len_mean	1.18e+03	
ep_rew_mean	485	
exploration rate	0.35	
time/		
episodes	28300	
fps	202	
time_elapsed	9652	
total timesteps	1953396	
train/		
learning_rate	0.0001	
loss	0.0557	
n_updates	475848	

EP=9,590 | STEP=1,955,041 | clip=43 | consec=1 | max_consec=36 | avg_clip50=50.59

EP=9,600 | STEP=1,957,826 | clip=49 | consec=11 | max_consec=36 | avg_clip50=49.57

EP=9,610 | STEP=1,960,548 | clip=44 | consec=8 | max_consec=36 | avg_clip50=48.29

custom/		
avg_clip50	48.9	
clipped_episode_r...	39.5	
consecutive_clipp...	17	
max_consecutive_c...	36	
rollout/		
ep_len_mean	1.18e+03	
ep_rew_mean	477	
exploration rate	0.346	
time/		
episodes	28400	
fps	202	
time_elapsed	9705	
total timesteps	1963164	
train/		
learning_rate	0.0001	
loss	0.027	
n_updates	478290	

EP=9,620 | STEP=1,963,316 | clip=28 | consec=18 | max_consec=36 | avg_clip50=48.45

EP=9,630 | STEP=1,965,798 | clip=60 | consec=7 | max_consec=36 | avg_clip50=45.67

EP=9,640 | STEP=1,968,563 | clip=28 | consec=17 | max_consec=36 | avg_clip50=44.37

EP=9,650 | STEP=1,971,178 | clip=34 | consec=27 | max_consec=36 | avg_clip50=43.33

custom/		
avg_clip50	43.9	
clipped_episode_r...	28	
consecutive_clipp...	30	

max_consecutive_c...	36
rollout/	
ep_len_mean	1.16e+03
ep_rew_mean	468
exploration rate	0.343
time/	
episodes	28500
fps	202
time_elapsed	9753
total timesteps	1972168
train/	
learning_rate	0.0001
loss	0.113
n_updates	480541

EP=9,660 | STEP=1,973,780 | clip=18 | consec=0 | max_consec=36 | avg_clip50=42.77
EP=9,670 | STEP=1,976,858 | clip=110 | consec=10 | max_consec=36 | avg_clip50=44.49
EP=9,680 | STEP=1,979,348 | clip=60 | consec=20 | max_consec=36 | avg_clip50=44.45

custom/	
avg_clip50	44
clipped_episode_r...	54
consecutive_clipp...	6
max_consecutive_c...	36
rollout/	
ep_len_mean	1.1e+03
ep_rew_mean	440
exploration rate	0.34
time/	
episodes	28600
fps	202
time_elapsed	9801
total timesteps	1981282
train/	
learning_rate	0.0001
loss	0.0855
n_updates	482820

EP=9,690 | STEP=1,981,744 | clip=22 | consec=8 | max_consec=36 | avg_clip50=43.46
EP=9,700 | STEP=1,984,683 | clip=14 | consec=0 | max_consec=36 | avg_clip50=44.89
EP=9,710 | STEP=1,987,389 | clip=60 | consec=3 | max_consec=36 | avg_clip50=46.37
EP=9,720 | STEP=1,990,208 | clip=64 | consec=6 | max_consec=36 | avg_clip50=44.46

custom/	
avg_clip50	45.5
clipped_episode_r...	57
consecutive_clipp...	9
max_consecutive_c...	36
rollout/	
ep_len_mean	1.13e+03
ep_rew_mean	450
exploration rate	0.337
time/	
episodes	28700
fps	202
time_elapsed	9854
total timesteps	1991136
train/	

learning_rate	0.0001
loss	0.381
n_updates	485283

EP=9,730	STEP=1,992,999	clip=75	consec=5	max_consec=36	avg_clip50=45.47
EP=9,740	STEP=1,995,709	clip=37	consec=1	max_consec=36	avg_clip50=47.82
EP=9,750	STEP=1,998,626	clip=80	consec=11	max_consec=36	avg_clip50=47.01

custom/	
avg_clip50	47
clipped_episode_r...	54.5
consecutive_clipp...	18
max_consecutive_c...	36
rollout/	
ep_len_mean	1.15e+03
ep_rew_mean	464
exploration rate	0.334
time/	
episodes	28800
fps	201
time_elapsed	9907
total timesteps	2000863
train/	
learning_rate	0.0001
loss	0.0382
n_updates	487715

EP=9,760	STEP=2,001,292	clip=41	consec=21	max_consec=36	avg_clip50=46.42
EP=9,770	STEP=2,003,920	clip=50	consec=31	max_consec=36	avg_clip50=45.85
EP=9,780	STEP=2,006,711	clip=54	consec=41	max_consec=41	avg_clip50=46.29
EP=9,790	STEP=2,009,435	clip=62	consec=51	max_consec=51	avg_clip50=45.91

custom/	
avg_clip50	44.7
clipped_episode_r...	36.5
consecutive_clipp...	54
max_consecutive_c...	54
rollout/	
ep_len_mean	1.14e+03
ep_rew_mean	461
exploration rate	0.331
time/	
episodes	28900
fps	201
time_elapsed	9956
total timesteps	2009951
train/	
learning_rate	0.0001
loss	0.316
n_updates	489987

EP=9,800	STEP=2,011,676	clip=60	consec=61	max_consec=61	avg_clip50=43.96
EP=9,810	STEP=2,014,201	clip=34	consec=3	max_consec=64	avg_clip50=43.84
EP=9,820	STEP=2,016,877	clip=55	consec=7	max_consec=64	avg_clip50=44.21

custom/	
avg_clip50	43.6
clipped_episode_r...	22
consecutive_clipp...	16
max_consecutive_c...	64
rollout/	
ep_len_mean	1.1e+03
ep_rew_mean	452
exploration rate	0.328
time/	
episodes	29000
fps	201
time_elapsed	10005
total timesteps	2019132
train/	
learning_rate	0.0001
loss	0.0549
n_updates	492282

EP=9,830 | STEP=2,019,507 | clip=58 | consec=17 | max_consec=64 | avg_clip50=43.71
EP=9,840 | STEP=2,022,125 | clip=63 | consec=6 | max_consec=64 | avg_clip50=42.57
EP=9,850 | STEP=2,025,022 | clip=72 | consec=16 | max_consec=64 | avg_clip50=45.80
EP=9,860 | STEP=2,027,390 | clip=36 | consec=9 | max_consec=64 | avg_clip50=44.83

custom/	
avg_clip50	45.5
clipped_episode_r...	13.5
consecutive_clipp...	0
max_consecutive_c...	64
rollout/	
ep_len_mean	1.09e+03
ep_rew_mean	444
exploration rate	0.325
time/	
episodes	29100
fps	201
time_elapsed	10054
total timesteps	2028194
train/	
learning_rate	0.0001
loss	0.0772
n_updates	494548

EP=9,870 | STEP=2,030,255 | clip=36 | consec=8 | max_consec=64 | avg_clip50=45.74
EP=9,880 | STEP=2,033,046 | clip=55 | consec=9 | max_consec=64 | avg_clip50=45.99
EP=9,890 | STEP=2,035,726 | clip=48 | consec=5 | max_consec=64 | avg_clip50=46.50

custom/	
avg_clip50	46.5
clipped_episode_r...	55
consecutive_clipp...	12
max_consecutive_c...	64
rollout/	
ep_len_mean	1.13e+03
ep_rew_mean	460
exploration rate	0.321
time/	

episodes	29200
fps	201
time_elapsed	10105
total timesteps	2037851
train/	
learning_rate	0.0001
loss	0.564
n_updates	496962

EP=9,900 | STEP=2,038,575 | clip=40 | consec=15 | max_consec=64 | avg_clip50=46.01

EP=9,910 | STEP=2,041,500 | clip=32 | consec=2 | max_consec=64 | avg_clip50=48.19

EP=9,920 | STEP=2,044,136 | clip=63 | consec=1 | max_consec=64 | avg_clip50=47.04

EP=9,930 | STEP=2,047,006 | clip=60 | consec=11 | max_consec=64 | avg_clip50=47.85

custom/	
avg_clip50	47.6
clipped_episode_r...	42
consecutive_clipp...	12
max_consecutive_c...	64
rollout/	
ep_len_mean	1.15e+03
ep_rew_mean	471
exploration rate	0.318
time/	
episodes	29300
fps	201
time_elapsed	10154
total timesteps	2047222
train/	
learning_rate	0.0001
loss	0.0634
n_updates	499305

EP=9,940 | STEP=2,049,471 | clip=54 | consec=2 | max_consec=64 | avg_clip50=47.42

EP=9,950 | STEP=2,052,339 | clip=58 | consec=12 | max_consec=64 | avg_clip50=47.60

EP=9,960 | STEP=2,055,179 | clip=57 | consec=6 | max_consec=64 | avg_clip50=47.07

custom/	
avg_clip50	47.7
clipped_episode_r...	57
consecutive_clipp...	11
max_consecutive_c...	64
rollout/	
ep_len_mean	1.15e+03
ep_rew_mean	478
exploration rate	0.315
time/	
episodes	29400
fps	201
time_elapsed	10204
total timesteps	2056449
train/	
learning_rate	0.0001
loss	0.103
n_updates	501612

EP=9,970 | STEP=2,058,033 | clip=54 | consec=16 | max_consec=64 | avg_clip50=48.3

6

EP=9,980 | STEP=2,060,553 | clip=72 | consec=3 | max_consec=64 | avg_clip50=46.71
EP=9,990 | STEP=2,063,109 | clip=49 | consec=5 | max_consec=64 | avg_clip50=46.51
EP=10,000 | STEP=2,065,649 | clip=56 | consec=8 | max_consec=64 | avg_clip50=45.3

6

custom/		
avg_clip50	45.4	
clipped_episode_r...	55.5	
consecutive_clipp...	8	
max_consecutive_c...	64	
rollout/		
ep_len_mean	1.13e+03	
ep_rew_mean	465	
exploration rate	0.312	
time/		
episodes	29500	
fps	201	
time_elapsed	10252	
total timesteps	2065649	
train/		
learning_rate	0.0001	
loss	0.213	
n_updates	503912	

EP=10,010 | STEP=2,068,298 | clip=30 | consec=6 | max_consec=64 | avg_clip50=45.4

3

EP=10,020 | STEP=2,071,275 | clip=25 | consec=4 | max_consec=64 | avg_clip50=45.2

1

EP=10,030 | STEP=2,073,971 | clip=36 | consec=14 | max_consec=64 | avg_clip50=46.

83

custom/		
avg_clip50	48.5	
clipped_episode_r...	40	
consecutive_clipp...	18	
max_consecutive_c...	64	
rollout/		
ep_len_mean	1.13e+03	
ep_rew_mean	477	
exploration rate	0.309	
time/		
episodes	29600	
fps	201	
time_elapsed	10303	
total timesteps	2075258	
train/		
learning_rate	0.0001	
loss	0.0741	
n_updates	506314	

EP=10,040 | STEP=2,076,560 | clip=48 | consec=24 | max_consec=64 | avg_clip50=48.

58

EP=10,050 | STEP=2,079,089 | clip=39 | consec=7 | max_consec=64 | avg_clip50=49.2

7

EP=10,060 | STEP=2,082,166 | clip=46 | consec=4 | max_consec=64 | avg_clip50=50.0

1

custom/		
avg_clip50	48.8	

	clipped_episode_r...	52	
	consecutive_clipp...	5	
	max_consecutive_c...	64	
	rollout/		
	ep_len_mean	1.12e+03	
	ep_rew_mean	473	
	exploration rate	0.306	
	time/		
	episodes	29700	
	fps	201	
	time_elapsed	10353	
	total timesteps	2084733	
	train/		
	learning_rate	0.0001	
	loss	0.038	
	n_updates	508683	

EP=10,070 | STEP=2,085,015 | clip=50 | consec=6 | max_consec=64 | avg_clip50=49.34

EP=10,080 | STEP=2,087,706 | clip=46 | consec=16 | max_consec=64 | avg_clip50=49.00

EP=10,090 | STEP=2,090,519 | clip=55 | consec=26 | max_consec=64 | avg_clip50=48.12

EP=10,100 | STEP=2,093,557 | clip=47 | consec=36 | max_consec=64 | avg_clip50=48.67

	custom/		
	avg_clip50	47.5	
	clipped_episode_r...	53	
	consecutive_clipp...	42	
	max_consecutive_c...	64	
	rollout/		
	ep_len_mean	1.15e+03	
	ep_rew_mean	486	
	exploration rate	0.302	
	time/		
	episodes	29800	
	fps	201	
	time_elapsed	10408	
	total timesteps	2094947	
	train/		
	learning_rate	0.0001	
	loss	0.0416	
	n_updates	511236	

EP=10,110 | STEP=2,096,145 | clip=33 | consec=46 | max_consec=64 | avg_clip50=47.85

EP=10,120 | STEP=2,098,924 | clip=48 | consec=7 | max_consec=64 | avg_clip50=47.58

EP=10,130 | STEP=2,101,322 | clip=66 | consec=2 | max_consec=64 | avg_clip50=46.24

EP=10,140 | STEP=2,104,216 | clip=52 | consec=4 | max_consec=64 | avg_clip50=46.92

	custom/		
	avg_clip50	46.9	
	clipped_episode_r...	52.5	
	consecutive_clipp...	4	
	max_consecutive_c...	64	
	rollout/		

	ep_len_mean	1.15e+03	
	ep_rew_mean	475	
	exploration rate	0.299	
	time/		
	episodes	29900	
	fps	201	
	time_elapsed	10459	
	total timesteps	2104359	
	train/		
	learning_rate	0.0001	
	loss	0.443	
	n_updates	513589	

EP=10,150 | STEP=2,107,151 | clip=74 | consec=14 | max_consec=64 | avg_clip50=46.94

EP=10,160 | STEP=2,110,029 | clip=58 | consec=9 | max_consec=64 | avg_clip50=47.23

EP=10,170 | STEP=2,112,514 | clip=34 | consec=2 | max_consec=64 | avg_clip50=46.06

	custom/		
	avg_clip50	47.2	
	clipped_episode_r...	77.5	
	consecutive_clipp...	7	
	max_consecutive_c...	64	
	rollout/		
	ep_len_mean	1.14e+03	
	ep_rew_mean	468	
	exploration rate	0.296	
	time/		
	episodes	30000	
	fps	201	
	time_elapsed	10512	
	total timesteps	2113892	
	train/		
	learning_rate	0.0001	
	loss	0.112	
	n_updates	515972	

EP=10,180 | STEP=2,115,385 | clip=32 | consec=12 | max_consec=64 | avg_clip50=49.88

EP=10,190 | STEP=2,118,155 | clip=31 | consec=22 | max_consec=64 | avg_clip50=47.95

EP=10,200 | STEP=2,120,476 | clip=50 | consec=4 | max_consec=64 | avg_clip50=45.30

EP=10,210 | STEP=2,123,467 | clip=36 | consec=14 | max_consec=64 | avg_clip50=46.80

	custom/		
	avg_clip50	47.2	
	clipped_episode_r...	57.5	
	consecutive_clipp...	15	
	max_consecutive_c...	64	
	rollout/		
	ep_len_mean	1.14e+03	
	ep_rew_mean	471	
	exploration rate	0.293	
	time/		
	episodes	30100	
	fps	201	

	time_elapsed	10564	
	total timesteps	2123788	
	train/		
	learning_rate	0.0001	
	loss	0.298	
	n_updates	518446	

EP=10,220 | STEP=2,125,997 | clip=21 | consec=4 | max_consec=64 | avg_clip50=48.02

EP=10,230 | STEP=2,128,902 | clip=54 | consec=14 | max_consec=64 | avg_clip50=46.56

EP=10,240 | STEP=2,131,445 | clip=50 | consec=24 | max_consec=64 | avg_clip50=47.31

	custom/		
	avg_clip50	48.5	
	clipped_episode_r...	37	
	consecutive_clipp...	29	
	max_consecutive_c...	64	
	rollout/		
	ep_len_mean	1.13e+03	
	ep_rew_mean	472	
	exploration rate	0.29	
	time/		
	episodes	30200	
	fps	200	
	time_elapsed	10613	
	total timesteps	2132719	
	train/		
	learning_rate	0.0001	
	loss	0.239	
	n_updates	520679	

EP=10,250 | STEP=2,134,161 | clip=52 | consec=34 | max_consec=64 | avg_clip50=49.20

EP=10,260 | STEP=2,136,942 | clip=62 | consec=6 | max_consec=64 | avg_clip50=48.59

EP=10,270 | STEP=2,139,442 | clip=2 | consec=0 | max_consec=64 | avg_clip50=48.72

	custom/		
	avg_clip50	47.5	
	clipped_episode_r...	54	
	consecutive_clipp...	4	
	max_consecutive_c...	64	
	rollout/		
	ep_len_mean	1.12e+03	
	ep_rew_mean	468	
	exploration rate	0.287	
	time/		
	episodes	30300	
	fps	200	
	time_elapsed	10662	
	total timesteps	2142270	
	train/		
	learning_rate	0.0001	
	loss	0.173	
	n_updates	523067	

EP=10,280 | STEP=2,142,320 | clip=72 | consec=5 | max_consec=64 | avg_clip50=47.82

EP=10,290 | STEP=2,144,787 | clip=48 | consec=6 | max_consec=64 | avg_clip50=47.6
9
EP=10,300 | STEP=2,147,003 | clip=44 | consec=4 | max_consec=64 | avg_clip50=46.4
7
EP=10,310 | STEP=2,149,543 | clip=41 | consec=14 | max_consec=64 | avg_clip50=44.
70

custom/		
avg_clip50	44.2	
clipped_episode_r...	47.5	
consecutive_clipp...	2	
max_consecutive_c...	64	
rollout/		
ep_len_mean	1.08e+03	
ep_rew_mean	463	
exploration rate	0.284	
time/		
episodes	30400	
fps	200	
time_elapsed	10708	
total timesteps	2150380	
train/		
learning_rate	0.0001	
loss	0.0407	
n_updates	525094	

EP=10,320 | STEP=2,151,862 | clip=8 | consec=0 | max_consec=64 | avg_clip50=44.74
EP=10,330 | STEP=2,154,866 | clip=54 | consec=10 | max_consec=64 | avg_clip50=45.
15
EP=10,340 | STEP=2,157,837 | clip=80 | consec=20 | max_consec=64 | avg_clip50=48.
38

custom/		
avg_clip50	50.6	
clipped_episode_r...	60	
consecutive_clipp...	28	
max_consecutive_c...	64	
rollout/		
ep_len_mean	1.11e+03	
ep_rew_mean	488	
exploration rate	0.281	
time/		
episodes	30500	
fps	200	
time_elapsed	10762	
total timesteps	2160528	
train/		
learning_rate	0.0001	
loss	0.26	
n_updates	527631	

EP=10,350 | STEP=2,160,909 | clip=50 | consec=30 | max_consec=64 | avg_clip50=51.
02
EP=10,360 | STEP=2,163,813 | clip=30 | consec=40 | max_consec=64 | avg_clip50=52.
34
EP=10,370 | STEP=2,166,635 | clip=57 | consec=7 | max_consec=64 | avg_clip50=53.1
7
EP=10,380 | STEP=2,169,310 | clip=49 | consec=17 | max_consec=64 | avg_clip50=52.
33

custom/	
avg_clip50	52.1
clipped_episode_r...	42.5
consecutive_clipp...	20
max_consecutive_c...	64
rollout/	
ep_len_mean	1.13e+03
ep_rew_mean	491
exploration rate	0.277
time/	
episodes	30600
fps	200
time_elapsed	10812
total timesteps	2170186
train/	
learning_rate	0.0001
loss	0.0413
n_updates	530046

EP=10,390 | STEP=2,172,097 | clip=60 | consec=6 | max_consec=64 | avg_clip50=49.66

EP=10,400 | STEP=2,174,521 | clip=26 | consec=16 | max_consec=64 | avg_clip50=48.51

EP=10,410 | STEP=2,177,174 | clip=58 | consec=5 | max_consec=64 | avg_clip50=48.40

custom/	
avg_clip50	47.6
clipped_episode_r...	28.5
consecutive_clipp...	13
max_consecutive_c...	64
rollout/	
ep_len_mean	1.16e+03
ep_rew_mean	496
exploration rate	0.274
time/	
episodes	30700
fps	200
time_elapsed	10862
total timesteps	2179487
train/	
learning_rate	0.0001
loss	0.0423
n_updates	532371

EP=10,420 | STEP=2,180,106 | clip=57 | consec=15 | max_consec=64 | avg_clip50=47.93

EP=10,430 | STEP=2,183,457 | clip=44 | consec=1 | max_consec=64 | avg_clip50=49.23

EP=10,440 | STEP=2,185,915 | clip=36 | consec=9 | max_consec=64 | avg_clip50=49.03

EP=10,450 | STEP=2,189,146 | clip=48 | consec=19 | max_consec=64 | avg_clip50=49.45

custom/	
avg_clip50	49.2
clipped_episode_r...	12.5
consecutive_clipp...	0
max_consecutive_c...	64
rollout/	

	ep_len_mean	1.17e+03	
	ep_rew_mean	489	
	exploration rate	0.271	
	time/		
	episodes	30800	
	fps	200	
	time_elapsed	10920	
	total timesteps	2190261	
	train/		
	learning_rate	0.0001	
	loss	0.424	
	n_updates	535065	

EP=10,460 | STEP=2,191,847 | clip=45 | consec=7 | max_consec=64 | avg_clip50=49.7
1

EP=10,470 | STEP=2,194,202 | clip=17 | consec=0 | max_consec=64 | avg_clip50=48.4
5

EP=10,480 | STEP=2,196,957 | clip=35 | consec=10 | max_consec=64 | avg_clip50=47.
97

	custom/		
	avg_clip50	49.8	
	clipped_episode_r...	65.5	
	consecutive_clipp...	19	
	max_consecutive_c...	64	
	rollout/		
	ep_len_mean	1.16e+03	
	ep_rew_mean	496	
	exploration rate	0.268	
	time/		
	episodes	30900	
	fps	200	
	time_elapsed	10970	
	total timesteps	2199574	
	train/		
	learning_rate	0.0001	
	loss	0.175	
	n_updates	537393	

EP=10,490 | STEP=2,199,774 | clip=68 | consec=20 | max_consec=64 | avg_clip50=50.
38

EP=10,500 | STEP=2,202,587 | clip=57 | consec=1 | max_consec=64 | avg_clip50=50.7
6

EP=10,510 | STEP=2,205,520 | clip=47 | consec=11 | max_consec=64 | avg_clip50=51.
49

EP=10,520 | STEP=2,207,803 | clip=33 | consec=4 | max_consec=64 | avg_clip50=51.3
7

	custom/		
	avg_clip50	49.8	
	clipped_episode_r...	77	
	consecutive_clipp...	9	
	max_consecutive_c...	64	
	rollout/		
	ep_len_mean	1.12e+03	
	ep_rew_mean	486	
	exploration rate	0.264	
	time/		
	episodes	31000	
	fps	200	

	time_elapsed	11021	
	total timesteps	2209032	
	train/		
	learning_rate	0.0001	
	loss	0.0949	
	n_updates	539757	

EP=10,530 | STEP=2,210,513 | clip=50 | consec=14 | max_consec=64 | avg_clip50=50.65

EP=10,540 | STEP=2,213,021 | clip=47 | consec=6 | max_consec=64 | avg_clip50=48.10

EP=10,550 | STEP=2,215,714 | clip=30 | consec=2 | max_consec=64 | avg_clip50=46.30

EP=10,560 | STEP=2,218,503 | clip=34 | consec=3 | max_consec=64 | avg_clip50=44.25

	custom/		
	avg_clip50	44.4	
	clipped_episode_r...	44	
	consecutive_clipp...	4	
	max_consecutive_c...	64	
	rollout/		
	ep_len_mean	1.11e+03	
	ep_rew_mean	480	
	exploration rate	0.261	
	time/		
	episodes	31100	
	fps	200	
	time_elapsed	11074	
	total timesteps	2218928	
	train/		
	learning_rate	0.0001	
	loss	0.0342	
	n_updates	542231	

EP=10,570 | STEP=2,221,258 | clip=48 | consec=5 | max_consec=64 | avg_clip50=45.44

EP=10,580 | STEP=2,223,880 | clip=46 | consec=5 | max_consec=64 | avg_clip50=45.37

EP=10,590 | STEP=2,226,976 | clip=40 | consec=15 | max_consec=64 | avg_clip50=46.45

	custom/		
	avg_clip50	44.8	
	clipped_episode_r...	34	
	consecutive_clipp...	21	
	max_consecutive_c...	64	
	rollout/		
	ep_len_mean	1.12e+03	
	ep_rew_mean	461	
	exploration rate	0.258	
	time/		
	episodes	31200	
	fps	200	
	time_elapsed	11127	
	total timesteps	2228847	
	train/		
	learning_rate	0.0001	
	loss	0.0795	
	n_updates	544711	

```

-----
EP=10,600 | STEP=2,229,975 | clip=56 | consec=25 | max_consec=64 | avg_clip50=46.
32
EP=10,610 | STEP=2,232,524 | clip=86 | consec=35 | max_consec=64 | avg_clip50=47.
69
EP=10,620 | STEP=2,235,389 | clip=33 | consec=45 | max_consec=64 | avg_clip50=47.
83
EP=10,630 | STEP=2,238,404 | clip=36 | consec=55 | max_consec=64 | avg_clip50=48.
25

```

```

-----
| custom/                |          |
|   avg_clip50            |    48.5  |
|   clipped_episode_r...  |    29    |
|   consecutive_clipp...  |    57    |
|   max_consecutive_c...  |    64    |
| rollout/                |          |
|   ep_len_mean           |  1.16e+03 |
|   ep_rew_mean           |    467    |
|   exploration rate      |    0.254  |
| time/                   |          |
|   episodes              |   31300   |
|   fps                   |    200    |
|   time_elapsed          |   11182   |
|   total timesteps       |  2239022  |
| train/                  |          |
|   learning_rate         |   0.0001  |
|   loss                  |   0.0549  |
|   n_updates             |   547255  |

```

```

-----
EP=10,640 | STEP=2,241,193 | clip=55 | consec=65 | max_consec=65 | avg_clip50=48.
75
EP=10,650 | STEP=2,244,069 | clip=25 | consec=7  | max_consec=67 | avg_clip50=49.3
8
EP=10,660 | STEP=2,246,647 | clip=52 | consec=8  | max_consec=67 | avg_clip50=48.2
7

```

```

-----
| custom/                |          |
|   avg_clip50            |    50.1  |
|   clipped_episode_r...  |   103    |
|   consecutive_clipp...  |     2    |
|   max_consecutive_c...  |    67    |
| rollout/                |          |
|   ep_len_mean           |  1.17e+03 |
|   ep_rew_mean           |    487    |
|   exploration rate      |    0.251  |
| time/                   |          |
|   episodes              |   31400   |
|   fps                   |    200    |
|   time_elapsed          |   11231   |
|   total timesteps       |  2248511  |
| train/                  |          |
|   learning_rate         |   0.0001  |
|   loss                  |   0.0872  |
|   n_updates             |   549627  |

```

```

-----
EP=10,670 | STEP=2,249,372 | clip=58 | consec=6  | max_consec=67 | avg_clip50=49.0
1
EP=10,680 | STEP=2,252,395 | clip=60 | consec=6  | max_consec=67 | avg_clip50=49.8
2
EP=10,690 | STEP=2,255,444 | clip=30 | consec=16 | max_consec=67 | avg_clip50=50.

```


12

EP=10,700 | STEP=2,258,422 | clip=54 | consec=7 | max_consec=67 | avg_clip50=51.27

custom/		
avg_clip50	51.7	
clipped_episode_r...	60	
consecutive_clipp...	9	
max_consecutive_c...	67	
rollout/		
ep_len_mean	1.18e+03	
ep_rew_mean	502	
exploration rate	0.248	
time/		
episodes	31500	
fps	200	
time_elapsed	11289	
total timesteps	2259241	
train/		
learning_rate	0.0001	
loss	0.0472	
n_updates	552310	

EP=10,710 | STEP=2,261,340 | clip=54 | consec=3 | max_consec=67 | avg_clip50=51.78

EP=10,720 | STEP=2,264,609 | clip=66 | consec=13 | max_consec=67 | avg_clip50=53.75

EP=10,730 | STEP=2,267,528 | clip=80 | consec=23 | max_consec=67 | avg_clip50=53.36

custom/		
avg_clip50	53.4	
clipped_episode_r...	47	
consecutive_clipp...	31	
max_consecutive_c...	67	
rollout/		
ep_len_mean	1.21e+03	
ep_rew_mean	520	
exploration rate	0.244	
time/		
episodes	31600	
fps	200	
time_elapsed	11347	
total timesteps	2269965	
train/		
learning_rate	0.0001	
loss	0.0507	
n_updates	554991	

EP=10,740 | STEP=2,270,602 | clip=58 | consec=33 | max_consec=67 | avg_clip50=53.56

EP=10,750 | STEP=2,273,546 | clip=60 | consec=43 | max_consec=67 | avg_clip50=53.97

EP=10,760 | STEP=2,276,668 | clip=36 | consec=53 | max_consec=67 | avg_clip50=56.04

EP=10,770 | STEP=2,279,605 | clip=52 | consec=63 | max_consec=67 | avg_clip50=54.41

custom/		
avg_clip50	53.8	

	clipped_episode_r...	41.5	
	consecutive_clipp...	67	
	max_consecutive_c...	67	
	rollout/		
	ep_len_mean	1.25e+03	
	ep_rew_mean	539	
	exploration rate	0.241	
	time/		
	episodes	31700	
	fps	199	
	time_elapsed	11404	
	total timesteps	2280681	
	train/		
	learning_rate	0.0001	
	loss	0.0551	
	n_updates	557670	

EP=10,780 | STEP=2,282,359 | clip=37 | consec=73 | max_consec=73 | avg_clip50=53.95

EP=10,790 | STEP=2,285,078 | clip=47 | consec=83 | max_consec=83 | avg_clip50=53.37

EP=10,800 | STEP=2,288,255 | clip=42 | consec=93 | max_consec=93 | avg_clip50=54.78

Objetivo conseguido: 100 episodios consecutivos superando el umbral.

Entrenamiento finalizado.

max_consec alcanzado: 100

Log JSON: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.json

Log CSV: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.csv

Checkpoint PyTorch guardado en: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_double/dqn_spaceinvaders_v0_policy_latest.pt

model.save() falló (no es crítico, se usa el .pt): IndexError('tuple index out of range')

In []: *# M6-bis: Dueling DQN con checkpoint periódico*

```

class PeriodicPolicyCheckpoint(BaseCallback):
    def __init__(self, save_path, every_steps=250_000, verbose=0):
        super().__init__(verbose)
        self.save_path = save_path
        self.every_steps = every_steps
        self.last_save = 0

    def _on_step(self):
        if self.num_timesteps - self.last_save >= self.every_steps:
            save_policy_checkpoint(self.model, self.save_path)
            self.last_save = self.num_timesteps
            print(f'[checkpoint periódico] guardado a {self.num_timesteps:,} steps')
        return True

paths = RUN_PATHS['dueling']
env_dueling = create_train_env(paths['monitor'])
model_dueling = build_model('dueling', env_dueling, paths['tensorboard'])

training_logger = SB3TrainingLogger(json_path=paths['json'], csv_path=paths['csv'])
callback = ConsecutiveRewardCallback(training_logger=training_logger, verbose=1)
ckpt_cb = PeriodicPolicyCheckpoint(paths['policy'], every_steps=250_000)

model_dueling.learn(

```

```
total_timesteps=TOTAL_TIMESTEPS,
callback=[callback, ckpt_cb],
reset_num_timesteps=True,
tb_log_name='DQN_dueling',
log_interval=100
)

save_policy_checkpoint(model_dueling, paths['policy'])
print('Checkpoint final guardado en:', paths['policy'])
try:
    model_dueling.save(str(paths['model_zip']))
except Exception as e:
    print('model.save() falló (no crítico):', repr(e))
env_dueling.close()
```

Se han truncado las últimas 5000 líneas del flujo de salida.

EP=3,370 | STEP=562,571 | clip=45 | consec=2 | max_consec=8 | avg_clip50=17.69

custom/		
avg_clip50	19.7	
clipped_episode_r...	8.5	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	756	
ep_rew_mean	202	
exploration rate	0.812	
time/		
episodes	10100	
fps	218	
time_elapsed	2581	
total timesteps	564298	
train/		
learning_rate	0.0001	
loss	0.191	
n_updates	128574	

EP=3,380 | STEP=564,367 | clip=10 | consec=0 | max_consec=8 | avg_clip50=19.07

EP=3,390 | STEP=565,911 | clip=12 | consec=0 | max_consec=8 | avg_clip50=19.03

EP=3,400 | STEP=567,605 | clip=5 | consec=0 | max_consec=8 | avg_clip50=17.88

EP=3,410 | STEP=569,465 | clip=12 | consec=0 | max_consec=8 | avg_clip50=17.98

custom/		
avg_clip50	18.6	
clipped_episode_r...	31.5	
consecutive_clipp...	1	
max_consecutive_c...	8	
rollout/		
ep_len_mean	743	
ep_rew_mean	186	
exploration rate	0.81	
time/		
episodes	10200	
fps	218	
time_elapsed	2608	
total timesteps	570005	
train/		
learning_rate	0.0001	
loss	0.0661	
n_updates	130001	

EP=3,420 | STEP=571,034 | clip=0 | consec=0 | max_consec=8 | avg_clip50=17.23

EP=3,430 | STEP=573,189 | clip=14 | consec=0 | max_consec=8 | avg_clip50=18.07

EP=3,440 | STEP=575,025 | clip=4 | consec=0 | max_consec=8 | avg_clip50=18.26

custom/		
avg_clip50	19	
clipped_episode_r...	23.5	
consecutive_clipp...	1	
max_consecutive_c...	8	
rollout/		
ep_len_mean	759	
ep_rew_mean	188	
exploration rate	0.808	
time/		

episodes	10300	
fps	218	
time_elapsed	2638	
total timesteps	576322	
train/		
learning_rate	0.0001	
loss	0.208	
n_updates	131580	

EP=3,450	STEP=576,894	clip=12	consec=0	max_consec=8	avg_clip50=19.44
EP=3,460	STEP=578,769	clip=46	consec=2	max_consec=8	avg_clip50=20.19
EP=3,470	STEP=580,562	clip=10	consec=0	max_consec=8	avg_clip50=21.39

custom/		
avg_clip50	21.7	
clipped_episode_r...	12	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	772	
ep_rew_mean	198	
exploration rate	0.806	
time/		
episodes	10400	
fps	218	
time_elapsed	2668	
total timesteps	582541	
train/		
learning_rate	0.0001	
loss	0.0626	
n_updates	133135	

EP=3,480	STEP=582,643	clip=18	consec=0	max_consec=8	avg_clip50=21.74
EP=3,490	STEP=584,728	clip=25	consec=1	max_consec=8	avg_clip50=22.80
EP=3,500	STEP=586,304	clip=4	consec=0	max_consec=8	avg_clip50=23.33
EP=3,510	STEP=588,325	clip=29	consec=1	max_consec=8	avg_clip50=23.68

custom/		
avg_clip50	24.2	
clipped_episode_r...	17.5	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	800	
ep_rew_mean	218	
exploration rate	0.804	
time/		
episodes	10500	
fps	218	
time_elapsed	2699	
total timesteps	589098	
train/		
learning_rate	0.0001	
loss	0.209	
n_updates	134774	

EP=3,520	STEP=590,299	clip=17	consec=0	max_consec=8	avg_clip50=23.79
EP=3,530	STEP=592,117	clip=28	consec=1	max_consec=8	avg_clip50=22.20
EP=3,540	STEP=593,673	clip=10	consec=0	max_consec=8	avg_clip50=20.28

custom/	
avg_clip50	20.8
clipped_episode_r...	30.5
consecutive_clipp...	2
max_consecutive_c...	8
rollout/	
ep_len_mean	789
ep_rew_mean	217
exploration rate	0.802
time/	
episodes	10600
fps	218
time_elapsed	2728
total timesteps	594975
train/	
learning_rate	0.0001
loss	0.0691
n_updates	136243

EP=3,550	STEP=595,480	clip=20	consec=0	max_consec=8	avg_clip50=19.88
EP=3,560	STEP=597,565	clip=35	consec=5	max_consec=8	avg_clip50=20.31
EP=3,570	STEP=599,229	clip=4	consec=0	max_consec=8	avg_clip50=20.73
EP=3,580	STEP=601,094	clip=32	consec=1	max_consec=8	avg_clip50=20.33

custom/	
avg_clip50	20.3
clipped_episode_r...	32.5
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	784
ep_rew_mean	213
exploration rate	0.8
time/	
episodes	10700
fps	218
time_elapsed	2756
total timesteps	601157
train/	
learning_rate	0.0001
loss	0.251
n_updates	137789

EP=3,590	STEP=602,934	clip=12	consec=0	max_consec=8	avg_clip50=21.14
EP=3,600	STEP=604,750	clip=42	consec=1	max_consec=8	avg_clip50=20.99
EP=3,610	STEP=606,496	clip=14	consec=0	max_consec=8	avg_clip50=18.82

custom/	
avg_clip50	18.9
clipped_episode_r...	29
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	764
ep_rew_mean	196
exploration rate	0.798
time/	
episodes	10800
fps	217
time_elapsed	2785

total timesteps	607149
train/	
learning_rate	0.0001
loss	0.0539
n_updates	139287

EP=3,620	STEP=607,985	clip=3	consec=0	max_consec=8	avg_clip50=17.34
EP=3,630	STEP=610,324	clip=14	consec=0	max_consec=8	avg_clip50=19.95
EP=3,640	STEP=612,330	clip=34	consec=1	max_consec=8	avg_clip50=20.53

custom/	
avg_clip50	21.5
clipped_episode_r...	22
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	795
ep_rew_mean	209
exploration rate	0.796
time/	
episodes	10900
fps	217
time_elapsed	2817
total timesteps	613882
train/	
learning_rate	0.0001
loss	0.0282
n_updates	140970

EP=3,650	STEP=614,408	clip=18	consec=0	max_consec=8	avg_clip50=20.86
EP=3,660	STEP=616,463	clip=20	consec=0	max_consec=8	avg_clip50=22.60
EP=3,670	STEP=618,534	clip=26	consec=3	max_consec=8	avg_clip50=24.11
EP=3,680	STEP=620,345	clip=33	consec=1	max_consec=8	avg_clip50=22.54

custom/	
avg_clip50	22.5
clipped_episode_r...	33
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	814
ep_rew_mean	212
exploration rate	0.793
time/	
episodes	11000
fps	217
time_elapsed	2848
total timesteps	620610
train/	
learning_rate	0.0001
loss	0.119
n_updates	142652

EP=3,690	STEP=622,443	clip=24	consec=1	max_consec=8	avg_clip50=24.23
EP=3,700	STEP=624,338	clip=17	consec=0	max_consec=8	avg_clip50=23.72
EP=3,710	STEP=626,163	clip=82	consec=1	max_consec=8	avg_clip50=23.72

custom/	
avg_clip50	22.6
clipped_episode_r...	9

consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	823
ep_rew_mean	227
exploration rate	0.791
time/	
episodes	11100
fps	217
time_elapsed	2876
total timesteps	626762
train/	
learning_rate	0.0001
loss	0.049
n_updates	144190

EP=3,720	STEP=628,011	clip=17	consec=0	max_consec=8	avg_clip50=23.46
EP=3,730	STEP=629,973	clip=24	consec=2	max_consec=8	avg_clip50=23.02
EP=3,740	STEP=631,851	clip=10	consec=0	max_consec=8	avg_clip50=20.42

custom/	
avg_clip50	21.3
clipped_episode_r...	48.5
consecutive_clipp...	2
max_consecutive_c...	8
rollout/	
ep_len_mean	819
ep_rew_mean	225
exploration rate	0.789
time/	
episodes	11200
fps	217
time_elapsed	2907
total timesteps	633401
train/	
learning_rate	0.0001
loss	0.0443
n_updates	145850

EP=3,750	STEP=633,744	clip=9	consec=0	max_consec=8	avg_clip50=20.89
EP=3,760	STEP=635,648	clip=16	consec=0	max_consec=8	avg_clip50=20.70
EP=3,770	STEP=637,412	clip=12	consec=0	max_consec=8	avg_clip50=21.05
EP=3,780	STEP=639,224	clip=13	consec=0	max_consec=8	avg_clip50=20.66

custom/	
avg_clip50	21.7
clipped_episode_r...	57.5
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	803
ep_rew_mean	220
exploration rate	0.787
time/	
episodes	11300
fps	217
time_elapsed	2937
total timesteps	639789
train/	
learning_rate	0.0001

loss	0.0664
n_updates	147447

EP=3,790	STEP=641,236	clip=10	consec=0	max_consec=8	avg_clip50=22.29
EP=3,800	STEP=642,889	clip=18	consec=0	max_consec=8	avg_clip50=20.99
EP=3,810	STEP=644,698	clip=35	consec=1	max_consec=8	avg_clip50=20.41

custom/	
avg_clip50	20.2
clipped_episode_r...	30
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	792
ep_rew_mean	210
exploration rate	0.785
time/	
episodes	11400
fps	217
time_elapsed	2963
total timesteps	645399
train/	
learning_rate	0.0001
loss	0.053
n_updates	148849

EP=3,820	STEP=646,451	clip=16	consec=0	max_consec=8	avg_clip50=20.31
EP=3,830	STEP=648,599	clip=6	consec=0	max_consec=8	avg_clip50=21.36
EP=3,840	STEP=650,933	clip=18	consec=0	max_consec=8	avg_clip50=21.65

custom/	
avg_clip50	23.1
clipped_episode_r...	17
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	811
ep_rew_mean	219
exploration rate	0.783
time/	
episodes	11500
fps	217
time_elapsed	2996
total timesteps	652722
train/	
learning_rate	0.0001
loss	0.0767
n_updates	150680

EP=3,850	STEP=652,818	clip=8	consec=0	max_consec=8	avg_clip50=22.87
EP=3,860	STEP=654,440	clip=62	consec=2	max_consec=8	avg_clip50=22.60
EP=3,870	STEP=655,960	clip=18	consec=0	max_consec=8	avg_clip50=20.74
EP=3,880	STEP=658,152	clip=3	consec=0	max_consec=8	avg_clip50=20.88

custom/	
avg_clip50	20.6
clipped_episode_r...	29
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	

ep_len_mean	791
ep_rew_mean	205
exploration rate	0.781
time/	
episodes	11600
fps	217
time_elapsed	3024
total timesteps	658694
train/	
learning_rate	0.0001
loss	0.0712
n_updates	152173

EP=3,890 | STEP=660,358 | clip=30 | consec=4 | max_consec=8 | avg_clip50=21.30
EP=3,900 | STEP=662,580 | clip=14 | consec=0 | max_consec=8 | avg_clip50=22.37
EP=3,910 | STEP=664,165 | clip=0 | consec=0 | max_consec=8 | avg_clip50=22.47

custom/	
avg_clip50	24.1
clipped_episode_r...	19
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	834
ep_rew_mean	230
exploration rate	0.778
time/	
episodes	11700
fps	217
time_elapsed	3054
total timesteps	665454
train/	
learning_rate	0.0001
loss	0.128
n_updates	153863

EP=3,920 | STEP=666,140 | clip=18 | consec=0 | max_consec=8 | avg_clip50=24.37
EP=3,930 | STEP=668,053 | clip=36 | consec=1 | max_consec=8 | avg_clip50=23.14
EP=3,940 | STEP=669,898 | clip=4 | consec=0 | max_consec=8 | avg_clip50=21.61

custom/	
avg_clip50	19.3
clipped_episode_r...	12
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	788
ep_rew_mean	208
exploration rate	0.776
time/	
episodes	11800
fps	217
time_elapsed	3081
total timesteps	671373
train/	
learning_rate	0.0001
loss	0.0867
n_updates	155343

EP=3,950 | STEP=671,444 | clip=24 | consec=1 | max_consec=8 | avg_clip50=19.49

EP=3,960	STEP=673,410	clip=19	consec=0	max_consec=8	avg_clip50=20.05
EP=3,970	STEP=675,400	clip=17	consec=0	max_consec=8	avg_clip50=20.20
EP=3,980	STEP=677,079	clip=29	consec=2	max_consec=8	avg_clip50=19.82

custom/	
avg_clip50	18.9
clipped_episode_r...	8
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	795
ep_rew_mean	212
exploration rate	0.774
time/	
episodes	11900
fps	217
time_elapsed	3109
total timesteps	677499
train/	
learning_rate	0.0001
loss	0.042
n_updates	156874

EP=3,990	STEP=678,815	clip=32	consec=1	max_consec=8	avg_clip50=19.56
EP=4,000	STEP=680,523	clip=30	consec=1	max_consec=8	avg_clip50=20.33
EP=4,010	STEP=682,624	clip=16	consec=0	max_consec=8	avg_clip50=21.27

custom/	
avg_clip50	21.2
clipped_episode_r...	31
consecutive_clipp...	3
max_consecutive_c...	8
rollout/	
ep_len_mean	779
ep_rew_mean	205
exploration rate	0.772
time/	
episodes	12000
fps	217
time_elapsed	3137
total timesteps	683822
train/	
learning_rate	0.0001
loss	0.0248
n_updates	158455

EP=4,020	STEP=684,401	clip=10	consec=0	max_consec=8	avg_clip50=21.42
EP=4,030	STEP=686,506	clip=57	consec=2	max_consec=8	avg_clip50=23.03
EP=4,040	STEP=688,146	clip=20	consec=0	max_consec=8	avg_clip50=22.98
EP=4,050	STEP=689,756	clip=32	consec=2	max_consec=8	avg_clip50=22.13

custom/	
avg_clip50	22.6
clipped_episode_r...	45
consecutive_clipp...	3
max_consecutive_c...	8
rollout/	
ep_len_mean	781
ep_rew_mean	215
exploration rate	0.77

time/	
episodes	12100
fps	217
time_elapsed	3166
total timesteps	690085
train/	
learning_rate	0.0001
loss	0.0823
n_updates	160021

EP=4,060	STEP=692,338	clip=14	consec=0	max_consec=8	avg_clip50=23.12
EP=4,070	STEP=694,134	clip=26	consec=3	max_consec=8	avg_clip50=22.23
EP=4,080	STEP=696,137	clip=49	consec=1	max_consec=8	avg_clip50=22.33

custom/	
avg_clip50	21.4
clipped_episode_r...	4.5
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	814
ep_rew_mean	230
exploration rate	0.768
time/	
episodes	12200
fps	217
time_elapsed	3197
total timesteps	696980
train/	
learning_rate	0.0001
loss	0.0537
n_updates	161744

EP=4,090	STEP=697,961	clip=29	consec=3	max_consec=8	avg_clip50=23.33
EP=4,100	STEP=700,155	clip=24	consec=1	max_consec=8	avg_clip50=26.92
EP=4,110	STEP=702,059	clip=9	consec=0	max_consec=8	avg_clip50=24.92

custom/	
avg_clip50	24.7
clipped_episode_r...	25
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	823
ep_rew_mean	234
exploration rate	0.766
time/	
episodes	12300
fps	218
time_elapsed	3226
total timesteps	703554
train/	
learning_rate	0.0001
loss	0.0304
n_updates	163388

EP=4,120	STEP=703,721	clip=5	consec=0	max_consec=8	avg_clip50=24.10
EP=4,130	STEP=705,668	clip=42	consec=1	max_consec=8	avg_clip50=24.28
EP=4,140	STEP=707,704	clip=18	consec=0	max_consec=8	avg_clip50=24.36
EP=4,150	STEP=709,863	clip=22	consec=2	max_consec=8	avg_clip50=23.23

custom/		
avg_clip50	23.3	
clipped_episode_r...	9.5	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	843	
ep_rew_mean	249	
exploration rate	0.763	
time/		
episodes	12400	
fps	218	
time_elapsed	3257	
total timesteps	710392	
train/		
learning_rate	0.0001	
loss	0.0489	
n_updates	165097	

EP=4,160 | STEP=711,693 | clip=56 | consec=2 | max_consec=8 | avg_clip50=23.16
EP=4,170 | STEP=713,636 | clip=11 | consec=0 | max_consec=8 | avg_clip50=24.89
EP=4,180 | STEP=715,517 | clip=14 | consec=0 | max_consec=8 | avg_clip50=24.47

custom/		
avg_clip50	23.1	
clipped_episode_r...	9.5	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	815	
ep_rew_mean	243	
exploration rate	0.761	
time/		
episodes	12500	
fps	218	
time_elapsed	3284	
total timesteps	716298	
train/		
learning_rate	0.0001	
loss	0.0528	
n_updates	166574	

EP=4,190 | STEP=716,940 | clip=23 | consec=1 | max_consec=8 | avg_clip50=22.60
EP=4,200 | STEP=718,542 | clip=17 | consec=0 | max_consec=8 | avg_clip50=20.87
EP=4,210 | STEP=720,442 | clip=18 | consec=0 | max_consec=8 | avg_clip50=21.92

custom/		
avg_clip50	21.1	
clipped_episode_r...	20.5	
consecutive_clipp...	1	
max_consecutive_c...	8	
rollout/		
ep_len_mean	784	
ep_rew_mean	229	
exploration rate	0.76	
time/		
episodes	12600	
fps	218	
time_elapsed	3310	

total timesteps	722162
train/	
learning_rate	0.0001
loss	0.28
n_updates	168040

EP=4,220	STEP=722,290	clip=16	consec=0	max_consec=8	avg_clip50=21.16
EP=4,230	STEP=723,875	clip=10	consec=0	max_consec=8	avg_clip50=19.37
EP=4,240	STEP=725,492	clip=14	consec=0	max_consec=8	avg_clip50=20.02
EP=4,250	STEP=727,342	clip=7	consec=0	max_consec=8	avg_clip50=20.06

custom/	
avg_clip50	17.4
clipped_episode_r...	19
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	740
ep_rew_mean	203
exploration rate	0.758
time/	
episodes	12700
fps	218
time_elapsed	3336
total timesteps	727798
train/	
learning_rate	0.0001
loss	0.245
n_updates	169449

EP=4,260	STEP=729,506	clip=23	consec=2	max_consec=8	avg_clip50=20.01
EP=4,270	STEP=731,599	clip=18	consec=0	max_consec=8	avg_clip50=22.01
EP=4,280	STEP=732,952	clip=18	consec=0	max_consec=8	avg_clip50=21.32

custom/	
avg_clip50	22.6
clipped_episode_r...	10
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	759
ep_rew_mean	215
exploration rate	0.756
time/	
episodes	12800
fps	218
time_elapsed	3365
total timesteps	734185
train/	
learning_rate	0.0001
loss	0.14
n_updates	171046

EP=4,290	STEP=734,914	clip=21	consec=3	max_consec=8	avg_clip50=23.26
EP=4,300	STEP=736,846	clip=12	consec=0	max_consec=8	avg_clip50=24.34
EP=4,310	STEP=738,567	clip=6	consec=0	max_consec=8	avg_clip50=22.60

custom/	
avg_clip50	22.4
clipped_episode_r...	61

consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	782
ep_rew_mean	222
exploration rate	0.753
time/	
episodes	12900
fps	218
time_elapsed	3394
total timesteps	740700
train/	
learning_rate	0.0001
loss	0.0669
n_updates	172674

EP=4,320 | STEP=740,721 | clip=16 | consec=0 | max_consec=8 | avg_clip50=22.35
EP=4,330 | STEP=742,632 | clip=8 | consec=0 | max_consec=8 | avg_clip50=24.59
EP=4,340 | STEP=744,779 | clip=28 | consec=1 | max_consec=8 | avg_clip50=24.13
EP=4,350 | STEP=746,670 | clip=13 | consec=0 | max_consec=8 | avg_clip50=23.66

custom/	
avg_clip50	24.3
clipped_episode_r...	54.5
consecutive_clipp...	3
max_consecutive_c...	8
rollout/	
ep_len_mean	830
ep_rew_mean	248
exploration rate	0.751
time/	
episodes	13000
fps	218
time_elapsed	3425
total timesteps	747468
train/	
learning_rate	0.0001
loss	0.0419
n_updates	174366

EP=4,360 | STEP=748,638 | clip=19 | consec=0 | max_consec=8 | avg_clip50=24.84
[checkpoint periodico] guardado a 750,000 steps
EP=4,370 | STEP=750,266 | clip=9 | consec=0 | max_consec=8 | avg_clip50=22.59
EP=4,380 | STEP=751,971 | clip=2 | consec=0 | max_consec=8 | avg_clip50=21.31

custom/	
avg_clip50	21.6
clipped_episode_r...	9.5
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	814
ep_rew_mean	228
exploration rate	0.749
time/	
episodes	13100
fps	218
time_elapsed	3454
total timesteps	753647
train/	

learning_rate	0.0001
loss	0.0951
n_updates	175911

EP=4,390	STEP=754,261	clip=33	consec=2	max_consec=8	avg_clip50=22.10
EP=4,400	STEP=756,631	clip=36	consec=1	max_consec=8	avg_clip50=23.17
EP=4,410	STEP=758,510	clip=14	consec=0	max_consec=8	avg_clip50=22.49
EP=4,420	STEP=760,709	clip=8	consec=0	max_consec=8	avg_clip50=24.16

custom/	
avg_clip50	24.2
clipped_episode_r...	7.5
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	845
ep_rew_mean	234
exploration rate	0.747
time/	
episodes	13200
fps	218
time_elapsed	3487
total timesteps	760709
train/	
learning_rate	0.0001
loss	0.182
n_updates	177677

EP=4,430	STEP=762,286	clip=25	consec=1	max_consec=8	avg_clip50=24.30
EP=4,440	STEP=763,910	clip=14	consec=0	max_consec=8	avg_clip50=21.60
EP=4,450	STEP=766,045	clip=16	consec=0	max_consec=8	avg_clip50=22.12

custom/	
avg_clip50	23.2
clipped_episode_r...	38.5
consecutive_clipp...	2
max_consecutive_c...	8
rollout/	
ep_len_mean	820
ep_rew_mean	226
exploration rate	0.745
time/	
episodes	13300
fps	218
time_elapsed	3516
total timesteps	767036
train/	
learning_rate	0.0001
loss	0.107
n_updates	179258

EP=4,460	STEP=768,072	clip=5	consec=0	max_consec=8	avg_clip50=23.40
EP=4,470	STEP=769,527	clip=6	consec=0	max_consec=8	avg_clip50=21.25
EP=4,480	STEP=771,222	clip=18	consec=0	max_consec=8	avg_clip50=21.93

custom/	
avg_clip50	22.3
clipped_episode_r...	25.5
consecutive_clipp...	1
max_consecutive_c...	8

rollout/	
ep_len_mean	795
ep_rew_mean	223
exploration rate	0.743
time/	
episodes	13400
fps	218
time_elapsed	3541
total timesteps	772589
train/	
learning_rate	0.0001
loss	0.035
n_updates	180647

EP=4,490 | STEP=773,109 | clip=61 | consec=3 | max_consec=8 | avg_clip50=23.62
EP=4,500 | STEP=774,823 | clip=24 | consec=2 | max_consec=8 | avg_clip50=21.16
EP=4,510 | STEP=776,589 | clip=18 | consec=0 | max_consec=8 | avg_clip50=19.60
EP=4,520 | STEP=778,052 | clip=32 | consec=1 | max_consec=8 | avg_clip50=21.00

custom/	
avg_clip50	20.6
clipped_episode_r...	42
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	738
ep_rew_mean	212
exploration rate	0.741
time/	
episodes	13500
fps	218
time_elapsed	3568
total timesteps	778416
train/	
learning_rate	0.0001
loss	0.191
n_updates	182103

EP=4,530 | STEP=779,710 | clip=22 | consec=1 | max_consec=8 | avg_clip50=20.66
EP=4,540 | STEP=781,459 | clip=12 | consec=0 | max_consec=8 | avg_clip50=19.39
EP=4,550 | STEP=783,559 | clip=38 | consec=1 | max_consec=8 | avg_clip50=20.00

custom/	
avg_clip50	20.1
clipped_episode_r...	20
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	745
ep_rew_mean	204
exploration rate	0.739
time/	
episodes	13600
fps	218
time_elapsed	3597
total timesteps	784624
train/	
learning_rate	0.0001
loss	0.0261
n_updates	183655

EP=4,560 | STEP=785,330 | clip=6 | consec=0 | max_consec=8 | avg_clip50=19.88
EP=4,570 | STEP=787,186 | clip=19 | consec=0 | max_consec=8 | avg_clip50=19.78
EP=4,580 | STEP=788,745 | clip=4 | consec=0 | max_consec=8 | avg_clip50=18.87

custom/	
avg_clip50	19.1
clipped_episode_r...	4
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	753
ep_rew_mean	198
exploration rate	0.737
time/	
episodes	13700
fps	218
time_elapsed	3622
total timesteps	790278
train/	
learning_rate	0.0001
loss	0.141
n_updates	185069

EP=4,590 | STEP=790,497 | clip=24 | consec=1 | max_consec=8 | avg_clip50=19.34
EP=4,600 | STEP=792,381 | clip=58 | consec=1 | max_consec=8 | avg_clip50=19.89
EP=4,610 | STEP=794,308 | clip=70 | consec=1 | max_consec=8 | avg_clip50=21.35
EP=4,620 | STEP=796,529 | clip=34 | consec=2 | max_consec=8 | avg_clip50=23.14

custom/	
avg_clip50	23.4
clipped_episode_r...	13.5
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	789
ep_rew_mean	215
exploration rate	0.735
time/	
episodes	13800
fps	218
time_elapsed	3654
total timesteps	797082
train/	
learning_rate	0.0001
loss	0.0517
n_updates	186770

EP=4,630 | STEP=798,471 | clip=8 | consec=0 | max_consec=8 | avg_clip50=25.78
EP=4,640 | STEP=800,514 | clip=30 | consec=1 | max_consec=8 | avg_clip50=28.32
EP=4,650 | STEP=802,553 | clip=16 | consec=0 | max_consec=8 | avg_clip50=28.38

custom/	
avg_clip50	28.2
clipped_episode_r...	49
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	797
ep_rew_mean	241

exploration rate	0.732
time/	
episodes	13900
fps	218
time_elapsed	3684
total timesteps	803703
train/	
learning_rate	0.0001
loss	0.0488
n_updates	188425

EP=4,660 | STEP=804,118 | clip=28 | consec=1 | max_consec=8 | avg_clip50=26.93
EP=4,670 | STEP=806,130 | clip=16 | consec=0 | max_consec=8 | avg_clip50=26.18
EP=4,680 | STEP=807,792 | clip=10 | consec=0 | max_consec=8 | avg_clip50=24.25
EP=4,690 | STEP=809,413 | clip=5 | consec=0 | max_consec=8 | avg_clip50=21.33

custom/	
avg_clip50	21.3
clipped_episode_r...	5
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	802
ep_rew_mean	248
exploration rate	0.73
time/	
episodes	14000
fps	218
time_elapsed	3711
total timesteps	809484
train/	
learning_rate	0.0001
loss	0.0668
n_updates	189870

EP=4,700 | STEP=811,527 | clip=31 | consec=3 | max_consec=8 | avg_clip50=21.48
EP=4,710 | STEP=813,298 | clip=26 | consec=1 | max_consec=8 | avg_clip50=21.94
EP=4,720 | STEP=814,998 | clip=30 | consec=1 | max_consec=8 | avg_clip50=19.96

custom/	
avg_clip50	20.4
clipped_episode_r...	23
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	781
ep_rew_mean	229
exploration rate	0.728
time/	
episodes	14100
fps	218
time_elapsed	3741
total timesteps	816038
train/	
learning_rate	0.0001
loss	0.0908
n_updates	191509

EP=4,730 | STEP=816,985 | clip=18 | consec=0 | max_consec=8 | avg_clip50=21.29
EP=4,740 | STEP=819,049 | clip=30 | consec=4 | max_consec=8 | avg_clip50=22.92

EP=4,750 | STEP=820,951 | clip=24 | consec=3 | max_consec=8 | avg_clip50=22.44

custom/		
avg_clip50	22.2	
clipped_episode_r...	29	
consecutive_clipp...	1	
max_consecutive_c...	8	
rollout/		
ep_len_mean	786	
ep_rew_mean	221	
exploration rate	0.726	
time/		
episodes	14200	
fps	218	
time_elapsed	3771	
total timesteps	822488	
train/		
learning_rate	0.0001	
loss	0.189	
n_updates	193121	

EP=4,760 | STEP=822,600 | clip=11 | consec=0 | max_consec=8 | avg_clip50=21.90
EP=4,770 | STEP=824,374 | clip=16 | consec=0 | max_consec=8 | avg_clip50=22.89
EP=4,780 | STEP=826,386 | clip=38 | consec=1 | max_consec=8 | avg_clip50=22.54
EP=4,790 | STEP=828,093 | clip=26 | consec=3 | max_consec=8 | avg_clip50=21.18

custom/		
avg_clip50	21.2	
clipped_episode_r...	18	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	797	
ep_rew_mean	218	
exploration rate	0.724	
time/		
episodes	14300	
fps	218	
time_elapsed	3801	
total timesteps	828909	
train/		
learning_rate	0.0001	
loss	0.255	
n_updates	194727	

EP=4,800 | STEP=830,182 | clip=16 | consec=0 | max_consec=8 | avg_clip50=21.63
EP=4,810 | STEP=831,798 | clip=12 | consec=0 | max_consec=8 | avg_clip50=21.39
EP=4,820 | STEP=833,374 | clip=24 | consec=1 | max_consec=8 | avg_clip50=21.05

custom/		
avg_clip50	20.9	
clipped_episode_r...	18	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	783	
ep_rew_mean	217	
exploration rate	0.722	
time/		
episodes	14400	

	fps		218	
	time_elapsed		3828	
	total timesteps		834794	
	train/			
	learning_rate		0.0001	
	loss		0.0526	
	n_updates		196198	

EP=4,830		STEP=835,342		clip=29		consec=1		max_consec=8		avg_clip50=20.29
EP=4,840		STEP=837,007		clip=6		consec=0		max_consec=8		avg_clip50=19.77
EP=4,850		STEP=839,339		clip=26		consec=2		max_consec=8		avg_clip50=20.81
EP=4,860		STEP=841,006		clip=6		consec=0		max_consec=8		avg_clip50=21.45

	custom/			
	avg_clip50		21.4	
	clipped_episode_r...		6	
	consecutive_clipp...		0	
	max_consecutive_c...		8	
	rollout/			
	ep_len_mean		784	
	ep_rew_mean		214	
	exploration rate		0.72	
	time/			
	episodes		14500	
	fps		218	
	time_elapsed		3856	
	total timesteps		841006	
	train/			
	learning_rate		0.0001	
	loss		0.048	
	n_updates		197751	

EP=4,870		STEP=842,893		clip=32		consec=1		max_consec=8		avg_clip50=23.00
EP=4,880		STEP=845,064		clip=61		consec=2		max_consec=8		avg_clip50=24.80
EP=4,890		STEP=846,624		clip=20		consec=0		max_consec=8		avg_clip50=25.02

	custom/			
	avg_clip50		25.4	
	clipped_episode_r...		21	
	consecutive_clipp...		3	
	max_consecutive_c...		8	
	rollout/			
	ep_len_mean		781	
	ep_rew_mean		224	
	exploration rate		0.718	
	time/			
	episodes		14600	
	fps		218	
	time_elapsed		3886	
	total timesteps		847373	
	train/			
	learning_rate		0.0001	
	loss		0.184	
	n_updates		199343	

EP=4,900		STEP=848,842		clip=18		consec=0		max_consec=8		avg_clip50=25.22
EP=4,910		STEP=850,833		clip=70		consec=2		max_consec=8		avg_clip50=26.57
EP=4,920		STEP=852,639		clip=8		consec=0		max_consec=8		avg_clip50=25.43

	custom/			
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avg_clip50	24.5
clipped_episode_r...	3
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	802
ep_rew_mean	242
exploration rate	0.716
time/	
episodes	14700
fps	218
time_elapsed	3916
total timesteps	853913
train/	
learning_rate	0.0001
loss	0.0388
n_updates	200978

EP=4,930	STEP=854,436	clip=32	consec=2	max_consec=8	avg_clip50=24.07
EP=4,940	STEP=856,775	clip=50	consec=4	max_consec=8	avg_clip50=26.90
EP=4,950	STEP=858,875	clip=32	consec=1	max_consec=8	avg_clip50=26.62
EP=4,960	STEP=860,547	clip=5	consec=0	max_consec=8	avg_clip50=25.30

custom/	
avg_clip50	25.5
clipped_episode_r...	20.5
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	822
ep_rew_mean	256
exploration rate	0.713
time/	
episodes	14800
fps	218
time_elapsed	3948
total timesteps	860718
train/	
learning_rate	0.0001
loss	0.0782
n_updates	202679

EP=4,970	STEP=862,388	clip=44	consec=2	max_consec=8	avg_clip50=25.91
EP=4,980	STEP=864,073	clip=12	consec=0	max_consec=8	avg_clip50=24.62
EP=4,990	STEP=866,199	clip=41	consec=1	max_consec=8	avg_clip50=24.42

custom/	
avg_clip50	24.4
clipped_episode_r...	26
consecutive_clipp...	2
max_consecutive_c...	8
rollout/	
ep_len_mean	834
ep_rew_mean	258
exploration rate	0.711
time/	
episodes	14900
fps	217
time_elapsed	3978
total timesteps	867177

train/	
learning_rate	0.0001
loss	0.0418
n_updates	204294

EP=5,000	STEP=868,081	clip=10	consec=0	max_consec=8	avg_clip50=22.55
EP=5,010	STEP=870,062	clip=28	consec=5	max_consec=8	avg_clip50=23.93
EP=5,020	STEP=872,107	clip=31	consec=3	max_consec=8	avg_clip50=24.06

custom/	
avg_clip50	26.6
clipped_episode_r...	38.5
consecutive_clipp...	6
max_consecutive_c...	8
rollout/	
ep_len_mean	846
ep_rew_mean	259
exploration rate	0.709
time/	
episodes	15000
fps	217
time_elapsed	4010
total timesteps	873987
train/	
learning_rate	0.0001
loss	0.0966
n_updates	205996

EP=5,030	STEP=874,299	clip=14	consec=0	max_consec=8	avg_clip50=26.95
EP=5,040	STEP=876,156	clip=35	consec=2	max_consec=8	avg_clip50=24.90
EP=5,050	STEP=877,903	clip=37	consec=1	max_consec=8	avg_clip50=25.04
EP=5,060	STEP=879,900	clip=38	consec=3	max_consec=8	avg_clip50=24.65

custom/	
avg_clip50	24.1
clipped_episode_r...	24
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	821
ep_rew_mean	242
exploration rate	0.707
time/	
episodes	15100
fps	217
time_elapsed	4039
total timesteps	880204
train/	
learning_rate	0.0001
loss	0.0636
n_updates	207550

EP=5,070	STEP=881,786	clip=4	consec=0	max_consec=8	avg_clip50=24.61
EP=5,080	STEP=883,484	clip=10	consec=0	max_consec=8	avg_clip50=23.15
EP=5,090	STEP=885,676	clip=16	consec=0	max_consec=8	avg_clip50=24.25

custom/	
avg_clip50	24.2
clipped_episode_r...	13.5
consecutive_clipp...	0

max_consecutive_c...	8
rollout/	
ep_len_mean	817
ep_rew_mean	245
exploration rate	0.705
time/	
episodes	15200
fps	217
time_elapsed	4069
total timesteps	886760
train/	
learning_rate	0.0001
loss	0.0629
n_updates	209189

EP=5,100 | STEP=887,385 | clip=40 | consec=1 | max_consec=8 | avg_clip50=24.32
EP=5,110 | STEP=889,352 | clip=6 | consec=0 | max_consec=8 | avg_clip50=23.56
EP=5,120 | STEP=891,367 | clip=42 | consec=2 | max_consec=8 | avg_clip50=22.93
EP=5,130 | STEP=893,415 | clip=12 | consec=0 | max_consec=8 | avg_clip50=23.93

custom/	
avg_clip50	23.9
clipped_episode_r...	12
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	808
ep_rew_mean	235
exploration rate	0.702
time/	
episodes	15300
fps	217
time_elapsed	4100
total timesteps	893468
train/	
learning_rate	0.0001
loss	0.115
n_updates	210866

EP=5,140 | STEP=895,479 | clip=54 | consec=2 | max_consec=8 | avg_clip50=24.69
EP=5,150 | STEP=897,961 | clip=28 | consec=1 | max_consec=8 | avg_clip50=26.59
EP=5,160 | STEP=900,088 | clip=17 | consec=0 | max_consec=8 | avg_clip50=27.48

custom/	
avg_clip50	28.2
clipped_episode_r...	14.5
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	850
ep_rew_mean	251
exploration rate	0.7
time/	
episodes	15400
fps	217
time_elapsed	4135
total timesteps	900848
train/	
learning_rate	0.0001
loss	0.0363

n_updates		212711				
EP=5,170	STEP=901,899	clip=12	consec=0	max_consec=8	avg_clip50=26.93	
EP=5,180	STEP=904,068	clip=23	consec=1	max_consec=8	avg_clip50=26.58	
EP=5,190	STEP=906,320	clip=32	consec=1	max_consec=8	avg_clip50=27.29	
custom/						
avg_clip50		26.3				
clipped_episode_r...		31				
consecutive_clipp...		4				
max_consecutive_c...		8				
rollout/						
ep_len_mean		890				
ep_rew_mean		265				
exploration rate		0.698				
time/						
episodes		15500				
fps		217				
time_elapsed		4168				
total timesteps		908100				
train/						
learning_rate		0.0001				
loss		0.0574				
n_updates		214524				
EP=5,200	STEP=908,380	clip=22	consec=1	max_consec=8	avg_clip50=25.96	
EP=5,210	STEP=910,428	clip=34	consec=4	max_consec=8	avg_clip50=25.75	
EP=5,220	STEP=912,341	clip=27	consec=1	max_consec=8	avg_clip50=26.88	
EP=5,230	STEP=914,298	clip=8	consec=0	max_consec=8	avg_clip50=27.39	
custom/						
avg_clip50		27				
clipped_episode_r...		24.5				
consecutive_clipp...		1				
max_consecutive_c...		8				
rollout/						
ep_len_mean		888				
ep_rew_mean		271				
exploration rate		0.695				
time/						
episodes		15600				
fps		217				
time_elapsed		4198				
total timesteps		914607				
train/						
learning_rate		0.0001				
loss		0.0676				
n_updates		216151				
EP=5,240	STEP=916,302	clip=36	consec=4	max_consec=8	avg_clip50=25.15	
EP=5,250	STEP=918,140	clip=26	consec=1	max_consec=8	avg_clip50=25.13	
EP=5,260	STEP=920,464	clip=23	consec=1	max_consec=8	avg_clip50=26.69	
custom/						
avg_clip50		25				
clipped_episode_r...		25.5				
consecutive_clipp...		1				
max_consecutive_c...		8				
rollout/						
ep len mean		859				

ep_rew_mean	260
exploration rate	0.693
time/	
episodes	15700
fps	217
time_elapsed	4229
total timesteps	921331
train/	
learning_rate	0.0001
loss	0.0901
n_updates	217832

EP=5,270	STEP=922,364	clip=25	consec=2	max_consec=8	avg_clip50=25.45
EP=5,280	STEP=924,423	clip=26	consec=3	max_consec=8	avg_clip50=26.09
EP=5,290	STEP=926,575	clip=34	consec=4	max_consec=8	avg_clip50=27.12

custom/	
avg_clip50	26.6
clipped_episode_r...	44.5
consecutive_clipp...	2
max_consecutive_c...	8
rollout/	
ep_len_mean	846
ep_rew_mean	258
exploration rate	0.691
time/	
episodes	15800
fps	217
time_elapsed	4263
total timesteps	928252
train/	
learning_rate	0.0001
loss	0.0357
n_updates	219562

EP=5,300	STEP=928,474	clip=28	consec=3	max_consec=8	avg_clip50=26.68
EP=5,310	STEP=929,988	clip=58	consec=1	max_consec=8	avg_clip50=23.59
EP=5,320	STEP=931,780	clip=52	consec=2	max_consec=8	avg_clip50=25.06
EP=5,330	STEP=933,716	clip=34	consec=2	max_consec=8	avg_clip50=24.29

custom/	
avg_clip50	24.2
clipped_episode_r...	48
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	826
ep_rew_mean	255
exploration rate	0.689
time/	
episodes	15900
fps	217
time_elapsed	4291
total timesteps	934234
train/	
learning_rate	0.0001
loss	0.0714
n_updates	221058

EP=5,340	STEP=935,613	clip=6	consec=0	max_consec=8	avg_clip50=23.70
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EP=5,350 | STEP=937,886 | clip=24 | consec=3 | max_consec=8 | avg_clip50=24.91
EP=5,360 | STEP=939,989 | clip=4 | consec=0 | max_consec=8 | avg_clip50=27.17

custom/		
avg_clip50	28.1	
clipped_episode_r...	19.5	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	834	
ep_rew_mean	260	
exploration rate	0.687	
time/		
episodes	16000	
fps	217	
time_elapsed	4323	
total timesteps	941303	
train/		
learning_rate	0.0001	
loss	0.0498	
n_updates	222825	

EP=5,370 | STEP=942,549 | clip=33 | consec=5 | max_consec=8 | avg_clip50=29.28
EP=5,380 | STEP=944,642 | clip=20 | consec=1 | max_consec=8 | avg_clip50=29.06
EP=5,390 | STEP=946,470 | clip=12 | consec=0 | max_consec=8 | avg_clip50=28.31

custom/		
avg_clip50	27.1	
clipped_episode_r...	16	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	838	
ep_rew_mean	260	
exploration rate	0.684	
time/		
episodes	16100	
fps	217	
time_elapsed	4355	
total timesteps	948189	
train/		
learning_rate	0.0001	
loss	0.0223	
n_updates	224547	

EP=5,400 | STEP=948,311 | clip=36 | consec=1 | max_consec=8 | avg_clip50=27.36
EP=5,410 | STEP=950,295 | clip=32 | consec=2 | max_consec=8 | avg_clip50=26.70
EP=5,420 | STEP=952,175 | clip=24 | consec=4 | max_consec=8 | avg_clip50=24.12
EP=5,430 | STEP=953,849 | clip=30 | consec=2 | max_consec=8 | avg_clip50=22.38

custom/		
avg_clip50	22.1	
clipped_episode_r...	17	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	850	
ep_rew_mean	256	
exploration rate	0.682	
time/		

episodes	16200	
fps	217	
time_elapsed	4384	
total timesteps	954423	
train/		
learning_rate	0.0001	
loss	0.1	
n_updates	226105	

EP=5,440	STEP=956,169	clip=45	consec=2	max_consec=8	avg_clip50=24.27
EP=5,450	STEP=958,254	clip=12	consec=0	max_consec=8	avg_clip50=26.36
EP=5,460	STEP=960,258	clip=32	consec=3	max_consec=8	avg_clip50=25.97

custom/		
avg_clip50	26.3	
clipped_episode_r...	43	
consecutive_clipp...	1	
max_consecutive_c...	8	
rollout/		
ep_len_mean	844	
ep_rew_mean	256	
exploration rate	0.68	
time/		
episodes	16300	
fps	217	
time_elapsed	4418	
total timesteps	961578	
train/		
learning_rate	0.0001	
loss	0.0851	
n_updates	227894	

EP=5,470	STEP=962,202	clip=66	consec=1	max_consec=8	avg_clip50=27.05
EP=5,480	STEP=964,310	clip=58	consec=5	max_consec=8	avg_clip50=29.88
EP=5,490	STEP=966,341	clip=23	consec=1	max_consec=8	avg_clip50=28.80
EP=5,500	STEP=968,234	clip=32	consec=1	max_consec=8	avg_clip50=26.31

custom/		
avg_clip50	26.3	
clipped_episode_r...	31.5	
consecutive_clipp...	1	
max_consecutive_c...	8	
rollout/		
ep_len_mean	844	
ep_rew_mean	263	
exploration rate	0.678	
time/		
episodes	16400	
fps	217	
time_elapsed	4449	
total timesteps	968304	
train/		
learning_rate	0.0001	
loss	0.272	
n_updates	229575	

EP=5,510	STEP=970,073	clip=15	consec=0	max_consec=8	avg_clip50=26.28
EP=5,520	STEP=972,204	clip=31	consec=2	max_consec=8	avg_clip50=27.57
EP=5,530	STEP=974,694	clip=20	consec=5	max_consec=8	avg_clip50=27.55

custom/	
avg_clip50	27.5
clipped_episode_r...	40.5
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	893
ep_rew_mean	293
exploration rate	0.675
time/	
episodes	16500
fps	217
time_elapsed	4483
total timesteps	975719
train/	
learning_rate	0.0001
loss	0.127
n_updates	231429

EP=5,540	STEP=976,851	clip=52	consec=2	max_consec=8	avg_clip50=28.17
EP=5,550	STEP=978,688	clip=28	consec=1	max_consec=8	avg_clip50=28.66
EP=5,560	STEP=980,804	clip=40	consec=4	max_consec=8	avg_clip50=29.55

custom/	
avg_clip50	29.7
clipped_episode_r...	42.5
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	881
ep_rew_mean	286
exploration rate	0.673
time/	
episodes	16600
fps	217
time_elapsed	4516
total timesteps	982670
train/	
learning_rate	0.0001
loss	0.0451
n_updates	233167

EP=5,570	STEP=983,097	clip=35	consec=3	max_consec=8	avg_clip50=29.56
EP=5,580	STEP=985,707	clip=58	consec=5	max_consec=8	avg_clip50=30.63
EP=5,590	STEP=987,739	clip=14	consec=0	max_consec=8	avg_clip50=30.21
EP=5,600	STEP=989,906	clip=24	consec=1	max_consec=8	avg_clip50=31.09

custom/	
avg_clip50	31.5
clipped_episode_r...	36.5
consecutive_clipp...	2
max_consecutive_c...	8
rollout/	
ep_len_mean	921
ep_rew_mean	302
exploration rate	0.67
time/	
episodes	16700
fps	217
time_elapsed	4551

total timesteps	990320
train/	
learning_rate	0.0001
loss	0.657
n_updates	235079

EP=5,610 | STEP=992,060 | clip=25 | consec=1 | max_consec=8 | avg_clip50=30.80
EP=5,620 | STEP=994,239 | clip=6 | consec=0 | max_consec=8 | avg_clip50=28.64
EP=5,630 | STEP=996,388 | clip=44 | consec=2 | max_consec=8 | avg_clip50=27.14

custom/	
avg_clip50	26.4
clipped_episode_r...	23
consecutive_clipp...	2
max_consecutive_c...	8
rollout/	
ep_len_mean	906
ep_rew_mean	285
exploration rate	0.668
time/	
episodes	16800
fps	217
time_elapsed	4584
total timesteps	997348
train/	
learning_rate	0.0001
loss	0.037
n_updates	236836

EP=5,640 | STEP=998,599 | clip=8 | consec=0 | max_consec=8 | avg_clip50=28.25
[checkpoint periodico] guardado a 1,000,000 steps
EP=5,650 | STEP=1,000,670 | clip=7 | consec=0 | max_consec=8 | avg_clip50=27.72
EP=5,660 | STEP=1,002,528 | clip=32 | consec=4 | max_consec=8 | avg_clip50=26.70

custom/	
avg_clip50	27.4
clipped_episode_r...	22.5
consecutive_clipp...	4
max_consecutive_c...	8
rollout/	
ep_len_mean	912
ep_rew_mean	283
exploration rate	0.666
time/	
episodes	16900
fps	217
time_elapsed	4618
total timesteps	1004462
train/	
learning_rate	0.0001
loss	0.0336
n_updates	238615

EP=5,670 | STEP=1,004,642 | clip=25 | consec=5 | max_consec=8 | avg_clip50=27.75
EP=5,680 | STEP=1,006,428 | clip=8 | consec=0 | max_consec=8 | avg_clip50=26.05
EP=5,690 | STEP=1,008,264 | clip=24 | consec=2 | max_consec=8 | avg_clip50=23.96
EP=5,700 | STEP=1,010,510 | clip=20 | consec=0 | max_consec=8 | avg_clip50=24.24

custom/	
avg_clip50	25.1

	clipped_episode_r...	32.5	
	consecutive_clipp...	1	
	max_consecutive_c...	8	
	rollout/		
	ep_len_mean	862	
	ep_rew_mean	258	
	exploration rate	0.663	
	time/		
	episodes	17000	
	fps	217	
	time_elapsed	4649	
	total timesteps	1011087	
	train/		
	learning_rate	0.0001	
	loss	0.332	
	n_updates	240271	

EP=5,710		STEP=1,012,946		clip=76		consec=2		max_consec=8		avg_clip50=27.10
EP=5,720		STEP=1,014,806		clip=6		consec=0		max_consec=8		avg_clip50=26.37
EP=5,730		STEP=1,016,551		clip=18		consec=0		max_consec=8		avg_clip50=27.00

	custom/		
	avg_clip50	27.1	
	clipped_episode_r...	12.5	
	consecutive_clipp...	0	
	max_consecutive_c...	8	
	rollout/		
	ep_len_mean	840	
	ep_rew_mean	260	
	exploration rate	0.661	
	time/		
	episodes	17100	
	fps	217	
	time_elapsed	4681	
	total timesteps	1017855	
	train/		
	learning_rate	0.0001	
	loss	0.067	
	n_updates	241963	

EP=5,740		STEP=1,018,665		clip=40		consec=3		max_consec=8		avg_clip50=28.36
EP=5,750		STEP=1,020,194		clip=37		consec=1		max_consec=8		avg_clip50=26.60
EP=5,760		STEP=1,022,175		clip=24		consec=1		max_consec=8		avg_clip50=25.22
EP=5,770		STEP=1,024,040		clip=11		consec=0		max_consec=8		avg_clip50=25.08

	custom/		
	avg_clip50	25.1	
	clipped_episode_r...	11	
	consecutive_clipp...	0	
	max_consecutive_c...	8	
	rollout/		
	ep_len_mean	821	
	ep_rew_mean	257	
	exploration rate	0.659	
	time/		
	episodes	17200	
	fps	217	
	time_elapsed	4710	
	total timesteps	1024184	
	train/		

	learning_rate	0.0001				
	loss	0.081				
	n_updates	243545				

EP=5,780	STEP=1,026,000	clip=22	consec=2	max_consec=8	avg_clip50=25.51	
EP=5,790	STEP=1,028,117	clip=32	consec=2	max_consec=8	avg_clip50=26.22	
EP=5,800	STEP=1,029,781	clip=14	consec=0	max_consec=8	avg_clip50=26.18	

	custom/					
	avg_clip50	24.8				
	clipped_episode_r...	33				
	consecutive_clipp...	1				
	max_consecutive_c...	8				
	rollout/					
	ep_len_mean	805				
	ep_rew_mean	257				
	exploration rate	0.657				
	time/					
	episodes	17300				
	fps	217				
	time_elapsed	4741				
	total timesteps	1030474				
	train/					
	learning_rate	0.0001				
	loss	0.105				
	n_updates	245118				

EP=5,810	STEP=1,031,638	clip=20	consec=0	max_consec=8	avg_clip50=24.71	
EP=5,820	STEP=1,033,676	clip=16	consec=0	max_consec=8	avg_clip50=24.84	
EP=5,830	STEP=1,035,779	clip=32	consec=3	max_consec=8	avg_clip50=25.67	

	custom/					
	avg_clip50	24.7				
	clipped_episode_r...	9.5				
	consecutive_clipp...	0				
	max_consecutive_c...	8				
	rollout/					
	ep_len_mean	818				
	ep_rew_mean	258				
	exploration rate	0.655				
	time/					
	episodes	17400				
	fps	217				
	time_elapsed	4774				
	total timesteps	1037536				
	train/					
	learning_rate	0.0001				
	loss	0.122				
	n_updates	246883				

EP=5,840	STEP=1,037,831	clip=26	consec=1	max_consec=8	avg_clip50=24.46	
EP=5,850	STEP=1,040,041	clip=14	consec=0	max_consec=8	avg_clip50=26.53	
EP=5,860	STEP=1,042,361	clip=81	consec=1	max_consec=8	avg_clip50=28.16	
EP=5,870	STEP=1,044,580	clip=54	consec=2	max_consec=8	avg_clip50=29.38	

	custom/					
	avg_clip50	28.5				
	clipped_episode_r...	31.5				
	consecutive_clipp...	3				
	max_consecutive_c...	8				

rollout/	
ep_len_mean	866
ep_rew_mean	271
exploration rate	0.652
time/	
episodes	17500
fps	217
time_elapsed	4808
total timesteps	1045074
train/	
learning_rate	0.0001
loss	0.0509
n_updates	248768

EP=5,880 | STEP=1,046,406 | clip=7 | consec=0 | max_consec=8 | avg_clip50=28.13
EP=5,890 | STEP=1,048,629 | clip=10 | consec=0 | max_consec=8 | avg_clip50=29.14
EP=5,900 | STEP=1,050,412 | clip=30 | consec=1 | max_consec=8 | avg_clip50=28.09

custom/	
avg_clip50	28.1
clipped_episode_r...	13.5
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	870
ep_rew_mean	276
exploration rate	0.65
time/	
episodes	17600
fps	217
time_elapsed	4838
total timesteps	1051186
train/	
learning_rate	0.0001
loss	0.045
n_updates	250296

EP=5,910 | STEP=1,052,650 | clip=50 | consec=5 | max_consec=8 | avg_clip50=28.21
EP=5,920 | STEP=1,054,592 | clip=18 | consec=0 | max_consec=8 | avg_clip50=26.24
EP=5,930 | STEP=1,056,875 | clip=4 | consec=0 | max_consec=8 | avg_clip50=28.69

custom/	
avg_clip50	27.1
clipped_episode_r...	27
consecutive_clipp...	3
max_consecutive_c...	8
rollout/	
ep_len_mean	885
ep_rew_mean	284
exploration rate	0.648
time/	
episodes	17700
fps	217
time_elapsed	4872
total timesteps	1058540
train/	
learning_rate	0.0001
loss	0.0516
n_updates	252134

EP=5,940	STEP=1,058,807	clip=38	consec=1	max_consec=8	avg_clip50=27.76
EP=5,950	STEP=1,061,008	clip=60	consec=1	max_consec=8	avg_clip50=28.90
EP=5,960	STEP=1,063,614	clip=43	consec=4	max_consec=8	avg_clip50=30.66
EP=5,970	STEP=1,065,871	clip=70	consec=1	max_consec=8	avg_clip50=34.02

custom/	
avg_clip50	33.8
clipped_episode_r...	44.5
consecutive_clipp...	3
max_consecutive_c...	8
rollout/	
ep_len_mean	896
ep_rew_mean	300
exploration rate	0.645
time/	
episodes	17800
fps	217
time_elapsed	4909
total timesteps	1066537
train/	
learning_rate	0.0001
loss	0.0716
n_updates	254134

EP=5,980	STEP=1,068,226	clip=27	consec=2	max_consec=8	avg_clip50=33.18
EP=5,990	STEP=1,070,255	clip=53	consec=1	max_consec=8	avg_clip50=33.26
EP=6,000	STEP=1,072,502	clip=25	consec=6	max_consec=8	avg_clip50=33.03

custom/	
avg_clip50	32.3
clipped_episode_r...	24.5
consecutive_clipp...	12
max_consecutive_c...	12
rollout/	
ep_len_mean	938
ep_rew_mean	314
exploration rate	0.642
time/	
episodes	17900
fps	217
time_elapsed	4943
total timesteps	1073864
train/	
learning_rate	0.0001
loss	0.146
n_updates	255965

EP=6,010	STEP=1,074,775	clip=10	consec=0	max_consec=13	avg_clip50=31.16
EP=6,020	STEP=1,076,830	clip=60	consec=5	max_consec=13	avg_clip50=29.87
EP=6,030	STEP=1,079,065	clip=38	consec=1	max_consec=13	avg_clip50=29.01
EP=6,040	STEP=1,081,408	clip=36	consec=5	max_consec=13	avg_clip50=30.54

custom/	
avg_clip50	30.5
clipped_episode_r...	36
consecutive_clipp...	5
max_consecutive_c...	13
rollout/	
ep_len_mean	948
ep_rew_mean	319

exploration rate	0.64
time/	
episodes	18000
fps	217
time_elapsed	4979
total timesteps	1081544
train/	
learning_rate	0.0001
loss	0.0214
n_updates	257885

EP=6,050	STEP=1,083,969	clip=46	consec=4	max_consec=13	avg_clip50=31.82
EP=6,060	STEP=1,086,274	clip=47	consec=4	max_consec=13	avg_clip50=32.54
EP=6,070	STEP=1,088,723	clip=41	consec=6	max_consec=13	avg_clip50=32.38

custom/	
avg_clip50	32.1
clipped_episode_r...	24.5
consecutive_clipp...	1
max_consecutive_c...	13
rollout/	
ep_len_mean	953
ep_rew_mean	308
exploration rate	0.637
time/	
episodes	18100
fps	217
time_elapsed	5015
total timesteps	1089478
train/	
learning_rate	0.0001
loss	0.0333
n_updates	259869

EP=6,080	STEP=1,090,880	clip=18	consec=0	max_consec=13	avg_clip50=32.67
EP=6,090	STEP=1,093,309	clip=17	consec=0	max_consec=13	avg_clip50=32.72
EP=6,100	STEP=1,095,781	clip=24	consec=4	max_consec=13	avg_clip50=34.16

custom/	
avg_clip50	34.5
clipped_episode_r...	44.5
consecutive_clipp...	1
max_consecutive_c...	13
rollout/	
ep_len_mean	965
ep_rew_mean	330
exploration rate	0.635
time/	
episodes	18200
fps	217
time_elapsed	5052
total timesteps	1097295
train/	
learning_rate	0.0001
loss	0.269
n_updates	261823

EP=6,110	STEP=1,097,910	clip=30	consec=4	max_consec=13	avg_clip50=33.82
EP=6,120	STEP=1,099,545	clip=7	consec=0	max_consec=13	avg_clip50=31.14
EP=6,130	STEP=1,101,464	clip=16	consec=0	max_consec=13	avg_clip50=30.74

EP=6,140 | STEP=1,103,731 | clip=40 | consec=1 | max_consec=13 | avg_clip50=29.84

custom/		
avg_clip50	29.8	
clipped_episode_r...	39.5	
consecutive_clipp...	1	
max_consecutive_c...	13	
rollout/		
ep_len_mean	938	
ep_rew_mean	313	
exploration rate	0.632	
time/		
episodes	18300	
fps	217	
time_elapsed	5083	
total timesteps	1103834	
train/		
learning_rate	0.0001	
loss	0.307	
n_updates	263458	

EP=6,150 | STEP=1,105,430 | clip=12 | consec=0 | max_consec=13 | avg_clip50=25.54

EP=6,160 | STEP=1,107,550 | clip=106 | consec=2 | max_consec=13 | avg_clip50=24.9
1

EP=6,170 | STEP=1,109,694 | clip=40 | consec=1 | max_consec=13 | avg_clip50=26.86

custom/		
avg_clip50	25.7	
clipped_episode_r...	11.5	
consecutive_clipp...	0	
max_consecutive_c...	13	
rollout/		
ep_len_mean	887	
ep_rew_mean	292	
exploration rate	0.63	
time/		
episodes	18400	
fps	217	
time_elapsed	5114	
total timesteps	1110599	
train/		
learning_rate	0.0001	
loss	0.344	
n_updates	265149	

EP=6,180 | STEP=1,111,902 | clip=36 | consec=5 | max_consec=13 | avg_clip50=28.43

EP=6,190 | STEP=1,114,229 | clip=26 | consec=2 | max_consec=13 | avg_clip50=29.62

EP=6,200 | STEP=1,116,329 | clip=24 | consec=1 | max_consec=13 | avg_clip50=30.77

custom/		
avg_clip50	34	
clipped_episode_r...	52.5	
consecutive_clipp...	1	
max_consecutive_c...	13	
rollout/		
ep_len_mean	884	
ep_rew_mean	287	
exploration rate	0.628	
time/		
episodes	18500	

	fps		217	
	time_elapsed		5151	
	total timesteps		1118433	
	train/			
	learning_rate		0.0001	
	loss		0.0678	
	n_updates		267108	

EP=6,210		STEP=1,118,659		clip=8		consec=0		max_consec=13		avg_clip50=31.68
EP=6,220		STEP=1,121,275		clip=70		consec=1		max_consec=13		avg_clip50=34.09
EP=6,230		STEP=1,123,047		clip=8		consec=0		max_consec=13		avg_clip50=31.63
EP=6,240		STEP=1,125,411		clip=26		consec=2		max_consec=13		avg_clip50=31.41

	custom/			
	avg_clip50		31.5	
	clipped_episode_r...		7.5	
	consecutive_clipp...		0	
	max_consecutive_c...		13	
	rollout/			
	ep_len_mean		911	
	ep_rew_mean		304	
	exploration rate		0.625	
	time/			
	episodes		18600	
	fps		217	
	time_elapsed		5184	
	total timesteps		1125637	
	train/			
	learning_rate		0.0001	
	loss		0.0832	
	n_updates		268909	

EP=6,250		STEP=1,127,650		clip=41		consec=2		max_consec=13		avg_clip50=32.68
EP=6,260		STEP=1,129,971		clip=52		consec=2		max_consec=13		avg_clip50=33.01
EP=6,270		STEP=1,131,913		clip=3		consec=0		max_consec=13		avg_clip50=29.57

	custom/			
	avg_clip50		31.1	
	clipped_episode_r...		6	
	consecutive_clipp...		0	
	max_consecutive_c...		13	
	rollout/			
	ep_len_mean		946	
	ep_rew_mean		321	
	exploration rate		0.623	
	time/			
	episodes		18700	
	fps		217	
	time_elapsed		5220	
	total timesteps		1133146	
	train/			
	learning_rate		0.0001	
	loss		0.0352	
	n_updates		270786	

EP=6,280		STEP=1,133,982		clip=24		consec=2		max_consec=13		avg_clip50=30.74
EP=6,290		STEP=1,136,180		clip=5		consec=0		max_consec=13		avg_clip50=30.35
EP=6,300		STEP=1,138,257		clip=14		consec=0		max_consec=13		avg_clip50=30.36

	custom/			
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avg_clip50	30.8
clipped_episode_r...	23.5
consecutive_clipp...	4
max_consecutive_c...	13
rollout/	
ep_len_mean	923
ep_rew_mean	315
exploration rate	0.62
time/	
episodes	18800
fps	217
time_elapsed	5255
total timesteps	1140769
train/	
learning_rate	0.0001
loss	0.0486
n_updates	272692

EP=6,310	STEP=1,140,808	clip=51	consec=5	max_consec=13	avg_clip50=30.80
EP=6,320	STEP=1,143,033	clip=30	consec=1	max_consec=13	avg_clip50=32.89
EP=6,330	STEP=1,145,200	clip=9	consec=0	max_consec=13	avg_clip50=33.46
EP=6,340	STEP=1,147,147	clip=56	consec=2	max_consec=13	avg_clip50=32.73

custom/	
avg_clip50	32.5
clipped_episode_r...	18.5
consecutive_clipp...	0
max_consecutive_c...	13
rollout/	
ep_len_mean	919
ep_rew_mean	316
exploration rate	0.618
time/	
episodes	18900
fps	216
time_elapsed	5290
total timesteps	1147813
train/	
learning_rate	0.0001
loss	0.13
n_updates	274453

EP=6,350	STEP=1,149,217	clip=18	consec=0	max_consec=13	avg_clip50=31.51
EP=6,360	STEP=1,151,373	clip=16	consec=0	max_consec=13	avg_clip50=30.73
EP=6,370	STEP=1,153,599	clip=46	consec=8	max_consec=13	avg_clip50=30.68

custom/	
avg_clip50	28.7
clipped_episode_r...	27.5
consecutive_clipp...	1
max_consecutive_c...	13
rollout/	
ep_len_mean	914
ep_rew_mean	317
exploration rate	0.615
time/	
episodes	19000
fps	216
time_elapsed	5328
total timesteps	1155293

train/						
learning_rate		0.0001				
loss		0.224				
n_updates		276323				

EP=6,380	STEP=1,155,767	clip=28	consec=4	max_consec=13	avg_clip50=30.12	
EP=6,390	STEP=1,158,267	clip=56	consec=2	max_consec=13	avg_clip50=32.18	
EP=6,400	STEP=1,160,370	clip=12	consec=0	max_consec=13	avg_clip50=32.41	
EP=6,410	STEP=1,162,518	clip=8	consec=0	max_consec=13	avg_clip50=31.66	

custom/						
avg_clip50		32.1				
clipped_episode_r...		28.5				
consecutive_clipp...		2				
max_consecutive_c...		13				
rollout/						
ep_len_mean		914				
ep_rew_mean		311				
exploration rate		0.613				
time/						
episodes		19100				
fps		216				
time_elapsed		5369				
total timesteps		1162962				
train/						
learning_rate		0.0001				
loss		0.0656				
n_updates		278240				

EP=6,420	STEP=1,164,706	clip=27	consec=1	max_consec=13	avg_clip50=30.88	
EP=6,430	STEP=1,166,954	clip=38	consec=7	max_consec=13	avg_clip50=31.19	
EP=6,440	STEP=1,168,942	clip=32	consec=1	max_consec=13	avg_clip50=29.19	

custom/						
avg_clip50		29				
clipped_episode_r...		27.5				
consecutive_clipp...		2				
max_consecutive_c...		13				
rollout/						
ep_len_mean		915				
ep_rew_mean		309				
exploration rate		0.61				
time/						
episodes		19200				
fps		216				
time_elapsed		5407				
total timesteps		1170281				
train/						
learning_rate		0.0001				
loss		0.0346				
n_updates		280070				

EP=6,450	STEP=1,170,857	clip=4	consec=0	max_consec=13	avg_clip50=28.99	
EP=6,460	STEP=1,173,353	clip=36	consec=2	max_consec=13	avg_clip50=30.90	
EP=6,470	STEP=1,175,315	clip=22	consec=1	max_consec=13	avg_clip50=31.03	
EP=6,480	STEP=1,177,515	clip=14	consec=0	max_consec=13	avg_clip50=30.58	

custom/						
avg_clip50		30.7				
clipped_episode_r...		16				

consecutive_clipp...	0
max_consecutive_c...	13
rollout/	
ep_len_mean	914
ep_rew_mean	307
exploration rate	0.608
time/	
episodes	19300
fps	216
time_elapsed	5444
total timesteps	1177696
train/	
learning_rate	0.0001
loss	0.122
n_updates	281923

EP=6,490 | STEP=1,179,858 | clip=54 | consec=5 | max_consec=13 | avg_clip50=31.89
EP=6,500 | STEP=1,182,209 | clip=42 | consec=8 | max_consec=13 | avg_clip50=34.10
EP=6,510 | STEP=1,184,279 | clip=34 | consec=1 | max_consec=13 | avg_clip50=31.85

custom/	
avg_clip50	31.7
clipped_episode_r...	44.5
consecutive_clipp...	1
max_consecutive_c...	13
rollout/	
ep_len_mean	926
ep_rew_mean	320
exploration rate	0.605
time/	
episodes	19400
fps	216
time_elapsed	5485
total timesteps	1185618
train/	
learning_rate	0.0001
loss	0.13
n_updates	283904

EP=6,520 | STEP=1,186,674 | clip=64 | consec=1 | max_consec=13 | avg_clip50=33.31
EP=6,530 | STEP=1,188,809 | clip=36 | consec=11 | max_consec=13 | avg_clip50=34.9
2
EP=6,540 | STEP=1,190,844 | clip=16 | consec=0 | max_consec=16 | avg_clip50=34.18

custom/	
avg_clip50	34.1
clipped_episode_r...	32.5
consecutive_clipp...	2
max_consecutive_c...	16
rollout/	
ep_len_mean	928
ep_rew_mean	338
exploration rate	0.603
time/	
episodes	19500
fps	215
time_elapsed	5524
total timesteps	1193055
train/	
learning_rate	0.0001

	loss	0.112				
	n_updates	285763				

EP=6,550	STEP=1,193,133	clip=54	consec=3	max_consec=16	avg_clip50=34.38	
EP=6,560	STEP=1,195,186	clip=12	consec=0	max_consec=16	avg_clip50=34.35	
EP=6,570	STEP=1,197,541	clip=30	consec=4	max_consec=16	avg_clip50=33.95	
EP=6,580	STEP=1,200,068	clip=32	consec=9	max_consec=16	avg_clip50=35.35	

	custom/					
	avg_clip50	34.5				
	clipped_episode_r...	21.5				
	consecutive_clipp...	2				
	max_consecutive_c...	16				
	rollout/					
	ep_len_mean	942				
	ep_rew_mean	349				
	exploration rate	0.6				
	time/					
	episodes	19600				
	fps	215				
	time_elapsed	5562				
	total timesteps	1200689				
	train/					
	learning_rate	0.0001				
	loss	0.0911				
	n_updates	287672				

EP=6,590	STEP=1,202,110	clip=24	consec=9	max_consec=16	avg_clip50=34.56	
EP=6,600	STEP=1,204,477	clip=33	consec=1	max_consec=16	avg_clip50=33.49	
EP=6,610	STEP=1,206,367	clip=56	consec=1	max_consec=16	avg_clip50=34.04	

	custom/					
	avg_clip50	35.2				
	clipped_episode_r...	50				
	consecutive_clipp...	3				
	max_consecutive_c...	16				
	rollout/					
	ep_len_mean	930				
	ep_rew_mean	340				
	exploration rate	0.598				
	time/					
	episodes	19700				
	fps	215				
	time_elapsed	5600				
	total timesteps	1208119				
	train/					
	learning_rate	0.0001				
	loss	0.0824				
	n_updates	289529				

EP=6,620	STEP=1,208,905	clip=34	consec=2	max_consec=16	avg_clip50=33.95	
EP=6,630	STEP=1,211,234	clip=55	consec=4	max_consec=16	avg_clip50=32.47	
EP=6,640	STEP=1,213,621	clip=44	consec=8	max_consec=16	avg_clip50=35.13	
EP=6,650	STEP=1,215,652	clip=12	consec=0	max_consec=16	avg_clip50=34.22	

	custom/					
	avg_clip50	34.2				
	clipped_episode_r...	12.5				
	consecutive_clipp...	0				
	max_consecutive_c...	16				

rollout/	
ep_len_mean	946
ep_rew_mean	339
exploration rate	0.595
time/	
episodes	19800
fps	215
time_elapsed	5639
total timesteps	1215829
train/	
learning_rate	0.0001
loss	0.0539
n_updates	291457

EP=6,660	STEP=1,218,264	clip=45	consec=2	max_consec=16	avg_clip50=37.09
EP=6,670	STEP=1,220,446	clip=49	consec=6	max_consec=16	avg_clip50=36.39
EP=6,680	STEP=1,222,452	clip=18	consec=0	max_consec=16	avg_clip50=34.58

custom/	
avg_clip50	34.6
clipped_episode_r...	15.5
consecutive_clipp...	0
max_consecutive_c...	16
rollout/	
ep_len_mean	943
ep_rew_mean	338
exploration rate	0.593
time/	
episodes	19900
fps	215
time_elapsed	5676
total timesteps	1223277
train/	
learning_rate	0.0001
loss	0.0259
n_updates	293319

EP=6,690	STEP=1,224,641	clip=54	consec=1	max_consec=16	avg_clip50=33.12
EP=6,700	STEP=1,226,677	clip=46	consec=3	max_consec=16	avg_clip50=34.01
EP=6,710	STEP=1,228,714	clip=28	consec=1	max_consec=16	avg_clip50=30.40

custom/	
avg_clip50	29.8
clipped_episode_r...	19.5
consecutive_clipp...	0
max_consecutive_c...	16
rollout/	
ep_len_mean	921
ep_rew_mean	327
exploration rate	0.59
time/	
episodes	20000
fps	215
time_elapsed	5712
total timesteps	1230250
train/	
learning_rate	0.0001
loss	0.0297
n_updates	295062

EP=6,720	STEP=1,230,951	clip=16	consec=0	max_consec=16	avg_clip50=30.32
EP=6,730	STEP=1,233,365	clip=26	consec=3	max_consec=16	avg_clip50=31.98
EP=6,740	STEP=1,235,414	clip=44	consec=1	max_consec=16	avg_clip50=31.29
EP=6,750	STEP=1,237,477	clip=32	consec=1	max_consec=16	avg_clip50=30.35

custom/	
avg_clip50	30.2
clipped_episode_r...	11
consecutive_clipp...	0
max_consecutive_c...	16
rollout/	
ep_len_mean	914
ep_rew_mean	317
exploration rate	0.588
time/	
episodes	20100
fps	215
time_elapsed	5750
total timesteps	1237801
train/	
learning_rate	0.0001
loss	0.0628
n_updates	296950

EP=6,760	STEP=1,239,821	clip=48	consec=2	max_consec=16	avg_clip50=31.72
EP=6,770	STEP=1,242,231	clip=17	consec=0	max_consec=16	avg_clip50=32.56
EP=6,780	STEP=1,244,464	clip=30	consec=2	max_consec=16	avg_clip50=31.63

custom/	
avg_clip50	31
clipped_episode_r...	60.5
consecutive_clipp...	1
max_consecutive_c...	16
rollout/	
ep_len_mean	927
ep_rew_mean	319
exploration rate	0.585
time/	
episodes	20200
fps	215
time_elapsed	5791
total timesteps	1245539
train/	
learning_rate	0.0001
loss	0.0297
n_updates	298884

EP=6,790	STEP=1,246,601	clip=49	consec=2	max_consec=16	avg_clip50=31.34
EP=6,800	STEP=1,249,430	clip=32	consec=3	max_consec=16	avg_clip50=34.54
[checkpoint periodico] guardado a 1,250,000 steps					
EP=6,810	STEP=1,251,506	clip=16	consec=0	max_consec=16	avg_clip50=33.87

custom/	
avg_clip50	33.5
clipped_episode_r...	21.5
consecutive_clipp...	2
max_consecutive_c...	16
rollout/	
ep_len_mean	964
ep_rew_mean	331

exploration rate	0.583
time/	
episodes	20300
fps	214
time_elapsed	5834
total timesteps	1253512
train/	
learning_rate	0.0001
loss	0.0737
n_updates	300877

EP=6,820	STEP=1,253,690	clip=18	consec=0	max_consec=16	avg_clip50=32.75
EP=6,830	STEP=1,256,467	clip=84	consec=1	max_consec=16	avg_clip50=36.33
EP=6,840	STEP=1,258,501	clip=68	consec=2	max_consec=16	avg_clip50=36.65
EP=6,850	STEP=1,261,076	clip=44	consec=7	max_consec=16	avg_clip50=36.09

custom/	
avg_clip50	35.9
clipped_episode_r...	39.5
consecutive_clipp...	1
max_consecutive_c...	16
rollout/	
ep_len_mean	987
ep_rew_mean	350
exploration rate	0.58
time/	
episodes	20400
fps	214
time_elapsed	5875
total timesteps	1261666
train/	
learning_rate	0.0001
loss	0.0467
n_updates	302916

EP=6,860	STEP=1,263,159	clip=34	consec=1	max_consec=16	avg_clip50=35.85
EP=6,870	STEP=1,265,036	clip=22	consec=2	max_consec=16	avg_clip50=35.09
EP=6,880	STEP=1,267,185	clip=20	consec=0	max_consec=16	avg_clip50=31.93

custom/	
avg_clip50	31.3
clipped_episode_r...	12.5
consecutive_clipp...	0
max_consecutive_c...	16
rollout/	
ep_len_mean	943
ep_rew_mean	336
exploration rate	0.578
time/	
episodes	20500
fps	214
time_elapsed	5908
total timesteps	1268340
train/	
learning_rate	0.0001
loss	0.0336
n_updates	304584

EP=6,890	STEP=1,269,027	clip=32	consec=3	max_consec=16	avg_clip50=30.99
EP=6,900	STEP=1,271,171	clip=10	consec=0	max_consec=16	avg_clip50=30.06

EP=6,910	STEP=1,273,311	clip=27	consec=3	max_consec=16	avg_clip50=29.23
EP=6,920	STEP=1,275,578	clip=54	consec=3	max_consec=16	avg_clip50=31.74

custom/	
avg_clip50	32.4
clipped_episode_r...	53
consecutive_clipp...	4
max_consecutive_c...	16
rollout/	
ep_len_mean	923
ep_rew_mean	336
exploration rate	0.575
time/	
episodes	20600
fps	214
time_elapsed	5946
total timesteps	1275965
train/	
learning_rate	0.0001
loss	0.0604
n_updates	306491

EP=6,930	STEP=1,277,526	clip=32	consec=1	max_consec=16	avg_clip50=30.92
EP=6,940	STEP=1,279,656	clip=14	consec=0	max_consec=16	avg_clip50=31.50
EP=6,950	STEP=1,281,639	clip=11	consec=0	max_consec=16	avg_clip50=29.86

custom/	
avg_clip50	30.7
clipped_episode_r...	47
consecutive_clipp...	5
max_consecutive_c...	16
rollout/	
ep_len_mean	869
ep_rew_mean	302
exploration rate	0.573
time/	
episodes	20700
fps	214
time_elapsed	5980
total timesteps	1282720
train/	
learning_rate	0.0001
loss	0.0711
n_updates	308179

EP=6,960	STEP=1,284,004	clip=54	consec=3	max_consec=16	avg_clip50=32.15
EP=6,970	STEP=1,286,001	clip=44	consec=3	max_consec=16	avg_clip50=31.62
EP=6,980	STEP=1,288,193	clip=12	consec=0	max_consec=16	avg_clip50=33.25

custom/	
avg_clip50	34.8
clipped_episode_r...	43.5
consecutive_clipp...	1
max_consecutive_c...	16
rollout/	
ep_len_mean	912
ep_rew_mean	333
exploration rate	0.57
time/	
episodes	20800

fps	214
time_elapsed	6019
total timesteps	1290475
train/	
learning_rate	0.0001
loss	0.102
n_updates	310118

EP=6,990	STEP=1,290,661	clip=25	consec=2	max_consec=16	avg_clip50=35.03
EP=7,000	STEP=1,292,471	clip=14	consec=0	max_consec=16	avg_clip50=34.42
EP=7,010	STEP=1,294,805	clip=11	consec=0	max_consec=16	avg_clip50=33.48
EP=7,020	STEP=1,297,246	clip=36	consec=6	max_consec=16	avg_clip50=33.33

custom/	
avg_clip50	32.6
clipped_episode_r...	16.5
consecutive_clipp...	0
max_consecutive_c...	16
rollout/	
ep_len_mean	909
ep_rew_mean	321
exploration rate	0.568
time/	
episodes	20900
fps	214
time_elapsed	6056
total timesteps	1297707
train/	
learning_rate	0.0001
loss	0.112
n_updates	311926

EP=7,030	STEP=1,299,220	clip=10	consec=0	max_consec=16	avg_clip50=31.63
EP=7,040	STEP=1,301,556	clip=14	consec=0	max_consec=16	avg_clip50=31.23
EP=7,050	STEP=1,303,744	clip=41	consec=1	max_consec=16	avg_clip50=33.33

custom/	
avg_clip50	32.8
clipped_episode_r...	30.5
consecutive_clipp...	6
max_consecutive_c...	16
rollout/	
ep_len_mean	927
ep_rew_mean	338
exploration rate	0.565
time/	
episodes	21000
fps	214
time_elapsed	6094
total timesteps	1305217
train/	
learning_rate	0.0001
loss	0.0463
n_updates	313804

EP=7,060	STEP=1,305,870	clip=32	consec=9	max_consec=16	avg_clip50=32.96
EP=7,070	STEP=1,307,519	clip=8	consec=0	max_consec=16	avg_clip50=30.30
EP=7,080	STEP=1,309,492	clip=8	consec=0	max_consec=16	avg_clip50=29.71
EP=7,090	STEP=1,311,613	clip=30	consec=1	max_consec=16	avg_clip50=28.71

custom/	
avg_clip50	27.5
clipped_episode_r...	33.5
consecutive_clipp...	3
max_consecutive_c...	16
rollout/	
ep_len_mean	884
ep_rew_mean	301
exploration rate	0.563
time/	
episodes	21100
fps	214
time_elapsed	6130
total timesteps	1312075
train/	
learning_rate	0.0001
loss	0.0559
n_updates	315518

EP=7,100	STEP=1,313,804	clip=32	consec=1	max_consec=16	avg_clip50=26.98
EP=7,110	STEP=1,315,673	clip=35	consec=1	max_consec=16	avg_clip50=26.69
EP=7,120	STEP=1,318,185	clip=74	consec=8	max_consec=16	avg_clip50=30.08

custom/	
avg_clip50	32.2
clipped_episode_r...	51
consecutive_clipp...	2
max_consecutive_c...	16
rollout/	
ep_len_mean	890
ep_rew_mean	309
exploration rate	0.561
time/	
episodes	21200
fps	213
time_elapsed	6169
total timesteps	1319734
train/	
learning_rate	0.0001
loss	0.0504
n_updates	317433

EP=7,130	STEP=1,320,681	clip=22	consec=6	max_consec=16	avg_clip50=33.22
EP=7,140	STEP=1,322,936	clip=24	consec=7	max_consec=16	avg_clip50=34.78
EP=7,150	STEP=1,325,666	clip=41	consec=1	max_consec=16	avg_clip50=38.11
EP=7,160	STEP=1,327,370	clip=21	consec=1	max_consec=16	avg_clip50=36.64

custom/	
avg_clip50	36.6
clipped_episode_r...	21
consecutive_clipp...	1
max_consecutive_c...	16
rollout/	
ep_len_mean	902
ep_rew_mean	317
exploration rate	0.558
time/	
episodes	21300
fps	213
time_elapsed	6207

total timesteps	1327370
train/	
learning_rate	0.0001
loss	0.397
n_updates	319342

EP=7,170	STEP=1,329,640	clip=7	consec=0	max_consec=16	avg_clip50=35.17
EP=7,180	STEP=1,331,886	clip=27	consec=1	max_consec=16	avg_clip50=33.26
EP=7,190	STEP=1,334,387	clip=11	consec=0	max_consec=16	avg_clip50=34.02

custom/	
avg_clip50	34.5
clipped_episode_r...	100
consecutive_clipp...	2
max_consecutive_c...	16
rollout/	
ep_len_mean	965
ep_rew_mean	356
exploration rate	0.555
time/	
episodes	21400
fps	213
time_elapsed	6252
total timesteps	1335830
train/	
learning_rate	0.0001
loss	0.304
n_updates	321457

EP=7,200	STEP=1,337,010	clip=42	consec=3	max_consec=16	avg_clip50=34.06
EP=7,210	STEP=1,339,295	clip=50	consec=3	max_consec=16	avg_clip50=36.66
EP=7,220	STEP=1,341,719	clip=40	consec=6	max_consec=16	avg_clip50=38.13

custom/	
avg_clip50	37.3
clipped_episode_r...	33.5
consecutive_clipp...	1
max_consecutive_c...	16
rollout/	
ep_len_mean	973
ep_rew_mean	357
exploration rate	0.553
time/	
episodes	21500
fps	213
time_elapsed	6292
total timesteps	1343436
train/	
learning_rate	0.0001
loss	0.169
n_updates	323358

EP=7,230	STEP=1,343,817	clip=50	consec=1	max_consec=16	avg_clip50=37.85
EP=7,240	STEP=1,346,488	clip=50	consec=4	max_consec=16	avg_clip50=38.51
EP=7,250	STEP=1,348,718	clip=30	consec=2	max_consec=16	avg_clip50=36.63
EP=7,260	STEP=1,351,036	clip=29	consec=5	max_consec=16	avg_clip50=37.86

custom/	
avg_clip50	38.4
clipped_episode_r...	52

consecutive_clipp...	2
max_consecutive_c...	16
rollout/	
ep_len_mean	988
ep_rew_mean	373
exploration rate	0.55
time/	
episodes	21600
fps	213
time_elapsed	6336
total timesteps	1351883
train/	
learning_rate	0.0001
loss	0.0461
n_updates	325470

EP=7,270	STEP=1,353,221	clip=50	consec=1	max_consec=16	avg_clip50=37.30
EP=7,280	STEP=1,355,655	clip=52	consec=4	max_consec=16	avg_clip50=39.37
EP=7,290	STEP=1,357,725	clip=33	consec=1	max_consec=16	avg_clip50=36.96

custom/	
avg_clip50	37.2
clipped_episode_r...	26
consecutive_clipp...	8
max_consecutive_c...	16
rollout/	
ep_len_mean	971
ep_rew_mean	372
exploration rate	0.547
time/	
episodes	21700
fps	213
time_elapsed	6376
total timesteps	1359578
train/	
learning_rate	0.0001
loss	0.0998
n_updates	327394

EP=7,300	STEP=1,360,159	clip=63	consec=11	max_consec=16	avg_clip50=38.13
EP=7,310	STEP=1,362,378	clip=41	consec=8	max_consec=16	avg_clip50=37.41
EP=7,320	STEP=1,364,701	clip=57	consec=3	max_consec=16	avg_clip50=37.68
EP=7,330	STEP=1,367,384	clip=31	consec=13	max_consec=16	avg_clip50=38.38

custom/	
avg_clip50	38.4
clipped_episode_r...	22
consecutive_clipp...	14
max_consecutive_c...	16
rollout/	
ep_len_mean	985
ep_rew_mean	386
exploration rate	0.545
time/	
episodes	21800
fps	213
time_elapsed	6417
total timesteps	1367582

train/	
learning_rate	0.0001
loss	0.0508
n_updates	329395

EP=7,340 | STEP=1,370,016 | clip=52 | consec=23 | max_consec=23 | avg_clip50=40.32

EP=7,350 | STEP=1,372,473 | clip=17 | consec=0 | max_consec=28 | avg_clip50=39.83

EP=7,360 | STEP=1,374,390 | clip=10 | consec=0 | max_consec=28 | avg_clip50=38.34

custom/	
avg_clip50	39.5
clipped_episode_r...	68.5
consecutive_clipp...	1
max_consecutive_c...	28
rollout/	
ep_len_mean	985
ep_rew_mean	381
exploration rate	0.542
time/	
episodes	21900
fps	213
time_elapsed	6459
total timesteps	1375874
train/	
learning_rate	0.0001
loss	0.319
n_updates	331468

EP=7,370 | STEP=1,377,161 | clip=80 | consec=3 | max_consec=28 | avg_clip50=39.86

EP=7,380 | STEP=1,379,674 | clip=36 | consec=5 | max_consec=28 | avg_clip50=39.45

EP=7,390 | STEP=1,381,593 | clip=22 | consec=3 | max_consec=28 | avg_clip50=37.15

custom/	
avg_clip50	36
clipped_episode_r...	45.5
consecutive_clipp...	6
max_consecutive_c...	28
rollout/	
ep_len_mean	991
ep_rew_mean	383
exploration rate	0.539
time/	
episodes	22000
fps	212
time_elapsed	6498
total timesteps	1383656
train/	
learning_rate	0.0001
loss	0.0636
n_updates	333413

EP=7,400 | STEP=1,383,964 | clip=42 | consec=8 | max_consec=28 | avg_clip50=37.40

EP=7,410 | STEP=1,386,078 | clip=42 | consec=1 | max_consec=28 | avg_clip50=38.26

EP=7,420 | STEP=1,388,085 | clip=46 | consec=1 | max_consec=28 | avg_clip50=35.90

EP=7,430 | STEP=1,390,335 | clip=12 | consec=0 | max_consec=28 | avg_clip50=34.52

custom/	
avg_clip50	35.5
clipped_episode_r...	77.5

consecutive_clipp...	2
max_consecutive_c...	28
rollout/	
ep_len_mean	971
ep_rew_mean	376
exploration rate	0.537
time/	
episodes	22100
fps	212
time_elapsed	6537
total timesteps	1391201
train/	
learning_rate	0.0001
loss	0.0431
n_updates	335300

EP=7,440 | STEP=1,392,754 | clip=26 | consec=6 | max_consec=28 | avg_clip50=36.53
EP=7,450 | STEP=1,395,018 | clip=40 | consec=5 | max_consec=28 | avg_clip50=36.39
EP=7,460 | STEP=1,397,445 | clip=24 | consec=1 | max_consec=28 | avg_clip50=36.64

custom/	
avg_clip50	38.2
clipped_episode_r...	28
consecutive_clipp...	7
max_consecutive_c...	28
rollout/	
ep_len_mean	957
ep_rew_mean	374
exploration rate	0.534
time/	
episodes	22200
fps	212
time_elapsed	6577
total timesteps	1398969
train/	
learning_rate	0.0001
loss	0.0354
n_updates	337242

EP=7,470 | STEP=1,399,966 | clip=24 | consec=11 | max_consec=28 | avg_clip50=38.84
EP=7,480 | STEP=1,402,393 | clip=48 | consec=3 | max_consec=28 | avg_clip50=39.92
EP=7,490 | STEP=1,404,729 | clip=60 | consec=2 | max_consec=28 | avg_clip50=38.36
EP=7,500 | STEP=1,406,758 | clip=5 | consec=0 | max_consec=28 | avg_clip50=37.79

custom/	
avg_clip50	37.8
clipped_episode_r...	5
consecutive_clipp...	0
max_consecutive_c...	28
rollout/	
ep_len_mean	956
ep_rew_mean	371
exploration rate	0.532
time/	
episodes	22300
fps	212
time_elapsed	6617
total timesteps	1406758
train/	

learning_rate	0.0001
loss	0.0278
n_updates	339189

EP=7,510	STEP=1,409,273	clip=74	consec=3	max_consec=28	avg_clip50=39.63
EP=7,520	STEP=1,411,393	clip=36	consec=6	max_consec=28	avg_clip50=36.58
EP=7,530	STEP=1,413,700	clip=53	consec=5	max_consec=28	avg_clip50=35.84

custom/	
avg_clip50	35.3
clipped_episode_r...	27
consecutive_clipp...	8
max_consecutive_c...	28
rollout/	
ep_len_mean	968
ep_rew_mean	367
exploration rate	0.529
time/	
episodes	22400
fps	212
time_elapsed	6656
total timesteps	1414659
train/	
learning_rate	0.0001
loss	0.033
n_updates	341164

EP=7,540	STEP=1,416,038	clip=16	consec=0	max_consec=28	avg_clip50=35.78
EP=7,550	STEP=1,418,444	clip=27	consec=1	max_consec=28	avg_clip50=36.80
EP=7,560	STEP=1,421,061	clip=28	consec=7	max_consec=28	avg_clip50=35.47

custom/	
avg_clip50	36
clipped_episode_r...	22.5
consecutive_clipp...	1
max_consecutive_c...	28
rollout/	
ep_len_mean	979
ep_rew_mean	364
exploration rate	0.526
time/	
episodes	22500
fps	212
time_elapsed	6697
total timesteps	1422700
train/	
learning_rate	0.0001
loss	0.0583
n_updates	343174

EP=7,570	STEP=1,423,196	clip=22	consec=2	max_consec=28	avg_clip50=35.51
EP=7,580	STEP=1,425,519	clip=41	consec=3	max_consec=28	avg_clip50=35.46
EP=7,590	STEP=1,427,952	clip=44	consec=2	max_consec=28	avg_clip50=36.20
EP=7,600	STEP=1,430,193	clip=16	consec=0	max_consec=28	avg_clip50=35.93

custom/	
avg_clip50	35.6
clipped_episode_r...	46
consecutive_clipp...	2
max_consecutive_c...	28

rollout/	
ep_len_mean	982
ep_rew_mean	364
exploration rate	0.524
time/	
episodes	22600
fps	212
time_elapsed	6739
total timesteps	1430746
train/	
learning_rate	0.0001
loss	0.0629
n_updates	345186

EP=7,610	STEP=1,432,471	clip=32	consec=1	max_consec=28	avg_clip50=35.37
EP=7,620	STEP=1,435,117	clip=52	consec=6	max_consec=28	avg_clip50=37.91
EP=7,630	STEP=1,437,383	clip=58	consec=2	max_consec=28	avg_clip50=38.80

custom/	
avg_clip50	38.7
clipped_episode_r...	49
consecutive_clipp...	1
max_consecutive_c...	28
rollout/	
ep_len_mean	992
ep_rew_mean	375
exploration rate	0.521
time/	
episodes	22700
fps	212
time_elapsed	6781
total timesteps	1438895
train/	
learning_rate	0.0001
loss	0.211
n_updates	347223

EP=7,640	STEP=1,439,660	clip=34	consec=5	max_consec=28	avg_clip50=38.73
EP=7,650	STEP=1,442,400	clip=54	consec=15	max_consec=28	avg_clip50=40.00
EP=7,660	STEP=1,445,182	clip=18	consec=0	max_consec=28	avg_clip50=41.97

custom/	
avg_clip50	40.8
clipped_episode_r...	57.5
consecutive_clipp...	4
max_consecutive_c...	28
rollout/	
ep_len_mean	1.01e+03
ep_rew_mean	391
exploration rate	0.518
time/	
episodes	22800
fps	212
time_elapsed	6823
total timesteps	1447202
train/	
learning_rate	0.0001
loss	0.0375
n_updates	349300

EP=7,670 | STEP=1,447,312 | clip=18 | consec=0 | max_consec=28 | avg_clip50=40.13
EP=7,680 | STEP=1,449,728 | clip=52 | consec=3 | max_consec=28 | avg_clip50=39.35
EP=7,690 | STEP=1,452,532 | clip=50 | consec=13 | max_consec=28 | avg_clip50=41.0
4
EP=7,700 | STEP=1,454,876 | clip=34 | consec=5 | max_consec=28 | avg_clip50=38.77

custom/	
avg_clip50	38.3
clipped_episode_r...	57
consecutive_clipp...	9
max_consecutive_c...	28
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	398
exploration rate	0.515
time/	
episodes	22900
fps	212
time_elapsed	6867
total timesteps	1456020
train/	
learning_rate	0.0001
loss	0.474
n_updates	351504

EP=7,710 | STEP=1,457,136 | clip=10 | consec=0 | max_consec=28 | avg_clip50=37.42
EP=7,720 | STEP=1,459,549 | clip=36 | consec=1 | max_consec=28 | avg_clip50=38.25
EP=7,730 | STEP=1,461,971 | clip=22 | consec=2 | max_consec=28 | avg_clip50=38.74

custom/	
avg_clip50	38.1
clipped_episode_r...	47
consecutive_clipp...	3
max_consecutive_c...	28
rollout/	
ep_len_mean	1.02e+03
ep_rew_mean	392
exploration rate	0.513
time/	
episodes	23000
fps	211
time_elapsed	6907
total timesteps	1463886
train/	
learning_rate	0.0001
loss	0.0521
n_updates	353471

EP=7,740 | STEP=1,464,151 | clip=31 | consec=5 | max_consec=28 | avg_clip50=37.63
EP=7,750 | STEP=1,466,539 | clip=54 | consec=3 | max_consec=28 | avg_clip50=38.39
EP=7,760 | STEP=1,468,802 | clip=22 | consec=13 | max_consec=28 | avg_clip50=38.4
3
EP=7,770 | STEP=1,470,776 | clip=46 | consec=2 | max_consec=28 | avg_clip50=36.61

custom/	
avg_clip50	36
clipped_episode_r...	38.5
consecutive_clipp...	5
max_consecutive_c...	28

rollout/	
ep_len_mean	984
ep_rew_mean	376
exploration rate	0.51
time/	
episodes	23100
fps	211
time_elapsed	6947
total timesteps	1471650
train/	
learning_rate	0.0001
loss	0.0809
n_updates	355412

EP=7,780 | STEP=1,473,072 | clip=12 | consec=0 | max_consec=28 | avg_clip50=35.62
EP=7,790 | STEP=1,475,729 | clip=64 | consec=10 | max_consec=28 | avg_clip50=38.0
5
EP=7,800 | STEP=1,478,309 | clip=39 | consec=6 | max_consec=28 | avg_clip50=37.90

custom/	
avg_clip50	37.6
clipped_episode_r...	26.5
consecutive_clipp...	4
max_consecutive_c...	28
rollout/	
ep_len_mean	985
ep_rew_mean	380
exploration rate	0.507
time/	
episodes	23200
fps	211
time_elapsed	6990
total timesteps	1480224
train/	
learning_rate	0.0001
loss	0.126
n_updates	357555

EP=7,810 | STEP=1,480,646 | clip=24 | consec=1 | max_consec=28 | avg_clip50=37.48
EP=7,820 | STEP=1,483,268 | clip=38 | consec=11 | max_consec=28 | avg_clip50=40.6
6
EP=7,830 | STEP=1,485,479 | clip=16 | consec=0 | max_consec=28 | avg_clip50=40.40
EP=7,840 | STEP=1,487,702 | clip=12 | consec=0 | max_consec=28 | avg_clip50=38.19

custom/	
avg_clip50	37.7
clipped_episode_r...	31
consecutive_clipp...	1
max_consecutive_c...	28
rollout/	
ep_len_mean	982
ep_rew_mean	379
exploration rate	0.504
time/	
episodes	23300
fps	211
time_elapsed	7031
total timesteps	1488187
train/	
learning_rate	0.0001

loss	0.306
n_updates	359546

EP=7,850	STEP=1,490,050	clip=26	consec=5	max_consec=28	avg_clip50=38.39
EP=7,860	STEP=1,492,283	clip=78	consec=5	max_consec=28	avg_clip50=39.48
EP=7,870	STEP=1,494,286	clip=6	consec=0	max_consec=28	avg_clip50=37.34

custom/	
avg_clip50	36.6
clipped_episode_r...	22
consecutive_clipp...	3
max_consecutive_c...	28
rollout/	
ep_len_mean	971
ep_rew_mean	387
exploration rate	0.502
time/	
episodes	23400
fps	211
time_elapsed	7068
total timesteps	1495369
train/	
learning_rate	0.0001
loss	0.0567
n_updates	361342

EP=7,880	STEP=1,496,321	clip=31	consec=4	max_consec=28	avg_clip50=36.60
EP=7,890	STEP=1,498,028	clip=8	consec=0	max_consec=28	avg_clip50=32.97

[checkpoint periodico] guardado a 1,500,000 steps

EP=7,900	STEP=1,500,563	clip=92	consec=2	max_consec=28	avg_clip50=35.64
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custom/	
avg_clip50	35.2
clipped_episode_r...	29.5
consecutive_clipp...	11
max_consecutive_c...	28
rollout/	
ep_len_mean	927
ep_rew_mean	368
exploration rate	0.5
time/	
episodes	23500
fps	211
time_elapsed	7106
total timesteps	1502559
train/	
learning_rate	0.0001
loss	0.0565
n_updates	363139

EP=7,910	STEP=1,502,799	clip=66	consec=12	max_consec=28	avg_clip50=34.99
EP=7,920	STEP=1,505,037	clip=9	consec=0	max_consec=28	avg_clip50=37.07
EP=7,930	STEP=1,507,361	clip=92	consec=4	max_consec=28	avg_clip50=39.42
EP=7,940	STEP=1,509,180	clip=12	consec=0	max_consec=28	avg_clip50=39.37

custom/	
avg_clip50	38
clipped_episode_r...	54
consecutive_clipp...	2

	max_consecutive_c...	28	
	rollout/		
	ep_len_mean	914	
	ep_rew_mean	365	
	exploration rate	0.497	
	time/		
	episodes	23600	
	fps	211	
	time_elapsed	7144	
	total timesteps	1510343	
	train/		
	learning_rate	0.0001	
	loss	0.0569	
	n_updates	365085	

EP=7,950		STEP=1,511,515		clip=50		consec=4		max_consec=28		avg_clip50=36.40
EP=7,960		STEP=1,513,324		clip=12		consec=0		max_consec=28		avg_clip50=34.10
EP=7,970		STEP=1,515,116		clip=44		consec=2		max_consec=28		avg_clip50=31.61

	custom/		
	avg_clip50	32.4	
	clipped_episode_r...	48.5	
	consecutive_clipp...	6	
	max_consecutive_c...	28	
	rollout/		
	ep_len_mean	893	
	ep_rew_mean	353	
	exploration rate	0.495	
	time/		
	episodes	23700	
	fps	211	
	time_elapsed	7180	
	total timesteps	1517252	
	train/		
	learning_rate	0.0001	
	loss	0.301	
	n_updates	366812	

EP=7,980		STEP=1,517,637		clip=70		consec=8		max_consec=28		avg_clip50=31.93
EP=7,990		STEP=1,520,265		clip=41		consec=1		max_consec=28		avg_clip50=36.28
EP=8,000		STEP=1,522,719		clip=20		consec=1		max_consec=28		avg_clip50=36.90
EP=8,010		STEP=1,525,005		clip=45		consec=4		max_consec=28		avg_clip50=39.11

	custom/		
	avg_clip50	39	
	clipped_episode_r...	47.5	
	consecutive_clipp...	6	
	max_consecutive_c...	28	
	rollout/		
	ep_len_mean	932	
	ep_rew_mean	364	
	exploration rate	0.492	
	time/		
	episodes	23800	
	fps	211	
	time_elapsed	7220	
	total timesteps	1525459	
	train/		
	learning_rate	0.0001	
	loss	0.0639	

n_updates	368864
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EP=8,020	STEP=1,527,685	clip=39	consec=5	max_consec=28	avg_clip50=40.71
EP=8,030	STEP=1,530,079	clip=52	consec=6	max_consec=28	avg_clip50=41.25
EP=8,040	STEP=1,532,127	clip=42	consec=4	max_consec=28	avg_clip50=37.98

custom/	
avg_clip50	36.3
clipped_episode_r...	34.5
consecutive_clipp...	10
max_consecutive_c...	28
rollout/	
ep_len_mean	970
ep_rew_mean	374
exploration rate	0.489
time/	
episodes	23900
fps	211
time_elapsed	7262
total timesteps	1533760
train/	
learning_rate	0.0001
loss	0.119
n_updates	370939

EP=8,050	STEP=1,534,543	clip=32	consec=1	max_consec=28	avg_clip50=36.95
EP=8,060	STEP=1,537,391	clip=40	consec=2	max_consec=28	avg_clip50=39.62
EP=8,070	STEP=1,539,795	clip=24	consec=2	max_consec=28	avg_clip50=40.09
EP=8,080	STEP=1,542,628	clip=64	consec=12	max_consec=28	avg_clip50=41.25

custom/	
avg_clip50	41.2
clipped_episode_r...	63.5
consecutive_clipp...	12
max_consecutive_c...	28
rollout/	
ep_len_mean	1.05e+03
ep_rew_mean	412
exploration rate	0.486
time/	
episodes	24000
fps	211
time_elapsed	7306
total timesteps	1542746
train/	
learning_rate	0.0001
loss	0.0924
n_updates	373186

EP=8,090	STEP=1,545,189	clip=51	consec=5	max_consec=28	avg_clip50=44.87
EP=8,100	STEP=1,547,993	clip=60	consec=15	max_consec=28	avg_clip50=47.33
EP=8,110	STEP=1,550,514	clip=60	consec=3	max_consec=28	avg_clip50=45.63

custom/	
avg_clip50	46.1
clipped_episode_r...	55.5
consecutive_clipp...	1
max_consecutive_c...	28

rollout/	
ep_len_mean	1.07e+03
ep_rew_mean	431
exploration rate	0.483
time/	
episodes	24100
fps	211
time_elapsed	7351
total timesteps	1551741
train/	
learning_rate	0.0001
loss	0.0306
n_updates	375435

EP=8,120	STEP=1,553,046	clip=46	consec=6	max_consec=28	avg_clip50=46.73
EP=8,130	STEP=1,555,418	clip=22	consec=5	max_consec=28	avg_clip50=43.90
EP=8,140	STEP=1,557,665	clip=15	consec=0	max_consec=28	avg_clip50=41.72

custom/	
avg_clip50	39.6
clipped_episode_r...	26
consecutive_clipp...	3
max_consecutive_c...	28
rollout/	
ep_len_mean	1.06e+03
ep_rew_mean	432
exploration rate	0.481
time/	
episodes	24200
fps	210
time_elapsed	7392
total timesteps	1559681
train/	
learning_rate	0.0001
loss	0.0575
n_updates	377420

EP=8,150	STEP=1,559,970	clip=66	consec=4	max_consec=28	avg_clip50=39.67
EP=8,160	STEP=1,562,496	clip=58	consec=5	max_consec=28	avg_clip50=39.53
EP=8,170	STEP=1,564,819	clip=24	consec=2	max_consec=28	avg_clip50=36.39
EP=8,180	STEP=1,567,390	clip=54	consec=8	max_consec=28	avg_clip50=38.33

custom/	
avg_clip50	38.4
clipped_episode_r...	69.5
consecutive_clipp...	1
max_consecutive_c...	28
rollout/	
ep_len_mean	1.03e+03
ep_rew_mean	412
exploration rate	0.478
time/	
episodes	24300
fps	210
time_elapsed	7435
total timesteps	1568029
train/	
learning_rate	0.0001
loss	0.0369
n_updates	379507

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-----
EP=8,190 | STEP=1,569,776 | clip=44 | consec=3 | max_consec=28 | avg_clip50=38.79
EP=8,200 | STEP=1,572,156 | clip=39 | consec=7 | max_consec=28 | avg_clip50=38.68
EP=8,210 | STEP=1,574,576 | clip=34 | consec=2 | max_consec=28 | avg_clip50=39.60
-----

```

```

| custom/
|   avg_clip50           | 42.1
|   clipped_episode_r... | 35
|   consecutive_clipp... | 8
|   max_consecutive_c... | 28
| rollout/
|   ep_len_mean          | 1.01e+03
|   ep_rew_mean          | 397
|   exploration rate     | 0.475
| time/
|   episodes             | 24400
|   fps                  | 210
|   time_elapsed         | 7477
|   total timesteps      | 1576361
| train/
|   learning_rate        | 0.0001
|   loss                 | 0.0397
|   n_updates            | 381590
-----

```

```

EP=8,220 | STEP=1,577,081 | clip=24 | consec=12 | max_consec=28 | avg_clip50=41.7
2
EP=8,230 | STEP=1,579,326 | clip=18 | consec=0 | max_consec=28 | avg_clip50=39.74
EP=8,240 | STEP=1,581,817 | clip=26 | consec=2 | max_consec=28 | avg_clip50=39.83
EP=8,250 | STEP=1,584,591 | clip=77 | consec=2 | max_consec=28 | avg_clip50=42.23
-----

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```

| custom/
|   avg_clip50           | 42.1
|   clipped_episode_r... | 54.5
|   consecutive_clipp... | 3
|   max_consecutive_c... | 28
| rollout/
|   ep_len_mean          | 1.03e+03
|   ep_rew_mean          | 405
|   exploration rate     | 0.472
| time/
|   episodes             | 24500
|   fps                  | 210
|   time_elapsed         | 7521
|   total timesteps      | 1585035
| train/
|   learning_rate        | 0.0001
|   loss                 | 0.0596
|   n_updates            | 383758
-----

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```

EP=8,260 | STEP=1,587,641 | clip=34 | consec=1 | max_consec=28 | avg_clip50=43.00
EP=8,270 | STEP=1,590,366 | clip=36 | consec=4 | max_consec=28 | avg_clip50=43.98
EP=8,280 | STEP=1,592,242 | clip=6 | consec=0 | max_consec=28 | avg_clip50=41.22
-----

```

```

| custom/
|   avg_clip50           | 41.2
|   clipped_episode_r... | 51
|   consecutive_clipp... | 5
|   max_consecutive_c... | 28
| rollout/
|   ep_len_mean          | 1.05e+03
-----

```

ep_rew_mean	408
exploration rate	0.469
time/	
episodes	24600
fps	210
time_elapsed	7565
total timesteps	1593804
train/	
learning_rate	0.0001
loss	0.0582
n_updates	385950

EP=8,290 | STEP=1,594,871 | clip=43 | consec=3 | max_consec=28 | avg_clip50=42.19
EP=8,300 | STEP=1,597,867 | clip=51 | consec=13 | max_consec=28 | avg_clip50=43.0
1
EP=8,310 | STEP=1,600,712 | clip=50 | consec=1 | max_consec=28 | avg_clip50=41.70
EP=8,320 | STEP=1,603,263 | clip=50 | consec=9 | max_consec=28 | avg_clip50=40.50

custom/	
avg_clip50	40.5
clipped_episode_r...	49.5
consecutive_clipp...	9
max_consecutive_c...	28
rollout/	
ep_len_mean	1.09e+03
ep_rew_mean	422
exploration rate	0.466
time/	
episodes	24700
fps	210
time_elapsed	7613
total timesteps	1603478
train/	
learning_rate	0.0001
loss	0.0228
n_updates	388369

EP=8,330 | STEP=1,606,106 | clip=54 | consec=19 | max_consec=28 | avg_clip50=45.2
1
EP=8,340 | STEP=1,608,668 | clip=43 | consec=1 | max_consec=28 | avg_clip50=43.93
EP=8,350 | STEP=1,611,210 | clip=58 | consec=4 | max_consec=28 | avg_clip50=41.84

custom/	
avg_clip50	42
clipped_episode_r...	38.5
consecutive_clipp...	4
max_consecutive_c...	28
rollout/	
ep_len_mean	1.11e+03
ep_rew_mean	425
exploration rate	0.463
time/	
episodes	24800
fps	210
time_elapsed	7661
total timesteps	1612610
train/	
learning_rate	0.0001
loss	0.0828
n_updates	390652

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-----
EP=8,360 | STEP=1,613,704 | clip=26 | consec=3 | max_consec=28 | avg_clip50=41.48
EP=8,370 | STEP=1,616,082 | clip=34 | consec=5 | max_consec=28 | avg_clip50=40.67
EP=8,380 | STEP=1,618,639 | clip=64 | consec=1 | max_consec=28 | avg_clip50=41.19
-----

```

```

| custom/
|   avg_clip50           | 42.2
|   clipped_episode_r... | 44.5
|   consecutive_clipp... | 3
|   max_consecutive_c... | 28
| rollout/
|   ep_len_mean          | 1.09e+03
|   ep_rew_mean          | 431
|   exploration rate     | 0.46
| time/
|   episodes             | 24900
|   fps                  | 210
|   time_elapsed         | 7706
|   total timesteps      | 1620990
| train/
|   learning_rate        | 0.0001
|   loss                 | 0.0399
|   n_updates            | 392747
-----

```

```

EP=8,390 | STEP=1,621,120 | clip=46 | consec=4 | max_consec=28 | avg_clip50=42.28
EP=8,400 | STEP=1,624,092 | clip=34 | consec=9 | max_consec=28 | avg_clip50=43.21
EP=8,410 | STEP=1,626,875 | clip=34 | consec=8 | max_consec=28 | avg_clip50=43.21
EP=8,420 | STEP=1,628,895 | clip=6  | consec=0 | max_consec=28 | avg_clip50=42.50
-----

```

```

| custom/
|   avg_clip50           | 42.6
|   clipped_episode_r... | 60
|   consecutive_clipp... | 3
|   max_consecutive_c... | 28
| rollout/
|   ep_len_mean          | 1.08e+03
|   ep_rew_mean          | 416
|   exploration rate     | 0.457
| time/
|   episodes             | 25000
|   fps                  | 210
|   time_elapsed         | 7750
|   total timesteps      | 1629867
| train/
|   learning_rate        | 0.0001
|   loss                 | 0.0538
|   n_updates            | 394966
-----

```

```

EP=8,430 | STEP=1,631,703 | clip=110 | consec=3 | max_consec=28 | avg_clip50=41.8
4
EP=8,440 | STEP=1,634,150 | clip=34 | consec=8 | max_consec=28 | avg_clip50=40.98
EP=8,450 | STEP=1,636,777 | clip=50 | consec=7 | max_consec=28 | avg_clip50=39.83
-----

```

```

| custom/
|   avg_clip50           | 41
|   clipped_episode_r... | 72
|   consecutive_clipp... | 13
|   max_consecutive_c... | 28
| rollout/
|   ep_len_mean          | 1.07e+03
-----

```

	ep_rew_mean	421	
	exploration rate	0.454	
	time/		
	episodes	25100	
	fps	210	
	time_elapsed	7794	
	total timesteps	1638595	
	train/		
	learning_rate	0.0001	
	loss	0.0342	
	n_updates	397148	

EP=8,460 | STEP=1,639,794 | clip=110 | consec=17 | max_consec=28 | avg_clip50=43.77

EP=8,470 | STEP=1,642,283 | clip=54 | consec=27 | max_consec=28 | avg_clip50=44.74

EP=8,480 | STEP=1,644,948 | clip=41 | consec=37 | max_consec=37 | avg_clip50=44.80

EP=8,490 | STEP=1,647,779 | clip=27 | consec=1 | max_consec=45 | avg_clip50=45.34

	custom/		
	avg_clip50	45.3	
	clipped_episode_r...	27	
	consecutive_clipp...	1	
	max_consecutive_c...	45	
	rollout/		
	ep_len_mean	1.11e+03	
	ep_rew_mean	432	
	exploration rate	0.451	
	time/		
	episodes	25200	
	fps	210	
	time_elapsed	7841	
	total timesteps	1647779	
	train/		
	learning_rate	0.0001	
	loss	0.124	
	n_updates	399444	

EP=8,500 | STEP=1,649,861 | clip=36 | consec=2 | max_consec=45 | avg_clip50=43.80

EP=8,510 | STEP=1,652,301 | clip=52 | consec=3 | max_consec=45 | avg_clip50=38.87

EP=8,520 | STEP=1,654,795 | clip=46 | consec=13 | max_consec=45 | avg_clip50=39.52

	custom/		
	avg_clip50	38.9	
	clipped_episode_r...	20	
	consecutive_clipp...	0	
	max_consecutive_c...	45	
	rollout/		
	ep_len_mean	1.08e+03	
	ep_rew_mean	422	
	exploration rate	0.449	
	time/		
	episodes	25300	
	fps	209	
	time_elapsed	7884	
	total timesteps	1655796	
	train/		
	learning_rate	0.0001	

loss	0.0739
n_updates	401448

EP=8,530	STEP=1,657,333	clip=58	consec=6	max_consec=45	avg_clip50=38.78
EP=8,540	STEP=1,659,501	clip=41	consec=1	max_consec=45	avg_clip50=37.00
EP=8,550	STEP=1,661,760	clip=50	consec=2	max_consec=45	avg_clip50=38.03

custom/	
avg_clip50	38.3
clipped_episode_r...	48
consecutive_clipp...	2
max_consecutive_c...	45
rollout/	
ep_len_mean	1.02e+03
ep_rew_mean	392
exploration rate	0.446
time/	
episodes	25400
fps	209
time_elapsed	7925
total timesteps	1663444
train/	
learning_rate	0.0001
loss	0.127
n_updates	403360

EP=8,560	STEP=1,664,010	clip=48	consec=4	max_consec=45	avg_clip50=38.94
EP=8,570	STEP=1,666,406	clip=38	consec=4	max_consec=45	avg_clip50=39.38
EP=8,580	STEP=1,668,897	clip=48	consec=5	max_consec=45	avg_clip50=39.55
EP=8,590	STEP=1,671,320	clip=59	consec=15	max_consec=45	avg_clip50=41.7

custom/	
avg_clip50	41.9
clipped_episode_r...	23.5
consecutive_clipp...	16
max_consecutive_c...	45
rollout/	
ep_len_mean	984
ep_rew_mean	392
exploration rate	0.443
time/	
episodes	25500
fps	209
time_elapsed	7968
total timesteps	1671869
train/	
learning_rate	0.0001
loss	0.0489
n_updates	405467

EP=8,600	STEP=1,674,073	clip=52	consec=25	max_consec=45	avg_clip50=43.9
EP=8,610	STEP=1,676,774	clip=47	consec=6	max_consec=45	avg_clip50=44.77
EP=8,620	STEP=1,679,417	clip=53	consec=16	max_consec=45	avg_clip50=45.2

custom/	
avg_clip50	44.3
clipped_episode_r...	54.5

consecutive_clipp...	1
max_consecutive_c...	45
rollout/	
ep_len_mean	1.02e+03
ep_rew_mean	425
exploration rate	0.44
time/	
episodes	25600
fps	209
time_elapsed	8013
total timesteps	1680712
train/	
learning_rate	0.0001
loss	0.197
n_updates	407677

EP=8,630	STEP=1,681,516	clip=37	consec=2	max_consec=45	avg_clip50=44.22
EP=8,640	STEP=1,683,758	clip=21	consec=12	max_consec=45	avg_clip50=43.37
EP=8,650	STEP=1,686,064	clip=46	consec=2	max_consec=45	avg_clip50=40.65
EP=8,660	STEP=1,688,778	clip=31	consec=1	max_consec=45	avg_clip50=40.64

custom/	
avg_clip50	40.6
clipped_episode_r...	31
consecutive_clipp...	1
max_consecutive_c...	45
rollout/	
ep_len_mean	1.03e+03
ep_rew_mean	427
exploration rate	0.438
time/	
episodes	25700
fps	209
time_elapsed	8055
total timesteps	1688851
train/	
learning_rate	0.0001
loss	0.0509
n_updates	409712

EP=8,670	STEP=1,691,243	clip=50	consec=3	max_consec=45	avg_clip50=39.54
EP=8,680	STEP=1,693,838	clip=26	consec=1	max_consec=45	avg_clip50=40.56
EP=8,690	STEP=1,696,640	clip=35	consec=3	max_consec=45	avg_clip50=42.89

custom/	
avg_clip50	44.2
clipped_episode_r...	53
consecutive_clipp...	2
max_consecutive_c...	45
rollout/	
ep_len_mean	1.06e+03
ep_rew_mean	433
exploration rate	0.435
time/	
episodes	25800
fps	209
time_elapsed	8100
total timesteps	1697757
train/	

	learning_rate		0.0001	
	loss		0.093	
	n_updates		411939	

EP=8,700		STEP=1,699,062		clip=72		consec=8		max_consec=45		avg_clip50=43.76
EP=8,710		STEP=1,701,616		clip=54		consec=1		max_consec=45		avg_clip50=43.43
EP=8,720		STEP=1,704,123		clip=55		consec=3		max_consec=45		avg_clip50=45.25

	custom/			
	avg_clip50		44.2	
	clipped_episode_r...		12	
	consecutive_clipp...		0	
	max_consecutive_c...		45	
	rollout/			
	ep_len_mean		1.05e+03	
	ep_rew_mean		425	
	exploration rate		0.432	
	time/			
	episodes		25900	
	fps		209	
	time_elapsed		8145	
	total timesteps		1706546	
	train/			
	learning_rate		0.0001	
	loss		0.059	
	n_updates		414136	

EP=8,730		STEP=1,706,796		clip=44		consec=1		max_consec=45		avg_clip50=44.60
EP=8,740		STEP=1,709,529		clip=50		consec=11		max_consec=45		avg_clip50=43.86
EP=8,750		STEP=1,711,663		clip=60		consec=1		max_consec=45		avg_clip50=42.73
EP=8,760		STEP=1,714,333		clip=60		consec=5		max_consec=45		avg_clip50=42.64

	custom/			
	avg_clip50		42.9	
	clipped_episode_r...		27	
	consecutive_clipp...		11	
	max_consecutive_c...		45	
	rollout/			
	ep_len_mean		1.08e+03	
	ep_rew_mean		434	
	exploration rate		0.429	
	time/			
	episodes		26000	
	fps		209	
	time_elapsed		8195	
	total timesteps		1716042	
	train/			
	learning_rate		0.0001	
	loss		0.0295	
	n_updates		416510	

EP=8,770		STEP=1,717,169		clip=47		consec=15		max_consec=45		avg_clip50=42.94
EP=8,780		STEP=1,719,509		clip=34		consec=5		max_consec=45		avg_clip50=42.69
EP=8,790		STEP=1,721,818		clip=52		consec=1		max_consec=45		avg_clip50=40.25
EP=8,800		STEP=1,724,235		clip=66		consec=11		max_consec=45		avg_clip50=42.35

	custom/			
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avg_clip50	42.8
clipped_episode_r...	61
consecutive_clipp...	12
max_consecutive_c...	45
rollout/	
ep_len_mean	1.05e+03
ep_rew_mean	427
exploration rate	0.426
time/	
episodes	26100
fps	209
time_elapsed	8239
total timesteps	1724537
train/	
learning_rate	0.0001
loss	0.0445
n_updates	418634

EP=8,810 | STEP=1,726,902 | clip=44 | consec=21 | max_consec=45 | avg_clip50=42.63

EP=8,820 | STEP=1,729,464 | clip=17 | consec=0 | max_consec=45 | avg_clip50=41.15

EP=8,830 | STEP=1,731,947 | clip=52 | consec=10 | max_consec=45 | avg_clip50=42.45

custom/	
avg_clip50	42.4
clipped_episode_r...	22.5
consecutive_clipp...	4
max_consecutive_c...	45
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	418
exploration rate	0.423
time/	
episodes	26200
fps	209
time_elapsed	8284
total timesteps	1733419
train/	
learning_rate	0.0001
loss	0.0339
n_updates	420854

EP=8,840 | STEP=1,734,442 | clip=47 | consec=8 | max_consec=45 | avg_clip50=43.63

EP=8,850 | STEP=1,736,634 | clip=6 | consec=0 | max_consec=45 | avg_clip50=40.85

EP=8,860 | STEP=1,738,934 | clip=50 | consec=8 | max_consec=45 | avg_clip50=40.06

EP=8,870 | STEP=1,741,266 | clip=15 | consec=0 | max_consec=45 | avg_clip50=39.00

custom/	
avg_clip50	39
clipped_episode_r...	15
consecutive_clipp...	0
max_consecutive_c...	45
rollout/	
ep_len_mean	1.01e+03
ep_rew_mean	401
exploration rate	0.42
time/	
episodes	26300
fps	209

	time_elapsed	8326	
	total timesteps	1741403	
	train/		
	learning_rate	0.0001	
	loss	0.0699	
	n_updates	422850	

EP=8,880 | STEP=1,743,714 | clip=42 | consec=10 | max_consec=45 | avg_clip50=38.51

EP=8,890 | STEP=1,746,421 | clip=38 | consec=20 | max_consec=45 | avg_clip50=39.84

EP=8,900 | STEP=1,749,002 | clip=61 | consec=4 | max_consec=45 | avg_clip50=43.56
[checkpoint periodico] guardado a 1,750,000 steps

	custom/		
	avg_clip50	44.2	
	clipped_episode_r...	57	
	consecutive_clipp...	11	
	max_consecutive_c...	45	
	rollout/		
	ep_len_mean	1.02e+03	
	ep_rew_mean	417	
	exploration rate	0.417	
	time/		
	episodes	26400	
	fps	209	
	time_elapsed	8374	
	total timesteps	1750713	
	train/		
	learning_rate	0.0001	
	loss	0.078	
	n_updates	425178	

EP=8,910 | STEP=1,751,350 | clip=58 | consec=14 | max_consec=45 | avg_clip50=43.68

EP=8,920 | STEP=1,753,576 | clip=42 | consec=1 | max_consec=45 | avg_clip50=43.20

EP=8,930 | STEP=1,756,016 | clip=20 | consec=0 | max_consec=45 | avg_clip50=41.98

EP=8,940 | STEP=1,758,874 | clip=62 | consec=7 | max_consec=45 | avg_clip50=43.28

	custom/		
	avg_clip50	43.5	
	clipped_episode_r...	51	
	consecutive_clipp...	8	
	max_consecutive_c...	45	
	rollout/		
	ep_len_mean	1.03e+03	
	ep_rew_mean	418	
	exploration rate	0.414	
	time/		
	episodes	26500	
	fps	208	
	time_elapsed	8419	
	total timesteps	1759394	
	train/		
	learning_rate	0.0001	
	loss	0.0384	
	n_updates	427348	

EP=8,950 | STEP=1,761,554 | clip=32 | consec=17 | max_consec=45 | avg_clip50=42.43

EP=8,960 | STEP=1,764,038 | clip=58 | consec=27 | max_consec=45 | avg_clip50=42.8
1
EP=8,970 | STEP=1,766,875 | clip=49 | consec=37 | max_consec=45 | avg_clip50=46.9
1

custom/		
avg_clip50	48.2	
clipped_episode_r...	37	
consecutive_clipp...	43	
max_consecutive_c...	45	
rollout/		
ep_len_mean	1.06e+03	
ep_rew_mean	452	
exploration rate	0.411	
time/		
episodes	26600	
fps	208	
time_elapsed	8466	
total timesteps	1768480	
train/		
learning_rate	0.0001	
loss	0.0448	
n_updates	429619	

EP=8,980 | STEP=1,769,175 | clip=8 | consec=0 | max_consec=45 | avg_clip50=47.79
EP=8,990 | STEP=1,771,731 | clip=56 | consec=1 | max_consec=45 | avg_clip50=47.01
EP=9,000 | STEP=1,774,319 | clip=80 | consec=5 | max_consec=45 | avg_clip50=47.93
EP=9,010 | STEP=1,777,119 | clip=65 | consec=15 | max_consec=45 | avg_clip50=49.3
7

custom/		
avg_clip50	48.6	
clipped_episode_r...	49	
consecutive_clipp...	17	
max_consecutive_c...	45	
rollout/		
ep_len_mean	1.08e+03	
ep_rew_mean	463	
exploration rate	0.408	
time/		
episodes	26700	
fps	208	
time_elapsed	8514	
total timesteps	1777684	
train/		
learning_rate	0.0001	
loss	0.0435	
n_updates	431920	

EP=9,020 | STEP=1,779,632 | clip=46 | consec=6 | max_consec=45 | avg_clip50=46.73
EP=9,030 | STEP=1,781,796 | clip=14 | consec=0 | max_consec=45 | avg_clip50=46.11
EP=9,040 | STEP=1,784,231 | clip=13 | consec=0 | max_consec=45 | avg_clip50=44.59

custom/		
avg_clip50	44.8	
clipped_episode_r...	26.5	
consecutive_clipp...	6	
max_consecutive_c...	45	
rollout/		
ep_len_mean	1.05e+03	

ep_rew_mean	457
exploration rate	0.405
time/	
episodes	26800
fps	208
time_elapsed	8557
total timesteps	1785860
train/	
learning_rate	0.0001
loss	0.0586
n_updates	433964

EP=9,050 | STEP=1,786,705 | clip=50 | consec=3 | max_consec=45 | avg_clip50=43.41
EP=9,060 | STEP=1,789,665 | clip=47 | consec=4 | max_consec=45 | avg_clip50=43.91
EP=9,070 | STEP=1,792,315 | clip=16 | consec=0 | max_consec=45 | avg_clip50=44.57
EP=9,080 | STEP=1,795,036 | clip=50 | consec=4 | max_consec=45 | avg_clip50=46.68

custom/	
avg_clip50	46.7
clipped_episode_r...	50.5
consecutive_clipp...	4
max_consecutive_c...	45
rollout/	
ep_len_mean	1.08e+03
ep_rew_mean	464
exploration rate	0.402
time/	
episodes	26900
fps	208
time_elapsed	8604
total timesteps	1795036
train/	
learning_rate	0.0001
loss	0.0725
n_updates	436258

EP=9,090 | STEP=1,797,442 | clip=52 | consec=2 | max_consec=45 | avg_clip50=47.29
EP=9,100 | STEP=1,800,128 | clip=44 | consec=12 | max_consec=45 | avg_clip50=48.4
5
EP=9,110 | STEP=1,802,057 | clip=32 | consec=2 | max_consec=45 | avg_clip50=44.99

custom/	
avg_clip50	46.2
clipped_episode_r...	38.5
consecutive_clipp...	7
max_consecutive_c...	45
rollout/	
ep_len_mean	1.05e+03
ep_rew_mean	453
exploration rate	0.399
time/	
episodes	27000
fps	208
time_elapsed	8648
total timesteps	1803531
train/	
learning_rate	0.0001
loss	0.228
n_updates	438382

EP=9,120 | STEP=1,804,734 | clip=25 | consec=12 | max_consec=45 | avg_clip50=46.14
EP=9,130 | STEP=1,807,242 | clip=40 | consec=6 | max_consec=45 | avg_clip50=45.78
EP=9,140 | STEP=1,809,787 | clip=60 | consec=2 | max_consec=45 | avg_clip50=46.39
EP=9,150 | STEP=1,812,323 | clip=34 | consec=6 | max_consec=45 | avg_clip50=45.19

custom/		
avg_clip50	45.2	
clipped_episode_r...	34.5	
consecutive_clipp...	6	
max_consecutive_c...	45	
rollout/		
ep_len_mean	1.07e+03	
ep_rew_mean	468	
exploration rate	0.396	
time/		
episodes	27100	
fps	208	
time_elapsed	8695	
total timesteps	1812424	
train/		
learning_rate	0.0001	
loss	0.0985	
n_updates	440605	

EP=9,160 | STEP=1,814,766 | clip=74 | consec=2 | max_consec=45 | avg_clip50=47.39
EP=9,170 | STEP=1,817,065 | clip=47 | consec=5 | max_consec=45 | avg_clip50=44.48
EP=9,180 | STEP=1,819,884 | clip=32 | consec=15 | max_consec=45 | avg_clip50=44.33

custom/		
avg_clip50	45	
clipped_episode_r...	39	
consecutive_clipp...	1	
max_consecutive_c...	45	
rollout/		
ep_len_mean	1.04e+03	
ep_rew_mean	454	
exploration rate	0.394	
time/		
episodes	27200	
fps	208	
time_elapsed	8739	
total timesteps	1820977	
train/		
learning_rate	0.0001	
loss	0.0467	
n_updates	442744	

EP=9,190 | STEP=1,822,584 | clip=53 | consec=7 | max_consec=45 | avg_clip50=44.64
EP=9,200 | STEP=1,825,233 | clip=20 | consec=0 | max_consec=45 | avg_clip50=45.54
EP=9,210 | STEP=1,827,818 | clip=54 | consec=10 | max_consec=45 | avg_clip50=45.37

custom/		
avg_clip50	48.8	
clipped_episode_r...	67.5	
consecutive_clipp...	18	
max_consecutive_c...	45	
rollout/		

ep_len_mean	1.07e+03
ep_rew_mean	465
exploration rate	0.391
time/	
episodes	27300
fps	208
time_elapsed	8786
total timesteps	1830032
train/	
learning_rate	0.0001
loss	0.125
n_updates	445007

EP=9,220 | STEP=1,830,652 | clip=73 | consec=20 | max_consec=45 | avg_clip50=49.35

EP=9,230 | STEP=1,833,167 | clip=67 | consec=7 | max_consec=45 | avg_clip50=48.20

EP=9,240 | STEP=1,836,060 | clip=41 | consec=17 | max_consec=45 | avg_clip50=48.09

EP=9,250 | STEP=1,838,583 | clip=30 | consec=27 | max_consec=45 | avg_clip50=46.59

custom/	
avg_clip50	46.3
clipped_episode_r...	31.5
consecutive_clipp...	28
max_consecutive_c...	45
rollout/	
ep_len_mean	1.09e+03
ep_rew_mean	457
exploration rate	0.388
time/	
episodes	27400
fps	208
time_elapsed	8832
total timesteps	1839012
train/	
learning_rate	0.0001
loss	0.0484
n_updates	447252

EP=9,260 | STEP=1,841,131 | clip=36 | consec=7 | max_consec=45 | avg_clip50=45.36

EP=9,270 | STEP=1,843,509 | clip=60 | consec=17 | max_consec=45 | avg_clip50=42.86

EP=9,280 | STEP=1,845,901 | clip=34 | consec=27 | max_consec=45 | avg_clip50=42.76

custom/	
avg_clip50	43.6
clipped_episode_r...	51.5
consecutive_clipp...	33
max_consecutive_c...	45
rollout/	
ep_len_mean	1.09e+03
ep_rew_mean	459
exploration rate	0.385
time/	
episodes	27500
fps	208
time_elapsed	8879
total timesteps	1847744

train/	
learning_rate	0.0001
loss	0.026
n_updates	449435

EP=9,290 | STEP=1,848,748 | clip=34 | consec=37 | max_consec=45 | avg_clip50=43.3
1

EP=9,300 | STEP=1,851,596 | clip=46 | consec=47 | max_consec=47 | avg_clip50=44.4
2

EP=9,310 | STEP=1,854,430 | clip=54 | consec=57 | max_consec=57 | avg_clip50=46.1
3

EP=9,320 | STEP=1,856,807 | clip=57 | consec=8 | max_consec=58 | avg_clip50=45.17

custom/	
avg_clip50	45.2
clipped_episode_r...	57
consecutive_clipp...	8
max_consecutive_c...	58
rollout/	
ep_len_mean	1.09e+03
ep_rew_mean	440
exploration rate	0.382
time/	
episodes	27600
fps	207
time_elapsed	8928
total timesteps	1857086
train/	
learning_rate	0.0001
loss	0.0513
n_updates	451771

EP=9,330 | STEP=1,860,431 | clip=124 | consec=18 | max_consec=58 | avg_clip50=50.
65

EP=9,340 | STEP=1,862,966 | clip=26 | consec=7 | max_consec=58 | avg_clip50=49.37

EP=9,350 | STEP=1,865,494 | clip=48 | consec=4 | max_consec=58 | avg_clip50=48.07

custom/	
avg_clip50	47.8
clipped_episode_r...	38
consecutive_clipp...	9
max_consecutive_c...	58
rollout/	
ep_len_mean	1.13e+03
ep_rew_mean	471
exploration rate	0.378
time/	
episodes	27700
fps	207
time_elapsed	8980
total timesteps	1867054
train/	
learning_rate	0.0001
loss	0.0344
n_updates	454263

EP=9,360 | STEP=1,868,194 | clip=60 | consec=14 | max_consec=58 | avg_clip50=47.9
7

EP=9,370 | STEP=1,871,114 | clip=48 | consec=24 | max_consec=58 | avg_clip50=51.1
7

EP=9,380 | STEP=1,873,860 | clip=53 | consec=8 | max_consec=58 | avg_clip50=46.40
EP=9,390 | STEP=1,876,486 | clip=38 | consec=18 | max_consec=58 | avg_clip50=46.5
2

custom/		
avg_clip50	46.4	
clipped_episode_r...	44	
consecutive_clipp...	20	
max_consecutive_c...	58	
rollout/		
ep_len_mean	1.15e+03	
ep_rew_mean	479	
exploration rate	0.375	
time/		
episodes	27800	
fps	207	
time_elapsed	9030	
total timesteps	1876996	
train/		
learning_rate	0.0001	
loss	0.401	
n_updates	456748	

EP=9,400 | STEP=1,879,197 | clip=68 | consec=6 | max_consec=58 | avg_clip50=46.95
EP=9,410 | STEP=1,881,535 | clip=6 | consec=0 | max_consec=58 | avg_clip50=45.98
EP=9,420 | STEP=1,884,530 | clip=72 | consec=4 | max_consec=58 | avg_clip50=45.08

custom/		
avg_clip50	47.2	
clipped_episode_r...	77	
consecutive_clipp...	10	
max_consecutive_c...	58	
rollout/		
ep_len_mean	1.14e+03	
ep_rew_mean	479	
exploration rate	0.372	
time/		
episodes	27900	
fps	207	
time_elapsed	9079	
total timesteps	1886376	
train/		
learning_rate	0.0001	
loss	0.0528	
n_updates	459093	

EP=9,430 | STEP=1,887,231 | clip=26 | consec=14 | max_consec=58 | avg_clip50=46.8
6
EP=9,440 | STEP=1,889,913 | clip=60 | consec=24 | max_consec=58 | avg_clip50=47.2
6
EP=9,450 | STEP=1,892,567 | clip=36 | consec=34 | max_consec=58 | avg_clip50=49.4
8
EP=9,460 | STEP=1,895,268 | clip=46 | consec=44 | max_consec=58 | avg_clip50=49.8
9

custom/		
avg_clip50	49.5	
clipped_episode_r...	23	
consecutive_clipp...	45	
max_consecutive_c...	58	

rollout/	
ep_len_mean	1.12e+03
ep_rew_mean	475
exploration rate	0.369
time/	
episodes	28000
fps	207
time_elapsed	9127
total timesteps	1895487
train/	
learning_rate	0.0001
loss	0.0809
n_updates	461371

EP=9,470 | STEP=1,897,745 | clip=58 | consec=4 | max_consec=58 | avg_clip50=48.77
EP=9,480 | STEP=1,900,672 | clip=48 | consec=7 | max_consec=58 | avg_clip50=48.35
EP=9,490 | STEP=1,903,949 | clip=29 | consec=17 | max_consec=58 | avg_clip50=50.6
5

custom/	
avg_clip50	50
clipped_episode_r...	52
consecutive_clipp...	22
max_consecutive_c...	58
rollout/	
ep_len_mean	1.16e+03
ep_rew_mean	498
exploration rate	0.365
time/	
episodes	28100
fps	207
time_elapsed	9181
total timesteps	1905490
train/	
learning_rate	0.0001
loss	0.0595
n_updates	463872

EP=9,500 | STEP=1,906,674 | clip=54 | consec=2 | max_consec=58 | avg_clip50=50.45
EP=9,510 | STEP=1,909,139 | clip=45 | consec=12 | max_consec=58 | avg_clip50=49.0
3
EP=9,520 | STEP=1,912,476 | clip=46 | consec=22 | max_consec=58 | avg_clip50=52.5
6
EP=9,530 | STEP=1,914,953 | clip=50 | consec=4 | max_consec=58 | avg_clip50=52.22

custom/	
avg_clip50	52.2
clipped_episode_r...	49.5
consecutive_clipp...	4
max_consecutive_c...	58
rollout/	
ep_len_mean	1.15e+03
ep_rew_mean	503
exploration rate	0.362
time/	
episodes	28200
fps	207
time_elapsed	9231
total timesteps	1914953
train/	

learning_rate	0.0001
loss	0.0339
n_updates	466238

EP=9,540 | STEP=1,917,313 | clip=55 | consec=14 | max_consec=58 | avg_clip50=49.35

EP=9,550 | STEP=1,920,285 | clip=58 | consec=24 | max_consec=58 | avg_clip50=48.68

EP=9,560 | STEP=1,922,812 | clip=37 | consec=9 | max_consec=58 | avg_clip50=49.95

custom/	
avg_clip50	47.5
clipped_episode_r...	52
consecutive_clipp...	13
max_consecutive_c...	58
rollout/	
ep_len_mean	1.14e+03
ep_rew_mean	493
exploration rate	0.359
time/	
episodes	28300
fps	207
time_elapsed	9277
total timesteps	1923772
train/	
learning_rate	0.0001
loss	0.0937
n_updates	468442

EP=9,570 | STEP=1,925,313 | clip=40 | consec=19 | max_consec=58 | avg_clip50=45.80

EP=9,580 | STEP=1,927,817 | clip=20 | consec=0 | max_consec=58 | avg_clip50=45.50

EP=9,590 | STEP=1,930,628 | clip=74 | consec=10 | max_consec=58 | avg_clip50=46.77

custom/	
avg_clip50	47.6
clipped_episode_r...	36
consecutive_clipp...	18
max_consecutive_c...	58
rollout/	
ep_len_mean	1.13e+03
ep_rew_mean	486
exploration rate	0.356
time/	
episodes	28400
fps	207
time_elapsed	9328
total timesteps	1933173
train/	
learning_rate	0.0001
loss	0.0697
n_updates	470793

EP=9,600 | STEP=1,933,619 | clip=44 | consec=20 | max_consec=58 | avg_clip50=47.59

EP=9,610 | STEP=1,936,051 | clip=16 | consec=0 | max_consec=58 | avg_clip50=47.58

EP=9,620 | STEP=1,938,677 | clip=55 | consec=10 | max_consec=58 | avg_clip50=48.88

EP=9,630 | STEP=1,941,366 | clip=22 | consec=20 | max_consec=58 | avg_clip50=48.9

custom/		
avg_clip50	49.9	
clipped_episode_r...	52.5	
consecutive_clipp...	22	
max_consecutive_c...	58	
rollout/		
ep_len_mean	1.1e+03	
ep_rew_mean	471	
exploration rate	0.353	
time/		
episodes	28500	
fps	207	
time_elapsed	9375	
total timesteps	1942151	
train/		
learning_rate	0.0001	
loss	0.0631	
n_updates	473037	

EP=9,640 | STEP=1,944,227 | clip=55 | consec=4 | max_consec=58 | avg_clip50=49.39
 EP=9,650 | STEP=1,947,008 | clip=25 | consec=5 | max_consec=58 | avg_clip50=47.58
 EP=9,660 | STEP=1,949,666 | clip=25 | consec=15 | max_consec=58 | avg_clip50=47.4
 1

custom/		
avg_clip50	47.2	
clipped_episode_r...	50	
consecutive_clipp...	21	
max_consecutive_c...	58	
rollout/		
ep_len_mean	1.13e+03	
ep_rew_mean	477	
exploration rate	0.35	
time/		
episodes	28600	
fps	207	
time_elapsed	9425	
total timesteps	1951370	
train/		
learning_rate	0.0001	
loss	0.0768	
n_updates	475342	

EP=9,670 | STEP=1,952,392 | clip=28 | consec=25 | max_consec=58 | avg_clip50=47.2
 1
 EP=9,680 | STEP=1,955,231 | clip=21 | consec=35 | max_consec=58 | avg_clip50=47.6
 9
 EP=9,690 | STEP=1,958,165 | clip=37 | consec=45 | max_consec=58 | avg_clip50=47.6
 5
 EP=9,700 | STEP=1,960,465 | clip=52 | consec=55 | max_consec=58 | avg_clip50=47.9
 7

custom/		
avg_clip50	48	
clipped_episode_r...	51.5	
consecutive_clipp...	55	
max_consecutive_c...	58	
rollout/		

	ep_len_mean	1.12e+03	
	ep_rew_mean	478	
	exploration rate	0.347	
	time/		
	episodes	28700	
	fps	206	
	time_elapsed	9473	
	total timesteps	1960665	
	train/		
	learning_rate	0.0001	
	loss	0.0664	
	n_updates	477666	

EP=9,710 | STEP=1,963,207 | clip=46 | consec=65 | max_consec=65 | avg_clip50=49.70

EP=9,720 | STEP=1,965,961 | clip=30 | consec=75 | max_consec=75 | avg_clip50=50.20

EP=9,730 | STEP=1,968,919 | clip=80 | consec=85 | max_consec=85 | avg_clip50=51.43

	custom/		
	avg_clip50	50	
	clipped_episode_r...	54.5	
	consecutive_clipp...	1	
	max_consecutive_c...	87	
	rollout/		
	ep_len_mean	1.14e+03	
	ep_rew_mean	486	
	exploration rate	0.344	
	time/		
	episodes	28800	
	fps	206	
	time_elapsed	9524	
	total timesteps	1970193	
	train/		
	learning_rate	0.0001	
	loss	0.052	
	n_updates	480048	

EP=9,740 | STEP=1,971,549 | clip=38 | consec=7 | max_consec=87 | avg_clip50=49.48

EP=9,750 | STEP=1,974,162 | clip=9 | consec=0 | max_consec=87 | avg_clip50=48.72

EP=9,760 | STEP=1,976,893 | clip=67 | consec=10 | max_consec=87 | avg_clip50=47.39

	custom/		
	avg_clip50	44.5	
	clipped_episode_r...	34.5	
	consecutive_clipp...	4	
	max_consecutive_c...	87	
	rollout/		
	ep_len_mean	1.12e+03	
	ep_rew_mean	473	
	exploration rate	0.341	
	time/		
	episodes	28900	
	fps	206	
	time_elapsed	9569	
	total timesteps	1978755	
	train/		
	learning_rate	0.0001	

loss	0.0383
n_updates	482188

EP=9,770 | STEP=1,979,074 | clip=48 | consec=1 | max_consec=87 | avg_clip50=44.10
 EP=9,780 | STEP=1,981,654 | clip=55 | consec=2 | max_consec=87 | avg_clip50=41.96
 EP=9,790 | STEP=1,984,192 | clip=48 | consec=4 | max_consec=87 | avg_clip50=42.30
 EP=9,800 | STEP=1,986,950 | clip=78 | consec=14 | max_consec=87 | avg_clip50=43.71

custom/	
avg_clip50	43.8
clipped_episode_r...	44
consecutive_clipp...	18
max_consecutive_c...	87
rollout/	
ep_len_mean	1.11e+03
ep_rew_mean	459
exploration rate	0.338
time/	
episodes	29000
fps	206
time_elapsed	9617
total timesteps	1988151
train/	
learning_rate	0.0001
loss	0.289
n_updates	484537

EP=9,810 | STEP=1,989,887 | clip=52 | consec=24 | max_consec=87 | avg_clip50=44.35
 EP=9,820 | STEP=1,992,772 | clip=78 | consec=34 | max_consec=87 | avg_clip50=48.98
 EP=9,830 | STEP=1,995,544 | clip=72 | consec=44 | max_consec=87 | avg_clip50=50.17

custom/	
avg_clip50	50
clipped_episode_r...	45.5
consecutive_clipp...	3
max_consecutive_c...	87
rollout/	
ep_len_mean	1.1e+03
ep_rew_mean	460
exploration rate	0.335
time/	
episodes	29100
fps	206
time_elapsed	9668
total timesteps	1997835
train/	
learning_rate	0.0001
loss	0.0754
n_updates	486958

EP=9,840 | STEP=1,998,041 | clip=50 | consec=4 | max_consec=87 | avg_clip50=50.03
 [checkpoint periodico] guardado a 2,000,000 steps
 EP=9,850 | STEP=2,000,795 | clip=44 | consec=14 | max_consec=87 | avg_clip50=51.00
 EP=9,860 | STEP=2,003,886 | clip=58 | consec=24 | max_consec=87 | avg_clip50=53.10

EP=9,870 | STEP=2,006,541 | clip=52 | consec=5 | max_consec=87 | avg_clip50=50.13

custom/		
avg_clip50	49.9	
clipped_episode_r...	18	
consecutive_clipp...	0	
max_consecutive_c...	87	
rollout/		
ep_len_mean	1.14e+03	
ep_rew_mean	493	
exploration rate	0.331	
time/		
episodes	29200	
fps	206	
time_elapsed	9719	
total timesteps	2007589	
train/		
learning_rate	0.0001	
loss	0.284	
n_updates	489397	

EP=9,880 | STEP=2,009,263 | clip=32 | consec=6 | max_consec=87 | avg_clip50=48.68

EP=9,890 | STEP=2,012,015 | clip=36 | consec=1 | max_consec=87 | avg_clip50=49.02

EP=9,900 | STEP=2,014,707 | clip=57 | consec=11 | max_consec=87 | avg_clip50=47.8

4

custom/		
avg_clip50	45.9	
clipped_episode_r...	37.5	
consecutive_clipp...	19	
max_consecutive_c...	87	
rollout/		
ep_len_mean	1.15e+03	
ep_rew_mean	496	
exploration rate	0.328	
time/		
episodes	29300	
fps	206	
time_elapsed	9769	
total timesteps	2016924	
train/		
learning_rate	0.0001	
loss	0.108	
n_updates	491730	

EP=9,910 | STEP=2,017,156 | clip=16 | consec=0 | max_consec=87 | avg_clip50=44.77

EP=9,920 | STEP=2,019,603 | clip=51 | consec=10 | max_consec=87 | avg_clip50=45.5

6

EP=9,930 | STEP=2,022,148 | clip=39 | consec=4 | max_consec=87 | avg_clip50=45.09

EP=9,940 | STEP=2,024,835 | clip=52 | consec=9 | max_consec=87 | avg_clip50=45.49

custom/		
avg_clip50	46.3	
clipped_episode_r...	68	
consecutive_clipp...	13	
max_consecutive_c...	87	
rollout/		
ep_len_mean	1.13e+03	
ep_rew_mean	480	
exploration rate	0.325	

time/	
episodes	29400
fps	206
time_elapsed	9818
total timesteps	2026271
train/	
learning_rate	0.0001
loss	0.0367
n_updates	494067

EP=9,950 | STEP=2,027,872 | clip=56 | consec=19 | max_consec=87 | avg_clip50=47.12

EP=9,960 | STEP=2,030,160 | clip=23 | consec=6 | max_consec=87 | avg_clip50=46.52

EP=9,970 | STEP=2,032,464 | clip=77 | consec=7 | max_consec=87 | avg_clip50=45.88

custom/	
avg_clip50	47.6
clipped_episode_r...	40
consecutive_clipp...	5
max_consecutive_c...	87
rollout/	
ep_len_mean	1.08e+03
ep_rew_mean	463
exploration rate	0.322
time/	
episodes	29500
fps	206
time_elapsed	9863
total timesteps	2034886
train/	
learning_rate	0.0001
loss	0.0493
n_updates	496221

EP=9,980 | STEP=2,035,047 | clip=40 | consec=6 | max_consec=87 | avg_clip50=47.63

EP=9,990 | STEP=2,037,638 | clip=52 | consec=16 | max_consec=87 | avg_clip50=46.02

EP=10,000 | STEP=2,040,294 | clip=42 | consec=26 | max_consec=87 | avg_clip50=43.87

EP=10,010 | STEP=2,042,880 | clip=37 | consec=36 | max_consec=87 | avg_clip50=45.66

custom/	
avg_clip50	46.4
clipped_episode_r...	28
consecutive_clipp...	40
max_consecutive_c...	87
rollout/	
ep_len_mean	1.07e+03
ep_rew_mean	456
exploration rate	0.319
time/	
episodes	29600
fps	206
time_elapsed	9912
total timesteps	2043945
train/	
learning_rate	0.0001
loss	0.0604
n_updates	498486

EP=10,020 | STEP=2,045,820 | clip=70 | consec=46 | max_consec=87 | avg_clip50=47.88
EP=10,030 | STEP=2,048,792 | clip=58 | consec=6 | max_consec=87 | avg_clip50=47.59
EP=10,040 | STEP=2,051,396 | clip=31 | consec=16 | max_consec=87 | avg_clip50=48.56

custom/	
avg_clip50	50.5
clipped_episode_r...	31
consecutive_clipp...	24
max_consecutive_c...	87
rollout/	
ep_len_mean	1.1e+03
ep_rew_mean	474
exploration rate	0.316
time/	
episodes	29700
fps	206
time_elapsed	9964
total timesteps	2053755
train/	
learning_rate	0.0001
loss	0.0495
n_updates	500938

EP=10,050 | STEP=2,054,149 | clip=52 | consec=26 | max_consec=87 | avg_clip50=50.48
EP=10,060 | STEP=2,056,260 | clip=25 | consec=2 | max_consec=87 | avg_clip50=48.33
EP=10,070 | STEP=2,059,004 | clip=40 | consec=12 | max_consec=87 | avg_clip50=47.53
EP=10,080 | STEP=2,061,657 | clip=21 | consec=22 | max_consec=87 | avg_clip50=47.97

custom/	
avg_clip50	48.1
clipped_episode_r...	77
consecutive_clipp...	26
max_consecutive_c...	87
rollout/	
ep_len_mean	1.11e+03
ep_rew_mean	484
exploration rate	0.313
time/	
episodes	29800
fps	206
time_elapsed	10013
total timesteps	2062900
train/	
learning_rate	0.0001
loss	0.0768
n_updates	503224

EP=10,090 | STEP=2,064,268 | clip=18 | consec=0 | max_consec=87 | avg_clip50=48.49
EP=10,100 | STEP=2,067,461 | clip=54 | consec=10 | max_consec=87 | avg_clip50=49.00
EP=10,110 | STEP=2,069,910 | clip=47 | consec=3 | max_consec=87 | avg_clip50=51.1

custom/		
avg_clip50	48.9	
clipped_episode_r...	13.5	
consecutive_clipp...	0	
max_consecutive_c...	87	
rollout/		
ep_len_mean	1.11e+03	
ep_rew_mean	485	
exploration rate	0.31	
time/		
episodes	29900	
fps	205	
time_elapsed	10063	
total timesteps	2072315	
train/		
learning_rate	0.0001	
loss	0.0536	
n_updates	505578	

EP=10,120 | STEP=2,072,475 | clip=60 | consec=1 | max_consec=87 | avg_clip50=49.35
 EP=10,130 | STEP=2,074,829 | clip=47 | consec=11 | max_consec=87 | avg_clip50=47.43
 EP=10,140 | STEP=2,077,348 | clip=58 | consec=21 | max_consec=87 | avg_clip50=47.67
 EP=10,150 | STEP=2,080,458 | clip=33 | consec=31 | max_consec=87 | avg_clip50=46.79

custom/		
avg_clip50	46.9	
clipped_episode_r...	55	
consecutive_clipp...	36	
max_consecutive_c...	87	
rollout/		
ep_len_mean	1.12e+03	
ep_rew_mean	490	
exploration rate	0.307	
time/		
episodes	30000	
fps	205	
time_elapsed	10116	
total timesteps	2082103	
train/		
learning_rate	0.0001	
loss	0.294	
n_updates	508025	

EP=10,160 | STEP=2,083,512 | clip=70 | consec=41 | max_consec=87 | avg_clip50=47.48
 EP=10,170 | STEP=2,086,070 | clip=16 | consec=0 | max_consec=87 | avg_clip50=49.02
 EP=10,180 | STEP=2,088,524 | clip=27 | consec=6 | max_consec=87 | avg_clip50=48.99
 EP=10,190 | STEP=2,091,507 | clip=32 | consec=16 | max_consec=87 | avg_clip50=49.40

custom/		
avg_clip50	49.4	

	clipped_episode_r...	32.5	
	consecutive_clipp...	16	
	max_consecutive_c...	87	
	rollout/		
	ep_len_mean	1.13e+03	
	ep_rew_mean	485	
	exploration rate	0.303	
	time/		
	episodes	30100	
	fps	205	
	time_elapsed	10167	
	total timesteps	2091677	
	train/		
	learning_rate	0.0001	
	loss	0.0451	
	n_updates	510419	

EP=10,200 | STEP=2,094,305 | clip=46 | consec=26 | max_consec=87 | avg_clip50=48.85

EP=10,210 | STEP=2,097,076 | clip=57 | consec=36 | max_consec=87 | avg_clip50=47.57

EP=10,220 | STEP=2,100,067 | clip=58 | consec=46 | max_consec=87 | avg_clip50=49.59

	custom/		
	avg_clip50	50.3	
	clipped_episode_r...	12	
	consecutive_clipp...	0	
	max_consecutive_c...	87	
	rollout/		
	ep_len_mean	1.14e+03	
	ep_rew_mean	493	
	exploration rate	0.3	
	time/		
	episodes	30200	
	fps	205	
	time_elapsed	10220	
	total timesteps	2101340	
	train/		
	learning_rate	0.0001	
	loss	0.0607	
	n_updates	512834	

EP=10,230 | STEP=2,102,646 | clip=50 | consec=5 | max_consec=87 | avg_clip50=50.77

EP=10,240 | STEP=2,105,440 | clip=60 | consec=15 | max_consec=87 | avg_clip50=50.40

EP=10,250 | STEP=2,107,896 | clip=57 | consec=5 | max_consec=87 | avg_clip50=48.97

EP=10,260 | STEP=2,110,639 | clip=58 | consec=2 | max_consec=87 | avg_clip50=49.41

	custom/		
	avg_clip50	49.4	
	clipped_episode_r...	57.5	
	consecutive_clipp...	2	
	max_consecutive_c...	87	
	rollout/		
	ep_len_mean	1.13e+03	
	ep_rew_mean	485	

	exploration rate	0.297	
	time/		
	episodes	30300	
	fps	205	
	time_elapsed	10273	
	total timesteps	2110930	
	train/		
	learning_rate	0.0001	
	loss	0.0416	
	n_updates	515232	

EP=10,270 | STEP=2,113,376 | clip=46 | consec=12 | max_consec=87 | avg_clip50=47.58

EP=10,280 | STEP=2,116,049 | clip=30 | consec=22 | max_consec=87 | avg_clip50=48.78

EP=10,290 | STEP=2,118,571 | clip=46 | consec=6 | max_consec=87 | avg_clip50=47.01

	custom/		
	avg_clip50	47.2	
	clipped_episode_r...	27	
	consecutive_clipp...	13	
	max_consecutive_c...	87	
	rollout/		
	ep_len_mean	1.11e+03	
	ep_rew_mean	478	
	exploration rate	0.294	
	time/		
	episodes	30400	
	fps	205	
	time_elapsed	10323	
	total timesteps	2120365	
	train/		
	learning_rate	0.0001	
	loss	0.049	
	n_updates	517591	

EP=10,300 | STEP=2,121,126 | clip=10 | consec=0 | max_consec=87 | avg_clip50=47.54

EP=10,310 | STEP=2,123,915 | clip=100 | consec=10 | max_consec=87 | avg_clip50=47.91

EP=10,320 | STEP=2,126,551 | clip=19 | consec=0 | max_consec=87 | avg_clip50=48.29

EP=10,330 | STEP=2,128,896 | clip=40 | consec=8 | max_consec=87 | avg_clip50=45.57

	custom/		
	avg_clip50	46.3	
	clipped_episode_r...	74.5	
	consecutive_clipp...	11	
	max_consecutive_c...	87	
	rollout/		
	ep_len_mean	1.1e+03	
	ep_rew_mean	476	
	exploration rate	0.291	
	time/		
	episodes	30500	
	fps	205	
	time_elapsed	10375	
	total timesteps	2129911	

train/	
learning_rate	0.0001
loss	0.0437
n_updates	519977

EP=10,340 | STEP=2,131,714 | clip=42 | consec=18 | max_consec=87 | avg_clip50=47.12

EP=10,350 | STEP=2,134,175 | clip=60 | consec=5 | max_consec=87 | avg_clip50=47.71

EP=10,360 | STEP=2,136,997 | clip=74 | consec=15 | max_consec=87 | avg_clip50=47.73

custom/	
avg_clip50	48.6
clipped_episode_r...	60
consecutive_clipp...	23
max_consecutive_c...	87
rollout/	
ep_len_mean	1.1e+03
ep_rew_mean	477
exploration rate	0.288
time/	
episodes	30600
fps	205
time_elapsed	10424
total timesteps	2139225
train/	
learning_rate	0.0001
loss	0.0419
n_updates	522306

EP=10,370 | STEP=2,139,888 | clip=56 | consec=25 | max_consec=87 | avg_clip50=47.90

EP=10,380 | STEP=2,142,634 | clip=54 | consec=4 | max_consec=87 | avg_clip50=50.18

EP=10,390 | STEP=2,145,722 | clip=50 | consec=14 | max_consec=87 | avg_clip50=52.02

EP=10,400 | STEP=2,148,512 | clip=60 | consec=9 | max_consec=87 | avg_clip50=53.01

custom/	
avg_clip50	54.2
clipped_episode_r...	77.5
consecutive_clipp...	12
max_consecutive_c...	87
rollout/	
ep_len_mean	1.14e+03
ep_rew_mean	507
exploration rate	0.284
time/	
episodes	30700
fps	205
time_elapsed	10479
total timesteps	2149535
train/	
learning_rate	0.0001
loss	0.0608
n_updates	524883

EP=10,410 | STEP=2,151,580 | clip=57 | consec=19 | max_consec=87 | avg_clip50=54.

80
EP=10,420 | STEP=2,154,621 | clip=47 | consec=29 | max_consec=87 | avg_clip50=55.
06
EP=10,430 | STEP=2,157,342 | clip=52 | consec=4 | max_consec=87 | avg_clip50=54.6
9

custom/		
avg_clip50	52.5	
clipped_episode_r...	45.5	
consecutive_clipp...	4	
max_consecutive_c...	87	
rollout/		
ep_len_mean	1.17e+03	
ep_rew_mean	522	
exploration rate	0.281	
time/		
episodes	30800	
fps	205	
time_elapsed	10533	
total timesteps	2159601	
train/		
learning_rate	0.0001	
loss	0.0386	
n_updates	527400	

EP=10,440 | STEP=2,159,950 | clip=57 | consec=5 | max_consec=87 | avg_clip50=52.6
5
EP=10,450 | STEP=2,162,584 | clip=19 | consec=0 | max_consec=87 | avg_clip50=51.4
8
EP=10,460 | STEP=2,165,657 | clip=61 | consec=10 | max_consec=87 | avg_clip50=50.
98
EP=10,470 | STEP=2,168,217 | clip=32 | consec=1 | max_consec=87 | avg_clip50=49.2
1

custom/		
avg_clip50	48.7	
clipped_episode_r...	50	
consecutive_clipp...	5	
max_consecutive_c...	87	
rollout/		
ep_len_mean	1.18e+03	
ep_rew_mean	519	
exploration rate	0.278	
time/		
episodes	30900	
fps	204	
time_elapsed	10585	
total timesteps	2169347	
train/		
learning_rate	0.0001	
loss	0.0499	
n_updates	529836	

EP=10,480 | STEP=2,170,912 | clip=74 | consec=11 | max_consec=87 | avg_clip50=49.
52
EP=10,490 | STEP=2,173,932 | clip=55 | consec=21 | max_consec=87 | avg_clip50=50.
73
EP=10,500 | STEP=2,176,535 | clip=78 | consec=31 | max_consec=87 | avg_clip50=50.
83

custom/	
avg_clip50	49
clipped_episode_r...	50
consecutive_clipp...	39
max_consecutive_c...	87
rollout/	
ep_len_mean	1.15e+03
ep_rew_mean	498
exploration rate	0.275
time/	
episodes	31000
fps	204
time_elapsed	10633
total timesteps	2178565
train/	
learning_rate	0.0001
loss	0.0567
n_updates	532141

EP=10,510 | STEP=2,179,124 | clip=60 | consec=41 | max_consec=87 | avg_clip50=48.77

EP=10,520 | STEP=2,181,804 | clip=34 | consec=8 | max_consec=87 | avg_clip50=49.81

EP=10,530 | STEP=2,184,701 | clip=68 | consec=18 | max_consec=87 | avg_clip50=50.95

EP=10,540 | STEP=2,187,450 | clip=26 | consec=7 | max_consec=87 | avg_clip50=50.68

custom/	
avg_clip50	51.1
clipped_episode_r...	47.5
consecutive_clipp...	11
max_consecutive_c...	87
rollout/	
ep_len_mean	1.14e+03
ep_rew_mean	505
exploration rate	0.271
time/	
episodes	31100
fps	204
time_elapsed	10686
total timesteps	2188522
train/	
learning_rate	0.0001
loss	0.0472
n_updates	534630

EP=10,550 | STEP=2,190,301 | clip=63 | consec=17 | max_consec=87 | avg_clip50=51.10

EP=10,560 | STEP=2,193,216 | clip=52 | consec=27 | max_consec=87 | avg_clip50=52.57

EP=10,570 | STEP=2,196,196 | clip=56 | consec=37 | max_consec=87 | avg_clip50=53.89

custom/	
avg_clip50	53.6
clipped_episode_r...	27.5
consecutive_clipp...	45
max_consecutive_c...	87
rollout/	

ep_len_mean	1.17e+03
ep_rew_mean	524
exploration rate	0.268
time/	
episodes	31200
fps	204
time_elapsed	10739
total timesteps	2198768
train/	
learning_rate	0.0001
loss	0.0709
n_updates	537191

EP=10,580 | STEP=2,199,140 | clip=56 | consec=47 | max_consec=87 | avg_clip50=53.78

EP=10,590 | STEP=2,201,809 | clip=42 | consec=57 | max_consec=87 | avg_clip50=53.14

EP=10,600 | STEP=2,204,634 | clip=44 | consec=67 | max_consec=87 | avg_clip50=53.26

EP=10,610 | STEP=2,207,348 | clip=54 | consec=77 | max_consec=87 | avg_clip50=52.90

custom/	
avg_clip50	53.6
clipped_episode_r...	71
consecutive_clipp...	79
max_consecutive_c...	87
rollout/	
ep_len_mean	1.19e+03
ep_rew_mean	534
exploration rate	0.265
time/	
episodes	31300
fps	204
time_elapsed	10790
total timesteps	2207991
train/	
learning_rate	0.0001
loss	0.399
n_updates	539497

EP=10,620 | STEP=2,210,192 | clip=80 | consec=87 | max_consec=87 | avg_clip50=53.35

EP=10,630 | STEP=2,212,713 | clip=36 | consec=97 | max_consec=97 | avg_clip50=52.27

Objetivo conseguido: 100 episodios consecutivos superando el umbral.
Entrenamiento finalizado.
max_consec alcanzado: 100
Log JSON: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.json
Log CSV: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.csv
Checkpoint final guardado en: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_dueling/dqn_spaceinvaders_v0_policy_latest.pt
model.save() falló (no crítico): IndexError('tuple index out of range')

```
In [ ]: # =====
# M7. Entrenamiento Propuesta 3: exploración ajustada
# =====
```

```
env_tuned = create_train_env(RUN_PATHS['tuned']['monitor'])
model_tuned = build_model('tuned', env_tuned, RUN_PATHS['tuned']['tensorboard'])
model_tuned = train_variant('tuned', model_tuned)
env_tuned.close()
```

Using cuda device

Wrapping the env in a VecTransposeImage.

Logging to /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_tuned/tensorboard/DQN_tuned_1

EP=0 | STEP=267 | clip=18 | consec=0 | max_consec=0 | avg_clip50=18.00

EP=10 | STEP=1,894 | clip=66 | consec=1 | max_consec=1 | avg_clip50=18.45

EP=20 | STEP=3,135 | clip=8 | consec=0 | max_consec=1 | avg_clip50=13.05

EP=30 | STEP=4,765 | clip=16 | consec=0 | max_consec=1 | avg_clip50=14.29

custom/		
avg_clip50	14.1	
clipped_episode_r...	15.5	
consecutive_clipp...	0	
max_consecutive_c...	1	
rollout/		
ep_len_mean	659	
ep_rew_mean	141	
exploration rate	0.995	
time/		
episodes	100	
fps	341	
time_elapsed	15	
total timesteps	5126	

EP=40 | STEP=6,186 | clip=16 | consec=0 | max_consec=1 | avg_clip50=13.74

EP=50 | STEP=7,696 | clip=8 | consec=0 | max_consec=1 | avg_clip50=13.15

EP=60 | STEP=9,261 | clip=4 | consec=0 | max_consec=2 | avg_clip50=11.81

custom/		
avg_clip50	13.1	
clipped_episode_r...	14	
consecutive_clipp...	0	
max_consecutive_c...	2	
rollout/		
ep_len_mean	660	
ep_rew_mean	136	
exploration rate	0.99	
time/		
episodes	200	
fps	333	
time_elapsed	30	
total timesteps	10233	

EP=70 | STEP=10,844 | clip=32 | consec=1 | max_consec=2 | avg_clip50=13.80

EP=80 | STEP=12,464 | clip=34 | consec=2 | max_consec=2 | avg_clip50=14.04

EP=90 | STEP=14,027 | clip=4 | consec=0 | max_consec=2 | avg_clip50=13.89

custom/		
avg_clip50	14.8	
clipped_episode_r...	21	
consecutive_clipp...	1	
max_consecutive_c...	3	
rollout/		
ep_len_mean	665	
ep_rew_mean	141	
exploration rate	0.985	
time/		
episodes	300	
fps	326	
time_elapsed	47	

	total timesteps	15452				
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EP=100	STEP=15,660	clip=24	consec=2	max_consec=3	avg_clip50=15.15
EP=110	STEP=17,107	clip=6	consec=0	max_consec=3	avg_clip50=15.14
EP=120	STEP=18,403	clip=13	consec=0	max_consec=3	avg_clip50=13.64
EP=130	STEP=20,065	clip=20	consec=1	max_consec=3	avg_clip50=12.83

custom/		
avg_clip50	12.9	
clipped_episode_r...	17.5	
consecutive_clipp...	0	
max_consecutive_c...	3	
rollout/		
ep_len_mean	657	
ep_rew_mean	135	
exploration rate	0.98	
time/		
episodes	400	
fps	329	
time_elapsed	61	
total timesteps	20359	

EP=140	STEP=21,723	clip=2	consec=0	max_consec=3	avg_clip50=13.47
EP=150	STEP=23,061	clip=9	consec=0	max_consec=3	avg_clip50=11.68
EP=160	STEP=24,659	clip=6	consec=0	max_consec=3	avg_clip50=12.61

custom/		
avg_clip50	13.6	
clipped_episode_r...	15	
consecutive_clipp...	0	
max_consecutive_c...	3	
rollout/		
ep_len_mean	664	
ep_rew_mean	137	
exploration rate	0.975	
time/		
episodes	500	
fps	318	
time_elapsed	80	
total timesteps	25714	
train/		
learning_rate	0.0001	
loss	0.0302	
n_updates	178	

EP=170	STEP=26,416	clip=8	consec=0	max_consec=3	avg_clip50=14.13
EP=180	STEP=28,258	clip=11	consec=0	max_consec=3	avg_clip50=15.15
EP=190	STEP=29,915	clip=2	consec=0	max_consec=3	avg_clip50=14.92

custom/		
avg_clip50	15.8	
clipped_episode_r...	3	
consecutive_clipp...	0	
max_consecutive_c...	3	
rollout/		
ep_len_mean	685	
ep_rew_mean	139	
exploration rate	0.969	
time/		
episodes	600	

fps	298
time_elapsed	105
total timesteps	31423
train/	
learning_rate	0.0001
loss	0.015
n_updates	1605

EP=200	STEP=31,578	clip=11	consec=0	max_consec=3	avg_clip50=15.83
EP=210	STEP=33,317	clip=42	consec=1	max_consec=3	avg_clip50=16.29
EP=220	STEP=35,104	clip=14	consec=0	max_consec=3	avg_clip50=16.11
EP=230	STEP=36,620	clip=18	consec=0	max_consec=3	avg_clip50=14.94

custom/	
avg_clip50	14.7
clipped_episode_r...	7
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	710
ep_rew_mean	149
exploration rate	0.963
time/	
episodes	700
fps	289
time_elapsed	127
total timesteps	36986
train/	
learning_rate	0.0001
loss	0.047
n_updates	2996

EP=240	STEP=38,324	clip=10	consec=0	max_consec=3	avg_clip50=15.30
EP=250	STEP=40,087	clip=22	consec=1	max_consec=3	avg_clip50=15.10
EP=260	STEP=41,556	clip=2	consec=0	max_consec=3	avg_clip50=14.03

custom/	
avg_clip50	13.2
clipped_episode_r...	8
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	717
ep_rew_mean	151
exploration rate	0.958
time/	
episodes	800
fps	280
time_elapsed	151
total timesteps	42468
train/	
learning_rate	0.0001
loss	0.0321
n_updates	4366

EP=270	STEP=42,957	clip=12	consec=0	max_consec=3	avg_clip50=13.15
EP=280	STEP=44,848	clip=24	consec=2	max_consec=3	avg_clip50=13.51
EP=290	STEP=46,375	clip=10	consec=0	max_consec=3	avg_clip50=12.98

custom/	
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avg_clip50	12.9
clipped_episode_r...	28.5
consecutive_clipp...	1
max_consecutive_c...	3
rollout/	
ep_len_mean	697
ep_rew_mean	139
exploration rate	0.953
time/	
episodes	900
fps	275
time_elapsed	174
total timesteps	47973
train/	
learning_rate	0.0001
loss	0.00471
n_updates	5743

EP=300	STEP=48,053	clip=30	consec=2	max_consec=3	avg_clip50=13.10
EP=310	STEP=49,489	clip=18	consec=0	max_consec=3	avg_clip50=13.36
EP=320	STEP=50,932	clip=11	consec=0	max_consec=3	avg_clip50=13.68
EP=330	STEP=52,675	clip=30	consec=1	max_consec=3	avg_clip50=13.44

custom/	
avg_clip50	13.4
clipped_episode_r...	20
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	694
ep_rew_mean	138
exploration rate	0.947
time/	
episodes	1000
fps	270
time_elapsed	196
total timesteps	53221
train/	
learning_rate	0.0001
loss	0.0188
n_updates	7055

EP=340	STEP=54,260	clip=12	consec=0	max_consec=3	avg_clip50=13.34
EP=350	STEP=55,733	clip=12	consec=0	max_consec=3	avg_clip50=13.71
EP=360	STEP=57,284	clip=18	consec=0	max_consec=3	avg_clip50=13.80

custom/	
avg_clip50	13.7
clipped_episode_r...	8
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	673
ep_rew_mean	133
exploration rate	0.942
time/	
episodes	1100
fps	267
time_elapsed	218
total timesteps	58230

train/	
learning_rate	0.0001
loss	0.0372
n_updates	8307

EP=370	STEP=58,710	clip=20	consec=1	max_consec=3	avg_clip50=13.03
EP=380	STEP=60,286	clip=11	consec=0	max_consec=3	avg_clip50=13.69
EP=390	STEP=61,898	clip=28	consec=1	max_consec=3	avg_clip50=14.69
EP=400	STEP=63,860	clip=44	consec=2	max_consec=3	avg_clip50=15.80

custom/	
avg_clip50	15.8
clipped_episode_r...	44
consecutive_clipp...	2
max_consecutive_c...	3
rollout/	
ep_len_mean	677
ep_rew_mean	148
exploration rate	0.937
time/	
episodes	1200
fps	264
time_elapsed	241
total timesteps	63910
train/	
learning_rate	0.0001
loss	0.0208
n_updates	9727

EP=410	STEP=65,407	clip=14	consec=0	max_consec=3	avg_clip50=15.06
EP=420	STEP=66,901	clip=19	consec=0	max_consec=3	avg_clip50=15.20
EP=430	STEP=68,666	clip=26	consec=1	max_consec=3	avg_clip50=15.04

custom/	
avg_clip50	14.9
clipped_episode_r...	0
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	678
ep_rew_mean	140
exploration rate	0.931
time/	
episodes	1300
fps	261
time_elapsed	264
total timesteps	69193
train/	
learning_rate	0.0001
loss	0.0444
n_updates	11048

EP=440	STEP=70,104	clip=7	consec=0	max_consec=3	avg_clip50=13.49
EP=450	STEP=71,969	clip=14	consec=0	max_consec=3	avg_clip50=12.75
EP=460	STEP=73,480	clip=4	consec=0	max_consec=3	avg_clip50=13.45

custom/	
avg_clip50	15
clipped_episode_r...	21
consecutive_clipp...	1

max_consecutive_c...	3
rollout/	
ep_len_mean	699
ep_rew_mean	150
exploration rate	0.926
time/	
episodes	1400
fps	260
time_elapsed	286
total timesteps	74665
train/	
learning_rate	0.0001
loss	0.0264
n_updates	12416

EP=470 | STEP=75,037 | clip=24 | consec=1 | max_consec=3 | avg_clip50=15.08
EP=480 | STEP=76,431 | clip=6 | consec=0 | max_consec=3 | avg_clip50=13.45
EP=490 | STEP=78,061 | clip=5 | consec=0 | max_consec=3 | avg_clip50=14.29
EP=500 | STEP=79,683 | clip=14 | consec=0 | max_consec=3 | avg_clip50=13.68

custom/	
avg_clip50	13.7
clipped_episode_r...	13.5
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	680
ep_rew_mean	132
exploration rate	0.921
time/	
episodes	1500
fps	258
time_elapsed	308
total timesteps	79781
train/	
learning_rate	0.0001
loss	0.0365
n_updates	13695

EP=510 | STEP=81,247 | clip=4 | consec=0 | max_consec=3 | avg_clip50=12.86
EP=520 | STEP=82,788 | clip=10 | consec=0 | max_consec=3 | avg_clip50=11.81
EP=530 | STEP=84,802 | clip=44 | consec=1 | max_consec=3 | avg_clip50=14.58

custom/	
avg_clip50	14.8
clipped_episode_r...	1.5
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	700
ep_rew_mean	143
exploration rate	0.915
time/	
episodes	1600
fps	256
time_elapsed	333
total timesteps	85527
train/	
learning_rate	0.0001
loss	0.0533

n_updates	15131
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EP=540	STEP=86,482	clip=20	consec=0	max_consec=3	avg_clip50=14.47
EP=550	STEP=88,115	clip=26	consec=1	max_consec=3	avg_clip50=14.75
EP=560	STEP=89,878	clip=44	consec=2	max_consec=3	avg_clip50=15.30

custom/	
avg_clip50	14.5
clipped_episode_r...	20
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	699
ep_rew_mean	135
exploration rate	0.91
time/	
episodes	1700
fps	256
time_elapsed	355
total timesteps	90977
train/	
learning_rate	0.0001
loss	0.0367
n_updates	16494

EP=570	STEP=91,296	clip=12	consec=0	max_consec=3	avg_clip50=14.65
EP=580	STEP=92,914	clip=5	consec=0	max_consec=3	avg_clip50=12.60
EP=590	STEP=94,395	clip=26	consec=1	max_consec=3	avg_clip50=12.18
EP=600	STEP=95,856	clip=4	consec=0	max_consec=3	avg_clip50=12.42

custom/	
avg_clip50	12.5
clipped_episode_r...	11
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	693
ep_rew_mean	136
exploration rate	0.905
time/	
episodes	1800
fps	254
time_elapsed	376
total timesteps	95971
train/	
learning_rate	0.0001
loss	0.0151
n_updates	17742

EP=610	STEP=97,868	clip=26	consec=3	max_consec=3	avg_clip50=14.14
EP=620	STEP=99,520	clip=40	consec=1	max_consec=3	avg_clip50=14.76
EP=630	STEP=100,982	clip=21	consec=1	max_consec=3	avg_clip50=14.54

custom/	
avg_clip50	14.7
clipped_episode_r...	14.5
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	687

ep_rew_mean	134
exploration rate	0.899
time/	
episodes	1900
fps	254
time_elapsed	398
total timesteps	101534
train/	
learning_rate	0.0001
loss	0.0384
n_updates	19133

EP=640	STEP=102,485	clip=9	consec=0	max_consec=3	avg_clip50=15.20
EP=650	STEP=104,078	clip=15	consec=0	max_consec=3	avg_clip50=15.18
EP=660	STEP=105,853	clip=11	consec=0	max_consec=3	avg_clip50=15.01

custom/	
avg_clip50	15.3
clipped_episode_r...	28.5
consecutive_clipp...	1
max_consecutive_c...	3
rollout/	
ep_len_mean	693
ep_rew_mean	148
exploration rate	0.894
time/	
episodes	2000
fps	253
time_elapsed	423
total timesteps	107278
train/	
learning_rate	0.0001
loss	0.0303
n_updates	20569

EP=670	STEP=107,745	clip=17	consec=0	max_consec=3	avg_clip50=16.02
EP=680	STEP=109,479	clip=18	consec=0	max_consec=3	avg_clip50=17.76
EP=690	STEP=111,139	clip=10	consec=0	max_consec=3	avg_clip50=18.09
EP=700	STEP=112,773	clip=13	consec=0	max_consec=3	avg_clip50=17.88

custom/	
avg_clip50	17.9
clipped_episode_r...	7
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	725
ep_rew_mean	165
exploration rate	0.888
time/	
episodes	2100
fps	252
time_elapsed	447
total timesteps	112947
train/	
learning_rate	0.0001
loss	0.0261
n_updates	21986

EP=710	STEP=114,374	clip=10	consec=0	max_consec=3	avg_clip50=16.51
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EP=720 | STEP=115,992 | clip=8 | consec=0 | max_consec=3 | avg_clip50=15.40
EP=730 | STEP=117,645 | clip=4 | consec=0 | max_consec=3 | avg_clip50=14.80

custom/	
avg_clip50	15.1
clipped_episode_r...	28
consecutive_clipp...	1
max_consecutive_c...	3
rollout/	
ep_len_mean	718
ep_rew_mean	165
exploration rate	0.883
time/	
episodes	2200
fps	252
time_elapsed	469
total timesteps	118412
train/	
learning_rate	0.0001
loss	0.024
n_updates	23352

EP=740 | STEP=119,236 | clip=5 | consec=0 | max_consec=3 | avg_clip50=14.24
EP=750 | STEP=120,691 | clip=3 | consec=0 | max_consec=3 | avg_clip50=14.33
EP=760 | STEP=122,453 | clip=4 | consec=0 | max_consec=3 | avg_clip50=15.01

custom/	
avg_clip50	15.3
clipped_episode_r...	19
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	708
ep_rew_mean	160
exploration rate	0.878
time/	
episodes	2300
fps	251
time_elapsed	492
total timesteps	123687
train/	
learning_rate	0.0001
loss	0.0377
n_updates	24671

EP=770 | STEP=123,986 | clip=10 | consec=0 | max_consec=3 | avg_clip50=15.46
EP=780 | STEP=125,411 | clip=8 | consec=0 | max_consec=3 | avg_clip50=14.41
EP=790 | STEP=126,908 | clip=8 | consec=0 | max_consec=3 | avg_clip50=14.03
EP=800 | STEP=128,299 | clip=12 | consec=0 | max_consec=3 | avg_clip50=13.73

custom/	
avg_clip50	13.7
clipped_episode_r...	9.5
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	668
ep_rew_mean	141
exploration rate	0.873
time/	

episodes	2400
fps	251
time_elapsed	511
total timesteps	128486
train/	
learning_rate	0.0001
loss	0.0869
n_updates	25871

EP=810 | STEP=129,812 | clip=19 | consec=0 | max_consec=3 | avg_clip50=13.10
EP=820 | STEP=131,258 | clip=20 | consec=0 | max_consec=3 | avg_clip50=12.53
EP=830 | STEP=132,729 | clip=14 | consec=0 | max_consec=3 | avg_clip50=12.02

custom/	
avg_clip50	11.6
clipped_episode_r...	22.5
consecutive_clipp...	1
max_consecutive_c...	3
rollout/	
ep_len_mean	640
ep_rew_mean	129
exploration rate	0.868
time/	
episodes	2500
fps	249
time_elapsed	532
total timesteps	133225
train/	
learning_rate	0.0001
loss	0.123
n_updates	27056

EP=840 | STEP=134,044 | clip=3 | consec=0 | max_consec=3 | avg_clip50=11.62
EP=850 | STEP=135,627 | clip=26 | consec=1 | max_consec=3 | avg_clip50=11.53
EP=860 | STEP=137,007 | clip=2 | consec=0 | max_consec=3 | avg_clip50=11.70

custom/	
avg_clip50	12.6
clipped_episode_r...	31.5
consecutive_clipp...	1
max_consecutive_c...	3
rollout/	
ep_len_mean	629
ep_rew_mean	125
exploration rate	0.863
time/	
episodes	2600
fps	249
time_elapsed	554
total timesteps	138289
train/	
learning_rate	0.0001
loss	0.128
n_updates	28322

EP=870 | STEP=138,530 | clip=8 | consec=0 | max_consec=3 | avg_clip50=12.39
EP=880 | STEP=140,026 | clip=24 | consec=1 | max_consec=3 | avg_clip50=12.95
EP=890 | STEP=141,653 | clip=10 | consec=0 | max_consec=3 | avg_clip50=14.40
EP=900 | STEP=143,184 | clip=12 | consec=0 | max_consec=3 | avg_clip50=14.73

custom/	
avg_clip50	14.7
clipped_episode_r...	14
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	640
ep_rew_mean	132
exploration rate	0.858
time/	
episodes	2700
fps	248
time_elapsed	576
total timesteps	143456
train/	
learning_rate	0.0001
loss	0.0933
n_updates	29613

EP=910 | STEP=144,803 | clip=5 | consec=0 | max_consec=3 | avg_clip50=14.64
EP=920 | STEP=146,331 | clip=6 | consec=0 | max_consec=3 | avg_clip50=13.10
EP=930 | STEP=147,748 | clip=12 | consec=0 | max_consec=3 | avg_clip50=13.25

custom/	
avg_clip50	12.3
clipped_episode_r...	1.5
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	646
ep_rew_mean	131
exploration rate	0.853
time/	
episodes	2800
fps	248
time_elapsed	596
total timesteps	148277
train/	
learning_rate	0.0001
loss	0.0835
n_updates	30819

EP=940 | STEP=149,147 | clip=57 | consec=1 | max_consec=3 | avg_clip50=12.82
EP=950 | STEP=150,818 | clip=37 | consec=2 | max_consec=3 | avg_clip50=12.66
EP=960 | STEP=152,217 | clip=11 | consec=0 | max_consec=3 | avg_clip50=11.58

custom/	
avg_clip50	12.5
clipped_episode_r...	18.5
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	656
ep_rew_mean	129
exploration rate	0.848
time/	
episodes	2900
fps	247
time_elapsed	620
total timesteps	153656

train/	
learning_rate	0.0001
loss	0.132
n_updates	32163

EP=970	STEP=154,099	clip=48	consec=1	max_consec=3	avg_clip50=13.56
EP=980	STEP=155,409	clip=10	consec=0	max_consec=3	avg_clip50=12.70
EP=990	STEP=156,785	clip=19	consec=0	max_consec=3	avg_clip50=11.98
EP=1,000	STEP=158,172	clip=2	consec=0	max_consec=3	avg_clip50=11.24

custom/	
avg_clip50	11
clipped_episode_r...	7.5
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	647
ep_rew_mean	119
exploration rate	0.843
time/	
episodes	3000
fps	247
time_elapsed	640
total timesteps	158487
train/	
learning_rate	0.0001
loss	0.0879
n_updates	33371

EP=1,010	STEP=159,853	clip=9	consec=0	max_consec=3	avg_clip50=12.40
EP=1,020	STEP=161,233	clip=4	consec=0	max_consec=3	avg_clip50=11.15
EP=1,030	STEP=162,560	clip=10	consec=0	max_consec=3	avg_clip50=11.31

custom/	
avg_clip50	10.7
clipped_episode_r...	11
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	642
ep_rew_mean	120
exploration rate	0.838
time/	
episodes	3100
fps	246
time_elapsed	661
total timesteps	163187
train/	
learning_rate	0.0001
loss	0.0462
n_updates	34546

EP=1,040	STEP=164,032	clip=30	consec=1	max_consec=3	avg_clip50=11.24
EP=1,050	STEP=165,422	clip=2	consec=0	max_consec=3	avg_clip50=11.23
EP=1,060	STEP=167,327	clip=9	consec=0	max_consec=3	avg_clip50=12.48

custom/	
avg_clip50	12.4
clipped_episode_r...	9
consecutive_clipp...	0

max_consecutive_c...	3
rollout/	
ep_len_mean	645
ep_rew_mean	123
exploration rate	0.833
time/	
episodes	3200
fps	246
time_elapsed	684
total timesteps	168614
train/	
learning_rate	0.0001
loss	0.0985
n_updates	35903

EP=1,070	STEP=168,846	clip=10	consec=0	max_consec=3	avg_clip50=12.68
EP=1,080	STEP=170,581	clip=33	consec=1	max_consec=3	avg_clip50=14.84
EP=1,090	STEP=172,267	clip=60	consec=1	max_consec=3	avg_clip50=16.33
EP=1,100	STEP=173,479	clip=6	consec=0	max_consec=3	avg_clip50=15.92

custom/	
avg_clip50	15.7
clipped_episode_r...	9
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	654
ep_rew_mean	136
exploration rate	0.828
time/	
episodes	3300
fps	245
time_elapsed	706
total timesteps	173806
train/	
learning_rate	0.0001
loss	0.0365
n_updates	37201

EP=1,110	STEP=174,822	clip=4	consec=0	max_consec=3	avg_clip50=13.56
EP=1,120	STEP=176,217	clip=4	consec=0	max_consec=3	avg_clip50=13.16
EP=1,130	STEP=178,067	clip=14	consec=0	max_consec=3	avg_clip50=12.53

custom/	
avg_clip50	12.3
clipped_episode_r...	16
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	663
ep_rew_mean	139
exploration rate	0.823
time/	
episodes	3400
fps	245
time_elapsed	728
total timesteps	178748
train/	
learning_rate	0.0001
loss	0.0194

n_updates	38436
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EP=1,140	STEP=179,428	clip=20	consec=1	max_consec=3	avg_clip50=11.37
EP=1,150	STEP=181,228	clip=7	consec=0	max_consec=3	avg_clip50=12.27
EP=1,160	STEP=182,748	clip=10	consec=0	max_consec=3	avg_clip50=12.94

custom/	
avg_clip50	14
clipped_episode_r...	24.5
consecutive_clipp...	2
max_consecutive_c...	3
rollout/	
ep_len_mean	670
ep_rew_mean	137
exploration rate	0.818
time/	
episodes	3500
fps	245
time_elapsed	752
total timesteps	184305
train/	
learning_rate	0.0001
loss	0.0791
n_updates	39826

EP=1,170	STEP=184,555	clip=32	consec=3	max_consec=3	avg_clip50=14.58
EP=1,180	STEP=185,994	clip=6	consec=0	max_consec=3	avg_clip50=13.62
EP=1,190	STEP=187,600	clip=8	consec=0	max_consec=3	avg_clip50=13.78
EP=1,200	STEP=189,436	clip=11	consec=0	max_consec=3	avg_clip50=14.76

custom/	
avg_clip50	14.4
clipped_episode_r...	7.5
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	684
ep_rew_mean	135
exploration rate	0.812
time/	
episodes	3600
fps	244
time_elapsed	776
total timesteps	189798
train/	
learning_rate	0.0001
loss	0.076
n_updates	41199

EP=1,210	STEP=191,105	clip=14	consec=0	max_consec=3	avg_clip50=14.88
EP=1,220	STEP=193,010	clip=34	consec=2	max_consec=3	avg_clip50=15.10
EP=1,230	STEP=194,681	clip=8	consec=0	max_consec=3	avg_clip50=16.81

custom/	
avg_clip50	17.1
clipped_episode_r...	14
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	723

ep_rew_mean	156
exploration rate	0.806
time/	
episodes	3700
fps	243
time_elapsed	802
total timesteps	195699
train/	
learning_rate	0.0001
loss	0.0499
n_updates	42674

EP=1,240 | STEP=196,337 | clip=6 | consec=0 | max_consec=3 | avg_clip50=17.07
EP=1,250 | STEP=197,936 | clip=14 | consec=0 | max_consec=3 | avg_clip50=16.19
EP=1,260 | STEP=199,666 | clip=57 | consec=1 | max_consec=3 | avg_clip50=17.94

custom/	
avg_clip50	18.5
clipped_episode_r...	51.5
consecutive_clipp...	2
max_consecutive_c...	3
rollout/	
ep_len_mean	732
ep_rew_mean	168
exploration rate	0.801
time/	
episodes	3800
fps	243
time_elapsed	827
total timesteps	201501
train/	
learning_rate	0.0001
loss	0.177
n_updates	44125

EP=1,270 | STEP=201,616 | clip=15 | consec=0 | max_consec=3 | avg_clip50=18.11
EP=1,280 | STEP=203,436 | clip=16 | consec=0 | max_consec=3 | avg_clip50=18.48
EP=1,290 | STEP=204,953 | clip=6 | consec=0 | max_consec=3 | avg_clip50=17.93
EP=1,300 | STEP=206,592 | clip=12 | consec=0 | max_consec=3 | avg_clip50=18.04

custom/	
avg_clip50	17.8
clipped_episode_r...	7.5
consecutive_clipp...	0
max_consecutive_c...	3
rollout/	
ep_len_mean	732
ep_rew_mean	171
exploration rate	0.795
time/	
episodes	3900
fps	243
time_elapsed	850
total timesteps	206990
train/	
learning_rate	0.0001
loss	0.0896
n_updates	45497

EP=1,310 | STEP=208,272 | clip=16 | consec=0 | max_consec=3 | avg_clip50=16.97

EP=1,320 | STEP=209,632 | clip=26 | consec=1 | max_consec=3 | avg_clip50=14.89
EP=1,330 | STEP=211,130 | clip=15 | consec=0 | max_consec=3 | avg_clip50=13.17

custom/		
avg_clip50	13.4	
clipped_episode_r...	10.5	
consecutive_clipp...	0	
max_consecutive_c...	3	
rollout/		
ep_len_mean	704	
ep_rew_mean	156	
exploration rate	0.79	
time/		
episodes	4000	
fps	242	
time_elapsed	873	
total timesteps	212194	
train/		
learning_rate	0.0001	
loss	0.0752	
n_updates	46798	

EP=1,340 | STEP=212,761 | clip=4 | consec=0 | max_consec=3 | avg_clip50=14.07
EP=1,350 | STEP=214,780 | clip=16 | consec=0 | max_consec=3 | avg_clip50=15.36
EP=1,360 | STEP=216,639 | clip=4 | consec=0 | max_consec=3 | avg_clip50=15.19

custom/		
avg_clip50	16.3	
clipped_episode_r...	3	
consecutive_clipp...	0	
max_consecutive_c...	3	
rollout/		
ep_len_mean	715	
ep_rew_mean	155	
exploration rate	0.784	
time/		
episodes	4100	
fps	242	
time_elapsed	899	
total timesteps	218225	
train/		
learning_rate	0.0001	
loss	0.0892	
n_updates	48306	

EP=1,370 | STEP=218,317 | clip=20 | consec=0 | max_consec=3 | avg_clip50=16.20
EP=1,380 | STEP=220,180 | clip=16 | consec=0 | max_consec=4 | avg_clip50=17.59
EP=1,390 | STEP=221,947 | clip=22 | consec=2 | max_consec=4 | avg_clip50=17.78
EP=1,400 | STEP=223,825 | clip=6 | consec=0 | max_consec=4 | avg_clip50=17.78

custom/		
avg_clip50	17.4	
clipped_episode_r...	8	
consecutive_clipp...	0	
max_consecutive_c...	4	
rollout/		
ep_len_mean	734	
ep_rew_mean	166	
exploration rate	0.778	
time/		

episodes	4200
fps	242
time_elapsed	926
total timesteps	224245
train/	
learning_rate	0.0001
loss	0.0416
n_updates	49811

EP=1,410 | STEP=225,489 | clip=12 | consec=0 | max_consec=4 | avg_clip50=17.54
EP=1,420 | STEP=227,076 | clip=24 | consec=2 | max_consec=4 | avg_clip50=17.76
EP=1,430 | STEP=228,874 | clip=22 | consec=1 | max_consec=4 | avg_clip50=17.31

custom/	
avg_clip50	17.2
clipped_episode_r...	19.5
consecutive_clipp...	0
max_consecutive_c...	4
rollout/	
ep_len_mean	759
ep_rew_mean	177
exploration rate	0.772
time/	
episodes	4300
fps	241
time_elapsed	952
total timesteps	230017
train/	
learning_rate	0.0001
loss	0.112
n_updates	51254

EP=1,440 | STEP=230,752 | clip=54 | consec=1 | max_consec=4 | avg_clip50=17.75
EP=1,450 | STEP=232,602 | clip=20 | consec=0 | max_consec=4 | avg_clip50=16.85
EP=1,460 | STEP=234,440 | clip=26 | consec=1 | max_consec=4 | avg_clip50=17.54
EP=1,470 | STEP=236,293 | clip=32 | consec=1 | max_consec=4 | avg_clip50=17.68

custom/	
avg_clip50	17.7
clipped_episode_r...	32.5
consecutive_clipp...	1
max_consecutive_c...	4
rollout/	
ep_len_mean	765
ep_rew_mean	177
exploration rate	0.766
time/	
episodes	4400
fps	241
time_elapsed	980
total timesteps	236293
train/	
learning_rate	0.0001
loss	0.183
n_updates	52823

EP=1,480 | STEP=238,344 | clip=4 | consec=0 | max_consec=4 | avg_clip50=17.68
EP=1,490 | STEP=240,366 | clip=9 | consec=0 | max_consec=4 | avg_clip50=17.22
EP=1,500 | STEP=242,201 | clip=52 | consec=2 | max_consec=4 | avg_clip50=18.51

custom/	
avg_clip50	19.7
clipped_episode_r...	21
consecutive_clipp...	5
max_consecutive_c...	5
rollout/	
ep_len_mean	794
ep_rew_mean	184
exploration rate	0.759
time/	
episodes	4500
fps	240
time_elapsed	1009
total timesteps	243067
train/	
learning_rate	0.0001
loss	0.0774
n_updates	54516

EP=1,510 | STEP=244,173 | clip=15 | consec=0 | max_consec=5 | avg_clip50=19.01
EP=1,520 | STEP=246,113 | clip=7 | consec=0 | max_consec=5 | avg_clip50=19.79
EP=1,530 | STEP=247,971 | clip=13 | consec=0 | max_consec=5 | avg_clip50=20.28

custom/	
avg_clip50	20.7
clipped_episode_r...	25.5
consecutive_clipp...	4
max_consecutive_c...	5
rollout/	
ep_len_mean	820
ep_rew_mean	192
exploration rate	0.753
time/	
episodes	4600
fps	240
time_elapsed	1037
total timesteps	249534
train/	
learning_rate	0.0001
loss	0.0331
n_updates	56133

EP=1,540 | STEP=249,956 | clip=24 | consec=1 | max_consec=5 | avg_clip50=20.66
EP=1,550 | STEP=251,863 | clip=35 | consec=1 | max_consec=5 | avg_clip50=20.26
EP=1,560 | STEP=253,552 | clip=24 | consec=1 | max_consec=5 | avg_clip50=19.04
EP=1,570 | STEP=255,137 | clip=38 | consec=1 | max_consec=5 | avg_clip50=17.76

custom/	
avg_clip50	17.8
clipped_episode_r...	38.5
consecutive_clipp...	1
max_consecutive_c...	5
rollout/	
ep_len_mean	799
ep_rew_mean	188
exploration rate	0.747
time/	
episodes	4700
fps	240
time_elapsed	1063

total timesteps	255285
train/	
learning_rate	0.0001
loss	0.119
n_updates	57571

EP=1,580	STEP=257,411	clip=16	consec=0	max_consec=5	avg_clip50=18.62
EP=1,590	STEP=259,462	clip=6	consec=0	max_consec=5	avg_clip50=19.44
EP=1,600	STEP=261,394	clip=10	consec=0	max_consec=5	avg_clip50=19.40

custom/	
avg_clip50	19.8
clipped_episode_r...	14.5
consecutive_clipp...	0
max_consecutive_c...	5
rollout/	
ep_len_mean	800
ep_rew_mean	193
exploration rate	0.741
time/	
episodes	4800
fps	239
time_elapsed	1092
total timesteps	262112
train/	
learning_rate	0.0001
loss	0.103
n_updates	59277

EP=1,610	STEP=263,311	clip=7	consec=0	max_consec=5	avg_clip50=20.18
EP=1,620	STEP=265,042	clip=12	consec=0	max_consec=5	avg_clip50=20.37
EP=1,630	STEP=266,580	clip=6	consec=0	max_consec=5	avg_clip50=18.51

custom/	
avg_clip50	16.6
clipped_episode_r...	11
consecutive_clipp...	0
max_consecutive_c...	5
rollout/	
ep_len_mean	766
ep_rew_mean	179
exploration rate	0.735
time/	
episodes	4900
fps	239
time_elapsed	1117
total timesteps	267617
train/	
learning_rate	0.0001
loss	0.078
n_updates	60654

EP=1,640	STEP=267,885	clip=8	consec=0	max_consec=5	avg_clip50=16.16
EP=1,650	STEP=269,804	clip=34	consec=1	max_consec=5	avg_clip50=15.74
EP=1,660	STEP=271,964	clip=4	consec=0	max_consec=5	avg_clip50=16.20
EP=1,670	STEP=273,733	clip=11	consec=0	max_consec=5	avg_clip50=16.37

custom/	
avg_clip50	16.4
clipped_episode_r...	16.5

consecutive_clipp...	0
max_consecutive_c...	5
rollout/	
ep_len_mean	788
ep_rew_mean	184
exploration rate	0.729
time/	
episodes	5000
fps	239
time_elapsed	1144
total timesteps	273849
train/	
learning_rate	0.0001
loss	0.0422
n_updates	62212

EP=1,680 | STEP=275,687 | clip=34 | consec=4 | max_consec=5 | avg_clip50=18.25
EP=1,690 | STEP=277,533 | clip=28 | consec=3 | max_consec=5 | avg_clip50=19.98
EP=1,700 | STEP=279,342 | clip=8 | consec=0 | max_consec=5 | avg_clip50=19.40

custom/	
avg_clip50	19.2
clipped_episode_r...	11.5
consecutive_clipp...	0
max_consecutive_c...	5
rollout/	
ep_len_mean	767
ep_rew_mean	176
exploration rate	0.723
time/	
episodes	5100
fps	239
time_elapsed	1172
total timesteps	280119
train/	
learning_rate	0.0001
loss	0.259
n_updates	63779

EP=1,710 | STEP=281,156 | clip=20 | consec=0 | max_consec=5 | avg_clip50=18.72
EP=1,720 | STEP=282,694 | clip=10 | consec=0 | max_consec=5 | avg_clip50=17.78
EP=1,730 | STEP=284,432 | clip=8 | consec=0 | max_consec=5 | avg_clip50=15.27

custom/	
avg_clip50	15.3
clipped_episode_r...	21
consecutive_clipp...	1
max_consecutive_c...	5
rollout/	
ep_len_mean	771
ep_rew_mean	170
exploration rate	0.717
time/	
episodes	5200
fps	238
time_elapsed	1195
total timesteps	285708
train/	
learning_rate	0.0001
loss	0.0755

n_updates		65176				
EP=1,740	STEP=286,304	clip=18	consec=0	max_consec=5	avg_clip50=15.26	
EP=1,750	STEP=288,110	clip=14	consec=0	max_consec=5	avg_clip50=16.42	
EP=1,760	STEP=290,415	clip=14	consec=0	max_consec=5	avg_clip50=17.82	
EP=1,770	STEP=292,416	clip=40	consec=1	max_consec=5	avg_clip50=19.98	
custom/						
avg_clip50		19.9				
clipped_episode_r...		8.5				
consecutive_clipp...		0				
max_consecutive_c...		5				
rollout/						
ep_len_mean		796				
ep_rew_mean		188				
exploration rate		0.71				
time/						
episodes		5300				
fps		238				
time_elapsed		1226				
total timesteps		292626				
train/						
learning_rate		0.0001				
loss		0.155				
n_updates		66906				
EP=1,780	STEP=294,400	clip=11	consec=0	max_consec=5	avg_clip50=21.16	
EP=1,790	STEP=296,370	clip=17	consec=0	max_consec=5	avg_clip50=20.66	
EP=1,800	STEP=298,182	clip=18	consec=0	max_consec=5	avg_clip50=20.84	
custom/						
avg_clip50		21.4				
clipped_episode_r...		46.5				
consecutive_clipp...		1				
max_consecutive_c...		5				
rollout/						
ep_len_mean		806				
ep_rew_mean		191				
exploration rate		0.704				
time/						
episodes		5400				
fps		238				
time_elapsed		1255				
total timesteps		299144				
train/						
learning_rate		0.0001				
loss		0.0163				
n_updates		68535				
EP=1,810	STEP=300,385	clip=27	consec=1	max_consec=5	avg_clip50=20.84	
EP=1,820	STEP=302,237	clip=4	consec=0	max_consec=5	avg_clip50=21.08	
EP=1,830	STEP=304,271	clip=10	consec=0	max_consec=5	avg_clip50=21.90	
custom/						
avg_clip50		21.9				
clipped_episode_r...		24.5				
consecutive_clipp...		1				
max_consecutive_c...		5				
rollout/						
ep_len_mean		834				

ep_rew_mean	215
exploration rate	0.697
time/	
episodes	5500
fps	238
time_elapsed	1283
total timesteps	305606
train/	
learning_rate	0.0001
loss	0.0612
n_updates	70151

EP=1,840	STEP=306,046	clip=14	consec=0	max_consec=5	avg_clip50=22.28
EP=1,850	STEP=308,310	clip=16	consec=0	max_consec=5	avg_clip50=22.69
EP=1,860	STEP=310,594	clip=22	consec=2	max_consec=5	avg_clip50=23.34
EP=1,870	STEP=312,709	clip=22	consec=1	max_consec=5	avg_clip50=22.65

custom/	
avg_clip50	22
clipped_episode_r...	17
consecutive_clipp...	0
max_consecutive_c...	5
rollout/	
ep_len_mean	861
ep_rew_mean	220
exploration rate	0.69
time/	
episodes	5600
fps	237
time_elapsed	1315
total timesteps	312979
train/	
learning_rate	0.0001
loss	0.0872
n_updates	71994

EP=1,880	STEP=314,862	clip=36	consec=1	max_consec=5	avg_clip50=22.65
EP=1,890	STEP=316,798	clip=6	consec=0	max_consec=5	avg_clip50=23.11
EP=1,900	STEP=318,900	clip=32	consec=2	max_consec=5	avg_clip50=23.70

custom/	
avg_clip50	22.2
clipped_episode_r...	24.5
consecutive_clipp...	1
max_consecutive_c...	5
rollout/	
ep_len_mean	864
ep_rew_mean	226
exploration rate	0.683
time/	
episodes	5700
fps	237
time_elapsed	1344
total timesteps	319724
train/	
learning_rate	0.0001
loss	0.052
n_updates	73680

EP=1,910	STEP=320,614	clip=4	consec=0	max_consec=5	avg_clip50=21.33
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EP=1,920 | STEP=322,423 | clip=23 | consec=1 | max_consec=5 | avg_clip50=20.89
EP=1,930 | STEP=324,331 | clip=32 | consec=3 | max_consec=5 | avg_clip50=20.75

custom/		
avg_clip50	19.8	
clipped_episode_r...	25	
consecutive_clipp...	1	
max_consecutive_c...	5	
rollout/		
ep_len_mean	849	
ep_rew_mean	215	
exploration rate	0.677	
time/		
episodes	5800	
fps	237	
time_elapsed	1370	
total timesteps	325895	
train/		
learning_rate	0.0001	
loss	0.0517	
n_updates	75223	

EP=1,940 | STEP=325,978 | clip=8 | consec=0 | max_consec=5 | avg_clip50=19.85
EP=1,950 | STEP=327,902 | clip=34 | consec=3 | max_consec=5 | avg_clip50=18.45
EP=1,960 | STEP=329,774 | clip=20 | consec=1 | max_consec=5 | avg_clip50=19.41
EP=1,970 | STEP=331,687 | clip=10 | consec=0 | max_consec=5 | avg_clip50=19.94

custom/		
avg_clip50	21.1	
clipped_episode_r...	42	
consecutive_clipp...	2	
max_consecutive_c...	5	
rollout/		
ep_len_mean	804	
ep_rew_mean	208	
exploration rate	0.671	
time/		
episodes	5900	
fps	237	
time_elapsed	1398	
total timesteps	332211	
train/		
learning_rate	0.0001	
loss	0.169	
n_updates	76802	

EP=1,980 | STEP=333,706 | clip=24 | consec=2 | max_consec=5 | avg_clip50=21.20
EP=1,990 | STEP=335,608 | clip=45 | consec=1 | max_consec=5 | avg_clip50=21.85
EP=2,000 | STEP=337,657 | clip=31 | consec=1 | max_consec=5 | avg_clip50=21.47

custom/		
avg_clip50	21.4	
clipped_episode_r...	12	
consecutive_clipp...	0	
max_consecutive_c...	5	
rollout/		
ep_len_mean	803	
ep_rew_mean	200	
exploration rate	0.665	
time/		

episodes	6000
fps	237
time_elapsed	1428
total timesteps	338684
train/	
learning_rate	0.0001
loss	0.168
n_updates	78420

EP=2,010 | STEP=339,591 | clip=25 | consec=2 | max_consec=5 | avg_clip50=21.00
EP=2,020 | STEP=341,261 | clip=4 | consec=0 | max_consec=5 | avg_clip50=19.62
EP=2,030 | STEP=343,396 | clip=27 | consec=2 | max_consec=5 | avg_clip50=18.52

custom/	
avg_clip50	19.4
clipped_episode_r...	20.5
consecutive_clipp...	2
max_consecutive_c...	5
rollout/	
ep_len_mean	818
ep_rew_mean	202
exploration rate	0.658
time/	
episodes	6100
fps	236
time_elapsed	1458
total timesteps	345190
train/	
learning_rate	0.0001
loss	0.0613
n_updates	80047

EP=2,040 | STEP=345,454 | clip=32 | consec=3 | max_consec=5 | avg_clip50=19.11
EP=2,050 | STEP=347,466 | clip=22 | consec=2 | max_consec=5 | avg_clip50=20.09
EP=2,060 | STEP=349,714 | clip=32 | consec=1 | max_consec=5 | avg_clip50=21.25
EP=2,070 | STEP=351,739 | clip=24 | consec=3 | max_consec=5 | avg_clip50=22.77

custom/	
avg_clip50	22.6
clipped_episode_r...	48
consecutive_clipp...	2
max_consecutive_c...	5
rollout/	
ep_len_mean	841
ep_rew_mean	204
exploration rate	0.651
time/	
episodes	6200
fps	236
time_elapsed	1490
total timesteps	352654
train/	
learning_rate	0.0001
loss	0.101
n_updates	81913

EP=2,080 | STEP=353,901 | clip=14 | consec=0 | max_consec=5 | avg_clip50=23.23
EP=2,090 | STEP=356,311 | clip=40 | consec=7 | max_consec=7 | avg_clip50=24.91
EP=2,100 | STEP=358,526 | clip=20 | consec=3 | max_consec=7 | avg_clip50=25.78

custom/	
avg_clip50	25.3
clipped_episode_r...	31
consecutive_clipp...	1
max_consecutive_c...	7
rollout/	
ep_len_mean	884
ep_rew_mean	235
exploration rate	0.644
time/	
episodes	6300
fps	236
time_elapsed	1524
total timesteps	360101
train/	
learning_rate	0.0001
loss	0.134
n_updates	83775

EP=2,110 | STEP=360,587 | clip=10 | consec=0 | max_consec=7 | avg_clip50=26.39
EP=2,120 | STEP=362,294 | clip=4 | consec=0 | max_consec=7 | avg_clip50=25.40
EP=2,130 | STEP=364,262 | clip=26 | consec=1 | max_consec=7 | avg_clip50=24.28
EP=2,140 | STEP=366,075 | clip=2 | consec=0 | max_consec=7 | avg_clip50=21.75

custom/	
avg_clip50	21.5
clipped_episode_r...	12.5
consecutive_clipp...	0
max_consecutive_c...	7
rollout/	
ep_len_mean	864
ep_rew_mean	232
exploration rate	0.637
time/	
episodes	6400
fps	236
time_elapsed	1552
total timesteps	366398
train/	
learning_rate	0.0001
loss	0.153
n_updates	85349

EP=2,150 | STEP=367,950 | clip=12 | consec=0 | max_consec=7 | avg_clip50=20.79
EP=2,160 | STEP=370,253 | clip=48 | consec=1 | max_consec=7 | avg_clip50=22.31
EP=2,170 | STEP=372,355 | clip=12 | consec=0 | max_consec=7 | avg_clip50=23.95

custom/	
avg_clip50	24.3
clipped_episode_r...	38
consecutive_clipp...	2
max_consecutive_c...	7
rollout/	
ep_len_mean	882
ep_rew_mean	249
exploration rate	0.63
time/	
episodes	6500
fps	235
time_elapsed	1584

total timesteps	373898
train/	
learning_rate	0.0001
loss	0.243
n_updates	87224

EP=2,180	STEP=374,539	clip=13	consec=0	max_consec=7	avg_clip50=24.85
EP=2,190	STEP=376,657	clip=15	consec=0	max_consec=7	avg_clip50=26.36
EP=2,200	STEP=378,742	clip=5	consec=0	max_consec=7	avg_clip50=27.60
EP=2,210	STEP=380,903	clip=44	consec=1	max_consec=7	avg_clip50=26.42

custom/	
avg_clip50	26.4
clipped_episode_r...	44.5
consecutive_clipp...	1
max_consecutive_c...	7
rollout/	
ep_len_mean	859
ep_rew_mean	244
exploration rate	0.623
time/	
episodes	6600
fps	235
time_elapsed	1615
total timesteps	380949
train/	
learning_rate	0.0001
loss	0.148
n_updates	88987

EP=2,220	STEP=382,902	clip=9	consec=0	max_consec=7	avg_clip50=26.08
EP=2,230	STEP=385,474	clip=36	consec=10	max_consec=10	avg_clip50=29.25
EP=2,240	STEP=387,932	clip=19	consec=0	max_consec=10	avg_clip50=30.17

custom/	
avg_clip50	29.6
clipped_episode_r...	18.5
consecutive_clipp...	0
max_consecutive_c...	10
rollout/	
ep_len_mean	924
ep_rew_mean	284
exploration rate	0.615
time/	
episodes	6700
fps	235
time_elapsed	1650
total timesteps	388732
train/	
learning_rate	0.0001
loss	0.039
n_updates	90932

EP=2,250	STEP=390,267	clip=39	consec=6	max_consec=10	avg_clip50=30.67
EP=2,260	STEP=392,724	clip=22	consec=16	max_consec=16	avg_clip50=30.65
EP=2,270	STEP=395,158	clip=24	consec=6	max_consec=16	avg_clip50=31.67

custom/	
avg_clip50	30.9
clipped_episode_r...	42

consecutive_clipp...	1
max_consecutive_c...	16
rollout/	
ep_len_mean	964
ep_rew_mean	298
exploration rate	0.607
time/	
episodes	6800
fps	235
time_elapsed	1687
total timesteps	397438
train/	
learning_rate	0.0001
loss	0.0415
n_updates	93109

EP=2,280 | STEP=397,594 | clip=24 | consec=2 | max_consec=16 | avg_clip50=30.62
EP=2,290 | STEP=400,027 | clip=4 | consec=0 | max_consec=16 | avg_clip50=30.77
EP=2,300 | STEP=402,504 | clip=8 | consec=0 | max_consec=16 | avg_clip50=29.98
EP=2,310 | STEP=404,438 | clip=12 | consec=0 | max_consec=16 | avg_clip50=28.16

custom/	
avg_clip50	28.4
clipped_episode_r...	35.5
consecutive_clipp...	1
max_consecutive_c...	16
rollout/	
ep_len_mean	978
ep_rew_mean	294
exploration rate	0.599
time/	
episodes	6900
fps	235
time_elapsed	1722
total timesteps	405225
train/	
learning_rate	0.0001
loss	0.0355
n_updates	95056

EP=2,320 | STEP=406,987 | clip=52 | consec=4 | max_consec=16 | avg_clip50=29.03
EP=2,330 | STEP=409,412 | clip=30 | consec=5 | max_consec=16 | avg_clip50=28.24
EP=2,340 | STEP=411,100 | clip=8 | consec=0 | max_consec=16 | avg_clip50=25.65

custom/	
avg_clip50	26.1
clipped_episode_r...	21
consecutive_clipp...	6
max_consecutive_c...	16
rollout/	
ep_len_mean	979
ep_rew_mean	286
exploration rate	0.591
time/	
episodes	7000
fps	235
time_elapsed	1755
total timesteps	413048
train/	
learning_rate	0.0001

	loss	0.0575				
	n_updates	97011				

EP=2,350	STEP=413,767	clip=28	consec=8	max_consec=16	avg_clip50=27.32	
EP=2,360	STEP=416,277	clip=28	consec=3	max_consec=16	avg_clip50=28.64	
EP=2,370	STEP=418,399	clip=14	consec=0	max_consec=16	avg_clip50=27.53	
EP=2,380	STEP=420,552	clip=15	consec=0	max_consec=16	avg_clip50=27.06	

	custom/					
	avg_clip50	26.6				
	clipped_episode_r...	25				
	consecutive_clipp...	1				
	max_consecutive_c...	16				
	rollout/					
	ep_len_mean	963				
	ep_rew_mean	275				
	exploration rate	0.583				
	time/					
	episodes	7100				
	fps	235				
	time_elapsed	1790				
	total timesteps	420953				
	train/					
	learning_rate	0.0001				
	loss	0.0521				
	n_updates	98988				

EP=2,390	STEP=422,944	clip=2	consec=0	max_consec=16	avg_clip50=29.62	
EP=2,400	STEP=425,042	clip=12	consec=0	max_consec=16	avg_clip50=27.28	
EP=2,410	STEP=426,948	clip=2	consec=0	max_consec=16	avg_clip50=26.07	

	custom/					
	avg_clip50	26.8				
	clipped_episode_r...	34				
	consecutive_clipp...	2				
	max_consecutive_c...	16				
	rollout/					
	ep_len_mean	952				
	ep_rew_mean	280				
	exploration rate	0.576				
	time/					
	episodes	7200				
	fps	234				
	time_elapsed	1823				
	total timesteps	428357				
	train/					
	learning_rate	0.0001				
	loss	0.0533				
	n_updates	100839				

EP=2,420	STEP=428,998	clip=14	consec=0	max_consec=16	avg_clip50=26.54	
EP=2,430	STEP=431,538	clip=24	consec=6	max_consec=16	avg_clip50=27.03	
EP=2,440	STEP=433,934	clip=18	consec=0	max_consec=16	avg_clip50=26.93	
EP=2,450	STEP=436,667	clip=34	consec=4	max_consec=16	avg_clip50=29.13	

	custom/					
	avg_clip50	29.1				
	clipped_episode_r...	33.5				
	consecutive_clipp...	4				
	max_consecutive_c...	16				

rollout/		
ep_len_mean	960	
ep_rew_mean	282	
exploration rate	0.568	
time/		
episodes	7300	
fps	234	
time_elapsed	1860	
total timesteps	436667	
train/		
learning_rate	0.0001	
loss	0.0441	
n_updates	102916	

EP=2,460		STEP=439,105		clip=43		consec=2		max_consec=16		avg_clip50=31.41
EP=2,470		STEP=441,571		clip=30		consec=5		max_consec=16		avg_clip50=32.76
EP=2,480		STEP=444,020		clip=56		consec=2		max_consec=16		avg_clip50=32.92

custom/		
avg_clip50	33.3	
clipped_episode_r...	43.5	
consecutive_clipp...	2	
max_consecutive_c...	16	
rollout/		
ep_len_mean	987	
ep_rew_mean	300	
exploration rate	0.559	
time/		
episodes	7400	
fps	234	
time_elapsed	1898	
total timesteps	445343	
train/		
learning_rate	0.0001	
loss	0.0325	
n_updates	105085	

EP=2,490		STEP=446,742		clip=56		consec=1		max_consec=16		avg_clip50=34.72
EP=2,500		STEP=449,133		clip=41		consec=1		max_consec=16		avg_clip50=33.09
EP=2,510		STEP=451,548		clip=29		consec=3		max_consec=16		avg_clip50=33.22

custom/		
avg_clip50	32.8	
clipped_episode_r...	24	
consecutive_clipp...	1	
max_consecutive_c...	16	
rollout/		
ep_len_mean	1.04e+03	
ep_rew_mean	329	
exploration rate	0.551	
time/		
episodes	7500	
fps	234	
time_elapsed	1935	
total timesteps	453739	
train/		
learning_rate	0.0001	
loss	0.0385	
n_updates	107184	

EP=2,520	STEP=454,274	clip=39	consec=3	max_consec=16	avg_clip50=33.18
EP=2,530	STEP=456,598	clip=32	consec=1	max_consec=16	avg_clip50=32.35
EP=2,540	STEP=458,518	clip=26	consec=1	max_consec=16	avg_clip50=29.04
EP=2,550	STEP=460,972	clip=36	consec=2	max_consec=16	avg_clip50=30.68

custom/	
avg_clip50	32
clipped_episode_r...	62.5
consecutive_clipp...	4
max_consecutive_c...	16
rollout/	
ep_len_mean	1.02e+03
ep_rew_mean	323
exploration rate	0.543
time/	
episodes	7600
fps	234
time_elapsed	1972
total timesteps	462025
train/	
learning_rate	0.0001
loss	0.0853
n_updates	109256

EP=2,560	STEP=464,090	clip=32	consec=12	max_consec=16	avg_clip50=32.99
EP=2,570	STEP=466,660	clip=24	consec=1	max_consec=16	avg_clip50=32.34
EP=2,580	STEP=469,100	clip=54	consec=1	max_consec=16	avg_clip50=32.17

custom/	
avg_clip50	34.4
clipped_episode_r...	30.5
consecutive_clipp...	4
max_consecutive_c...	16
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	321
exploration rate	0.533
time/	
episodes	7700
fps	234
time_elapsed	2013
total timesteps	471268
train/	
learning_rate	0.0001
loss	0.056
n_updates	111566

EP=2,590	STEP=471,790	clip=23	consec=7	max_consec=16	avg_clip50=35.11
EP=2,600	STEP=474,483	clip=56	consec=3	max_consec=16	avg_clip50=34.64
EP=2,610	STEP=477,256	clip=40	consec=3	max_consec=16	avg_clip50=33.16
EP=2,620	STEP=479,977	clip=76	consec=2	max_consec=16	avg_clip50=34.26

custom/	
avg_clip50	33.4
clipped_episode_r...	8.5
consecutive_clipp...	0
max_consecutive_c...	16
rollout/	
ep_len_mean	1.07e+03
ep_rew_mean	332

exploration rate	0.525
time/	
episodes	7800
fps	233
time_elapsed	2052
total timesteps	480184
train/	
learning_rate	0.0001
loss	0.0622
n_updates	113795

EP=2,630	STEP=482,279	clip=24	consec=5	max_consec=16	avg_clip50=34.66
EP=2,640	STEP=484,595	clip=20	consec=0	max_consec=16	avg_clip50=31.20
EP=2,650	STEP=487,131	clip=22	consec=10	max_consec=16	avg_clip50=30.60

custom/	
avg_clip50	30
clipped_episode_r...	55
consecutive_clipp...	1
max_consecutive_c...	16
rollout/	
ep_len_mean	1.08e+03
ep_rew_mean	318
exploration rate	0.516
time/	
episodes	7900
fps	233
time_elapsed	2088
total timesteps	488447
train/	
learning_rate	0.0001
loss	0.271
n_updates	115861

EP=2,660	STEP=489,741	clip=80	consec=2	max_consec=16	avg_clip50=30.62
EP=2,670	STEP=491,871	clip=10	consec=0	max_consec=16	avg_clip50=28.27
EP=2,680	STEP=494,694	clip=74	consec=2	max_consec=16	avg_clip50=31.35

custom/	
avg_clip50	33.1
clipped_episode_r...	23
consecutive_clipp...	1
max_consecutive_c...	16
rollout/	
ep_len_mean	1.05e+03
ep_rew_mean	322
exploration rate	0.508
time/	
episodes	8000
fps	233
time_elapsed	2127
total timesteps	496792
train/	
learning_rate	0.0001
loss	0.243
n_updates	117947

EP=2,690	STEP=497,097	clip=38	consec=2	max_consec=16	avg_clip50=33.44
EP=2,700	STEP=499,862	clip=52	consec=3	max_consec=16	avg_clip50=34.66
EP=2,710	STEP=502,588	clip=34	consec=2	max_consec=16	avg_clip50=35.93

EP=2,720 | STEP=505,190 | clip=13 | consec=0 | max_consec=16 | avg_clip50=36.49

custom/		
avg_clip50	34.8	
clipped_episode_r...	15.5	
consecutive_clipp...	0	
max_consecutive_c...	16	
rollout/		
ep_len_mean	1.05e+03	
ep_rew_mean	324	
exploration rate	0.499	
time/		
episodes	8100	
fps	233	
time_elapsed	2167	
total timesteps	505783	
train/		
learning_rate	0.0001	
loss	0.0473	
n_updates	120195	

EP=2,730 | STEP=507,710 | clip=44 | consec=4 | max_consec=16 | avg_clip50=34.99

EP=2,740 | STEP=510,346 | clip=38 | consec=6 | max_consec=16 | avg_clip50=35.91

EP=2,750 | STEP=512,405 | clip=9 | consec=0 | max_consec=16 | avg_clip50=33.95

custom/		
avg_clip50	34.4	
clipped_episode_r...	47.5	
consecutive_clipp...	6	
max_consecutive_c...	16	
rollout/		
ep_len_mean	1.07e+03	
ep_rew_mean	351	
exploration rate	0.491	
time/		
episodes	8200	
fps	233	
time_elapsed	2205	
total timesteps	514283	
train/		
learning_rate	0.0001	
loss	0.0295	
n_updates	122320	

EP=2,760 | STEP=515,508 | clip=58 | consec=10 | max_consec=16 | avg_clip50=34.34

EP=2,770 | STEP=517,862 | clip=41 | consec=6 | max_consec=16 | avg_clip50=35.83

EP=2,780 | STEP=520,107 | clip=27 | consec=4 | max_consec=16 | avg_clip50=35.45

EP=2,790 | STEP=522,568 | clip=18 | consec=0 | max_consec=16 | avg_clip50=34.46

custom/		
avg_clip50	34.5	
clipped_episode_r...	18	
consecutive_clipp...	0	
max_consecutive_c...	16	
rollout/		
ep_len_mean	1.07e+03	
ep_rew_mean	352	
exploration rate	0.482	
time/		
episodes	8300	

fps	233
time_elapsed	2243
total timesteps	522731
train/	
learning_rate	0.0001
loss	0.0509
n_updates	124432

EP=2,800 | STEP=525,129 | clip=6 | consec=0 | max_consec=16 | avg_clip50=36.33
EP=2,810 | STEP=527,978 | clip=29 | consec=10 | max_consec=16 | avg_clip50=34.71
EP=2,820 | STEP=530,775 | clip=62 | consec=7 | max_consec=16 | avg_clip50=35.47

custom/	
avg_clip50	35
clipped_episode_r...	65.5
consecutive_clipp...	3
max_consecutive_c...	16
rollout/	
ep_len_mean	1.07e+03
ep_rew_mean	360
exploration rate	0.473
time/	
episodes	8400
fps	232
time_elapsed	2284
total timesteps	532001
train/	
learning_rate	0.0001
loss	0.0404
n_updates	126750

EP=2,830 | STEP=533,368 | clip=26 | consec=1 | max_consec=16 | avg_clip50=35.15
EP=2,840 | STEP=535,801 | clip=79 | consec=2 | max_consec=16 | avg_clip50=36.40
EP=2,850 | STEP=538,388 | clip=15 | consec=0 | max_consec=16 | avg_clip50=37.70

custom/	
avg_clip50	38.4
clipped_episode_r...	41
consecutive_clipp...	9
max_consecutive_c...	16
rollout/	
ep_len_mean	1.07e+03
ep_rew_mean	368
exploration rate	0.465
time/	
episodes	8500
fps	232
time_elapsed	2324
total timesteps	540894
train/	
learning_rate	0.0001
loss	0.0232
n_updates	128973

EP=2,860 | STEP=541,071 | clip=14 | consec=0 | max_consec=16 | avg_clip50=38.07
EP=2,870 | STEP=543,642 | clip=80 | consec=7 | max_consec=16 | avg_clip50=37.64
EP=2,880 | STEP=546,572 | clip=22 | consec=17 | max_consec=17 | avg_clip50=39.26
EP=2,890 | STEP=549,008 | clip=20 | consec=0 | max_consec=19 | avg_clip50=38.38

custom/	
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avg_clip50	39.2
clipped_episode_r...	8
consecutive_clipp...	0
max_consecutive_c...	19
rollout/	
ep_len_mean	1.1e+03
ep_rew_mean	377
exploration rate	0.456
time/	
episodes	8600
fps	232
time_elapsed	2364
total timesteps	549928
train/	
learning_rate	0.0001
loss	0.295
n_updates	131231

EP=2,900	STEP=551,738	clip=40	consec=3	max_consec=19	avg_clip50=38.26
EP=2,910	STEP=553,957	clip=19	consec=0	max_consec=19	avg_clip50=35.90
EP=2,920	STEP=556,396	clip=22	consec=5	max_consec=19	avg_clip50=34.41

custom/	
avg_clip50	33.1
clipped_episode_r...	28.5
consecutive_clipp...	7
max_consecutive_c...	19
rollout/	
ep_len_mean	1.06e+03
ep_rew_mean	359
exploration rate	0.447
time/	
episodes	8700
fps	232
time_elapsed	2403
total timesteps	558449
train/	
learning_rate	0.0001
loss	0.214
n_updates	133362

EP=2,930	STEP=559,014	clip=46	consec=9	max_consec=19	avg_clip50=33.53
EP=2,940	STEP=562,048	clip=34	consec=19	max_consec=19	avg_clip50=35.72
EP=2,950	STEP=565,163	clip=50	consec=29	max_consec=29	avg_clip50=36.13
EP=2,960	STEP=567,882	clip=32	consec=2	max_consec=33	avg_clip50=37.89

custom/	
avg_clip50	38.2
clipped_episode_r...	19
consecutive_clipp...	0
max_consecutive_c...	33
rollout/	
ep_len_mean	1.12e+03
ep_rew_mean	376
exploration rate	0.437
time/	
episodes	8800
fps	232
time_elapsed	2448
total timesteps	568591

train/	
learning_rate	0.0001
loss	0.0362
n_updates	135897

EP=2,970	STEP=570,360	clip=59	consec=1	max_consec=33	avg_clip50=38.73
EP=2,980	STEP=573,010	clip=52	consec=4	max_consec=33	avg_clip50=39.80
EP=2,990	STEP=575,406	clip=78	consec=5	max_consec=33	avg_clip50=38.35

custom/	
avg_clip50	37.3
clipped_episode_r...	20
consecutive_clipp...	0
max_consecutive_c...	33
rollout/	
ep_len_mean	1.09e+03
ep_rew_mean	361
exploration rate	0.429
time/	
episodes	8900
fps	231
time_elapsed	2487
total timesteps	576992
train/	
learning_rate	0.0001
loss	0.0534
n_updates	137997

EP=3,000	STEP=577,939	clip=70	consec=4	max_consec=33	avg_clip50=38.95
EP=3,010	STEP=580,441	clip=54	consec=5	max_consec=33	avg_clip50=38.67
EP=3,020	STEP=583,514	clip=68	consec=3	max_consec=33	avg_clip50=40.03
EP=3,030	STEP=585,948	clip=37	consec=2	max_consec=33	avg_clip50=37.17

custom/	
avg_clip50	37
clipped_episode_r...	36
consecutive_clipp...	3
max_consecutive_c...	33
rollout/	
ep_len_mean	1.12e+03
ep_rew_mean	383
exploration rate	0.42
time/	
episodes	9000
fps	231
time_elapsed	2528
total timesteps	586221
train/	
learning_rate	0.0001
loss	0.0642
n_updates	140305

EP=3,040	STEP=589,063	clip=20	consec=0	max_consec=33	avg_clip50=40.00
EP=3,050	STEP=591,826	clip=49	consec=4	max_consec=33	avg_clip50=39.57
EP=3,060	STEP=594,239	clip=17	consec=0	max_consec=33	avg_clip50=40.62

custom/	
avg_clip50	40.4
clipped_episode_r...	54
consecutive_clipp...	2

max_consecutive_c...	33
rollout/	
ep_len_mean	1.11e+03
ep_rew_mean	397
exploration rate	0.41
time/	
episodes	9100
fps	231
time_elapsed	2572
total timesteps	595925
train/	
learning_rate	0.0001
loss	0.0502
n_updates	142731

EP=3,070 | STEP=596,986 | clip=26 | consec=3 | max_consec=33 | avg_clip50=40.21
EP=3,080 | STEP=599,441 | clip=30 | consec=2 | max_consec=33 | avg_clip50=41.02
EP=3,090 | STEP=602,285 | clip=44 | consec=3 | max_consec=33 | avg_clip50=38.47

custom/	
avg_clip50	36.8
clipped_episode_r...	29
consecutive_clipp...	3
max_consecutive_c...	33
rollout/	
ep_len_mean	1.12e+03
ep_rew_mean	384
exploration rate	0.401
time/	
episodes	9200
fps	231
time_elapsed	2612
total timesteps	604598
train/	
learning_rate	0.0001
loss	0.0593
n_updates	144899

EP=3,100 | STEP=604,871 | clip=58 | consec=5 | max_consec=33 | avg_clip50=36.78
EP=3,110 | STEP=607,705 | clip=30 | consec=15 | max_consec=33 | avg_clip50=38.27
EP=3,120 | STEP=610,195 | clip=34 | consec=5 | max_consec=33 | avg_clip50=36.92
EP=3,130 | STEP=612,553 | clip=30 | consec=2 | max_consec=33 | avg_clip50=37.44

custom/	
avg_clip50	37
clipped_episode_r...	55.5
consecutive_clipp...	2
max_consecutive_c...	33
rollout/	
ep_len_mean	1.09e+03
ep_rew_mean	384
exploration rate	0.392
time/	
episodes	9300
fps	231
time_elapsed	2655
total timesteps	613953
train/	
learning_rate	0.0001
loss	0.303

n_updates	147238
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EP=3,140	STEP=615,658	clip=6	consec=0	max_consec=33	avg_clip50=38.58
EP=3,150	STEP=618,438	clip=7	consec=0	max_consec=33	avg_clip50=41.41
EP=3,160	STEP=621,421	clip=60	consec=5	max_consec=33	avg_clip50=41.80
EP=3,170	STEP=623,675	clip=6	consec=0	max_consec=33	avg_clip50=41.82

custom/	
avg_clip50	41.8
clipped_episode_r...	5.5
consecutive_clipp...	0
max_consecutive_c...	33
rollout/	
ep_len_mean	1.11e+03
ep_rew_mean	394
exploration rate	0.383
time/	
episodes	9400
fps	230
time_elapsed	2700
total timesteps	623675
train/	
learning_rate	0.0001
loss	0.313
n_updates	149668

EP=3,180	STEP=626,097	clip=20	consec=0	max_consec=33	avg_clip50=42.67
EP=3,190	STEP=628,876	clip=24	consec=1	max_consec=33	avg_clip50=42.60
EP=3,200	STEP=631,947	clip=58	consec=8	max_consec=33	avg_clip50=42.18

custom/	
avg_clip50	41.6
clipped_episode_r...	75.5
consecutive_clipp...	2
max_consecutive_c...	33
rollout/	
ep_len_mean	1.13e+03
ep_rew_mean	421
exploration rate	0.373
time/	
episodes	9500
fps	230
time_elapsed	2745
total timesteps	633234
train/	
learning_rate	0.0001
loss	0.0227
n_updates	152058

EP=3,210	STEP=634,837	clip=26	consec=8	max_consec=33	avg_clip50=40.87
EP=3,220	STEP=638,142	clip=77	consec=5	max_consec=33	avg_clip50=44.74
EP=3,230	STEP=641,013	clip=24	consec=9	max_consec=33	avg_clip50=46.07

custom/	
avg_clip50	45.6
clipped_episode_r...	67.5
consecutive_clipp...	17
max_consecutive_c...	33
rollout/	
ep_len_mean	1.16e+03

ep_rew_mean	441
exploration rate	0.363
time/	
episodes	9600
fps	230
time_elapsed	2791
total timesteps	643404
train/	
learning_rate	0.0001
loss	0.0706
n_updates	154600

EP=3,240 | STEP=643,781 | clip=56 | consec=19 | max_consec=33 | avg_clip50=46.76
EP=3,250 | STEP=646,835 | clip=27 | consec=5 | max_consec=33 | avg_clip50=47.33
EP=3,260 | STEP=649,582 | clip=18 | consec=0 | max_consec=33 | avg_clip50=47.06
EP=3,270 | STEP=652,975 | clip=84 | consec=10 | max_consec=33 | avg_clip50=48.30

custom/	
avg_clip50	48.9
clipped_episode_r...	48.5
consecutive_clipp...	14
max_consecutive_c...	33
rollout/	
ep_len_mean	1.23e+03
ep_rew_mean	471
exploration rate	0.352
time/	
episodes	9700
fps	230
time_elapsed	2841
total timesteps	654379
train/	
learning_rate	0.0001
loss	0.0421
n_updates	157344

EP=3,280 | STEP=655,852 | clip=28 | consec=20 | max_consec=33 | avg_clip50=47.43
EP=3,290 | STEP=658,698 | clip=42 | consec=7 | max_consec=33 | avg_clip50=46.33
EP=3,300 | STEP=662,141 | clip=35 | consec=9 | max_consec=33 | avg_clip50=46.49

custom/	
avg_clip50	46.6
clipped_episode_r...	38
consecutive_clipp...	5
max_consecutive_c...	33
rollout/	
ep_len_mean	1.25e+03
ep_rew_mean	471
exploration rate	0.342
time/	
episodes	9800
fps	230
time_elapsed	2888
total timesteps	664398
train/	
learning_rate	0.0001
loss	0.0801
n_updates	159849

EP=3,310 | STEP=664,791 | clip=14 | consec=0 | max_consec=33 | avg_clip50=46.32

EP=3,320	STEP=667,783	clip=78	consec=10	max_consec=33	avg_clip50=45.27
EP=3,330	STEP=670,756	clip=18	consec=0	max_consec=33	avg_clip50=45.67
EP=3,340	STEP=673,471	clip=14	consec=0	max_consec=33	avg_clip50=44.70

custom/	
avg_clip50	45.8
clipped_episode_r...	35.5
consecutive_clipp...	3
max_consecutive_c...	33
rollout/	
ep_len_mean	1.23e+03
ep_rew_mean	452
exploration rate	0.332
time/	
episodes	9900
fps	229
time_elapsed	2934
total timesteps	674523
train/	
learning_rate	0.0001
loss	0.0518
n_updates	162380

EP=3,350	STEP=676,243	clip=40	consec=1	max_consec=33	avg_clip50=44.46
EP=3,360	STEP=678,982	clip=38	consec=7	max_consec=33	avg_clip50=45.19
EP=3,370	STEP=682,112	clip=14	consec=0	max_consec=33	avg_clip50=45.05

custom/	
avg_clip50	45.6
clipped_episode_r...	27
consecutive_clipp...	7
max_consecutive_c...	33
rollout/	
ep_len_mean	1.21e+03
ep_rew_mean	450
exploration rate	0.323
time/	
episodes	10000
fps	229
time_elapsed	2980
total timesteps	684325
train/	
learning_rate	0.0001
loss	0.049
n_updates	164831

EP=3,380	STEP=685,039	clip=58	consec=10	max_consec=33	avg_clip50=45.59
EP=3,390	STEP=688,391	clip=38	consec=6	max_consec=33	avg_clip50=48.28
EP=3,400	STEP=691,521	clip=77	consec=2	max_consec=33	avg_clip50=49.73
EP=3,410	STEP=694,210	clip=42	consec=9	max_consec=33	avg_clip50=49.96

custom/	
avg_clip50	50.2
clipped_episode_r...	35.5
consecutive_clipp...	10
max_consecutive_c...	33
rollout/	
ep_len_mean	1.22e+03
ep_rew_mean	475
exploration rate	0.312

time/	
episodes	10100
fps	229
time_elapsed	3028
total timesteps	694700
train/	
learning_rate	0.0001
loss	0.0454
n_updates	167424

EP=3,420 | STEP=696,852 | clip=31 | consec=3 | max_consec=33 | avg_clip50=46.50
EP=3,430 | STEP=699,710 | clip=64 | consec=13 | max_consec=33 | avg_clip50=46.78
EP=3,440 | STEP=703,068 | clip=78 | consec=23 | max_consec=33 | avg_clip50=47.81

custom/	
avg_clip50	45.4
clipped_episode_r...	20
consecutive_clipp...	0
max_consecutive_c...	33
rollout/	
ep_len_mean	1.23e+03
ep_rew_mean	472
exploration rate	0.302
time/	
episodes	10200
fps	229
time_elapsed	3076
total timesteps	705004
train/	
learning_rate	0.0001
loss	0.199
n_updates	170000

EP=3,450 | STEP=706,172 | clip=74 | consec=4 | max_consec=33 | avg_clip50=45.81
EP=3,460 | STEP=709,225 | clip=117 | consec=14 | max_consec=33 | avg_clip50=46.46
EP=3,470 | STEP=712,204 | clip=26 | consec=7 | max_consec=33 | avg_clip50=48.47
EP=3,480 | STEP=715,624 | clip=46 | consec=17 | max_consec=33 | avg_clip50=49.71

custom/	
avg_clip50	49.7
clipped_episode_r...	54.5
consecutive_clipp...	19
max_consecutive_c...	33
rollout/	
ep_len_mean	1.27e+03
ep_rew_mean	481
exploration rate	0.291
time/	
episodes	10300
fps	228
time_elapsed	3130
total timesteps	716322
train/	
learning_rate	0.0001
loss	0.0588
n_updates	172830

EP=3,490 | STEP=718,736 | clip=60 | consec=27 | max_consec=33 | avg_clip50=49.40
EP=3,500 | STEP=721,644 | clip=53 | consec=3 | max_consec=33 | avg_clip50=48.69
EP=3,510 | STEP=724,328 | clip=54 | consec=1 | max_consec=33 | avg_clip50=47.23

custom/	
avg_clip50	48.1
clipped_episode_r...	69
consecutive_clipp...	8
max_consecutive_c...	33
rollout/	
ep_len_mean	1.26e+03
ep_rew_mean	475
exploration rate	0.281
time/	
episodes	10400
fps	228
time_elapsed	3178
total timesteps	726296
train/	
learning_rate	0.0001
loss	0.0783
n_updates	175323

EP=3,520 | STEP=727,164 | clip=46 | consec=11 | max_consec=33 | avg_clip50=47.03
EP=3,530 | STEP=730,346 | clip=50 | consec=21 | max_consec=33 | avg_clip50=47.90
EP=3,540 | STEP=733,328 | clip=60 | consec=6 | max_consec=33 | avg_clip50=46.60
EP=3,550 | STEP=736,198 | clip=36 | consec=5 | max_consec=33 | avg_clip50=46.88

custom/	
avg_clip50	47
clipped_episode_r...	33
consecutive_clipp...	8
max_consecutive_c...	33
rollout/	
ep_len_mean	1.25e+03
ep_rew_mean	478
exploration rate	0.27
time/	
episodes	10500
fps	228
time_elapsed	3229
total timesteps	737324
train/	
learning_rate	0.0001
loss	0.0405
n_updates	178080

EP=3,560 | STEP=739,223 | clip=38 | consec=6 | max_consec=33 | avg_clip50=47.65
EP=3,570 | STEP=741,719 | clip=95 | consec=1 | max_consec=33 | avg_clip50=47.06
EP=3,580 | STEP=744,713 | clip=64 | consec=5 | max_consec=33 | avg_clip50=45.56

custom/	
avg_clip50	46.5
clipped_episode_r...	52
consecutive_clipp...	2
max_consecutive_c...	33
rollout/	
ep_len_mean	1.2e+03
ep_rew_mean	470
exploration rate	0.26
time/	
episodes	10600
fps	228

	time_elapsed		3276	
	total timesteps		747165	
	train/			
	learning_rate		0.0001	
	loss		0.0748	
	n_updates		180541	

EP=3,590		STEP=747,615		clip=60		consec=4		max_consec=33		avg_clip50=46.41
EP=3,600		STEP=750,527		clip=58		consec=2		max_consec=33		avg_clip50=47.24
EP=3,610		STEP=753,597		clip=100		consec=12		max_consec=33		avg_clip50=48.54
EP=3,620		STEP=756,519		clip=30		consec=9		max_consec=33		avg_clip50=50.38

	custom/			
	avg_clip50		52.2	
	clipped_episode_r...		77	
	consecutive_clipp...		14	
	max_consecutive_c...		33	
	rollout/			
	ep_len_mean		1.21e+03	
	ep_rew_mean		480	
	exploration rate		0.249	
	time/			
	episodes		10700	
	fps		227	
	time_elapsed		3327	
	total timesteps		758137	
	train/			
	learning_rate		0.0001	
	loss		0.0439	
	n_updates		183284	

EP=3,630		STEP=759,553		clip=54		consec=19		max_consec=33		avg_clip50=49.75
EP=3,640		STEP=762,681		clip=48		consec=29		max_consec=33		avg_clip50=49.91
EP=3,650		STEP=765,211		clip=52		consec=4		max_consec=33		avg_clip50=48.45
EP=3,660		STEP=768,179		clip=55		consec=14		max_consec=33		avg_clip50=48.36

	custom/			
	avg_clip50		48.4	
	clipped_episode_r...		55	
	consecutive_clipp...		14	
	max_consecutive_c...		33	
	rollout/			
	ep_len_mean		1.2e+03	
	ep_rew_mean		484	
	exploration rate		0.239	
	time/			
	episodes		10800	
	fps		227	
	time_elapsed		3379	
	total timesteps		768483	
	train/			
	learning_rate		0.0001	
	loss		0.0826	
	n_updates		185870	

EP=3,670		STEP=771,485		clip=48		consec=24		max_consec=33		avg_clip50=50.68
EP=3,680		STEP=774,368		clip=49		consec=4		max_consec=33		avg_clip50=51.07
EP=3,690		STEP=777,461		clip=10		consec=0		max_consec=33		avg_clip50=50.41

	custom/			
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avg_clip50	52.8
clipped_episode_r...	72
consecutive_clipp...	3
max_consecutive_c...	33
rollout/	
ep_len_mean	1.23e+03
ep_rew_mean	504
exploration rate	0.228
time/	
episodes	10900
fps	226
time_elapsed	3436
total timesteps	779640
train/	
learning_rate	0.0001
loss	0.0429
n_updates	188659

EP=3,700	STEP=780,363	clip=54	consec=6	max_consec=33	avg_clip50=52.97
EP=3,710	STEP=783,309	clip=12	consec=0	max_consec=33	avg_clip50=52.26
EP=3,720	STEP=786,198	clip=54	consec=10	max_consec=33	avg_clip50=50.00
EP=3,730	STEP=789,235	clip=58	consec=7	max_consec=33	avg_clip50=51.62

custom/	
avg_clip50	50.8
clipped_episode_r...	40
consecutive_clipp...	10
max_consecutive_c...	33
rollout/	
ep_len_mean	1.22e+03
ep_rew_mean	512
exploration rate	0.218
time/	
episodes	11000
fps	226
time_elapsed	3487
total timesteps	790012
train/	
learning_rate	0.0001
loss	0.0417
n_updates	191252

EP=3,740	STEP=791,932	clip=44	consec=3	max_consec=33	avg_clip50=51.65
EP=3,750	STEP=794,783	clip=75	consec=4	max_consec=33	avg_clip50=51.92
EP=3,760	STEP=797,312	clip=14	consec=0	max_consec=33	avg_clip50=51.20

custom/	
avg_clip50	51.1
clipped_episode_r...	46.5
consecutive_clipp...	8
max_consecutive_c...	33
rollout/	
ep_len_mean	1.21e+03
ep_rew_mean	507
exploration rate	0.208
time/	
episodes	11100
fps	226
time_elapsed	3535
total timesteps	800084

train/	
learning_rate	0.0001
loss	0.0665
n_updates	193770

EP=3,770	STEP=800,335	clip=24	consec=10	max_consec=33	avg_clip50=50.44
EP=3,780	STEP=803,335	clip=54	consec=2	max_consec=33	avg_clip50=49.26
EP=3,790	STEP=806,085	clip=26	consec=9	max_consec=33	avg_clip50=47.88
EP=3,800	STEP=809,015	clip=46	consec=8	max_consec=33	avg_clip50=46.85

custom/	
avg_clip50	46.9
clipped_episode_r...	42
consecutive_clipp...	10
max_consecutive_c...	33
rollout/	
ep_len_mean	1.2e+03
ep_rew_mean	493
exploration rate	0.198
time/	
episodes	11200
fps	225
time_elapsed	3583
total timesteps	809907
train/	
learning_rate	0.0001
loss	0.261
n_updates	196226

EP=3,810	STEP=812,101	clip=42	consec=18	max_consec=33	avg_clip50=48.52
EP=3,820	STEP=815,118	clip=88	consec=5	max_consec=33	avg_clip50=49.45
EP=3,830	STEP=817,616	clip=46	consec=2	max_consec=33	avg_clip50=46.88

custom/	
avg_clip50	46.6
clipped_episode_r...	74.5
consecutive_clipp...	3
max_consecutive_c...	33
rollout/	
ep_len_mean	1.17e+03
ep_rew_mean	475
exploration rate	0.189
time/	
episodes	11300
fps	225
time_elapsed	3630
total timesteps	819282
train/	
learning_rate	0.0001
loss	0.0416
n_updates	198570

EP=3,840	STEP=820,210	clip=77	consec=7	max_consec=33	avg_clip50=47.19
EP=3,850	STEP=823,571	clip=46	consec=17	max_consec=33	avg_clip50=48.79
EP=3,860	STEP=826,225	clip=39	consec=6	max_consec=33	avg_clip50=46.81
EP=3,870	STEP=829,210	clip=58	consec=16	max_consec=33	avg_clip50=47.64

custom/	
avg_clip50	48.1
clipped_episode_r...	52

consecutive_clipp...	17
max_consecutive_c...	33
rollout/	
ep_len_mean	1.2e+03
ep_rew_mean	487
exploration rate	0.179
time/	
episodes	11400
fps	225
time_elapsed	3678
total timesteps	829520
train/	
learning_rate	0.0001
loss	0.0404
n_updates	201129

EP=3,880	STEP=831,949	clip=58	consec=26	max_consec=33	avg_clip50=49.40
EP=3,890	STEP=835,054	clip=53	consec=36	max_consec=36	avg_clip50=51.41
EP=3,900	STEP=837,719	clip=52	consec=2	max_consec=43	avg_clip50=48.29

custom/	
avg_clip50	49.9
clipped_episode_r...	71.5
consecutive_clipp...	8
max_consecutive_c...	43
rollout/	
ep_len_mean	1.19e+03
ep_rew_mean	487
exploration rate	0.169
time/	
episodes	11500
fps	225
time_elapsed	3728
total timesteps	839554
train/	
learning_rate	0.0001
loss	0.0221
n_updates	203638

EP=3,910	STEP=840,713	clip=26	consec=12	max_consec=43	avg_clip50=49.46
EP=3,920	STEP=843,538	clip=55	consec=4	max_consec=43	avg_clip50=48.75
EP=3,930	STEP=846,392	clip=70	consec=7	max_consec=43	avg_clip50=48.35
EP=3,940	STEP=849,573	clip=40	consec=17	max_consec=43	avg_clip50=48.14

custom/	
avg_clip50	47.5
clipped_episode_r...	30
consecutive_clipp...	18
max_consecutive_c...	43
rollout/	
ep_len_mean	1.22e+03
ep_rew_mean	496
exploration rate	0.159
time/	
episodes	11600
fps	224
time_elapsed	3777
total timesteps	849888
train/	
learning_rate	0.0001

loss	0.126
n_updates	206221

EP=3,950	STEP=852,611	clip=54	consec=27	max_consec=43	avg_clip50=49.09
EP=3,960	STEP=855,330	clip=22	consec=2	max_consec=43	avg_clip50=47.89
EP=3,970	STEP=858,365	clip=72	consec=8	max_consec=43	avg_clip50=49.76

custom/	
avg_clip50	50.1
clipped_episode_r...	37.5
consecutive_clipp...	12
max_consecutive_c...	43
rollout/	
ep_len_mean	1.21e+03
ep_rew_mean	494
exploration rate	0.149
time/	
episodes	11700
fps	224
time_elapsed	3826
total timesteps	859846
train/	
learning_rate	0.0001
loss	0.224
n_updates	208711

EP=3,980	STEP=861,393	clip=57	consec=18	max_consec=43	avg_clip50=51.79
EP=3,990	STEP=864,686	clip=80	consec=28	max_consec=43	avg_clip50=53.39
EP=4,000	STEP=868,029	clip=56	consec=38	max_consec=43	avg_clip50=56.50
EP=4,010	STEP=871,005	clip=36	consec=48	max_consec=48	avg_clip50=60.19

custom/	
avg_clip50	60.5
clipped_episode_r...	55
consecutive_clipp...	52
max_consecutive_c...	52
rollout/	
ep_len_mean	1.26e+03
ep_rew_mean	539
exploration rate	0.137
time/	
episodes	11800
fps	224
time_elapsed	3885
total timesteps	872107
train/	
learning_rate	0.0001
loss	0.0534
n_updates	211776

EP=4,020	STEP=873,736	clip=44	consec=2	max_consec=55	avg_clip50=56.32
EP=4,030	STEP=876,546	clip=30	consec=8	max_consec=55	avg_clip50=54.75
EP=4,040	STEP=879,652	clip=58	consec=18	max_consec=55	avg_clip50=54.19
EP=4,050	STEP=882,283	clip=51	consec=6	max_consec=55	avg_clip50=49.83

custom/	
avg_clip50	49.8
clipped_episode_r...	51
consecutive_clipp...	6
max_consecutive_c...	55

rollout/	
ep_len_mean	1.23e+03
ep_rew_mean	532
exploration rate	0.126
time/	
episodes	11900
fps	224
time_elapsed	3937
total timesteps	882482
train/	
learning_rate	0.0001
loss	0.0431
n_updates	214370

EP=4,060 | STEP=885,763 | clip=58 | consec=16 | max_consec=55 | avg_clip50=49.61
EP=4,070 | STEP=888,794 | clip=18 | consec=0 | max_consec=55 | avg_clip50=51.89
EP=4,080 | STEP=891,812 | clip=22 | consec=10 | max_consec=55 | avg_clip50=51.13

custom/	
avg_clip50	50.9
clipped_episode_r...	54.5
consecutive_clipp...	14
max_consecutive_c...	55
rollout/	
ep_len_mean	1.26e+03
ep_rew_mean	535
exploration rate	0.116
time/	
episodes	12000
fps	223
time_elapsed	3991
total timesteps	893249
train/	
learning_rate	0.0001
loss	0.309
n_updates	217062

EP=4,090 | STEP=894,918 | clip=61 | consec=20 | max_consec=55 | avg_clip50=51.01
EP=4,100 | STEP=898,263 | clip=40 | consec=1 | max_consec=55 | avg_clip50=53.68
EP=4,110 | STEP=901,892 | clip=58 | consec=11 | max_consec=55 | avg_clip50=54.03
EP=4,120 | STEP=905,597 | clip=76 | consec=21 | max_consec=55 | avg_clip50=57.08

custom/	
avg_clip50	57.1
clipped_episode_r...	75.5
consecutive_clipp...	21
max_consecutive_c...	55
rollout/	
ep_len_mean	1.32e+03
ep_rew_mean	545
exploration rate	0.103
time/	
episodes	12100
fps	223
time_elapsed	4052
total timesteps	905715
train/	
learning_rate	0.0001
loss	0.0468
n_updates	220178

EP=4,130 | STEP=908,905 | clip=34 | consec=31 | max_consec=55 | avg_clip50=60.16
EP=4,140 | STEP=911,825 | clip=34 | consec=41 | max_consec=55 | avg_clip50=59.03
EP=4,150 | STEP=914,611 | clip=52 | consec=6 | max_consec=55 | avg_clip50=56.70

custom/	
avg_clip50	55.7
clipped_episode_r...	52
consecutive_clipp...	11
max_consecutive_c...	55
rollout/	
ep_len_mean	1.32e+03
ep_rew_mean	547
exploration rate	0.0931
time/	
episodes	12200
fps	223
time_elapsed	4103
total timesteps	916044
train/	
learning_rate	0.0001
loss	0.11
n_updates	222760

EP=4,160 | STEP=917,678 | clip=74 | consec=16 | max_consec=55 | avg_clip50=55.54
EP=4,170 | STEP=921,189 | clip=47 | consec=7 | max_consec=55 | avg_clip50=52.90
EP=4,180 | STEP=924,192 | clip=78 | consec=7 | max_consec=55 | avg_clip50=51.11

custom/	
avg_clip50	53.4
clipped_episode_r...	86
consecutive_clipp...	16
max_consecutive_c...	55
rollout/	
ep_len_mean	1.37e+03
ep_rew_mean	565
exploration rate	0.0814
time/	
episodes	12300
fps	222
time_elapsed	4162
total timesteps	927850
train/	
learning_rate	0.0001
loss	0.0356
n_updates	225712

EP=4,190 | STEP=927,995 | clip=78 | consec=17 | max_consec=55 | avg_clip50=54.26
EP=4,200 | STEP=931,209 | clip=42 | consec=5 | max_consec=55 | avg_clip50=56.55
EP=4,210 | STEP=934,620 | clip=78 | consec=15 | max_consec=55 | avg_clip50=57.26
EP=4,220 | STEP=937,441 | clip=26 | consec=25 | max_consec=55 | avg_clip50=55.76

custom/	
avg_clip50	59
clipped_episode_r...	60.5
consecutive_clipp...	31
max_consecutive_c...	55
rollout/	
ep_len_mean	1.33e+03
ep_rew_mean	551

exploration rate	0.0696
time/	
episodes	12400
fps	222
time_elapsed	4221
total timesteps	939764
train/	
learning_rate	0.0001
loss	0.557
n_updates	228690

EP=4,230 | STEP=940,575 | clip=58 | consec=35 | max_consec=55 | avg_clip50=57.30
EP=4,240 | STEP=943,393 | clip=42 | consec=45 | max_consec=55 | avg_clip50=53.42
EP=4,250 | STEP=946,620 | clip=57 | consec=2 | max_consec=55 | avg_clip50=54.44
EP=4,260 | STEP=949,890 | clip=58 | consec=12 | max_consec=55 | avg_clip50=56.47

custom/	
avg_clip50	55.9
clipped_episode_r...	29
consecutive_clipp...	13
max_consecutive_c...	55
rollout/	
ep_len_mean	1.32e+03
ep_rew_mean	564
exploration rate	0.0592
time/	
episodes	12500
fps	222
time_elapsed	4273
total timesteps	950335
train/	
learning_rate	0.0001
loss	0.0546
n_updates	231333

EP=4,270 | STEP=952,575 | clip=57 | consec=22 | max_consec=55 | avg_clip50=57.57
EP=4,280 | STEP=955,204 | clip=56 | consec=8 | max_consec=55 | avg_clip50=55.95
EP=4,290 | STEP=958,042 | clip=54 | consec=18 | max_consec=55 | avg_clip50=57.35

custom/	
avg_clip50	55.4
clipped_episode_r...	60
consecutive_clipp...	24
max_consecutive_c...	55
rollout/	
ep_len_mean	1.24e+03
ep_rew_mean	558
exploration rate	0.0497
time/	
episodes	12600
fps	222
time_elapsed	4321
total timesteps	959943
train/	
learning_rate	0.0001
loss	0.0373
n_updates	233735

EP=4,300 | STEP=960,984 | clip=52 | consec=28 | max_consec=55 | avg_clip50=56.94
EP=4,310 | STEP=964,053 | clip=49 | consec=38 | max_consec=55 | avg_clip50=55.47

EP=4,320 | STEP=967,213 | clip=49 | consec=48 | max_consec=55 | avg_clip50=56.34
EP=4,330 | STEP=969,855 | clip=34 | consec=58 | max_consec=58 | avg_clip50=55.68

custom/	
avg_clip50	56.9
clipped_episode_r...	54.5
consecutive_clipp...	60
max_consecutive_c...	60
rollout/	
ep_len_mean	1.22e+03
ep_rew_mean	558
exploration rate	0.039
time/	
episodes	12700
fps	221
time_elapsed	4375
total timesteps	970664
train/	
learning_rate	0.0001
loss	0.0309
n_updates	236415

EP=4,340 | STEP=973,404 | clip=58 | consec=68 | max_consec=68 | avg_clip50=57.28
EP=4,350 | STEP=976,161 | clip=52 | consec=78 | max_consec=78 | avg_clip50=57.32
EP=4,360 | STEP=979,197 | clip=54 | consec=88 | max_consec=88 | avg_clip50=56.60
EP=4,370 | STEP=982,292 | clip=60 | consec=7 | max_consec=90 | avg_clip50=55.41

custom/	
avg_clip50	55.7
clipped_episode_r...	44
consecutive_clipp...	8
max_consecutive_c...	90
rollout/	
ep_len_mean	1.23e+03
ep_rew_mean	557
exploration rate	0.0272
time/	
episodes	12800
fps	221
time_elapsed	4435
total timesteps	982612
train/	
learning_rate	0.0001
loss	0.12
n_updates	239402

EP=4,380 | STEP=985,062 | clip=62 | consec=6 | max_consec=90 | avg_clip50=56.46
EP=4,390 | STEP=988,243 | clip=52 | consec=16 | max_consec=90 | avg_clip50=55.21
EP=4,400 | STEP=991,256 | clip=78 | consec=26 | max_consec=90 | avg_clip50=54.12

custom/	
avg_clip50	52.8
clipped_episode_r...	31
consecutive_clipp...	31
max_consecutive_c...	90
rollout/	
ep_len_mean	1.25e+03
ep_rew_mean	547
exploration rate	0.017
time/	

episodes	12900
fps	221
time_elapsed	4488
total timesteps	992883
train/	
learning_rate	0.0001
loss	0.023
n_updates	241970

EP=4,410 | STEP=994,287 | clip=60 | consec=36 | max_consec=90 | avg_clip50=53.26
EP=4,420 | STEP=997,761 | clip=76 | consec=46 | max_consec=90 | avg_clip50=54.85
EP=4,430 | STEP=1,000,682 | clip=80 | consec=56 | max_consec=90 | avg_clip50=56.15
EP=4,440 | STEP=1,003,729 | clip=57 | consec=66 | max_consec=90 | avg_clip50=55.60

custom/	
avg_clip50	55.5
clipped_episode_r...	60
consecutive_clipp...	69
max_consecutive_c...	90
rollout/	
ep_len_mean	1.26e+03
ep_rew_mean	550
exploration rate	0.01
time/	
episodes	13000
fps	220
time_elapsed	4549
total timesteps	1004500
train/	
learning_rate	0.0001
loss	0.0728
n_updates	244874

EP=4,450 | STEP=1,006,546 | clip=49 | consec=76 | max_consec=90 | avg_clip50=54.54
EP=4,460 | STEP=1,009,754 | clip=66 | consec=86 | max_consec=90 | avg_clip50=55.91
EP=4,470 | STEP=1,012,849 | clip=50 | consec=96 | max_consec=96 | avg_clip50=56.52

Objetivo conseguido: 100 episodios consecutivos superando el umbral.
Entrenamiento finalizado.
max_consec alcanzado: 100
Log JSON: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.json
Log CSV: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.csv
Checkpoint PyTorch guardado en: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_tuned/dqn_spaceinvaders_v0_policy_latest.pt
model.save() falló (no es crítico, se usa el .pt): IndexError('tuple index out of range')

Test uniforme de todas las variantes

Mismo protocolo que el test del modelo base: **120 episodios con política determinista** sobre el entorno de evaluación **sin clipping** de recompensa (se reporta la puntuación real del juego). Los resultados se guardan en `test_results.json` dentro de la carpeta

de cada variante, de modo que esta celda puede re-ejecutarse sin repetir tests ya realizados.

```
In [ ]: # =====
# M8. Test (120 episodios deterministas) de cada variante
# =====

TEST_EPISODES_VARIANTS = 120

def evaluate_variant(run_name, n_episodes=TEST_EPISODES_VARIANTS):
    paths = RUN_PATHS[run_name]
    eval_env = create_eval_env()
    model = build_model(run_name, eval_env, for_eval=True)

    checkpoint = th.load(paths['policy'], map_location=model.device)
    model.policy.load_state_dict(checkpoint['policy_state_dict'])

    rewards = []
    for ep in range(n_episodes):
        obs = eval_env.reset()
        done = False
        total_reward = 0.0
        while not done:
            action, _ = model.predict(obs, deterministic=True)
            obs, reward, done, infos = eval_env.step(action)
            total_reward += float(reward[0])
            done = bool(done[0])
        rewards.append(total_reward)
        if (ep + 1) % 20 == 0:
            print(f' {run_name}: episodio {ep + 1}/{n_episodes} | media parcial

    eval_env.close()
    return rewards

test_results = {}

for run_name in RUN_NAMES:
    paths = RUN_PATHS[run_name]

    if paths['test_json'].exists():
        with open(paths['test_json'], 'r') as f:
            test_results[run_name] = json.load(f)
        print(run_name, '-> resultados de test ya guardados, se reutilizan.')
        continue

    if not paths['policy'].exists():
        print(run_name, '-> sin checkpoint entrenado todavía, se omite.')
        continue

    print('Testeando variante:', RUN_LABELS[run_name])
    rewards = evaluate_variant(run_name)
    test_results[run_name] = {
        'rewards': [float(r) for r in rewards],
        'mean': float(np.mean(rewards)),
        'std': float(np.std(rewards)),
        'max': float(np.max(rewards)),
        'min': float(np.min(rewards)),
    }
```

```

with open(paths['test_json'], 'w') as f:
    json.dump(test_results[run_name], f, indent=2)

print()
for run_name, res in test_results.items():
    print(f"{RUN_LABELS[run_name]:28s} | media={res['mean']:7.1f} | std={res['st

```

Testeando variante: DQN base

Using cuda device

```

base: episodio 20/120 | media parcial: 629.0
base: episodio 40/120 | media parcial: 641.4
base: episodio 60/120 | media parcial: 656.5
base: episodio 80/120 | media parcial: 660.5
base: episodio 100/120 | media parcial: 648.1
base: episodio 120/120 | media parcial: 650.6

```

Testeando variante: Double DQN

Using cuda device

```

double: episodio 20/120 | media parcial: 618.2
double: episodio 40/120 | media parcial: 590.2
double: episodio 60/120 | media parcial: 606.6
double: episodio 80/120 | media parcial: 605.6
double: episodio 100/120 | media parcial: 600.1
double: episodio 120/120 | media parcial: 598.4

```

Testeando variante: Dueling DQN

Using cuda device

```

dueling: episodio 20/120 | media parcial: 576.2
dueling: episodio 40/120 | media parcial: 603.0
dueling: episodio 60/120 | media parcial: 592.1
dueling: episodio 80/120 | media parcial: 612.8
dueling: episodio 100/120 | media parcial: 614.2
dueling: episodio 120/120 | media parcial: 622.0

```

Testeando variante: DQN exploración ajustada

Using cuda device

```

tuned: episodio 20/120 | media parcial: 618.2
tuned: episodio 40/120 | media parcial: 610.4
tuned: episodio 60/120 | media parcial: 616.3
tuned: episodio 80/120 | media parcial: 607.5
tuned: episodio 100/120 | media parcial: 600.2
tuned: episodio 120/120 | media parcial: 591.4

```

DQN base	media= 650.6	std= 182.0	max= 1295
Double DQN	media= 598.4	std= 147.3	max= 1280
Dueling DQN	media= 622.0	std= 171.0	max= 1430
DQN exploración ajustada	media= 591.4	std= 180.7	max= 1400

Gráficas comparativas de las propuestas

Se generan cuatro gráficas comparando las variantes: **(1)** curvas de aprendizaje (media móvil de 50 episodios), **(2)** progreso hacia el criterio de éxito, **(3)** schedules de exploración y **(4)** recompensa media en test; más una tabla resumen con métricas de eficiencia de muestras (steps necesarios para alcanzar niveles de rendimiento).

```

In [ ]: # =====
# M9. Comparativa: curvas, test y tabla resumen
# =====

if 'test_results' not in globals():
    test_results = {}

```

```

dfs = {}
for run_name in RUN_NAMES:
    csv_path = RUN_PATHS[run_name]['csv']
    if csv_path.exists():
        dfs[run_name] = pd.read_csv(csv_path)
    else:
        print('Sin log de entrenamiento todavía:', run_name)

# --- Gráfica 1: curvas de aprendizaje ---
plt.figure(figsize=(11, 5))
for run_name, df_run in dfs.items():
    plt.plot(df_run['total_steps'], df_run['avg_clip50'], label=RUN_LABELS[run_n
plt.axhline(CONSECUTIVE_THRESHOLD, linestyle='--', color='gray', label='Umbral (
plt.title('Curva de aprendizaje: recompensa clipped media (últimos 50 episodios)
plt.xlabel('Steps')
plt.ylabel('avg_clip50')
plt.legend()
plt.grid(True)
plt.show()

# --- Gráfica 2: progreso hacia el criterio de éxito ---
plt.figure(figsize=(11, 5))
for run_name, df_run in dfs.items():
    plt.plot(df_run['total_steps'], df_run['max_consecutive_clipped_over_20'], l
plt.axhline(TARGET_CONSECUTIVE_EPISODES, linestyle='--', color='gray', label='Ob
plt.title('Máximo de episodios consecutivos con recompensa clipped > 20')
plt.xlabel('Steps')
plt.ylabel('max_consec')
plt.legend()
plt.grid(True)
plt.show()

# --- Gráfica 3: schedules de exploración ---
plt.figure(figsize=(11, 4))
for run_name, df_run in dfs.items():
    plt.plot(df_run['total_steps'], df_run['exploration_rate'], label=RUN_LABELS
plt.title('Evolución de epsilon por variante')
plt.xlabel('Steps')
plt.ylabel('epsilon')
plt.legend()
plt.grid(True)
plt.show()

# --- Gráfica 4: recompensa media en test ---
if len(test_results) > 0:
    names = [RUN_LABELS[k] for k in test_results]
    means = [test_results[k]['mean'] for k in test_results]
    stds = [test_results[k]['std'] for k in test_results]

    plt.figure(figsize=(9, 5))
    plt.bar(names, means, yerr=stds, capsize=6)
    plt.title('Recompensa media en test (120 episodios, política determinista)')
    plt.ylabel('Recompensa media (puntuación real)')
    plt.xticks(rotation=15)
    plt.grid(True, axis='y')
    plt.show()
else:
    print('Aún no hay resultados de test que graficar.')

```

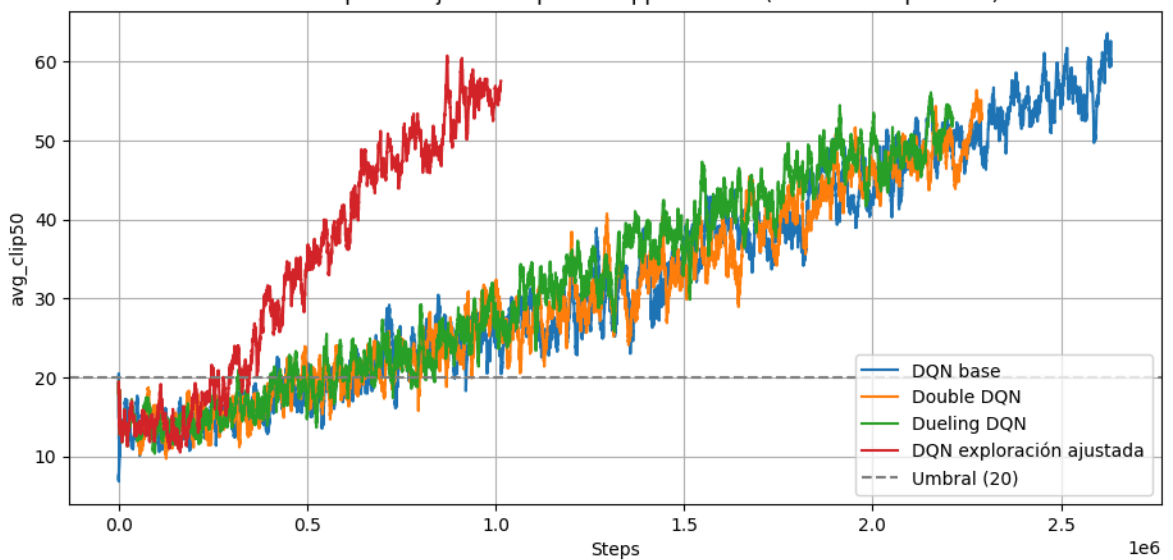


```
# --- Tabla resumen ---
def steps_to_reach(df_run, threshold):
    hits = df_run[df_run['avg_clip50'] >= threshold]
    return int(hits['total_steps'].iloc[0]) if len(hits) else None

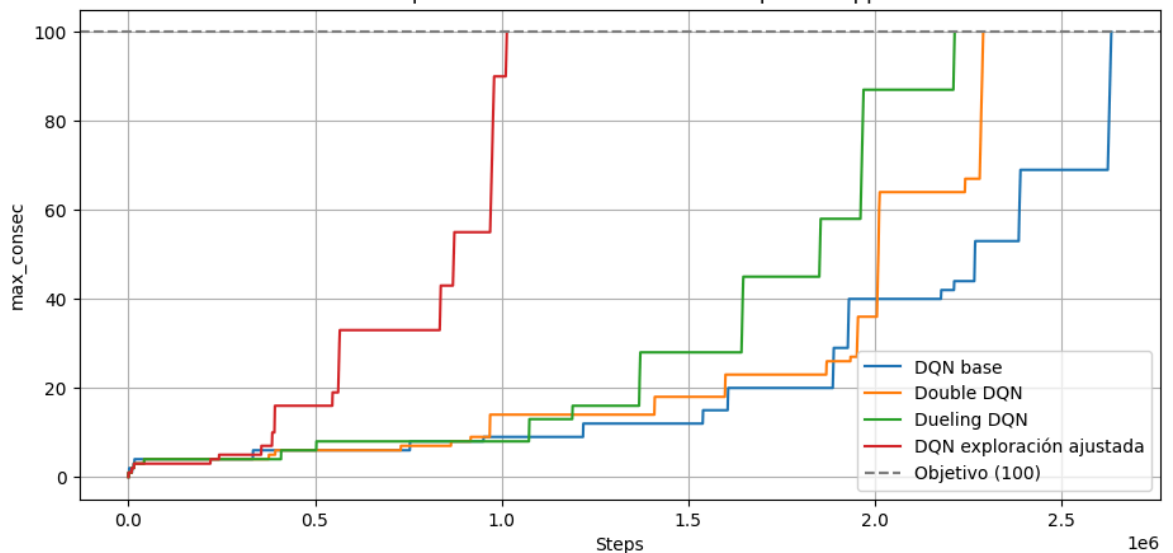
rows = []
for run_name, df_run in dfs.items():
    res = test_results.get(run_name, {})
    rows.append({
        'variante': RUN_LABELS[run_name],
        'steps_totales': int(df_run['total_steps'].iloc[-1]),
        'episodios': len(df_run),
        'steps_hasta_avg20': steps_to_reach(df_run, 20),
        'steps_hasta_avg40': steps_to_reach(df_run, 40),
        'mejor_avg_clip50': float(df_run['avg_clip50'].max()),
        'max_consec': int(df_run['max_consecutive_clipped_over_20'].max()),
        'test_media': res.get('mean'),
        'test_std': res.get('std'),
        'test_max': res.get('max'),
    })

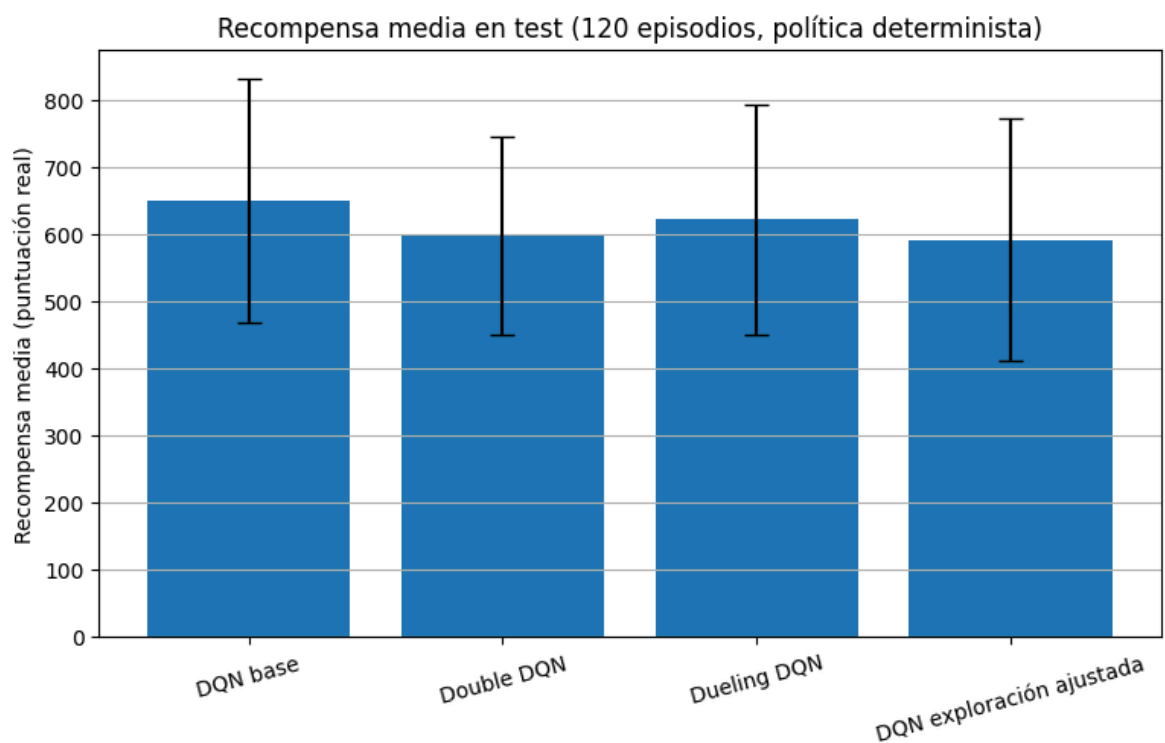
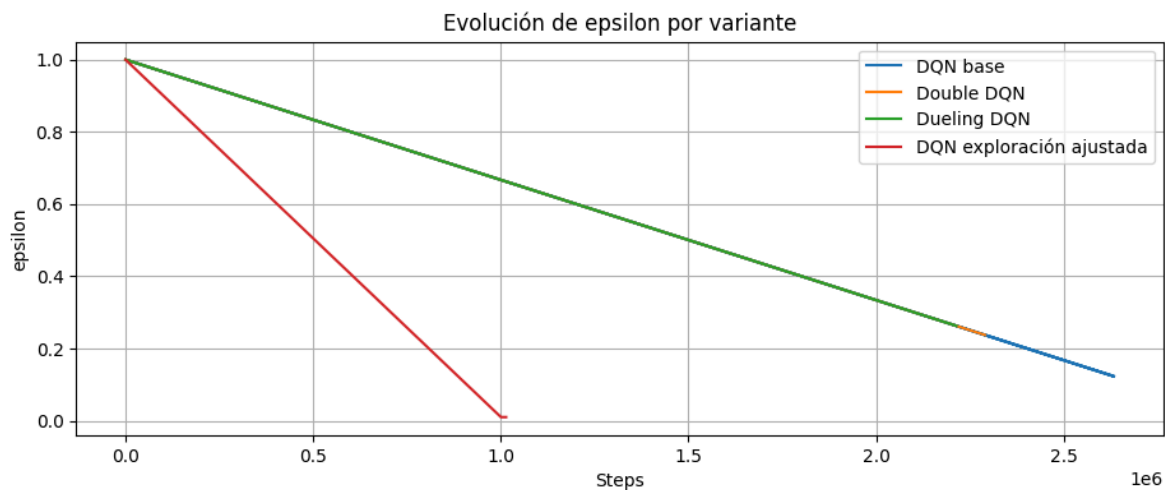
summary = pd.DataFrame(rows)
display(summary)
```

Curva de aprendizaje: recompensa clipped media (últimos 50 episodios)



Máximo de episodios consecutivos con recompensa clipped > 20





	variante	steps_totales	episodios	steps_hasta_avg20	steps_hasta_avg40	mejor_avg_100
0	DQN base	2633424	11961	338	1557345	1770016
1	Double DQN	2290139	10808	411061	1294769	1557345
2	Dueling DQN	2213601	10634	314826	1370016	1557345
3	DQN exploración ajustada	1014048	4475	245690	544860	1557345

PARTE 3. Justificación de los parámetros seleccionados y de los resultados

obtenidos

3.1. Entorno y preprocesado

El preprocesado replica el pipeline clásico de DQN para Atari (Mnih et al., 2015), aplicado automáticamente por `make_atari_env`:

Elemento	Valor	Justificación
Escala de grises + resize	84×84	El color no aporta información decisiva en Space Invaders, y reducir de 210×160×3 a 84×84×1 divide el coste computacional de la red por ~9
Pila de frames	4	Una imagen estática no contiene velocidades ni direcciones (proyectiles, desplazamiento de los invasores); el stack de 4 frames restaura aproximadamente la propiedad de Markov del estado
Frame-skip (<code>MaxAndSkipEnv</code>)	4	Repite cada acción durante 4 frames y toma el máximo píxel a píxel de los 2 últimos (elimina el parpadeo de sprites típico de Atari); acelera la simulación ~4× sin pérdida de control efectivo
Clipping de recompensa	[-1, 1] en entrenamiento	Homogeneiza la escala de los targets y estabiliza el gradiente (técnica estándar del paper de Nature)
<code>FireResetEnv</code>	—	Ejecuta FIRE automáticamente al inicio del episodio, necesario para que la partida arranque

Aclaración sobre la métrica `clip` de los logs: es `recompensa_real / 10`, una convención interna del proyecto para seguir el entrenamiento con números comparables a la práctica anterior; **no** es el clipping ± 1 de Atari (que sí se aplica internamente durante el entrenamiento). En **test** se usa el entorno sin clipping y se reporta la **puntuación real** del juego, que es lo que exige el enunciado.

3.2. Red neuronal

Arquitectura convolucional del DQN de Nature (Mnih et al., 2015), implementada como `features_extractor` de SB3 (`CustomCNN84`) con `net_arch=[]` para que la única capa posterior al extractor sea la cabeza lineal Q:

Capa	Configuración	Salida	Parámetros
Conv1	32 filtros 8×8, stride 4, ReLU	20×20×32	8.224
Conv2	64 filtros 4×4, stride 2, ReLU	9×9×64	32.832
Conv3	64 filtros 3×3, stride 1, ReLU	7×7×64	36.928
Flatten + Dense	3136 → 512, ReLU	512	1.606.144
Cabeza Q	512 → 6 (una salida por acción)	6	3.078

En total $\approx 1,69$ M de parámetros. Los strides grandes sustituyen al pooling (reducen resolución preservando información espacial aprendible), y la red produce las 6 Q-values en una sola pasada, lo que permite el `argmax` de la política greedy con una única inferencia.

3.3. Hiperparámetros del DQN base

Parámetro	Valor	Justificación
<code>learning_rate</code>	1e-4	Valor de referencia para Adam en DQN-Atari; con lr mayores la pérdida de Huber oscila y desestabiliza los targets
<code>buffer_size</code>	100.000	Compromiso entre diversidad de experiencias y RAM de Colab ($\sim 2,8$ GB en uint8 con frames $84 \times 84 \times 4$)
<code>learning_starts</code>	50.000	Llena el buffer con experiencia (mayoritariamente aleatoria) antes de la primera actualización, evitando sobreajustar a las primeras transiciones
<code>batch_size</code>	32	Valor del paper original: batches pequeños con muchas actualizaciones funcionan bien en DQN
<code>gamma</code>	0,99	Horizonte efectivo de ~ 100 steps, suficiente para conectar causalmente un disparo con su impacto
<code>train_freq</code>	4	Una actualización de gradiente por cada 4 steps de entorno (ratio del paper)
<code>target_update_interval</code>	10.000	La red target congelada estabiliza los targets del TD-error
$\epsilon: 1.0 \rightarrow 0.001$ (lineal, 30 % de 10M)	schedule lineal	Exploración intensa al principio, explotación progresiva
Criterio de parada	100 episodios consecutivos con clip > 20	Exige consistencia: no basta con un episodio bueno aislado

3.4. Resultados del DQN base

- El entrenamiento se detuvo por el **criterio de éxito** tras $\sim 2,63$ M steps y ~ 11.930 episodios: 100 episodios consecutivos con recompensa clipped > 20.
- Mejor media móvil `avg_clip50` = **63,6** (equivalente a ~ 636 puntos reales); ϵ final $\approx 0,123$.
- Test (120 episodios, política determinista): media = 662,25 puntos**, máximo = 1.405. El requisito del listado del grupo —**alcanzar más de 20 puntos con reward clipping durante más de 100 episodios consecutivos**— se cumple por partida doble: durante el entrenamiento el agente encadenó 100 episodios consecutivos con recompensa clipped > 20 (es precisamente el criterio de parada implementado),

y en modo test la media equivale a ~66 puntos clipped, más del triple del umbral. Las cuatro configuraciones de la Parte 2 satisfacen igualmente el requisito: todas alcanzaron los 100 episodios consecutivos y obtienen medias de test de 59-65 puntos clipped.

- La curva de aprendizaje muestra tres fases: (i) hasta ~0,5M steps, mejora lenta mientras domina la exploración; (ii) entre 0,5M y 2M, crecimiento sostenido de `avg_clip50` (de ~10 a ~50) a medida que ϵ decae; (iii) a partir de ~2M, consistencia suficiente para encadenar los 100 episodios del criterio.
- Un detalle relevante: el agente alcanzó el criterio **explorando todavía al 12 %**. La política greedy subyacente era mejor de lo que reflejaban los episodios de entrenamiento, y por eso el test determinista (662 puntos de media) supera a la media móvil de entrenamiento (~600). Esta observación es precisamente la que motiva la Propuesta 3.

3.5. Discusión de las tres propuestas

Propuesta 1 — Double DQN. *Qué esperamos:* al eliminar el sesgo de sobreestimación del `max`, los targets son más realistas y el aprendizaje más estable. En la literatura (van Hasselt et al., 2016) Double DQN iguala o supera a DQN en la gran mayoría de juegos de Atari, con ganancias moderadas en Space Invaders. *Resultado obtenido:* Double DQN alcanzó el criterio de éxito en **2,29M steps frente a los 2,63M del base (-13 %)**, y llegó antes a un nivel de juego alto (`avg_clip50` ≥ 40 en 1,29M steps frente a 1,56M, -17 %). En el test de 120 episodios obtuvo una media de **598,4 puntos con la desviación típica más baja de las cuatro configuraciones (147,3, un 19 % menos dispersión que el base)**. Es decir: aprendió algo más rápido y juega de forma más *consistente*, exactamente el efecto que predice la corrección del sesgo de sobreestimación (van Hasselt et al., 2016). Su media de test queda por debajo de la del base (650,6), pero el base entrenó un 15 % más de steps y la diferencia (~52 puntos) es del orden de 3 errores estándar con una sola semilla, por lo que no puede considerarse concluyente.

Propuesta 2 — Dueling DQN. *Qué esperamos:* mejor eficiencia de muestras, porque cada actualización mejora la estimación de `V(s)` compartida por todas las acciones. Wang et al. (2016) reportan una de las mayores mejoras relativas precisamente en Space Invaders, donde abundan los estados en los que la acción concreta importa poco. *Resultado obtenido:* Dueling fue la más eficiente de las dos mejoras algorítmicas: criterio en **2,21M steps (-16 % vs. base)** y el arranque más rápido de las variantes con schedule estándar (`avg_clip50` ≥ 20 en 315k steps, frente a 411k de Double). En test logró la **segunda mejor media (622,0 $\pm$ 171,0)** y el **mejor episodio individual de todo el estudio (1.430 puntos)**. El resultado es coherente con Wang et al. (2016), que reportan ganancias de eficiencia especialmente grandes en Space Invaders: al aprender `V(s)` compartido entre acciones, cada actualización aprovecha más la experiencia en los (muchos) estados donde la acción concreta importa poco.

Propuesta 3 — Exploración ajustada. *Qué esperamos:* una subida más temprana de la curva de aprendizaje al empezar a explotar antes (el annealing termina en ~1M steps en lugar de seguir al 12 % de aleatoriedad a los 2,6M). Los target updates más frecuentes

(5k) aceleran la propagación de lo aprendido con un leve coste de estabilidad. El riesgo de esta propuesta es converger prematuramente a una política subóptima por falta de exploración tardía. *Resultado obtenido*: el efecto más espectacular del estudio: criterio de éxito en **1,01M steps, 2,6 veces menos que el base**, y nivel alto de juego ($\text{avg_clip50} \geq 40$) en solo 545k steps frente a 1,56M (2,9×). La curva de aprendizaje (gráfica 1 de M9) se despegaba de las demás desde ~250k steps. Como contrapartida, su media de test es la más baja ($591,4 \pm 180,7$), con dos explicaciones honestas: (i) al parar tan pronto acumuló ~246k actualizaciones de gradiente frente a ~645k del base — su política final está menos refinada; y (ii) el criterio de parada no es del todo comparable entre schedules: esta variante encadenó sus 100 episodios jugando con $\epsilon = 0,01$ (sin apenas ruido en contra), mientras que base, Double y Dueling lo lograron explorando aún al 12-26 %, un mérito adicional que el criterio no captura.

3.6. Comparativa global y conclusiones

1. Eficiencia de muestras. Las tres propuestas superan al DQN base: exploración ajustada (1,01M steps hasta el criterio) $\ll$ Dueling (2,21M) $\approx$ Double (2,29M) $<$ base (2,63M). Con el umbral $\text{avg_clip50} \geq 40$, el orden se mantiene (545k / 1,37M / 1,29M / 1,56M). Nota metodológica: el valor `steps_hasta_avg20 = 338` del base es un artefacto de la media móvil de 50 episodios en el arranque (pocos episodios en la ventana), por lo que usamos el umbral 40 como métrica robusta de eficiencia.

2. Rendimiento final. El base obtiene la mejor media de test (650,6), seguido de Dueling (622,0), Double (598,4) y tuned (591,4). Pero estas diferencias deben leerse con cautela: el error estándar de cada media es de ± 13 -17 puntos (std ~147-182 sobre 120 episodios), hay una única semilla, y el diseño experimental iguala la *competencia* alcanzada (todas paran al mismo criterio), no el presupuesto: el base entrenó entre un 15 % y un 160 % más que las demás. La lectura correcta no es "el base es mejor", sino que **las tres mejoras alcanzan un rendimiento final comparable con un coste de entrenamiento sustancialmente menor**. Las cuatro configuraciones superan holgadamente el requisito mínimo del enunciado.

3. Estabilidad. Ninguna variante muestra colapsos ni oscilaciones patológicas en sus curvas. Double DQN es la más consistente en test (std 147,3 frente a 171-182 del resto), en línea con la teoría: targets menos sobreestimados producen una política de calidad más uniforme. Dueling combina consistencia razonable con los mejores episodios pico (máx. 1.430).

4. Coherencia con la literatura. El orden esperado en eficiencia (Dueling $\geq$ Double $\geq$ DQN) se reproduce, incluida la ventaja de Dueling en Space Invaders que reportan Wang et al. (2016). La sorpresa aparente —que ninguna mejora supere al base en media de test— desaparece al considerar el presupuesto: con parada por criterio de competencia, las mejoras convierten su ventaja en *ahorro de cómputo* en lugar de en puntuación final. Un experimento complementario natural sería la comparación a presupuesto fijo (mismos steps para todas), donde la literatura predice que las mejoras superarían también la puntuación del base.

Conclusión. Se entrega como mejor modelo el **DQN base (media de test 650,6 con política determinista)**, si bien el resultado central del estudio es que las tres propuestas alcanzan una competencia equivalente con hasta $2,6\times$ menos interacciones con el entorno — en un problema real, donde la experiencia es cara, la variante de exploración ajustada sería la elección preferible.

3.7. Limitaciones y trabajo futuro

- **Una sola semilla (42) por variante**, por limitación de cómputo: las diferencias pequeñas entre variantes no son estadísticamente concluyentes. Lo riguroso serían 3–5 semillas por variante y reportar media $\pm$ desviación entre semillas.
- Las mejoras se han evaluado **por separado** para aislar su efecto; combinarlas (al estilo Rainbow, Hessel et al., 2018), junto con Prioritized Experience Replay y n-step returns, sería la continuación natural.
- La **parada temprana** al alcanzar el criterio favorece comparar eficiencia de muestras, pero no el rendimiento asintótico: con más presupuesto las conclusiones podrían cambiar.
- El test usa una única configuración (120 episodios deterministas con la misma semilla de entorno); una evaluación más robusta usaría semillas de test distintas de la de entrenamiento.

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